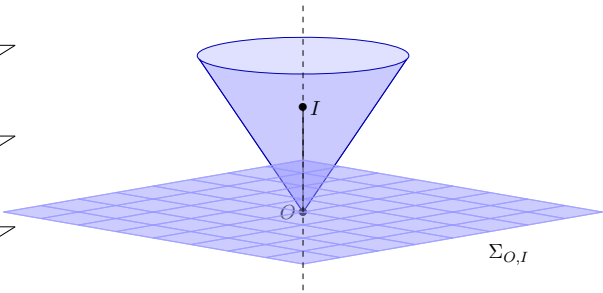
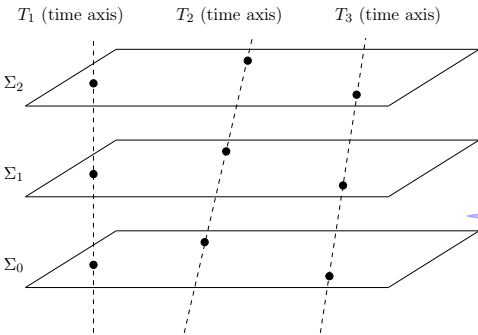
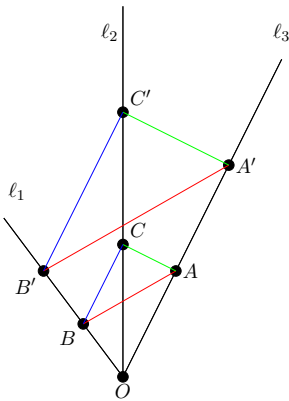
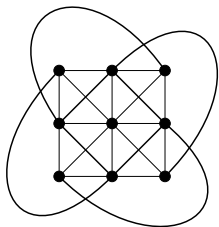


Notes on Synthetic Geometry

Jeffrey Ketland

2026-09-26



Contents

Preface	vii
I Background	1
1 Some Philosophical Background	3
§1.1 Mathematical Platonism and Nominalism	3
§1.2 Problems for Nominalism: Syntax and Applicability	4
§1.3 Field's Programme	4
§1.4 Readings	5
2 Metatheory of Applied Mathematics	7
§2.1 Feynman on Applied Mathematics	7
§2.2 Canonical Form	9
§2.3 The Axioms of AM	10
§2.4 Arithmetic and Analysis in AM	13
3 Conservativeness	17
§3.1 Urelement Arithmetic	17
§3.2 Field's Conservativeness Theorem(s)	18
§3.3 The Cardinality Theorem	21
§3.4 Non-Conservativeness I: Arithmetic	21
4 Analytic Geometry	25
§4.1 Cartesian Coordinates	25
§4.2 The Cartesian Plane	27
§4.3 Coordinate Geometry of the Cartesian Plane	31
5 Geometrical Structures on $\mathbb{R}^n$	33
§5.1 Standard Geometrical Relations on $\mathbb{R}^n$	33
§5.2 Automorphisms and Symmetry	36
6 Synthetic Geometry	41
§6.1 Euclid's Postulates	41

§6.2	Concepts of synthetic geometry	43
§6.3	Axioms	44
§6.4	Analytic Representation	45
II	Affine and Projective Planes	47
7	Incidence Geometry	49
§7.1	Hilbert's Incidence Axioms	49
§7.2	Points, Lines and Incidence	50
§7.3	Incidence Structures	50
8	Affine Planes	53
§8.1	Affine Planes	53
§8.2	Parallelism and Parallelograms	55
§8.3	Parallel Projection	56
§8.4	Affine Planes: Examples	59
§8.5	Affine Plane Combinatorics	61
§8.6	Vectors in Affine Planes	61
9	Projective Planes	63
§9.1	Projective planes	63
§9.2	Projective Planes: Examples	63
§9.3	Correspondence between affine and projective planes	64
§9.4	The Classification Problem	65
10	Collinearity and Parallelity Planes	67
§10.1	Collinearity Planes	67
§10.2	Szmielew's Definition of "Affine Plane"	68
§10.3	Parallelity Planes	68
11	Coordinate Affine and Projective Planes	71
§11.1	Coordinate Planes over Division Rings and Fields	71
12	The Groups Aut, Dil, Tr and Ht	73
§12.1	The Automorphism Group of an Affine Plane	73
§12.2	Dilations, Translations and Homotheties	74
13	Desarguesian and Pappian Planes	77
§13.1	Desargues's Axiom	77
§13.2	Pappus's Axiom	79
§13.3	Pappus Implies Desargues	79
§13.4	Finite Desarguesian Planes are Pappian	79
14	Coordinatization of Affine Planes	81

§14.1	Geometrical Addition and Multiplication	81
§14.2	Line Representation Theorems	82
§14.3	Lines are Isomorphic	83
§14.4	Coordinatization of Affine Planes	83
§14.5	The Unity of Algebra and Geometry	83
15	Coordinatization of Projective Planes	85
III	Betweenness Geometry	87
16	Betweenness	89
§16.1	Pasch and Hilbert	89
§16.2	Betweenness Structures	92
§16.3	Betweenness and Order	96
§16.4	Ordered Affine Planes	97
§16.5	Adding Geometrical Axioms	98
17	Axiomatic Betweenness Geometry	99
§17.1	The Language of Axiomatic Betweenness Geometry	99
§17.2	A Simple Conservativeness Lemma	100
§17.3	The Axiom System Bet	101
§17.4	Coplanarity, Cohyperplanarity,	103
§17.5	n -Frames	104
§17.6	Dimension Axioms in $L(\mathbf{B})$	105
§17.7	Szmielew's and Huntington's Axioms for One-Dimensional Betweenness	105
§17.8	Linear Order From Bet	107
18	Pasch's Axiom	109
§18.1	Pasch's Axiom in $L(\mathbf{B})$	109
§18.2	Dense Linear Order	112
19	Desargues's Axiom	115
§19.1	Desargues Axiom in $L(\mathbf{B})$	115
20	Euclid's Axiom and Playfair's Axiom	117
§20.1	Euclid in $L(\mathbf{B})$	117
21	Pappus's Axioms	119
§21.1	Pappus's Axiom in $L(\mathbf{B})$	119
22	The Continuity Axiom(s)	121
§22.1	Continuity	121
23	The Axiom System $\text{BG}_0(n)$	123

§23.1	The Axioms of $\text{BG}_0(n)$	123
§23.2	The Axioms of $\text{BG}(n)$	124
§23.3	Internal Point Theorem	125
§23.4	Core Theorems of $\text{BG}_0(n)$	126
§23.5	The Soundness Theorem for $\text{BG}_0(n)$	126
24	Proportionality	129
§24.1	Theory of Ratio Configurations and Proportion	129
§24.2	Parallel Copying and Proportionateness	130
§24.3	Encoding Geometry into Algebra	133
§24.4	Encoding Algebra into Geometry: The Greek Lemmas	134
§24.5	Line Isomorphism	136
25	Representation Theorems for $\text{BG}_0(n)$ and $\text{BG}(n)$	137
§25.1	Introduction	137
§25.2	Line Representation Theorem for $\text{BG}_0(n)$	137
§25.3	Line Representation Theorem for $\text{BG}(n)$	138
§25.4	Representation Theorem for $\text{BG}_0(n)$	139
§25.5	Representation Theorem for $\text{BG}(n)$	141
26	Abstraction: Reals	143
§26.1	Extension by Abstraction	143
§26.2	Abstraction Axioms for Reals: First-Order Case	144
§26.3	Reals: Second-Order Case	146
27	Abstraction: Vectors	149
§27.1	Abstraction Axioms for Vectors	149
28	The Completeness of Elementary Geometry	153
§28.1	The Cartesian Translation	153
§28.2	The Cartesian Interpretation Theorem	156
29	The Incompleteness of Second-Order Geometry	159
§29.1	PA_2 is Interpretable in $\text{BG}(4)$	159
§29.2	The Incompleteness of $\text{BG}(4)$	160
IV	Euclidean Geometry	163
30	Congruence and Orthogonality: Euclidean Geometry	165
§30.1	Introduction	165
§30.2	Euclid Proposition I	165
§30.3	Euclidean Geometry	166
§30.4	Tarski-Style Axiomatization	166
§30.5	The Second-Order Theory	170

§30.6	Coordinatization	170
V	Spacetime Geometries	173
31	Galilean Spacetime	175
§31.1	Introduction	175
§31.2	Axiomatization	176
§31.3	Coordinatization	176
32	Aristotelian and Newtonian Spacetimes	177
§32.1	Introduction	177
§32.2	Informal Pictures of Aristotelian and Newtonian Spacetime	178
§32.3	Standard Coordinate Structures	179
§32.4	Axiomatizing Aristotelian Spacetime	180
§32.5	Representation Theorem for Aristotelian Spacetime	181
§32.6	Newtonian Spacetime and the Notion of “Rest”	183
§32.7	Representation Theorem for Newtonian Spacetime	183
§32.8	Two Theorems Concerning Newtonian Spacetime	184
33	Minkowski Spacetime	189
34	Six Main Synthetic Geometries	191
§34.1	Definitions	191
§34.2	Soundness	192
§34.3	Representation Theorems for First-Order Geometries	192
§34.4	Representation Theorems for Second-Order Geometries	193
35	Coordinate Systems and Coordination Transformations	195
VI	Extension With Set Theory	197
36	Extension With Set Theory	199
§36.1	Adding Set Theory to Geometry	199
37	Internalization	203
§37.1	Internalization I	203
§37.2	Internalization II	208
38	Conservativeness and Representation	211
§38.1	Conservativeness	211
§38.2	Representation	212
§38.3	Mereological Synthetic Spacetime Theory	213
§38.4	The Conservativeness/Representation Problem	215

§38.5	A No-Go Theorem for Nominalization	217
APPENDICES		217
Appendix A	Basic Implementation	220
Appendix B	Algebra	221
§B.1	Groups, Rings and Fields	221
Appendix C	The Completeness of Elementary Algebra	223
Appendix D	Affine Spaces	225
Appendix E	Coordinate Structures on Real-Closed Fields	227
Appendix F	Geometrical Axioms	229
§F.1	Pasch's Axiom	229
§F.2	Playfair's Axiom (a version of Euclid)	230
§F.3	Desargues's Axiom	230
§F.4	Pappus's Axiom	230
Appendix G	Geometrical Axiom Systems	231
§G.1	Betweenness (Ordered Affine) Geometry	231
§G.2	Euclidean Geometry	232
§G.3	Synthetic Galilean Spacetime Geometry	232
§G.4	Synthetic Minkowski Spacetime Geometry	233
Indexes		235
Index		235
Index		237
Bibliography		239

Preface

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Edinburgh, February 2026

Detailed Table of Contents

Preface	vii
I Background	1
1 Some Philosophical Background	3
§1.1 Mathematical Platonism and Nominalism	3
§1.2 Problems for Nominalism: Syntax and Applicability	4
§1.3 Field's Programme	4
§1.4 Readings	5
2 Metatheory of Applied Mathematics	7
§2.1 Feynman on Applied Mathematics	7
§2.2 Canonical Form	9
§2.3 The Axioms of AM	10
§2.4 Arithmetic and Analysis in AM	13
3 Conservativeness	17
§3.1 Urelement Arithmetic	17
§3.2 Field's Conservativeness Theorem(s)	18
§3.3 The Cardinality Theorem	21
§3.4 Non-Conservativeness I: Arithmetic	21
4 Analytic Geometry	25
§4.1 Cartesian Coordinates	25
§4.2 The Cartesian Plane	27
§4.3 Coordinate Geometry of the Cartesian Plane	31
5 Geometrical Structures on $\mathbb{R}^n$	33
§5.1 Standard Geometrical Relations on $\mathbb{R}^n$	33
§5.2 Automorphisms and Symmetry	36
6 Synthetic Geometry	41
§6.1 Euclid's Postulates	41

§6.2	Concepts of synthetic geometry	43
§6.3	Axioms	44
§6.4	Analytic Representation	45
II Affine and Projective Planes		47
7	Incidence Geometry	49
§7.1	Hilbert's Incidence Axioms	49
§7.2	Points, Lines and Incidence	50
§7.3	Incidence Structures	50
8	Affine Planes	53
§8.1	Affine Planes	53
§8.2	Parallelism and Parallelograms	55
§8.3	Parallel Projection	56
§8.4	Affine Planes: Examples	59
§8.5	Affine Plane Combinatorics	61
§8.6	Vectors in Affine Planes	61
9	Projective Planes	63
§9.1	Projective planes	63
§9.2	Projective Planes: Examples	63
§9.3	Correspondence between affine and projective planes	64
§9.4	The Classification Problem	65
10	Collinearity and Parallelity Planes	67
§10.1	Collinearity Planes	67
§10.2	Szmielew's Definition of "Affine Plane"	68
§10.3	Parallelity Planes	68
11	Coordinate Affine and Projective Planes	71
§11.1	Coordinate Planes over Division Rings and Fields	71
12	The Groups Aut, Dil, Tr and Ht	73
§12.1	The Automorphism Group of an Affine Plane	73
§12.2	Dilations, Translations and Homotheties	74
13	Desarguesian and Pappian Planes	77
§13.1	Desargues's Axiom	77
§13.2	Pappus's Axiom	79
§13.3	Pappus Implies Desargues	79
§13.4	Finite Desarguesian Planes are Pappian	79
14	Coordinatization of Affine Planes	81

§14.1	Geometrical Addition and Multiplication	81
§14.2	Line Representation Theorems	82
§14.3	Lines are Isomorphic	83
§14.4	Coordinatization of Affine Planes	83
§14.5	The Unity of Algebra and Geometry	83
15	Coordinatization of Projective Planes	85
III	Betweenness Geometry	87
16	Betweenness	89
§16.1	Pasch and Hilbert	89
§16.2	Betweenness Structures	92
§16.3	Betweenness and Order	96
§16.4	Ordered Affine Planes	97
§16.5	Adding Geometrical Axioms	98
17	Axiomatic Betweenness Geometry	99
§17.1	The Language of Axiomatic Betweenness Geometry	99
§17.2	A Simple Conservativeness Lemma	100
§17.3	The Axiom System <i>Bet</i>	101
§17.4	Coplanarity, Cohyperplanarity,	103
§17.5	n -Frames	104
§17.6	Dimension Axioms in $L(\mathbf{B})$	105
§17.7	Szmielew's and Huntington's Axioms for One-Dimensional Betweenness	105
§17.8	Linear Order From <i>Bet</i>	107
18	Pasch's Axiom	109
§18.1	Pasch's Axiom in $L(\mathbf{B})$	109
§18.2	Dense Linear Order	112
19	Desargues's Axiom	115
§19.1	Desargues Axiom in $L(\mathbf{B})$	115
20	Euclid's Axiom and Playfair's Axiom	117
§20.1	Euclid in $L(\mathbf{B})$	117
21	Pappus's Axioms	119
§21.1	Pappus's Axiom in $L(\mathbf{B})$	119
22	The Continuity Axiom(s)	121
§22.1	Continuity	121
23	The Axiom System $BG_0(n)$	123

§23.1	The Axioms of $\mathbf{BG}_0(n)$	123
§23.2	The Axioms of $\mathbf{BG}(n)$	124
§23.3	Internal Point Theorem	125
§23.4	Core Theorems of $\mathbf{BG}_0(n)$	126
§23.5	The Soundness Theorem for $\mathbf{BG}_0(n)$	126
24	Proportionality	129
§24.1	Theory of Ratio Configurations and Proportion	129
§24.2	Parallel Copying and Proportionateness	130
§24.3	Encoding Geometry into Algebra	133
§24.4	Encoding Algebra into Geometry: The Greek Lemmas	134
§24.5	Line Isomorphism	136
25	Representation Theorems for $\mathbf{BG}_0(n)$ and $\mathbf{BG}(n)$	137
§25.1	Introduction	137
§25.2	Line Representation Theorem for $\mathbf{BG}_0(n)$	137
§25.3	Line Representation Theorem for $\mathbf{BG}(n)$	138
§25.4	Representation Theorem for $\mathbf{BG}_0(n)$	139
§25.5	Representation Theorem for $\mathbf{BG}(n)$	141
26	Abstraction: Reals	143
§26.1	Extension by Abstraction	143
§26.2	Abstraction Axioms for Reals: First-Order Case	144
§26.3	Reals: Second-Order Case	146
27	Abstraction: Vectors	149
§27.1	Abstraction Axioms for Vectors	149
28	The Completeness of Elementary Geometry	153
§28.1	The Cartesian Translation	153
§28.2	The Cartesian Interpretation Theorem	156
29	The Incompleteness of Second-Order Geometry	159
§29.1	$\mathbf{PA}_2$ is Interpretable in $\mathbf{BG}(4)$	159
§29.2	The Incompleteness of $\mathbf{BG}(4)$	160
IV	Euclidean Geometry	163
30	Congruence and Orthogonality: Euclidean Geometry	165
§30.1	Introduction	165
§30.2	Euclid Proposition I	165
§30.3	Euclidean Geometry	166
§30.4	Tarski-Style Axiomatization	166
§30.5	The Second-Order Theory	170

§30.6	Coordinatization	170
V	Spacetime Geometries	173
31	Galilean Spacetime	175
§31.1	Introduction	175
§31.2	Axiomatization	176
§31.3	Coordinatization	176
32	Aristotelian and Newtonian Spacetimes	177
§32.1	Introduction	177
§32.2	Informal Pictures of Aristotelian and Newtonian Spacetime	178
§32.3	Standard Coordinate Structures	179
§32.4	Axiomatizing Aristotelian Spacetime	180
§32.5	Representation Theorem for Aristotelian Spacetime	181
§32.6	Newtonian Spacetime and the Notion of “Rest”	183
§32.7	Representation Theorem for Newtonian Spacetime	183
§32.8	Two Theorems Concerning Newtonian Spacetime	184
33	Minkowski Spacetime	189
34	Six Main Synthetic Geometries	191
§34.1	Definitions	191
§34.2	Soundness	192
§34.3	Representation Theorems for First-Order Geometries	192
§34.4	Representation Theorems for Second-Order Geometries	193
35	Coordinate Systems and Coordination Transformations	195
VI	Extension With Set Theory	197
36	Extension With Set Theory	199
§36.1	Adding Set Theory to Geometry	199
37	Internalization	203
§37.1	Internalization I	203
§37.2	Internalization II	208
38	Conservativeness and Representation	211
§38.1	Conservativeness	211
§38.2	Representation	212
§38.3	Mereological Synthetic Spacetime Theory	213
§38.4	The Conservativeness/Representation Problem	215

§38.5	A No-Go Theorem for Nominalization	217
APPENDICES		217
Appendix A	Basic Implementation	220
Appendix B	Algebra	221
§B.1	Groups, Rings and Fields	221
Appendix C	The Completeness of Elementary Algebra	223
Appendix D	Affine Spaces	225
Appendix E	Coordinate Structures on Real-Closed Fields	227
Appendix F	Geometrical Axioms	229
§F.1	Pasch's Axiom	229
§F.2	Playfair's Axiom (a version of Euclid)	230
§F.3	Desargues's Axiom	230
§F.4	Pappus's Axiom	230
Appendix G	Geometrical Axiom Systems	231
§G.1	Betweenness (Ordered Affine) Geometry	231
§G.2	Euclidean Geometry	232
§G.3	Synthetic Galilean Spacetime Geometry	232
§G.4	Synthetic Minkowski Spacetime Geometry	233
Indexes		235
Index		235
Index		237
Bibliography		239

Part I

Background

Chapter 1

Some Philosophical Background

1.1 Mathematical Platonism and Nominalism

It is best to let the pre-eminent twentieth century representatives of both mathematical platonism and nominalism define what is meant by these philosophical doctrines.¹ The first is the logician Kurt Gödel, certainly a mathematical platonist, who defined it as follows:²

Mathematical Platonism. [Platonism is] the view that mathematics describes a non-sensual reality, which exists independently both of the acts and [of] the dispositions of the human mind and is only perceived, and probably perceived very incompletely, by the human mind. (Gödel (1951): 323. Page reference to Gödel (1995))

On the other hand, philosopher of mathematics Hartry Field is a mathematical nominalist, and defines it as follows:

Nominalism. Nominalism is the doctrine that there are no abstract entities. The term ‘abstract entity’ may not be entirely clear, but one thing that does seem clear is that such alleged entities as numbers, functions, and sets are abstract—that is, they would be abstract if they existed. In defending nominalism, therefore, I am denying that numbers, functions, sets, or any similar entities exist. (Field (1980): 1)

These definitions are ontological (“what there is”), not epistemological (“how what is claimed to be so is justified”), and do not advert in any way to the *justification* of the relevant claims or doctrine. However, note:

¹ John Burgess has, and with good reason, protested at the modern usage of the term “Platonism”, for being historically inaccurate. I’m happy to Humpty-Dumpty here, and use the term “Platonism”, while admitting its historical inaccuracy. In fact, I would go as far as saying (with James Robert Brown) that Plato discovered or invented Platonism as understood in its modern sense.

² My own view, I should report, is more or less the same as Gödel’s.

Philosophically, the justification for the axiom system of set theory that Zermelo introduced requires a Platonistic account of the nature of mathematics, according to which the objects of mathematics exist in an abstract realm outside of space and time independent of human ideas and constructions; concomitantly, mathematical truths hold whether or not they can be established by human beings. This accords with mathematical practice to the extent that mathematicians believe their work is a matter of discovery and verification, not of invention. (Feferman 2009, “Modernism in Mathematics: Review of *PLATO’S GHOST: The Modernist Transformation of Mathematics*, by Jeremy Gray”, *American Scientist*)

1.2 Problems for Nominalism: Syntax and Applicability

If we set aside all a lot of muddled water, nominalism faces two core problems:

The Syntax Problem for Nominalism (Poincaré, Hilbert, Quine, Church):

Nominalists refer to syntactical objects (sentences, formulas, derivations, theories) and presuppose—as true—a large body of syntactical axioms and facts. Syntactical objects are themselves abstract objects (expressions or finite sequences, of arbitrary length); there are infinitely many of them. And the large body of axioms and facts mentioned presupposes a minimum level, lying somewhere in the foundational hierarchy around the level of PRA or Q.

The Applicability Problem for Nominalism (Frege, Gödel, Quine, Putnam):

Mundane quantitative sentences, like “The height of Nelson’s Column is 52 m”, state that concrete things are related to abstract ones. This is already inconsistent with nominalism. At a more sophisticated level, Maxwell’s equation $\nabla \cdot \mathbf{B} = 0$ implies the existence of a *physical vector field* on spacetime, taking values in a vector space isomorphic to $\mathbb{R}^3$. Reasoning with $\nabla \cdot \mathbf{B} = 0$, and the other Maxwell’s equations, requires a body of mathematics somewhere at the level of ordinary mathematical analysis. If nominalism is true, Maxwell’s equations are false.

1.3 Field’s Programme

Over the years, especially the last half century, many attempts have been developed with the aim of vindicating nominalism. An extremely interesting attempt to vindicate mathematical nominalism was given by Hartry Field in his monograph *Science Without Numbers* (Field (1980)). This approach has sometimes been called *Field’s programme*. Boiled down to its essentials, this programme rests on two main theses:

Conservativeness Thesis: Given the ambient mathematical base theory App.Math for applied mathematics, and a “mathematics-free” theory N restricted to urelements, the mathematicized extension $\text{App.Math} + N$ is *conservative* with respect to mathematics-free sentences. That is, for any such “mathematics-free” sentence φ ,

$$\text{App.Math} + N \vdash \varphi \Rightarrow N \vdash \varphi. \quad (1.3.1)$$

Nominalization Thesis: Mathematically-formulated scientific theories can be “nominalized”. Given a mathematically-formulated theory $\text{App.Math} + P$, there exists a “mathematics-free” theory N such that:

$$\text{App.Math} + P \equiv_{df} \text{App.Math} + N. \quad (1.3.2)$$

In Hartry Field’s Field (1980), the “mathematics-free physical basis” is given by a *synthetic geometry, of points and regions*, called N above. The basic assumptions of geometrical nominalism then boil down to:

- Conservativeness. $\text{App.Math} + N$ conservatively extends N .
- Nominalization. The standard mathematicized *physical theory*, of spacetime points, regions and fields can be “nominalized”, as an equivalent synthetic theory based on N .

Are these claims true?

One of the main aims of these notes is to explain the underlying ideas involved and examine these two claims. In particular, I give a fairly standard formulation of App.Math , which I shall call AM . I shall also set out, in quite a lot of detail, a number of detailed versions of synthetic geometries N —a pair of which will be called $\text{BG}_0(4)$ and $\text{BG}(4)$, and two other more sophisticated pairs, called $\text{Gal}_0(1,3)$ and $\text{Gal}(4)$ (for Galilean spacetime geometry), and $\text{Mink}_0(1,3)$ and $\text{Mink}(4)$ (for Minkowski spacetime geometry). As it turns out, $\text{BG}_0(4)$ and $\text{BG}(4)$ have rather different metamathematical properties. (As do the corresponding pairs too.) And it is the *second-order* synthetic geometry $\text{BG}(4)$ which the *physicist* wants. Or, rather, the physicist who writes down the Standard Model Lagrangian on Minkowski spacetime is presupposing the second-order synthetic geometry $\text{Mink}(4)$, and not its first-order version.

1.4 Readings

Witten: <https://www.youtube.com/watch?v=03isFuQ2q2A>

Tao: <https://www.quantamagazine.org/what-makes-for-good-mathematics-20240201/>

Penrose: <https://www.youtube.com/watch?v=ujvS2K06dg4>

Chapter 2

Metatheory of Applied Mathematics

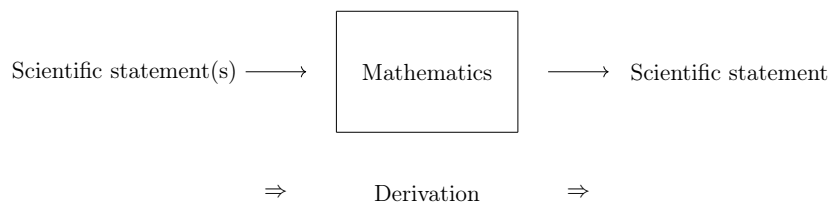
2.1 Feynman on Applied Mathematics

In his famous 1964 Messenger Lectures at Cornell, Richard Feynman stated:

Mathematics then is a way of going from one set of statements to another. It is evidently useful in physics. Because we have all these different ways that we could speak of things, . . . and to connect all the various statements. So that, as a matter of fact, the total amount that the physicist knows is very little. He has only to remember the rules for getting from one place to another. And he is able to do that then. In other words, all the various statements—about equal times, the forces in the direction of the radius and so on—are all *inter-connected by reasoning*. (Feynman (1965), Lecture 2, “The Relation of Mathematics and Physics”)

As Feynman describes it, “*mathematics then is a way of going from one set of statements to another*”, and these statements of physics are “all *inter-connected by reasoning*”. Let’s depict this schematically:

Figure 2.1: **Applied mathematics (Feynman)**

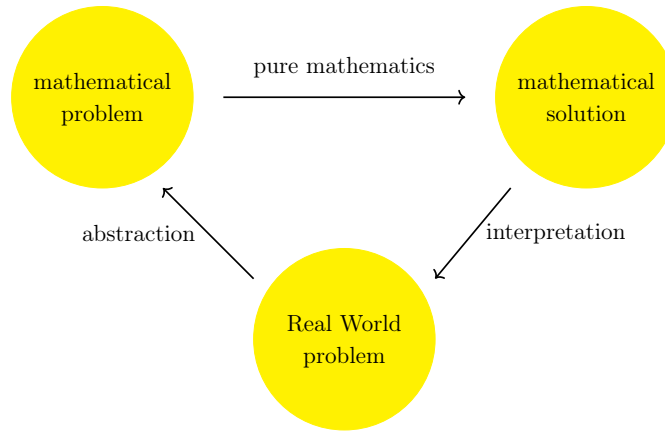


A related analysis is sketched by the mathematical logician Herbert Enderton in his classic textbook Enderton (1977) on set theory:

100 Mathematical concepts are useful in solving problems from the Real World to the extent that the concepts accurately reflect the essential features of those problems. The process of solving a problem mathematical has three parts (Fig 29). We begin with a Real World problem. Then we need to model the Real World problem by a mathematical problem. This typically involves simplifying or idealizing some aspects of the original problem. (For example, we might decide to ignore air resistance or friction.) The middle step in the process consists of finding a mathematical solution to the mathematical problem. The final step is to interpret the mathematical solution in terms of the original problem. The middle step in this process is called “pure mathematics”, and the whole process is called “applied mathematics”. (Enderton (1977): 126)

Enderton’s Fig 29 on page 127 is this:¹

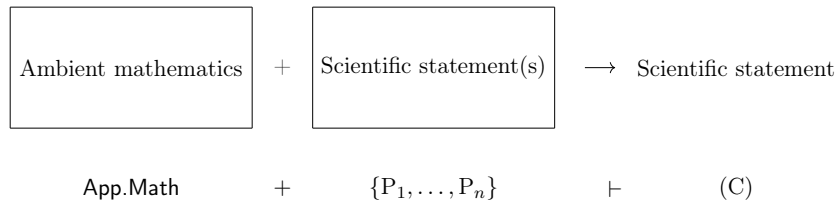
¹ In his Fig 29, the “abstraction” arrow seems to be inadvertently flipped.

Figure 2.2: **Applied mathematics (Enderton)**


These are, of course, rather intuitive or informal sketches by Feynman and Enderton. Nonetheless, I am certain that the gist of what they assert will be accepted by every professional scientist, physicist, mathematician, engineer and so on in the world.

2.2 Canonical Form

On Feynman's account, applied mathematics provides a kind of inferential machine, that converts scientific statements to scientific statements. Before continuing, I wish to render this in a more logically-oriented manner:

Figure 2.3: **Canonical Form (Reasoning in Applied mathematics)**


To give some very simple examples:

0.510 (P1)	There are 100 food bowls.
(P2)	There are 101 dalmatians.
(P3)	Each dalmatian uses exactly one food bowl.
(C)	At least two dalmatians use the same food bowl.

0.510 (P1)	There are 23 strawberries.
(P2)	There are 3 children.
(P3)	The strawberries are partitioned amongst the children.
(C)	At least two children receive different numbers of strawberries.
0.510 (P1)	Bill's age is twice Alfie's.
(P2)	Charlie's age is three times Alfie's.
(P3)	The ages of Alfie, Bill and Charlie sum to 12 years.
(C)	Alfie is 2 years old.

Each of these is an *applied mathematics inference*, and fits precisely the pattern given by Feynman. These inferences are not valid in *predicate logic* alone. They require an ambient applied mathematics base theory, which I'll call **App.Math**. When this is explicitly included, these inferences become *deductively valid* inferences. The general pattern is what I call “the canonical form” of the application:

Figure 2.4: Canonical form

$$\overbrace{\text{App.Math}}^{\text{applied base}} + \overbrace{\text{MA}}^{\text{modelling assumptions}} \vdash \overbrace{\varphi}^{\text{scientific conclusion}} \quad (2.2.1)$$

Below, in §2.3, we define **App.Math** more precisely.

2.3 The Axioms of AM

In constructing an axiomatic theory T , we usually make use of other axiomatic theories which are *presupposed* in the following sense: all the primitive notions in the presupposed theory are included in the system of primitive notions of T , and all the axioms of those theories are included in the axiom system of T . Mathematical theories presuppose as a rule mathematical logic and usually also set theory (to a larger or smaller extent). In developing geometry in this book we presuppose mathematical logic, set theory and the arithmetic of the real numbers (which can either be treated as an independent theory or can be constructed as a portion of set theory). An axiomatic treatment of these theories can be found in various special works. (Borsuk & Szmielew (1960): 6–7)

We shall build an ambient theory based on set (class) theory.

Definition 2.3.1 (Sorts and sort variables). The system of sorts is

$$\text{SORTS} := \{\text{atom}, \text{class}, \text{global}\}. \quad (2.3.1)$$

We assume **atom** and **class** are disjoint, and that both are subsorts of **global**.

Table 2.1: Sort variables

14 Sort	Sort variables
atom	$a, a_1, a_2, \dots$
class	$X, Y, Z, X_1, X_2, \dots$
global	$x, y, z, x_1, x_2, \dots$

We impose *no* sort restrictions on atomic sentences. Hence $a = X$, $X = x$, etc., are well-formed. Also, $a \in a$, $X \in a$, etc., are well-formed.

Definition 2.3.2. An *application signature*

$$\sigma = \overbrace{\{c_i\}}^{\text{constants}} \cup \overbrace{\{P_i\}}^{\text{predicates}} \cup \overbrace{\{F_i\}}^{\text{function symbols}} \quad (2.3.2)$$

over the sorts $\{\text{atom}, \text{class}, \text{global}\}$ also specifies a *sort declaration* for each primitive symbol. This is best explained with examples:

Table 2.2: Sort declarations (examples)

46 Primitive	Sort declaration
Yoko	$\text{yoko} :: \text{atom}$
Mother	$\text{mother} :: \text{atom} \Rightarrow \text{atom}$
The betweenness relation	$\text{B} :: \text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{bool}$
The length in cm function	$\text{len}_{\text{cm}} :: \text{atom} \Rightarrow \text{class}$
The electric field function	$\text{E} :: \text{atom} \Rightarrow \text{class}$

For each primitive symbol, the sort declaration determines a *sort axiom*. Again, this is best explained by examples:

Table 2.3: Sort axioms (examples)

47 Primitive	Sort axiom
Yoko	$\text{atom}(\text{yoko})$
Mother	$y = \text{mother}(x) \rightarrow (\text{atom}(x) \wedge \text{atom}(y)).$
Betweenness relation B	$\text{B}(x, y, z) \rightarrow (\text{atom}(x) \wedge \text{atom}(y) \wedge \text{atom}(z))$
The length in cm function	$y = \text{len}_{\text{cm}}(x) \rightarrow (\text{atom}(x) \wedge \text{class}(y))$
The electric field function	$y = \text{E}(x) \rightarrow (\text{atom}(x) \wedge \text{class}(y))$

Given an application signature, σ , the corresponding first-order language is defined entirely standardly, and denoted $L(\sigma)$. The sorts impose no restriction on the syntax of $L(\sigma)$. The first-order language which results from adding the binary predicate $\in$ is denoted $L_{\in}(\sigma)$.

Definition 2.3.3 (Urelement signature). A signature σ over the sorts is called an *urelement signature* just if, for each constant c in σ , its sort is **atom**, and for each predicate P in σ , its sort is **atom** $\Rightarrow \dots \Rightarrow$ **atom** $\Rightarrow$ **bool**, and for each function symbol F in σ , its sort is **atom** $\times \dots \times$ **atom** $\Rightarrow$ **atom**.

In order to formulate the axioms, we need a series of roughly fifty definitions. These are relegated to Appendix A. However, it is worth drawing attention to the following seven definitions:

Table 2.4: Seven important definitions

35 Definiendum	Definiens
atom (x)	$\exists a (a = x)$.
class (x)	$\exists X (X = x)$.
El (x)	$\exists X (x \in X)$.
set (x)	class (x) $\wedge$ El (x).
$X = \{x \mid \varphi(x)\}$	$\forall x (x \in X \leftrightarrow (\text{El}(x) \wedge \varphi(x)))$.
U	$\{x \mid x = x\}$.
Atom	$\{x \mid \text{atom}(x)\}$.

In addition, to simplify the axioms, we introduce specialized dedicated variables for sets, relations and functions:

Table 2.5: Specialized dedicated variables

35 Quantification	Definiens
$\forall A \varphi(A)$	$\forall X (\text{set}(X) \rightarrow \varphi(X))$.
$\forall R \varphi(R)$	$\forall X (\text{Rel}(X) \rightarrow \varphi(X))$.
$\forall f \varphi(f)$	$\forall X (\text{Fun}(X) \rightarrow \varphi(X))$.

The axioms for ambient applied mathematics are then as follows:

Definition 2.3.4 (Base Theory: AM). Let an application signature $\sigma = \{c_i\} \cup \{P_i\} \cup \{F_i\}$ over the sorts **atom**, **class**, **global** be given. Let $L_{\in}(\sigma)$ be the corresponding three-sorted first-order language (with $\in$). Let S be the sort axioms for σ . The axioms of AM in $L_{\in}(\sigma)$ are the sort axioms S for σ , along with:

Table 2.6: Base theory for applied mathematics: AM

Theory	AM
Partition	$\text{atom}(x) \leftrightarrow \neg \text{class}(x)$.
Atoms	$\text{atom}(x) \rightarrow (\text{empty}(x) \wedge \text{El}(x))$.
Extensionality	$\forall x (x \in X \leftrightarrow x \in Y) \rightarrow X = Y$.
Comprehension	$\exists X \forall x (x \in X \leftrightarrow (\text{El}(x) \wedge \varphi(x)))$.
Atoms set	$\text{Atom} \in U$.
Empty set	$\emptyset \in U$.
Pairing	$x, y \in U \rightarrow \{x, y\} \in U$.
Union	$X \in U \rightarrow \bigcup X \in U$.
Separation	$X \in U \wedge Y \subseteq X \rightarrow Y \in U$.
Power	$X \in U \rightarrow \mathcal{P}(X) \in U$.
Infinity	$(\exists X \in U) \text{Inductive}(X)$.
Choice (global)	$\exists f \text{Choice.fun}(f, X)$.
Replacement	$\text{dom}(f) \in U \rightarrow \text{ran}(f) \in U$.
Foundation	$(\forall A \neq \emptyset) (\exists B \in A) (B \cap A = \emptyset)$.

The axiom scheme Comprehension is also called *Class Comprehension*. A comprehension instance, Comp_φ , is obtained for any $L_{\in(\sigma)}$ -formula $\varphi(x)$, so long as the class variable X is not free in $\varphi(x)$. This formula is called *the defining formula* of the instance of comprehension. This formula may have additional variables free, known as *parameters*; if $\varphi(x)$ does have such parameters, then the instance should universally closed.

These axioms have been chunked into three subgroups.

Definition 2.3.5. AM_0 is the restricted subsystem of AM, containing just Partition, Atoms, Extensionality and Class Comprehension.

The subsystem AM_0 is a fairly weak subsystem, whose consistency is easily provable. It behaves rather like a form of monadic second-order logic wrt atoms (urelements). For mathematics-free sentences, one can show that AM_0 is conservative over predicate logic. However, it exhibits huge speed-up over the subsystem without class axioms.

Definition 2.3.6. AM_p is the restricted subsystem of AM, omitting Replacement and Foundation.

The subsystem AM_p is a powerful subsystem, with a series of dedicated set-existence axioms, and stronger than ZC set theory, and very probably adequate for theoretical physics.

2.4 Arithmetic and Analysis in AM

Although the details are long and messy, we can interpret second-order arithmetic, and second-order real number arithmetic into AM.

2.4.1 Arithmetic

Definition 2.4.1 (First-order Peano arithmetic, PA). Let σ be the one-sorted signature $\{0, S, +, \times\}$, with 0 a constant, S a unary function symbol, and $+$ and $\times$ binary function symbols. Let $L(\sigma)$ (or $L(0, S, +, \times)$) be the usual first-order language over σ . Let PA (i.e., Peano arithmetic) be the axiom system in $L(\sigma)$ consisting of the following axioms. First, the two successor axioms, and the recursion equations for $+$ and $\times$,

Table 2.7: Individual Axioms of PA

$$\begin{array}{ll} \forall n (Sn \neq 0). & \forall n_1 \forall n_2 (Sn_1 = Sn_2 \rightarrow n_1 = n_2). \\ \forall n (n + 0 = n) & \forall n_1 \forall n_2 (n_1 + Sn_2 = S(n_1 + n_2)). \\ \forall n (n \times 0 = 0) & \forall n_1 \forall n_2 (n_1 \times Sn_2 = n_1 \times n_2 + n_1). \end{array}$$

along with all instances Ind_φ of induction:

$$\text{Ind}_\varphi \quad (\varphi(0) \wedge \forall n (\varphi(n) \rightarrow \varphi(Sn))) \rightarrow \forall n \varphi(n) \quad (2.4.1)$$

for any formula $\varphi(n)$. Note that $\varphi(n)$ may hide implicit free variables, called parameters, and then in that case, Ind_φ is the universal closure of (2.4.1). We write:

$$\text{Ind}(L(\sigma)) \quad := \quad \{\text{Ind}_\varphi \mid \varphi \in L(0, S, +, \times)\} \quad (2.4.2)$$

We call the generic scheme (2.4.1) the *Induction scheme*.

Definition 2.4.2 (Second-order Peano arithmetic, PA_2). PA_2 is second-order Peano arithmetic—that is, the theory in the (monadic) second-order language $L_2(\sigma)$, with the axioms of PA in Table 2.7, along with,

$$\text{Comprehension scheme} \quad \exists X \forall n (n \in X \leftrightarrow \varphi(n)) \quad (2.4.3)$$

$$\text{Extensionality} \quad \forall n (n \in X \leftrightarrow n \in Y) \rightarrow X = Y \quad (2.4.4)$$

along with the second-order induction axiom:

$$\text{Induction axiom} \quad (0 \in X \wedge \forall n (n \in X \rightarrow Sn \in X)) \rightarrow \forall n (n \in X) \quad (2.4.5)$$

Definition 2.4.3 (Standard implementation of arithmetic in set theory). $\emptyset$ is defined to be $\{x \mid x \neq x\}$. Sx is defined to be $x \cup \{x\}$. A class is called “inductive” just if it contains $\emptyset$ and closed under S . ω is defined to be the least inductive set. $\mathbb{N}$ is defined to be ω . 0 is defined to be $\emptyset$. Then $+$ and $\times$, binary operations on $\mathbb{N}$, are defined via Dedekind’s Recursion Theorem.

Then, with the above suitable definitions of $\mathbb{N}, 0, \mathbb{S}, +, \times$, we have:

Theorem 2.4.4. $\text{AM} \vdash \text{PA}_2$.

For example,

$$\text{AM} \vdash (\forall x_1 \in \mathbb{N}) (\forall x_2 \in \mathbb{N}) (\mathbb{S}x_1 = \mathbb{S}x_2 \rightarrow x_1 = x_2) \quad (2.4.6)$$

Indeed, we can prove all the higher-order arithmetics PA_n 's in AM.

2.4.2 Analysis

Next, we sketch how (mathematician's) analysis is implemented in AM.

Then, with suitable definitions of $\mathbb{R}, 0, 1, +, \times, \leq$, we have:

Theorem 2.4.5. $\text{AM} \vdash \text{ALG}$.

For example,

$$\text{AM} \vdash (\forall x \in \mathbb{R}) (x \neq 0 \rightarrow (\exists y \in \mathbb{R}) (x \times y = 1)) \quad (2.4.7)$$

Definition 2.4.6. Let $\sigma = \emptyset$, the empty application signature. We define four $L_{\in}(\emptyset)$ -formulas

$$\begin{array}{ll} \text{Ord.Fld}(Z, x, y, f, g, R) & x \in Z \wedge y \in Z \wedge f : Z^2 \rightarrow Z \wedge g : Z^2 \rightarrow Z \wedge R \subseteq Z^2 \\ & \wedge x \neq y \wedge \dots \wedge \forall z_1 \forall z_2 \forall z_3 (R(x, z_1) \rightarrow R(g(z_2, z_1), (g(z_3, z_1)))) \\ \text{UB}(Z, X, R, z_1) & z_1 \in Z \wedge (\forall z_2 \in X) R(z_2, z_1). \\ \text{Ord.Comp}(Z, R) & (\forall X \subseteq Z) (X \neq \emptyset \wedge (\exists z_1 \in X) \text{UB}(Z, X, R, z_1) \rightarrow \exists z_1 \forall z_2 [\text{UB}(Z, X, R, z_2) \rightarrow R(z_1, z_2)]). \\ \text{ALG}(X) & \exists Z \exists x \exists y \exists f \exists g \exists R [X = (Z, x, y, f, g, R) \wedge \text{Ord.Fld}(Z, x, y, f, g, R) \wedge \text{Ord.Comp}(Z, R)]. \end{array}$$

Hence, $\text{ALG}(X)$ says “ X is an ordered field”.

Theorem 2.4.7. $\text{AM} \vdash \text{ALG}(X) \wedge \text{ALG}(Y) \rightarrow (X \cong Y)$.

I said that within AM we can develop all of core mathematics, and, in particular, all the relevant results assumed or developed in theoretical physics. Let's give some examples:

Example 2.4.8. My first example is a standard result in analysis:

$$\text{AM} \vdash (a \in \mathbb{R} \wedge b \in \mathbb{R} \wedge a < b \wedge f \in C[(a, b), \mathbb{R}] \wedge f(a) > 0 \wedge f(b) < 0) \rightarrow (\exists x \in (a, b)) (f(x) = 0) \quad (2.4.8)$$

□

Example 2.4.9. My second example is a standard result in vector analysis:

$$\text{AM} \vdash V \in C^\infty[\mathbb{R}^3, \mathbb{R}^3] \rightarrow (\nabla \cdot (\nabla \times V) = 0) \quad (2.4.9)$$

□

Example 2.4.10. My third example is a standard result in electromagnetic theory:

$$\text{AM} \vdash [\text{EM}(\mathbf{E}, \mathbf{B}, \mathbf{J}, \rho) \wedge \mathbf{J} = 0 \wedge \rho = 0] \rightarrow [(\nabla^2 - \partial_t^2)\mathbf{B} = 0] \quad (2.4.10)$$

□

Chapter 3

Conservativeness

3.1 Urelement Arithmetic

Ordinarily, using a standard set-theoretic (or type-theoretic) foundation like ZFC, we *reduce* $\mathbb{N}$ to a system of sets. However, when we look at this from the point of view of *the theory of application*, that “mixes up” or “entangles” arithmetic with set theory. But we wish to *study* this entangling and see what makes it tick. So, we shall first *disentangle*. We formulate what I call *urelement arithmetic* (or *sui generis arithmetic*).

Let $\sigma = \{\text{atom}, \text{nat}, 0, \mathbf{S}, +, \times\}$ be an urelement signature (see Definition 2.3.3), with nat a unary urelement predicate, 0 a urelement constant, and so on. This is called the first-order signature of “urelement arithmetic”.

We now define PA^{atom} , *first-order urelement Peano arithmetic* in $L(\sigma)$.

Definition 3.1.1 (First-order urelement Peano arithmetic, PA^{atom}). The axioms of PA^{atom} are the following eleven:

Table 3.1: Axioms of PA^{atom}

$\forall a (\text{nat}(a) \rightarrow \text{atom}(a))$	
$\text{nat}(0)$	$\forall a (\text{nat}(a) \leftrightarrow \text{nat}(\mathbf{S}a)).$
$\forall a_1 \forall a_2 ((\text{nat}(a_1) \wedge \text{nat}(a_2)) \leftrightarrow \text{nat}(a_1 + a_2))$	$\forall a_1 \forall a_2 ((\text{nat}(a_1) \wedge \text{nat}(a_2)) \leftrightarrow \text{nat}(a_1 \times a_2)).$
$\forall a (\mathbf{S}a \neq 0).$	$\forall a_1 \forall a_2 (\mathbf{S}a_1 = \mathbf{S}a_2 \rightarrow a_1 = a_2).$
$\forall a (a + 0 = a)$	$\forall a_1 \forall a_2 (a_1 + \mathbf{S}a_2 = \mathbf{S}(a_1 + a_2)).$
$\forall a (a \times 0 = 0)$	$\forall a_1 \forall a_2 (a_1 \times \mathbf{S}a_2 = a_1 \times a_2 + a_1).$

along with the *induction scheme*, the set $\text{Ind}(L(\sigma))$ of induction axioms Ind_φ in $L(\sigma)$. These take the form (of universal closures of):

$$\text{Ind}_\varphi \quad (\varphi(0) \wedge \forall a (\text{nat}(a) \rightarrow (\varphi(a) \rightarrow \varphi(\mathbf{S}a)))) \rightarrow \forall a (\text{nat}(a) \rightarrow \varphi(a)) \quad (3.1.1)$$

where $\varphi(a)$ is a formula of $L(\sigma)$ (possibly with implicit parameters).

We next define $\text{PA}_2^{\text{atom}}$, *second-order urelement Peano arithmetic* in $L_2(\sigma)$.

Definition 3.1.2 (Second-order urelement Peano arithmetic, $\text{PA}_2^{\text{atom}}$). The axioms of $\text{PA}_2^{\text{atom}}$ are the above axioms of PA^{atom} given in Table 3.1, along with Extensionality for sets of “nats” and Comprehension for sets of “nats”,

$$\text{Comprehension scheme} \quad \exists X \forall a (\text{nat}(a) \rightarrow (a \in X \leftrightarrow \varphi(a))) \quad (3.1.2)$$

$$\text{Extensionality} \quad (\forall X, Y \subseteq \mathbb{N}) [\forall a ((a \in X \leftrightarrow a \in Y) \rightarrow X = Y] \quad (3.1.3)$$

and the single *Induction axiom*:

$$(0 \in X \wedge \forall a (\text{nat}(a) \rightarrow (a \in X \rightarrow \text{Sa} \in X))) \rightarrow \forall a (\text{nat}(a) \rightarrow a \in X) \quad (3.1.4)$$

Next, we consider Gödel's incompleteness results. Let $\text{Con}(\text{PA})$ be some standard consistency sentence for PA. Then we have the Second Incompleteness Theorem:

Theorem 3.1.3. $\text{PA} \not\vdash \text{Con}(\text{PA})$.

Since urelement Peano arithmetic PA^{atom} is more or less a notational copy of PA, the same proof establishes a similar result for urelement Peano arithmetic:

Theorem 3.1.4. $\text{PA}^{\text{atom}} \not\vdash \text{Con}(\text{PA}^{\text{atom}})$.

Next, we have the standard result that second-order arithmetic proves the consistency of first-order arithmetic:

Theorem 3.1.5. $\text{PA}_2 \vdash \text{Con}(\text{PA})$.

Proof. Let CT be the standard “compositional truth theory” for Peano arithmetic. By known results, we have $\text{PA}_2 \vdash \text{CT}$ and $\text{CT} \vdash \text{Con}(\text{PA})$. Hence, $\text{PA}_2 \vdash \text{Con}(\text{PA})$. $\square$

Theorem 3.1.6. $\text{PA}_2^{\text{atom}} \vdash \text{Con}(\text{PA}^{\text{atom}})$.

Proof. We transcribe the proof of the above result to $\text{PA}_2^{\text{atom}}$. $\square$

3.2 Field's Conservativeness Theorem(s)

Throughout this section, σ is always an urelement signature (see Definition 2.3.3). A theory in $L(\sigma)$ is then an urelement theory—a mathematics-free theory.

Lemma 3.2.1. Let M be a σ -structure such that $\text{dom}(M)$ is a set of urelements. Then:

- 0.57 (i) There is an expansion $M^+ \models \text{AM}_0$.
- (ii) There is an expansion $M^{++} \models \text{AM}$.

Proof. For (i), let $A = \text{dom}(M)$. So, A is a set of urelements. We now define M^+ as follows.

First, we define the extensions (in M^+) of the sorts **atom**, **class** and **global**, and we define the extension (in M^+) of the membership predicate $\in$:

$$\text{atom}^{M^+} := A \quad (3.2.1)$$

$$\text{class}^{M^+} := \mathcal{P}(A) \quad (3.2.2)$$

$$\text{global}^{M^+} := A \cup \mathcal{P}(A) \quad (3.2.3)$$

$$\in^{M^+} := \{(x, y) \in A \times \mathcal{P}(A) \mid x \in y\} \quad (3.2.4)$$

Since A is a set of urelements, $\text{atom}^{M^+} \cap \text{class}^{M^+} = \emptyset$. And we also have $\text{global}^{M^+} = \text{atom}^{M^+} \cup \text{class}^{M^+}$. It follows from these two facts that **Partition** is true in M^+ . It also follows that **Atoms** is true in M^+ . Because $\text{class}^{M^+} = \mathcal{P}(A)$, it follows that **Extensionality** and **Comprehension** are true in M^+ . Hence, $M^+ \models \text{AM}_0$.

Second, we set, for any constant c , for any predicate P , any function symbol F ,

$$c^{M^+} := c^M \quad (3.2.5)$$

$$P^{M^+} := P^M \quad (3.2.6)$$

$$F^{M^+} := F^M \quad (3.2.7)$$

Thus, M^+ is an *expansion* of M .

For (ii), let $A = \text{dom}(M)$. So, A is a set of urelements. We now define M^{++} as follows, by analogy with (i) above.

First, we define the extensions (in M^+) of the sorts **atom**, **class** and **global**, and we define the extension (in M^+) of the membership predicate $\in$. But here we need to first define $V_{\alpha+1}(A)$, where α is an inaccessible.

$$\text{atom}^{M^+} := A \quad (3.2.8)$$

$$\text{class}^{M^+} := V_{\alpha+1}(A) - A \quad (3.2.9)$$

$$\text{global}^{M^+} := V_{\alpha+1}(A) \quad (3.2.10)$$

$$\in^{M^+} := \{(x, y) \in V_{\alpha+1}(A)^2 \mid x \in y\} \quad (3.2.11)$$

Again, since A is a set of urelements, $\text{atom}^{M^+} \cap \text{class}^{M^+} = \emptyset$. And we also have $\text{global}^{M^+} = \text{atom}^{M^+} \cup \text{class}^{M^+}$. It follows from these facts that **Partition** and **Atoms** are true in M^+ . Because $\text{class}^{M^+} = V_{\alpha+1}(A) - A$ and $\in^{M^+}$ is the *standard* $\in$ relation (restricted to $V_{\alpha+1}(A)$), it follows that **Extensionality** and **Comprehension** are true in M^+ .

The *sets* in the model M^{++} are the classes which are *elements*. In short, $\text{set}^{M^+} = V_{\alpha}(A) - A$. By specialized background considerations known to set theorists, the specific

set theoretical axioms are true in the “standard model”, M^{++} (cf., $(V_\alpha, V_{\alpha+1}, \in)$ is a standard model for Morse-Kelley). Hence, $M^{++} \models \text{AM}$.

Second, we repeat the previous definitions of the denotations for non-logical primitives: for any constant c , for any predicate P , any function symbol F ,

$$c^{M^{++}} := c^M \quad (3.2.12)$$

$$P^{M^{++}} := P^M \quad (3.2.13)$$

$$F^{M^{++}} := F^M \quad (3.2.14)$$

Thus, M^{++} is an *expansion* of M . □

We next assume that there exist at least $\aleph_0$ many urelements.

Theorem 3.2.2 (Field's conservativeness theorem (AM_0)). Let σ be an urelement signature. Let $\varphi \in L(\sigma)$. Then:

$$\text{If } \text{AM}_0 \vdash \varphi \text{ then } \vdash \varphi \quad (3.2.15)$$

Proof. Suppose $\not\vdash \varphi$. By the Completeness Theorem, there exists a countable model $M \models \neg\varphi$, whose domain can be taken to be a set of urelements. By Lemma 3.2.1(i), we have an expansion $M^+ \models \text{AM}_0$. Since M^+ is an expansion, then $M^+ \models \theta$ iff $M \models \theta$, for any $L(\sigma)$ -sentence θ . Thus, $M^+ \models \neg\varphi$. Hence, $\text{AM}_0 \not\vdash \varphi$. Contraposition now gives the claim. □

Theorem 3.2.3 (Field's conservativeness theorem (AM)). Let φ be an $L(\sigma)$ -sentence. Then:

$$\text{If } \text{AM} \vdash \varphi \text{ then } \vdash \varphi \quad (3.2.16)$$

Proof. The proof is the same as the AM_0 case. Suppose $\not\vdash \varphi$. By the Completeness Theorem, we have a countable model $M \models \neg\varphi$ whose domain can be taken to be a set of urelements. By Lemma 3.2.1(ii), we have an expansion $M^{++} \models \text{AM}$ such that $M^{++} \models \neg\varphi$. Hence, $\text{AM} \not\vdash \varphi$. Contraposition now gives the claim. □

Theorem 3.2.4 (Field's conservativeness theorem (AM)). Let Σ be a set of $L(\sigma)$ -sentences. Let φ be an $L(\sigma)$ -sentence. Then:

$$\text{If } \text{AM} + \Sigma \vdash \varphi \text{ then } \Sigma \vdash \varphi \quad (3.2.17)$$

Proof. The proof is a modification of that for Theorem 3.2.3. Suppose $\Sigma \not\vdash \varphi$. By the Completeness Theorem, we have a countable model $M \models \Sigma$ and $M \models \neg\varphi$, whose domain can be taken to be a set of urelements. By Lemma 3.2.1(ii), we have an expansion $M^{++} \models \text{AM}$ such that $M^{++} \models \Sigma$ and $M^{++} \models \neg\varphi$. Hence, $\text{AM} + \Sigma \not\vdash \varphi$. Contraposition now gives the claim. □

3.3 The Cardinality Theorem

In virtue of the sorting, $\{\text{atom}, \text{class}, \text{global}\}$, the applied mathematics base theory AM implies the existence of at least one atom:

Theorem 3.3.1. $\text{AM} \vdash \exists x \text{atom}(x)$.

Proof. $\text{AM} \vdash \exists a (a = a)$. Hence, $\text{AM} \vdash \exists x (x = a)$. Hence, $\text{AM} \vdash \exists x \text{atom}(x)$. $\square$

However, AM does not imply the existence of more than one atom:

Theorem 3.3.2. $\text{AM} \not\vdash \exists_{\geq 2} x \text{atom}(x)$.

Proof. Let a be an urelement. Via Lemma 3.2.1(ii), define a model $M(a)$ where $\text{atom}^{M(a)} = \{a\}$. We have $M(a) \models \text{AM}$ and $M(a) \models \exists_1 x \text{atom}(x)$. Thus, $M(a) \not\models \exists_{\geq 2} x \text{atom}(x)$. From this, we have: $\text{AM} \not\vdash \exists_{\geq 2} x \text{atom}(x)$. $\square$

Theorem 3.3.3 (Cardinality Theorem). Let σ be an urelement signature. Let Σ be a consistent set of $L(\sigma)$ -sentences. Then:

$$\text{AM} + \Sigma \not\vdash (|\text{Atom}| > \aleph_0) \quad (3.3.1)$$

Proof. Since Σ is consistent, let $M \models \Sigma$ be a countable model of Σ with $\text{dom}(M)$ a set of urelements. Construct M^{++} as per Lemma 3.2.1(ii). Then $M^{++} \models (|\text{Atom}| \leq \aleph_0)$. Hence, $M^{++} \models \text{AM} + \Sigma + (|\text{Atom}| \leq \aleph_0)$. Hence, $\text{AM} + \Sigma \not\vdash (|\text{Atom}| > \aleph_0)$. $\square$

In short, if Σ is a consistent mathematics-free theory, then no matter how much mathematics we throw at Σ , we cannot prove that there are more than $\aleph_0$ atoms/urelements (i.e., *non-mathematical* objects).

3.4 Non-Conservativeness I: Arithmetic

We can now apply the results in the previous section to urelement arithmetic, PA^{atom} :

Theorem 3.4.1. $\text{AM} + \text{PA}^{\text{atom}}$ is conservative over PA^{atom} .

Proof. This is immediate from Theorem 3.2.4, Field’s conservativeness theorem, since PA^{atom} is an urelement theory. $\square$

This conclusion is what vindicates, at least in part, Field’s instrumentalist position as regards “adding more mathematics” to a “mathematics-free” or “nominalistic” theory. Entirely analogous results will be given below—Theorem 38.1.1 and Theorem 38.3.1—in connection with geometry.

Next we discuss a kind of *non-conservativeness*. Theorem 3.4.1 states, informally, “adding set theory to urelement arithmetic is conservative”. However, there is a subtlety here as regards the exact specification of the axioms of this extension. Does the induction

scheme of PA^{atom} remain restricted? Or is the induction scheme extended to the richer language, containing $x \in y$? We shall obtain a very different result from Theorem 3.4.1 when the induction scheme of PA^{atom} is extended.

Recall that $\sigma = \{\text{atom}, \text{nat}, 0, S, +, \times\}$ is the first-order signature of “urelement arithmetic” (i.e., σ is an urelement signature (see Definition 2.3.3), with nat a unary urelement predicate, 0 a urelement constant, and so on). $L(\sigma)$ is the first-order language over σ , and $L_{\in}(\sigma)$ is the extended language with the $\in$ predicate. $\text{Ind}(L(\sigma))$ is the set of induction axioms Ind_{φ} in the sublanguage $L(\sigma)$. I.e.,

$$\text{Ind}_{\varphi} \quad (\varphi(0) \wedge \forall a (\text{nat}(a) \rightarrow (\varphi(a) \rightarrow \varphi(Sa)))) \rightarrow \forall a (\text{nat}(a) \rightarrow \varphi(a)) \quad (3.4.1)$$

where $\varphi(a)$ is a formula of $L(\sigma)$

Now let $\text{Ind}(L_{\in}(\sigma))$ be the set of induction axioms in the *extended language* with the $\in$ predicate. For vividness, we write $\text{Ind}(L_{\in}(\text{nat}, 0, S, +, \times))$ instead. We can now show that these additional induction axioms “boost” *first-order* urelement arithmetic PA^{atom} to *second-order* urelement arithmetic PA^{atom} :

Theorem 3.4.2. $\text{AM} + \text{PA}^{\text{atom}} + \text{Ind}(L_{\in}(\text{nat}, 0, S, +, \times)) \vdash \text{PA}_2^{\text{atom}}$.

Proof. We claim that the axioms of $\text{PA}_2^{\text{atom}}$ are provable in $\text{AM} + \text{PA}^{\text{atom}} + \text{Ind}(L_{\in}(\text{nat}, 0, S, +, \times))$. The ambient mathematics AM already provides us with Comprehension and Extensionality for “nat” sets. Next let $\varphi(a)$ be the formula $a \in X$. Then the corresponding induction axiom Ind_{φ} from $\text{Ind}(L_{\in}(\text{nat}, 0, S, +, \times))$ is:

$$(0 \in X \wedge \forall a (\text{nat}(a) \rightarrow (a \in X \rightarrow Sa \in X))) \rightarrow \forall a (\text{nat}(a) \rightarrow a \in X) \quad (3.4.2)$$

This is clearly the Induction axiom of $\text{PA}_2^{\text{atom}}$. Thus, we can prove all the axioms of $\text{PA}_2^{\text{atom}}$. □

By Theorem 3.2.4, $\text{AM} + \text{PA}^{\text{atom}}$ is conservative over PA^{atom} . What of $\text{AM} + \text{PA}^{\text{atom}} + \text{Ind}(L_{\in}(\text{nat}, 0, S, +, \times))$?

Theorem 3.4.3. $\text{AM} + \text{PA}^{\text{atom}} + \text{Ind}(L_{\in}(\text{nat}, 0, S, +, \times))$ is non-conservative over PA^{atom} .

Proof. By Gödel’s Second Incompleteness Theorem (i.e., Theorem 3.1.4), we have:

$$\text{PA}^{\text{atom}} \not\vdash \text{Con}(\text{PA}^{\text{atom}}). \quad (3.4.3)$$

Note that $\text{Con}(\text{PA}^{\text{atom}})$ is an $L(\text{nat}, 0, S, +, \times)$ -sentence—i.e., an *urelement sentence*. By Theorem 3.4.2 above, $\text{AM} + \text{PA}^{\text{atom}} + \text{Ind}(L_{\in}(\text{nat}, 0, S, +, \times)) \vdash \text{PA}_2^{\text{atom}}$. And by Theorem 3.1.6, we have: $\text{PA}_2^{\text{atom}} \vdash \text{Con}(\text{PA}^{\text{atom}})$. So,

$$\text{AM} + \text{PA}^{\text{atom}} + \overbrace{\text{Ind}(L_{\in}(\text{nat}, 0, S, +, \times))}^{\text{Extended induction}} \vdash \text{Con}(\text{PA}^{\text{atom}}). \quad (3.4.4)$$

From (3.4.3) and (3.4.4), we conclude that $\text{AM} + \text{PA}^{\text{atom}} + \text{Ind}(L_{\in}(\text{nat}, 0, S, +, \times))$ is non-conservative over PA^{atom} . $\square$

We can now summarize the results as follows:

Table 3.2: Arithmetical conservativeness results

0.510 (1)	The extension of urelement arithmetic PA^{atom} with AM, but with <i>restricted induction</i> , is a <i>conservative</i> extension.
(2)	The extension of urelement arithmetic PA^{atom} with AM, and with <i>extended induction</i> (i.e., allowing $\in$ in the induction axiom) is a <i>non-conservative</i> extension.

Chapter 4

Analytic Geometry

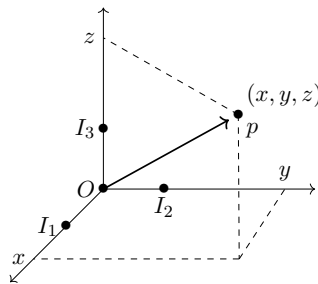
4.1 Cartesian Coordinates

How do *real numbers* get entangled with *physics*? In essentially two ways:

- (Coordinates) Real numbers serve as *coordinates* of points in space (or spacetime).
- (Quantities) Real numbers serve as *values of quantities*: lengths, temperatures, fields, etc.

We focus on the first. The most basic geometrical concept in physics is *the Cartesian coordinate system*, say for three-dimensional space:

Figure 4.1: **Cartesian coordinate system**



A Cartesian coordinate system is specified as follows:

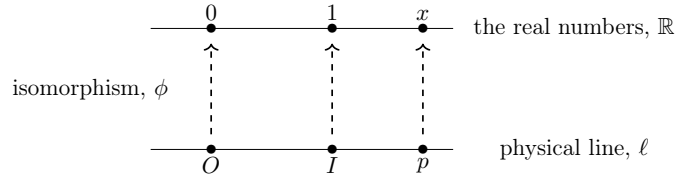
1. One selects an “origin” point O , arbitrarily.
2. One selects three new “unit points” I_1, I_2, I_3 such that the (unit) segments OI_1, OI_2 and OI_3 are equal in length and mutually orthogonal. The lines determined by OI_1, OI_2 and OI_3 are called “coordinate axes”.

3. The four points O, I_1, I_2, I_3 determine a function (a “coordinate system”) Φ , which labels each point p in space with a triple $(x, y, z) \in \mathbb{R}^3$ of real numbers. That is,

$$\Phi(p) = (x, y, z). \quad (4.1.1)$$

But how is this justified? Moreover, if we focus on a single physical line, ℓ , the same process, somehow, shows how each point $p \in \ell$ gets labeled by a real number $x \in \mathbb{R}$.

Figure 4.2: **A physical line ℓ is “isomorphic” to $\mathbb{R}$**



Physicists do, routinely, *identify* the physical line ℓ with $\mathbb{R}$. And physicists also, routinely, *identify* three-dimensional physical space with $\mathbb{R}^3$:

Throughout these lectures we will deal only with the dynamics of a single particle. For the present discussion, we’ll think about a particle moving in $\mathbb{R}^3$. (Tong, Lectures on quantum mechanics, p. 2)

This is the kind of statement which intrigued me as a physics student. How is this justified? Somehow the physical line ℓ is “isomorphic” to the mathematical line $\mathbb{R}$. But what does this mean? Can it be proved, perhaps? Do we need to specify certain properties of the physical line and its points to do this? What are these properties?

In these notes, we shall state precisely what these properties are. These properties are encoded into certain *axiom systems of synthetic geometry*. The four primary examples are:

Table 4.1: **Four axiom systems of synthetic geometry**

BG(4)	the synthetic geometry of betweenness geometry, with its the <i>betweenness relation</i> in four dimensions.
EG(3)	the synthetic geometry of Euclidean space, with its the <i>betweenness relation</i> and the <i>congruence relation</i> in three dimensions.
Gal(1, 3)	the synthetic geometry of Galilean spacetime, with its <i>betweenness relation</i> , the <i>simultaneity relation</i> , and restricted <i>congruence</i> , in four dimensions.
Mink(1, 3)	the synthetic geometry of Minkowski spacetime, with its <i>betweenness relation</i> , and the <i>light-signal relation</i> in four dimensions.

Our main results will lead up to certain “internal” coordinatization meta-theorems, such as:

$$\underbrace{\text{AM}}_{\text{Applied maths base}} + \underbrace{\text{BG}(4)}_{\text{synthetic axiom system}} \vdash O \neq I \rightarrow \left[\underbrace{(\ell, O, I, \dots)}_{\text{physical line}} \cong \underbrace{(\mathbb{R}, 0, 1, \dots)}_{\text{field of real numbers}} \right] \quad (4.1.2)$$

and

$$\underbrace{\text{AM}}_{\text{Applied maths base}} + \underbrace{\text{BG}(4)}_{\text{synthetic axiom system}} \vdash \left[\underbrace{(\mathbb{P}, B)}_{\text{synthetic structure}} \cong \underbrace{(\mathbb{R}^4, B_{\mathbb{R}^4})}_{\text{coordinate structure}} \right] \quad (4.1.3)$$

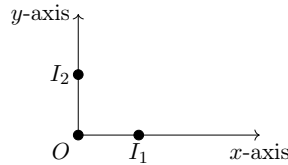
Then, having isolated the underlying physical geometry $\text{BG}(4)$, it follows that, *if* $\text{BG}(4)$ *is physically true (of points, lines and betweenness in the physical world)*, then these isomorphisms hold also in the physical world. In my view, this is the correct explanation of the existence of such isomorphisms.

Is $\text{BG}(4)$ physically true? That is an entirely separate matter, to be determined by experiment.

4.2 The Cartesian Plane

Above we sketched the construction of the three-dimensional Cartesian coordinate system, which is central in physics. In three dimensions, points get labelled by triples $(x, y, z) \in \mathbb{R}^3$.¹ Let’s consider the Cartesian plane of secondary school. The Cartesian plane is two-dimensional, and we identify points with coordinate pairs $(x, y) \in \mathbb{R}^2$:

Figure 4.3: **Cartesian plane**



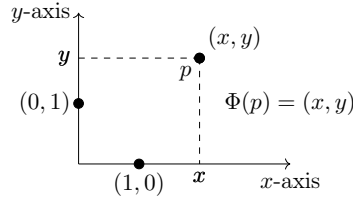
Here, the coordinate system Φ is determined (uniquely) by three distinct points, the origin O , and the “unit points” I_1 and I_2 . In particular, with this coordinate system Φ , we get:

¹ A nice textbook describing the Cartesian coordinate geometry of the plane is ...

$$\Phi(O) = (0, 0) \quad \Phi(I_1) = (1, 0) \quad \Phi(I_2) = (0, 1). \quad (4.2.1)$$

And, for any an arbitrary point p :

Figure 4.4: **Cartesian plane**



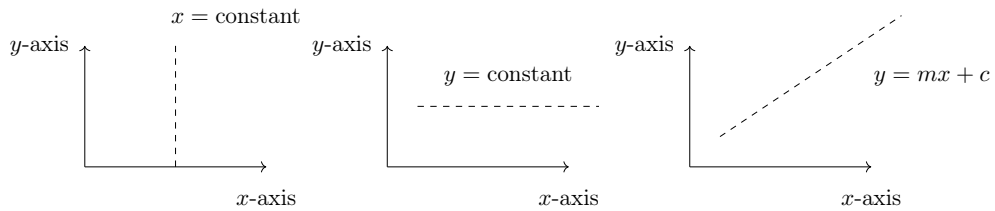
In this plane, as we know, there are various kinds of *straight lines*. We call them “straight” because when we plot them physically, on a visual graph on paper, the black-board or computer screen, they literally look “straight”.

In geometry, a straight line is a certain *set* of points. And, as we know, each straight line is defined by some *equation*. Usually, the equation one has seen in secondary school:

$$y = mx + c. \quad (4.2.2)$$

But this actually omit all “vertical” lines, parallel to the y -axis. Is there a more general way to classify all of these lines? There are “*vertical*” ones, with equation $x = \text{constant}$; and “*sloped/non-vertical*” ones, with equation $y = mx + c$; and “*horizontal*” ones, with equation $y = \text{constant}$ (that is, $m = 0$):

Figure 4.5: **Straight Lines in the Cartesian plane**



We shall devise a way to label these lines via the basic template:

$$ax + by + c = 0 \quad (4.2.3)$$

We can capture the required classes of line using the following choices of *parameters*, a, b, c :

a	b	c	$ax + by + c = 0$	Line
1	0	c	$x + c = 0$	$x = \text{constant.}$
0	1	c	$y + c = 0$	$y = \text{constant.}$
m	-1	c	$mx - y + c = 0$	$y = mx + c.$

Definition 4.2.1. We define, for any $a, b, c \in \mathbb{R}$, with a, b not both zero,

$$\ell_{a,b,c} := \{(x, y) \in \mathbb{R}^2 \mid ax + by + c = 0\} \quad (4.2.4)$$

This is *the line with parameters a, b, c* . These parameters are sometimes called *homogeneous coordinates* for the line.

Example 4.2.2. Examples are as follows:

$$\ell_{1,0,-c} = \text{the line } \{(x, y) \in \mathbb{R}^2 \mid x = c\} \quad (4.2.5)$$

$$\ell_{m,-1,c} = \text{the line } \{(x, y) \in \mathbb{R}^2 \mid y = mx + c\} \quad (4.2.6)$$

□

Note that the homogeneous coordinates of a line are not unique. The “degree of freedom” in a set of equivalent homogeneous coordinates is given by the following lemma:

Lemma 4.2.3. Let $k \in \mathbb{R}$, with $k \neq 0$. Then:

$$\ell_{a,b,c} = \ell_{ka,kb,kc}. \quad (4.2.7)$$

Definition 4.2.4. We then define the *set of all these lines*:

$$\mathcal{L}_{\mathbb{R}^2} := \{\ell_{a,b,c} \mid a, b, c \in \mathbb{R} \wedge (a, b) \neq (0, 0)\} \quad (4.2.8)$$

So, a *line* ℓ is defined to be an element of $\mathcal{L}_{\mathbb{R}^2}$.

Definition 4.2.5. Lines ℓ and ℓ' are said to be *parallel* just if they are either equal, or they do not intersect.

Lemma 4.2.6. The following results about parallelism hold:

- (1) All lines of the form $\ell_{1,0,c}$, for any $c \in \mathbb{R}$, are mutually parallel (vertically).
- (2) All lines $\ell_{m,-1,c}$, with fixed slope $m \in \mathbb{R}$ are mutually parallel.

Lemma 4.2.7. The following results about intersection hold:

- (1) Any two lines of the form $\ell_{1,0,c}$ and $\ell_{m,-1,c'}$ intersect, at exactly one point.
- (2) Any two sloped lines $\ell_{m,-1,c}$, $\ell_{m',-1,c'}$, of different slopes, m and m' , intersect at exactly one point.

Definition 4.2.8. We can now assemble the carrier set $\mathbb{R}^2$ of points and the system $\mathcal{L}_{\mathbb{R}^2}$ of lines and define the following structure:

$$\mathbb{A}^2(\mathbb{R}) := (\mathbb{R}^2, \mathcal{L}_{\mathbb{R}^2}). \quad (4.2.9)$$

The structure $\mathbb{A}^2(\mathbb{R})$ is called “*the affine plane over (the field) $\mathbb{R}$* ”.²

As we shall explain below, $\mathbb{A}^2(\mathbb{R})$ is a certain kind of *incidence structure*: see §7.2, and also Definition 7.3.1 and Definition 7.3.2. Roughly, an incidence structure is a pair $(\mathbb{P}, \mathcal{L})$, of “points” and “lines”, which satisfy certain quite simple conditions, as given in Definition 7.3.2.

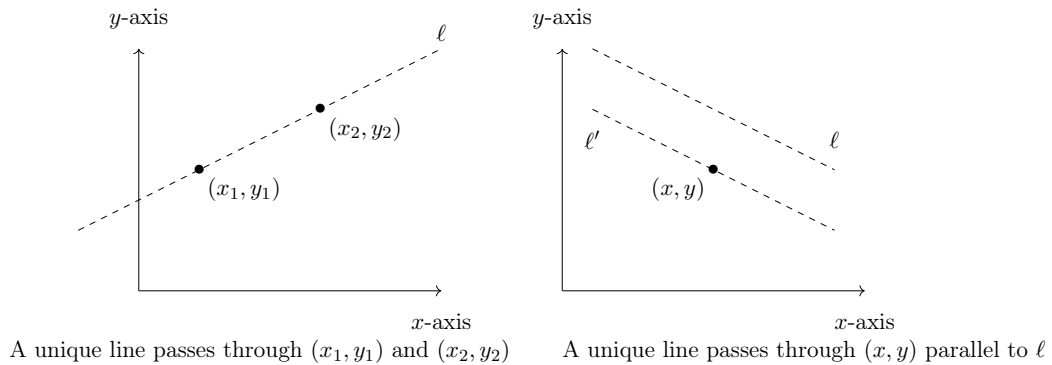
It is clear as well that $\mathbb{A}^2(\mathbb{R})$ satisfies the following three properties:

Table 4.2: **Affine Plane Properties**

(A1: Line uniqueness)	Given two distinct points (x_1, y_1) and (x_2, y_2) , there exists exactly one line ℓ through both.
(A2: Playfair)	Given a point (x, y) and any line ℓ , there exists exactly one line ℓ' which contains (x, y) and is parallel to ℓ .
(A3a: Non-triviality)	Each line contains at least two points.
(A3b: Non-triviality)	There exists at least two distinct lines.

The first two may be depicted as follows:

Figure 4.6: **Line uniqueness and Playfair in the Cartesian plane**



² Robert Goldblatt uses the notation $\alpha_{\mathbb{R}}$ to refer to this structure (Goldblatt (1987)). The notation “ $\mathbb{A}^2(\mathbb{R})$ ” generalizes along two dimensions. First, 2 can be replaced by any positive integer $n \geq 2$. Second, $\mathbb{R}$ can be replaced by any field, F (or division ring, D).

Below we treat these properties abstractly, as three defining conditions for an affine plane. See Definition 8.1.1 in §8.1: an affine plane is an incidence structure $(\mathbb{P}, \mathcal{L})$, of “points” and “lines”, which satisfy (A1)–(A3).

4.3 Coordinate Geometry of the Cartesian Plane

Equipped with the machinery of labelling points with coordinates (x, y) we are able to describe in coordinate language the following examples of point configurations:

112 (1)	Triangles
(2)	Quadrangles
(3)	Parallelograms
(4)	Polygons
112 (4)	Circles
(5)	Ellipses
(6)	Parabolas
(7)	Hyperbolas
(8)	General curves

Chapter 5

Geometrical Structures on $\mathbb{R}^n$

5.1 Standard Geometrical Relations on $\mathbb{R}^n$

In Chapter 4, §4.2, we described the Cartesian coordinate plane as a structure or “package”:

$$\mathbb{A}^2(\mathbb{R}) := (\mathbb{R}^2, \mathcal{L}_{\mathbb{R}^2}), \quad (5.1.1)$$

The “package” $\mathbb{A}^2(\mathbb{R})$ is an ordered pair, consisting of “points and lines”. The “domain” $\mathbb{R}^2$ of the package is the set of “points”, $(x, y) \in \mathbb{R}^2$; and $\mathcal{L}_{\mathbb{R}^2}$ is the set of “lines” in $\mathbb{R}^2$. As we shall see presently, this is an example of an *incidence structure*.¹ We have called this structure $\mathbb{A}^2(\mathbb{R})$, which is short for “the affine plane over (the field) $\mathbb{R}$ ”.²

In a sense, $\mathbb{A}^2(\mathbb{R})$ is one of the simplest geometrical structures on the “bare” point set

$$\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}. \quad (5.1.2)$$

The structure $\mathbb{A}^2(\mathbb{R})$ carries no “*betweenness*” information, or “*congruence*” information, or “*metrical*” information. It is simply a system of point & lines—an incidence structure where we have forgotten betweenness, congruence and distance.

However, there is a series of special operations and relations which can be defined on $\mathbb{R}^n$, given the built-in ordered field structure of $\mathbb{R}$, and these come to play a special role in geometry.

Definition 5.1.1. Let n be a positive integer. The standard Euclidean inner product $\langle \cdot, \cdot \rangle_n$ and norm $\|\cdot\|_n$ on $\mathbb{R}^n$ are defined as follows.³ For $\vec{x}, \vec{y} \in \mathbb{R}^n$,

¹ See §7.2, and also Definition 7.3.1 and Definition 7.3.2.

² See Definition 8.1.1 in §8.1 for the general definition of an affine planes.

³ We use the abbreviation $\vec{x} = (x_1, \dots, x_n)$, for n -tuples in $\mathbb{R}^n$. Similarly, for $\vec{y}, \vec{z}, \dots$

$$\langle \vec{x}, \vec{y} \rangle_n := \sum_{i=1}^n x_i y_i \quad (5.1.3)$$

$$\|\vec{x}\|_n := \sqrt{\langle \vec{x}, \vec{x} \rangle_n}. \quad (5.1.4)$$

The dimensional “ $\dots_n$ ” subscript is somewhat pedantic, since it ensures the right dimensionality and is often omitted. Thus:

$$\langle \vec{x}, \vec{y} \rangle := \sum_{i=1}^n x_i y_i \quad (5.1.5)$$

$$\|\vec{x}\| := \sqrt{\langle \vec{x}, \vec{x} \rangle}. \quad (5.1.6)$$

Definition 5.1.2. We can also define the special Minkowski inner product on $\mathbb{R}^n$ ($n \geq 2$):

$$\langle \vec{x}, \vec{y} \rangle^{\text{mink}} := x_n y_n - \sum_{i=1}^{n-1} (x_i y_i) \quad (5.1.7)$$

Definition 5.1.3 (Euclidean metric space on $\mathbb{R}^n$). For each $n \geq 1$, the standard Euclidean metrics $\Delta : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ are defined as follows:

$$\Delta(\vec{x}, \vec{y}) := \|\vec{x} - \vec{y}\|. \quad (5.1.8)$$

Strictly, there is one such metric Δ_n for each dimension $n \geq 1$.

Definition 5.1.4 (Betweenness, congruence, orthogonality). The following relations are the standard betweenness relation $B_{\mathbb{R}^n}$, the standard congruence relation $\equiv_{\mathbb{R}^n}$, and the standard orthogonality relation. For $\vec{x}, \vec{y}, \vec{z}, \vec{u} \in \mathbb{R}^n$:

$$B_{\mathbb{R}^n}(\vec{x}, \vec{y}, \vec{z}) := (\exists \alpha \in [0, 1])(\vec{y} - \vec{x} = \alpha(\vec{z} - \vec{x})) \quad (5.1.9)$$

$$\vec{x}\vec{y} \equiv_{\mathbb{R}^n} \vec{z}\vec{u} := \Delta(\vec{x}, \vec{y}) = \Delta(\vec{z}, \vec{u}) \quad (5.1.10)$$

$$\vec{x}\vec{y} \perp_{\mathbb{R}^n} \vec{z}\vec{u} := \langle \vec{x} - \vec{y}, \vec{z} - \vec{u} \rangle = 0 \quad (5.1.11)$$

Lemma 5.1.5. For the one dimensional case, i.e., $B_{\mathbb{R}}$, we have two alternative but equivalent definitions of betweenness $B_{\mathbb{R}}$:

$$B_{\mathbb{R}}(x, y, z) \leftrightarrow (x \leq y \leq z) \vee (z \leq y \leq x) \quad (5.1.12)$$

$$B_{\mathbb{R}}(x, y, z) \leftrightarrow \|x - y\| + \|y - z\| = \|x - z\|. \quad (5.1.13)$$

Lemma 5.1.6. The second of these in fact generalizes to $n > 1$ if we have a metric function Δ :

$$B_{\mathbb{R}^n}(\mathbf{x}, \mathbf{y}, \mathbf{z}) \leftrightarrow \Delta(\mathbf{x}, \mathbf{y}) + \Delta_n(\mathbf{y}, \mathbf{z}) = \Delta_n(\mathbf{x}, \mathbf{z}). \quad (5.1.14)$$

When we consider coordinate structures for *spacetime*, a few more relations turn out to be significant:

Definition 5.1.7 (Simultaneity, sim-congruence, light-signal relations). The following relations are *the standard simultaneity relation* $\sim_{\mathbb{R}^n}$, *the standard sim-congruence relation* $\equiv_{\mathbb{R}^n}$, and *the standard lightsignal relation* $\lambda_{\mathbb{R}^n}$ on $\mathbb{R}^n$. For $\vec{x}, \vec{y}, \vec{z}, \vec{u} \in \mathbb{R}^n$:

$$\vec{x} \sim_{\mathbb{R}^n} \vec{y} := x_n = y_n \quad (5.1.15)$$

$$\vec{x}\vec{y} \equiv_{\mathbb{R}^n} \vec{z}\vec{u} := \Delta(\vec{x}, \vec{y}) = \Delta(\vec{z}, \vec{u}) \ \& \ \vec{x} \sim_{\mathbb{R}^n} \vec{y} \sim_{\mathbb{R}^n} \vec{z} \sim_{\mathbb{R}^n} \vec{u} \quad (5.1.16)$$

$$\vec{x} \lambda_{\mathbb{R}^n} \vec{y} := \sum_{i=1}^{n-1} (x_i - y_i)^2 = (x_n - y_n)^2 \quad (5.1.17)$$

Definition 5.1.8. The *standard coordinate structures* over $\mathbb{R}$ are:

$\mathbb{B}^n$	Betweenness geometry in n dimensions	$:= (\mathbb{R}^n, B_{\mathbb{R}^n}).$
$\mathbb{E}^n$	Euclidean space in n dimensions	$:= (\mathbb{R}^n, B_{\mathbb{R}^n}, \equiv_{\mathbb{R}^n}).$
$\mathbb{E}_{\text{metric}}^n$	Euclidean metric space in n dimensions	$:= (\mathbb{R}^n, \Delta_n)$
$\mathbb{G}^n$	Galilean spacetime in n dimensions	$:= (\mathbb{R}^n, B_{\mathbb{R}^n}, \sim_{\mathbb{R}^n}, \equiv_{\mathbb{R}^n}).$
$\mathbb{M}^n$	Minkowski spacetime in n dimensions	$:= (\mathbb{R}^n, B_{\mathbb{R}^n}, \lambda_{\mathbb{R}^n}).$

From the perspective of physics, one's central interest is $\mathbb{G}^4$, the *standard coordinate structure for four-dimensional Galilean spacetime* and $\mathbb{M}^4$, the *standard coordinate structure for four-dimensional Minkowski spacetime*. Note that neither $\mathbb{G}^4$ nor $\mathbb{M}^4$ carries a metric or distance function. In the Chapters below we provide (second-order) categorical axiomatizations of $\mathbb{B}^4$, $\mathbb{E}^3$ and synthetic Galilean spacetime $\mathbb{G}^4$ and synthetic Minkowski spacetime $\mathbb{M}^4$.⁴

Regarding the definitions of $\mathbb{B}^n$, $\mathbb{E}^n$, $\mathbb{G}^n$ and $\mathbb{M}^n$. These still make sense if we replace $\mathbb{R}$ in the definition by a *Euclidean ordered field* F (an ordered field where all non-negative elements are squares). Cf. Szczerba & Tarski (1979), Definition 1.5, p. 160, who call a space $\mathbb{B}^n(F)$ a “Cartesian affine space” over F .

⁴ I mention here the remarkable fact, due to A. Robb, that the betweenness relation $B_{\mathbb{R}^4}$ is definable in terms of the light-signal relation $\lambda_{\mathbb{R}^4}$.

5.2 Automorphisms and Symmetry

In this section, we describe the symmetry (automorphism) groups of the structures defined above.

Definition 5.2.1 (General linear group). Let n be a positive integer. We say that A is an element of *the general linear group of dimension n* —in symbols, $A \in \text{GL}(n)$ —if and only if A is an $(n \times n)$ matrix, with real entries, and such that $\det A \neq 0$.

Definition 5.2.2 (Orthogonal group). We say that $A \in \text{GL}(n)$ is an element of *the orthogonal group of dimension n* —in symbols, $A \in \text{O}(n)$ —if and only if $A^T A = 1$.

Definition 5.2.3 (Extended Galilean matrix). We say that A is an *extended Galilean matrix*—in symbols, $A \in \text{GM}^e(n)$ —if and only if $A \in \text{GL}(n)$ and has the (block matrix) form

$$A = \begin{pmatrix} \alpha_1 R & \vec{v} \\ 0 & \alpha_2 \end{pmatrix} \quad (5.2.1)$$

where $A \in \text{O}(3)$, $\vec{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$ and $\alpha_1, \alpha_2 \in \mathbb{R} - \{0\}$. The $\text{O}(3)$ matrix R is called the *rotation* of A ; the 3-vector $\vec{v}$ is called the *(relative) velocity* of A ; the constant α_1 is called the *spatial scaling factor* of A and the constant α_2 is the *temporal scaling factor* of A .

Definition 5.2.4 (Lorentz matrix). We say that Λ is an *Lorentz matrix*—in symbols, $\Lambda \in \text{Lor}(n)$ —if and only if $\Lambda \in \text{GL}(n)$ such that its components Λ_{ij} satisfy:

$$\sum_{j=1}^n \Lambda_{ij} \Lambda_{jk} = \eta_{ik} \quad (5.2.2)$$

Or, equivalently:

$$\Lambda^T \Lambda = \eta. \quad (5.2.3)$$

We next define five Lie groups.

Definition 5.2.5 (Affine group). Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Then we say $h \in \text{Aff}(n)$ just if, for some $A \in \text{GL}(n)$ and some $\vec{d} \in \mathbb{R}^n$,

$$(\forall \vec{x} \in \mathbb{R}^n) [h(\vec{x}) = A\vec{x} + \vec{d}] \quad (5.2.4)$$

Definition 5.2.6 (Similitude group). Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Then we say $h \in \text{Sim}(n)$ just if, for some $R \in \text{O}(n)$, some non-zero real λ , and some $\vec{d} \in \mathbb{R}^n$,

$$(\forall \vec{x} \in \mathbb{R}^n) [h(\vec{x}) = \lambda R\vec{x} + \vec{d}] \quad (5.2.5)$$

Definition 5.2.7 (Euclidean group). Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Then we say $h \in E(n)$ just if, for some $R \in O(n)$ and some $\vec{d} \in \mathbb{R}^n$.

$$(\forall \vec{x} \in \mathbb{R}^n) [h(\vec{x}) = R\vec{x} + \vec{d}] \quad (5.2.6)$$

Definition 5.2.8 (Extended Galilean group). Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$. We say that $h \in \text{Gal}^e(1, 3)$ just if, for some $A \in \text{GM}^e(1, 3)$ and some $\vec{d} \in \mathbb{R}^4$,

$$(\forall \vec{x} \in \mathbb{R}^4) [h(\vec{x}) = A\vec{x} + \vec{d}] \quad (5.2.7)$$

Definition 5.2.9 (Poincaré-Weyl group). Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$. We say that $h \in \text{PW}(1, 3)$ just if, for some $\Lambda \in \text{Lor}(1, 3)$, some $\alpha \in \mathbb{R}^+$, and some $\vec{d} \in \mathbb{R}^4$.

$$(\forall \vec{x} \in \mathbb{R}^4) [h(\vec{x}) = \alpha\Lambda\vec{x} + \vec{d}] \quad (5.2.8)$$

Theorem 5.2.10. Let $n \geq 2$. For the coordinate structures $\mathbb{B}^n$, $\mathbb{E}^n$, $\mathbb{E}_{\text{metric}}^n$, $\mathbb{G}^{1,3}$ and $\mathbb{M}^{1,3}$ (Definition 5.1.8), the automorphism groups are given by:

$$\text{Aut}(\mathbb{B}^n) \cong \text{Aff}(n) \quad (5.2.9)$$

$$\text{Aut}(\mathbb{E}^n) \cong \text{Sim}(n) \quad (5.2.10)$$

$$\text{Aut}(\mathbb{E}_{\text{metric}}^n) \cong E(n) \quad (5.2.11)$$

$$\text{Aut}(\mathbb{G}^{1,3}) \cong \text{Gal}^e(1, 3) \quad (5.2.12)$$

$$\text{Aut}(\mathbb{M}^{1,3}) \cong \text{PW}(1, 3) \quad (5.2.13)$$

Proof. I give a brief summary. For the first, the proof relies on the requirement that straight lines get mapped to straight lines, and parallel lines get mapped to parallel lines. The outcome is that any such mapping h must be an affine transformation generated by a $GL(n)$ matrix A and a translation $\vec{d}$. So, the automorphism group is what is usually called $\text{Aff}(n)$, the *affine group* in n dimensions. For the third, the symmetry group is the isometry group of the metric space $\mathbb{E}\mathbb{G}_{\text{metric}}^n$ —thus, what’s usually called the *Euclidean group* $E(n)$: rotations, inversions, reflections and translations (reflections and inversions are $O(n)$ matrices with determinant -1). For the second, which is less familiar, the symmetries include rotations, inversions, reflections and translations again, but also include *scalings* too:

$$\vec{x} \mapsto \lambda\vec{x} \quad (5.2.14)$$

The latter are sometimes called *similitudes* or *dilations* (the non-zero factor λ represents this scaling). Although the metric distance between two points is not invariant, nonetheless *metric equalities* are invariant. Imagine a rubber sheet, pinned at some central point, say, O , and imagine “stretching” it uniformly and radially from O , by some

factor. The *distance* between two points on the sheet is not invariant under the stretching: $\Delta(\mathbf{x}, \mathbf{y}) \mapsto |\lambda| \Delta(\mathbf{x}, \mathbf{y})$; but equality between distances of points (i.e., congruence) is invariant.

To show $\mathcal{G}^e(1, 3) \subseteq \text{Aut}(\mathbb{G}^{(1,3)})$, we verify that each extended Galilean transformation is a symmetry of $\mathbb{G}^{(1,3)}$. Since $\mathbb{B}\mathbb{G}^4$ is a reduct of $\mathbb{G}^{(1,3)}$, and each extended Galilean transformation is affine, it follows that betweenness is invariant. The special form of extended Galilean matrices then ensures that simultaneity and sim-congruence are invariant.

To show that $\text{Aut}(\mathbb{G}^{(1,3)}) \subseteq \mathcal{G}^e(1, 3)$ is more involved. Since $\mathbb{B}\mathbb{G}^4$ is a reduct of $\mathbb{G}^{(1,3)}$, it follows that any symmetry h of $\mathbb{G}^{(1,3)}$ must be affine, and so there exists a $GL(4)$ matrix A , and displacement $\vec{d} \in \mathbb{R}^4$ such that, for any $\vec{x} \in \mathbb{R}^4$,

$$h(\vec{x}) = A\vec{x} + \vec{d} \quad (5.2.15)$$

To determine the sixteen components A_{ij} of A , one must then examine the conditions that simultaneity and sim-congruence be invariant. By examining certain choices of points, the invariance of simultaneity enforces that A must have the form

$$A = \begin{pmatrix} C & \vec{v} \\ 0 & \alpha_2 \end{pmatrix} \quad (5.2.16)$$

where C is 3×3 matrix, and α_2 is a non-zero constant. The invariance of sim-congruence enforces that the upper 3×3 block C must be a multiple $\alpha_1 R$, of an $O(3)$ matrix R by a non-zero real factor α_1 :

$$A = \begin{pmatrix} \alpha_1 R & \vec{v} \\ 0 & \alpha_2 \end{pmatrix} \quad (5.2.17)$$

But this is an extended Galilean matrix. Consequently, $\text{Aut}(\mathbb{G}^{(1,3)}) \subseteq \mathcal{G}^e(1, 3)$.

Together, these results imply that $\text{Aut}(\mathbb{G}^{(1,3)}) = \mathcal{G}^e(1, 3)$. □

The constants α_1, α_2 in any *extended* Galilean matrix A determine *scalings* of the spatial, and temporal coordinates, respectively. So, given some A in the extended Galilean matrix group, and any $(\vec{x}, t) \in \mathbb{R}^4$,

$$A(\vec{x}, t) = (\alpha_1 R\vec{x} + \vec{v}t, \alpha_2 t) \quad (5.2.18)$$

Let's set the relative rotation R to be $\mathbb{I}$ and set the relative velocity $\vec{v}$ to be zero:

$$A(\vec{x}, t) = (\alpha_1 \vec{x}, \alpha_2 t) \quad (5.2.19)$$

Thus, the spatial coordinates are scaled by α_1 and the temporal coordinate is scaled by α_2 . Instead, let us set these scalings α_1, α_2 at 1, and consider the image $(\vec{x}', t')$ of the point with coordinates $(\vec{x}, t)$ under an extended Galilean transformation:

$$\vec{x}' = R\vec{x} + \vec{v}t + \vec{d} \quad (5.2.20)$$

$$t' = t + d_t \quad (5.2.21)$$

These are the usual Galilean transformations as given in physics textbooks, in usually simplified form: e.g., French (1968): 68; or Sears et al. (1979): 252; or Longair (1984): 87; or Rindler (1969): 3. The conventional Galilean group $\mathcal{G}(1, 3)$ is normally understood to be this 10-parameter Lie group: the ten parameters are these: four parameters for the spatial and temporal translations, $\mathbf{d}$; three parameters (i.e., determined by the three Euler angles) for the rotation matrix R ; three parameters for the velocity $\vec{v}$.

Chapter 6

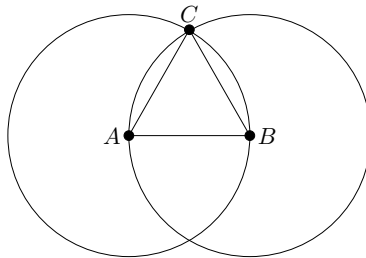
Synthetic Geometry

6.1 Euclid's Postulates

Synthetic geometry is, in a simple slogan, “geometry without coordinates”.

Theorem 6.1.1 (Euclid, Bk I, Prop 1). Let A, B be distinct points and AB the line segment. Then there exists a point C such that $\triangle ABC$ is an equilateral triangle.

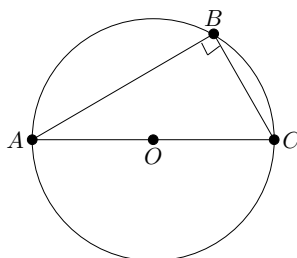
Figure 6.1: **Euclid Bk 1, Proposition 1**



Theorem 6.1.2 (Thales's theorem). Let A, B and C be distinct points on a circle where the line AC is a diameter. Then the angle $\angle ABC$ is a right angle.¹

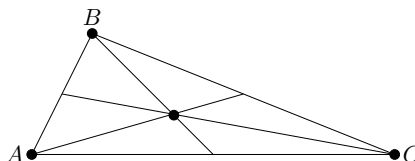
¹ This appears as Prop 31 of Euclid Bk 1: “In a circle the angle in the semicircle is right, that in a greater segment less than a right angle, and that in a less segment greater than a right angle; further the angle of the greater segment is greater than a right angle, and the angle of the less segment is less than a right angle”.

Figure 6.2: Thales's Theorem



Theorem 6.1.3 (Centroid theorem). Let $\triangle ABC$ be a triangle. Let P_1, P_2, P_3 be the midpoints of BC , AC , and AB . So, P_1 is “opposite” A , P_2 is “opposite” B ; and P_3 is “opposite” C . Then the lines AP_1 , BP_2 and CP_3 all intersect at the same point. The lines AP_1 , BP_2 and CP_3 are called *medians* and the intersection point is called *the centroid* of $\triangle ABC$.

Figure 6.3: Centroid Theorem



We need to go back to the beginning: Euclid's *The Elements*, Euclid (1956). The following are Sir Thomas Heath's classic English translation, with its very detailed commentary (see Euclid (1956)).

Euclidean geometry

Let the following be postulated:

1. To draw a straight line from any point to any point.
2. To produce (extend) a finite straight line continuously in a straight line.
3. To describe a circle with any centre and distance.
4. That all right angles are equal to one another.
5. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which the angles are less than two right angles. (Euclid (1956): 154–155)

Euclidean geometry

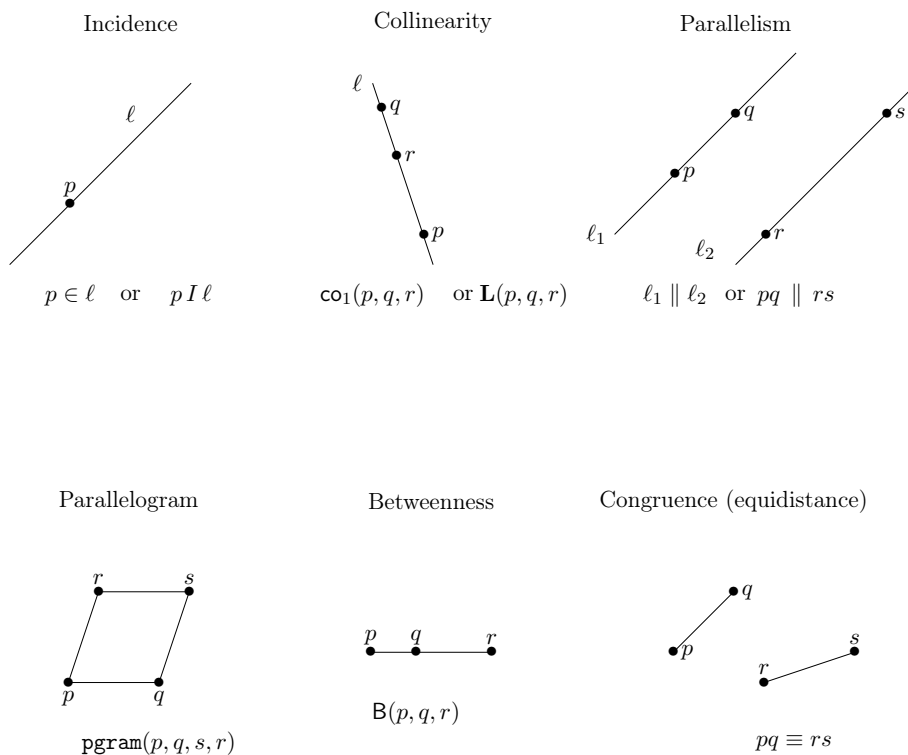
Common notions:

1. Things that are equal to the same thing are also equal to one another.
2. If equals are added to equals, then the wholes are equal.
3. If equals are subtracted from equals, then the differences are equal.
4. Things that coincide with one another are equal to one another.
5. The whole is greater than the part. (Euclid (1956): 155)

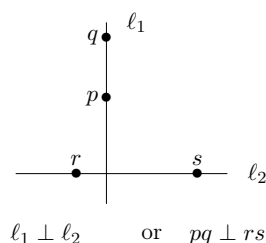
6.2 Concepts of synthetic geometry

Synthetic geometry is coordinate free. Its axioms involve concepts which do not refer to coordinates. The following figure illustrates eight important such concepts:

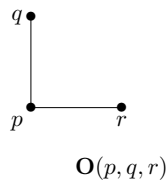
Figure 6.4: Concepts of synthetic geometry



Perpendicularity (orthogonality)



Orthonormality



6.3 Axioms

A system of axioms for synthetic geometry is then a system of axioms—i.e., sentences—deploying the above concepts—i.e., sentences in which predicates for the above concepts are treated as primitives. From the point of view of *physics*, we would search for a single axiom system, based on a some special choice of primitives, whose axioms and theorems are *physically true*. However, since the middle of the nineteenth century, this aim has been replaced by the study of a large variety of axiom systems which are not interpreted physically (although they are usually visualized), and which are, in some sense, studied for their own sake.

What are these “suitable axioms”? For these, we consult David Hilbert’s justly celebrated monograph, Hilbert (1899).² Hilbert’s axiom system is intended for three-dimensional Euclidean geometry. In their original and final form, these are organized into five axiom groups,

- | | |
|----------|------------------------------|
| 0.55 (1) | Incidence axioms. |
| (2) | Order (betweenness) axioms. |
| (3) | Congruence axioms. |
| (4) | Euclid’s Parallel postulate. |
| (5) | Continuity axioms. |

² One should mention too the systems developed in Borsuk & Szmielew (1960), Coxeter (1969) and Hartshorne (2000).

Table 6.1: I. AXIOMS OF INCIDENCE (Hilbert (1899): 3-4)

- I, 1. For every two points A, B , there exists a line that contains each of the points A, B .
- I, 2. For every two points A, B , there exists no more than one line that contains each of the points A, B .
- I, 3. There exists at least two points on a line. There exists at least three points that do not lie on a line.
- I, 4. For any three points A, B, C that do not lie on the same line there exists a plane α that contains each of the points A, B, C . For every plane there exists a point which it contains.
- I, 5. For any three points A, B, C that do not lie on one and the same line there exists no more than one plane that contains each of the points A, B, C .
- I, 6. If two points A, B of a line a lie in a plane α then every point of a lies in plane α .
- I, 7. If two planes α, β have a point A in common then they have at least one more point B in common.
- I, 8. There exists at least four points which do not lie in a plane.

Table 6.2: II. AXIOMS OF ORDER (Hilbert (1899): 5-6)

- II, 1. If a point B lies between a point A and a point C then the points A, B, C are three distinct points of a line, and B then also lies between C and A .
- II, 2. For two points A and C , there always exists at least one point B on the line AC such that C lies between A and B .
- II, 3. Of any three points on a line, there exists no more than one that lies between the other two.
- II, 4. Let A, B, C be three points that do not lie on a line and let a be a line in the plane ABC which does not meet any of the points A, B, C . If the line a passes through a point of the segment AB , it also passes through a point of the segment AC , or through a point of the segment BC .

We are next going to look at incidence geometry, and then betweenness geometry.

6.4 Analytic Representation

We shall give various axiom systems governing $\mathcal{B}$, and $\equiv$ (and $\sim$, $\equiv\sim$, λ and a few others) such that certain *equivalences* can be proved:

$$\text{synthetic condition} \qquad \text{analytic condition} \qquad (6.4.1)$$

$$\mathbf{B}(p, q, r) \leftrightarrow B_{\mathbb{R}^2}(\Phi(p), \Phi(q), \Phi(r)) \qquad (6.4.2)$$

$$pq \equiv rs \leftrightarrow \Delta_{\Phi}(p, q) = \Delta_{\Phi}(r, s). \qquad (6.4.3)$$

On the left, we have a synthetic (“coordinate-free”) condition relating points $p \in \mathbb{P}$. On the right, we have an equivalent analytic condition relating the same points, relative to a certain class of coordinate systems Φ , which map points to tuples of reals.

Part II

Affine and Projective Planes

Chapter 7

Incidence Geometry

7.1 Hilbert's Incidence Axioms

The most basic way of developing synthetic geometry is “incidence geometry”. In incidence geometry, we examine the incidence relationships between points and lines.

Figure 7.1: Incidence

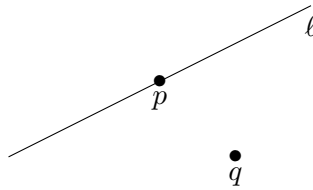


Figure 7.1 depicts the case where the point p lies on the line ℓ , while the point q does not lie on the line ℓ .

We give Hilbert's first three incidence axioms for a *plane*:

Table 7.1: **Hilbert's (plane) incidence axioms**

- I, 1. For every two points A and B there exists a line a that contains each of the points A, B .
I, 2. For every two points A and B there exists no more than one line that contains each of the points A, B .
...
I, 3. There exist at least two points on a line. There exist at least three points that do not lie on a line. (Hilbert (1899): 3-4)

7.2 Points, Lines and Incidence

In general, an incidence structure is a two-sorted relational structure:

$$(\mathbb{P}, \mathcal{L}, I) \tag{7.2.1}$$

where $\mathbb{P}$ is the carrier set of “points”, and $\mathcal{L}$ is the carrier set of “lines” and I is the *binary incidence relation*, where $I \subseteq \mathbb{P} \times \mathcal{L}$. The relationship $(p, \ell) \in I$ is usually abbreviated as:

$$p I \ell \tag{7.2.2}$$

This is short for “point p is incident of line $\mathcal{L}$ ”.

However, it is, in general, simpler to regard each line ℓ as a *set of points*. Then the incidence relation is simply the membership relation $\in$; and $\mathcal{L}$ is then some subset of $\mathcal{P}(\mathbb{P})$. Then an incidence structure takes the form of a second-order structure:

$$(\mathbb{P}, \mathcal{L}) \tag{7.2.3}$$

In this Chapter, our aim is to sketch the main parts of *the theory of affine incidence geometry*. In particular, the theory of affine (incidence) planes. And our eventual goal is to state, and sketch the main ideas behind the proof of, “The Fundamental Theorem of Synthetic Affine Geometry”:¹

Fundamental Theorem of Synthetic Affine Geometry. Let $(\mathbb{P}, \mathcal{L})$ be a Desarguesian affine plane, let $\ell \in \mathcal{L}$ and let O and I be any two distinct points on ℓ . Then $(\ell, +, \times)$ is a division ring, where $+$ and $\times$ are defined relative to O and I . Furthermore, $(\ell, +, \times)$ is a field if Pappus’ Theorem holds in $(\mathbb{P}, \mathcal{L})$. (Bennett (1995): 72)

With this established, we can then import this result into developing the two main representation theorems given in Chapter ?? for the formalized theories $\mathbf{BG}_0(n)$ and $\mathbf{EG}_0(n)$.

7.3 Incidence Structures

We now introduce the topic of incidence structures:²

¹ For the reader who has not studied affine geometry, I strongly recommend studying the textbook Bennett (1995) by Mary Bennett. Its level is around what in the UK would be a second or third year course in a mathematics degree. Annoyingly, it is out of print.

² The material in this section is covered in some depth in Behnke et al. (1984), Chapter 3, “Affine and Projective Planes”, and in Bennett (1995). Also, in Goldblatt (1987), Chapter 2.

Definition 7.3.1. An *incidence structure* is a two-sorted relational structure:

$$(\mathbb{P}, \mathcal{L}, I) \tag{7.3.1}$$

where $\mathbb{P}$ is the carrier set (of “points”) and $\mathcal{L}$ is the carrier set (of “lines”) and I is a binary relation $\subseteq \mathbb{P} \times \mathcal{L}$ (“the incidence relation”), where

$$(p, \ell) \in I \tag{7.3.2}$$

is abbreviated as:

$$p I \ell \tag{7.3.3}$$

and is read: “point p is incident of line ℓ ”.

We use “ p ”, “ q ”, “ r ”, “ s ”, . . . (and similar with subscripts, and primes) to quantify over points and we use “ ℓ ” (and similar with subscripts, and primes) to quantify over lines. “ $p I \ell$ ” is read: “point p is incident on line ℓ ”.

Definition 7.3.2. A *set-theoretical incidence structure* is a pair,

$$(\mathbb{P}, \mathcal{L}) \tag{7.3.4}$$

where $\mathcal{L}$ is some subset of $\mathcal{P}(\mathbb{P})$.

If $(\mathbb{P}, \mathcal{L})$ is a set-theoretical incidence structure, then $(\mathbb{P}, \mathcal{L}, \in)$ is an incidence structure. So, for any set-theoretical incidence structure $(\mathbb{P}, \mathcal{L})$, each line $\ell \in \mathcal{L}$ is a *set* of points—i.e., $\ell \subseteq \mathbb{P}$ —and then the incidence relation I is simply the membership relation $\in$. Henceforth, we stick to set-theoretical incidence structures, and call them simply “incidence structures”.

Definition 7.3.3 (Parallelism for incidence structures). Let $(\mathbb{P}, \mathcal{L})$ be an incidence structure and let $\ell, \ell' \in \mathcal{L}$. Then we say that ℓ is *parallel* to ℓ' if and only if either $\ell = \ell'$ or $\ell \cap \ell' = \emptyset$.³ When ℓ is *parallel* to ℓ' , we write:

$$\ell \parallel \ell' \tag{7.3.5}$$

³ This definition is intended for *planes*, with those being understood as *two dimensional* incidence geometries. So, notice that this definition is deficient for affine spaces of higher dimension than two. In that case, one needs an additional “coplanarity” clause: ℓ is *parallel* to ℓ' if and only if the lines ℓ, ℓ' are *coplanar*, and either $\ell = \ell'$ or $\ell \cap \ell' = \emptyset$.

Definition 7.3.4 (Copunctual). Let $(\mathbb{P}, \mathcal{L})$ be an incidence structure. A pair ℓ_1, ℓ_2 of lines are called *copunctual* if they intersect. In general, a set of lines is called *copunctual* if all lines in that set intersect at the same point (see Behnke et al. (1984): 66.)

Definition 7.3.5 (Pencils). Let $(\mathbb{P}, \mathcal{L})$ be an incidence structure. The set of lines parallel to a given line ℓ is called the *parallel pencil of ℓ* , denoted $\text{pencil}_{\parallel}(\ell)$ (see Behnke et al. (1984): 67.) This is also sometimes called the *parallel class* of ℓ (See Blumenthal (1961): 58). The set of lines passing through a given point p is called the *incidence pencil of ℓ* , denoted $\text{pencil}_I(p)$. A *pencil*, in general, is either *parallel pencil* or an *incidence pencil*: i.e., a set of lines all of which are parallel, or all of which pass through the same single point (so they are all copunctual). (See Behnke et al. (1984): 114 (footnote 1).)

Before defining affine planes, we briefly mention linear spaces:

Definition 7.3.6 (Linear space (geometry)). In geometry, a *linear space* is an incidence structure $(\mathbb{P}, \mathcal{L})$ such that:

- (L1) Two distinct points are incident with exactly one line.
- (L2) Every line is incident to at least two distinct points.

I will sometimes refer to (L1) as “the line uniqueness axiom”.

Chapter 8

Affine Planes

8.1 Affine Planes

For the definition of the concept of an affine plane, I follow the definition from Bennett (1995): 18.¹

Definition 8.1.1. An *affine plane* is an incidence structure $(\mathbb{P}, \mathcal{L})$ such that the following conditions hold:

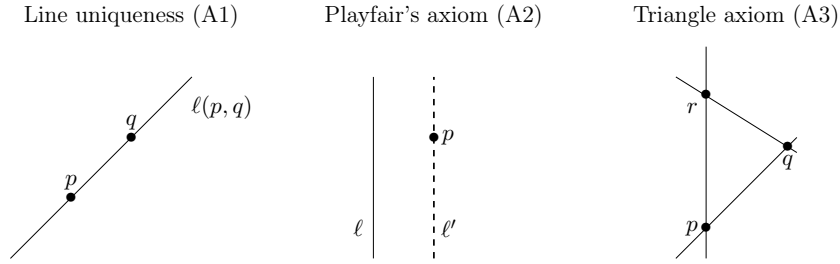
- (A1) For any two distinct points $p, q \in \mathbb{P}$, there is a unique line $\ell \in \mathcal{L}$ such that $p \in \ell$ and $q \in \ell$. This line is denoted $\ell(p, q)$.
- (A2) Given a point $p \in \mathbb{P}$ and a line $\ell \in \mathcal{L}$, there is a unique line $\ell' \in \mathcal{L}$ such that $p \in \ell'$ and $\ell' \parallel \ell$.
- (A3a) On any line, there are at least two points.
- (A3b) There are at least two lines.

I shall call (A1) “*the line uniqueness axiom*”. And Axiom (A2) is often called *Playfair’s axiom*. This is one of a great many equivalents of Euclid’s Fifth Postulate (or The Parallel Postulate). I shall refer to the conjunction of the pair of axioms (A3a), (A3b) “*the triangle axiom*”.² As Bennett notes (p. 18), “Axioms A1 and A2 are essentially Euclid’s first and fifth postulates”.

¹ An equivalent definition appears in Behnke et al. (1984): 66; and in Blumenthal (1961): 55; and in Hughes & Piper (1973): 82. Add Goldblatt and Harshorne.

² There are some equivalents of the conjunction of the pair of axioms (A3a) and (A3b). For example: “there are at least four points such that no three of them are collinear” and “there are at least three non-collinear points”.

Figure 8.1: **Affine plane axioms**



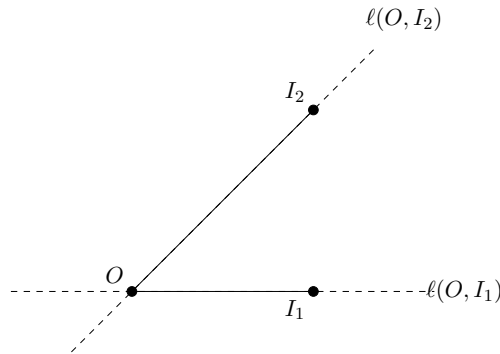
As Bennett notes (p. 18):

Lemma 8.1.2. Any two distinct lines are either parallel or intersect in a *single* point.

Proof. Let $\ell_1 \neq \ell_2$ be two distinct lines and not parallel. Since they're not parallel, $\ell_1 \cap \ell_2 \neq \emptyset$. Suppose there are two distinct points $p_1 \neq p_2$, each in both ℓ_1 and ℓ_2 . So, $\{p_1, p_2\} \subseteq \ell_1 \cap \ell_2$. By line uniqueness (A1), $\ell_1 = \ell(p_1, p_2)$ and $\ell_2 = \ell(p_1, p_2)$. Hence, $\ell_1 = \ell_2$. This is a contradiction. So, $\ell_1 \cap \ell_2 = \{p\}$, for some point p , which is therefore the intersection of ℓ_1 and ℓ_2 . $\square$

Definition 8.1.3. Let $(\mathbb{P}, \mathcal{L})$ be an affine plane. A *2-frame* in $(\mathbb{P}, \mathcal{L})$ is a triangle (O, I_1, I_2) of three non-collinear points in $\mathbb{P}$.

Figure 8.2: **A 2-frame**



The point O is called *the origin* of the 2-frame. The points I_1 and I_2 are called *the unit points* of the 2-frame. The lines $\ell(O, I_2)$ and $\ell(O, I_1)$ are called *the axes* of the 2-frame.

8.2 Parallelism and Parallelograms

Lemma 8.2.1. $\parallel$ is an equivalence relation.

Proof. It is clear that $\ell \parallel \ell$, since $\ell = \ell$. So $\parallel$ is reflexive. Supposing $\ell \parallel \ell'$, it is clear that $\ell' \parallel \ell$ too. So $\parallel$ is symmetric. To verify transitivity, let's suppose:

$$\ell \parallel \ell' \tag{8.2.1}$$

$$\ell' \parallel \ell'' \tag{8.2.2}$$

We claim: $\ell \parallel \ell''$.

If $\ell = \ell'$, the claim follows trivially. If $\ell' = \ell''$ the claim follows trivially. So, we can additionally suppose $\ell \neq \ell'$ and $\ell' \neq \ell''$. So, from the two parallelism assumptions,

$$\ell \cap \ell' = \emptyset \tag{8.2.3}$$

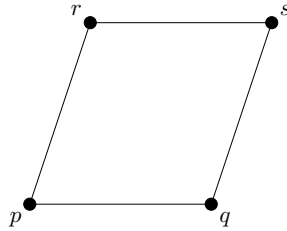
$$\ell' \cap \ell'' = \emptyset \tag{8.2.4}$$

Now, towards a contradiction, suppose $\ell \cap \ell'' \neq \emptyset$. Hence, there is some $p \in \mathbb{P}$ such that $p \in \ell$ and $p \in \ell''$. Now from our assumptions, we infer $p \notin \ell'$. By Playfair's Axiom, there is a *unique line* ℓ''' such that ℓ''' contains p and is parallel to ℓ' . Since $p \in \ell$ and $\ell \parallel \ell'$, it follows that $\ell''' = \ell$. Since $p \in \ell''$ and $\ell''' \parallel \ell'$, it follows that $\ell''' = \ell''$. Hence, $\ell = \ell''$, which is a contradiction. Thus, $\ell \cap \ell'' = \emptyset$, and from this it follows that $\ell \parallel \ell''$. $\square$

The above lemma requires just Playfair's Axiom, (A2).

Definition 8.2.2. In an affine plane, a (vector) *parallelogram* is an ordered 4-tuple (p, q, r, s) of mutually non-collinear distinct points such that $\ell(p, q) \parallel \ell(r, s)$ and $\ell(p, r) \parallel \ell(q, s)$.

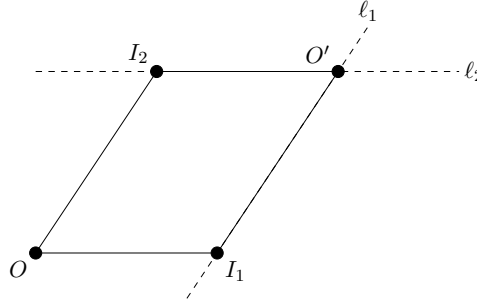
Figure 8.3: **A Parallelogram**



Remark 8.2.3. One can also talk of a *cyclic* parallelogram, which is an ordered 4-tuple (p, q, r, s) of mutually non-collinear distinct points such that (p, q, s, r) is a vector parallelogram. But these notions are inter-definable. $\square$

Lemma 8.2.4. Let a 2-frame (O, I_1, I_2) be given in an affine plane. There exists a unique point O' such that (O, I_1, I_2, O') is a parallelogram.

Proof. By Playfair, there exists a unique line ℓ_1 passing through I_1 and parallel to $\ell(O, I_2)$. By Playfair, there exists a unique line ℓ_2 passing through I_2 and parallel to $\ell(O, I_1)$. ℓ_1 and ℓ_2 are not parallel. Hence, these lines intersect at a unique point, denoted O' .



By construction, all four points are distinct, and $\ell(O, I_1) \parallel \ell(I_2, O')$ and $\ell(O, I_2) \parallel \ell(I_1, O')$. Hence, (O, I_1, I_2, O') is a parallelogram

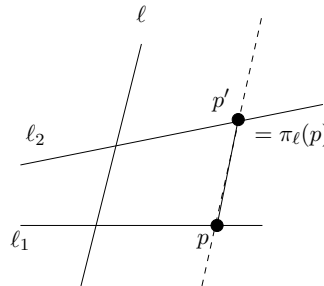
□

8.3 Parallel Projection

We next explain the notion of “parallel projection”, which yields a function π_ℓ mapping points on a line to points on another line, where the direction of the mapping is parallel to a given line, ℓ . See Szemielew (1983): ??? for a similar exposition, although notationally different.

Lemma 8.3.1 (Parallel projection lemma). Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. Let $\ell_1, \ell_2 \in \mathcal{L}$ be distinct lines. Let $\ell \in \mathcal{L}$ be a line not parallel to either ℓ_1 or ℓ_2 . Let $p \in \ell_1$ with $p \notin \ell_2$. There exists a unique point $p' \in \ell_2$ such that $\ell(p, p') \parallel \ell$.

Figure 8.4: Parallel projection



Definition 8.3.2 (Parallel projection). Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. Let $\ell_1, \ell_2 \in \mathcal{L}$ be distinct lines. Let $\ell \in \mathcal{L}$ be a line not parallel to either ℓ_1 or ℓ_2 . Let $p \in \ell_1$. This determines a unique point p' as above. Let

$$\pi_\ell(p) := \begin{cases} p' & \text{if } p \notin \ell_2 \\ p & \text{if } p \in \ell_2 \end{cases} \quad (8.3.1)$$

This defines a *function* π_ℓ on ℓ_1 :

$$\pi_\ell : \ell_1 \rightarrow \ell_2 \quad (8.3.2)$$

This function π_ℓ is called *the parallel projection from ℓ_1 to ℓ_2 parallel to ℓ* .

We next prove a couple of lemmas to the effect that parallel projection $\pi_\ell : \ell_1 \rightarrow \ell_2$ is a bijection.

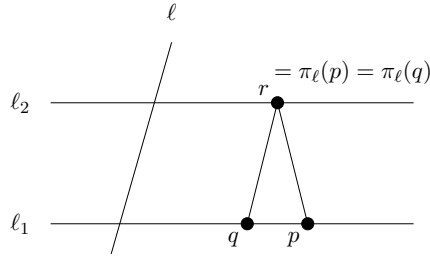
Lemma 8.3.3. Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. Let $\ell_1, \ell_2 \in \mathcal{L}$ be distinct lines. Let $\ell \in \mathcal{L}$ be a line not parallel to either ℓ_1 or ℓ_2 . Then $\pi_\ell : \ell_1 \rightarrow \ell_2$ is a surjection.

Proof. By “reversing” the above construction, so to speak, it follows that for any $p' \in \ell_2$, there is some $p \in \ell_1$ such that $\pi_\ell(p) = p'$. Hence, π_ℓ is surjective. $\square$

Lemma 8.3.4. Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. Let $\ell_1, \ell_2 \in \mathcal{L}$ be distinct *parallel* lines. Let $\ell \in \mathcal{L}$ be a line not parallel to either ℓ_1 or ℓ_2 . Then $\pi_\ell : \ell_1 \rightarrow \ell_2$ is an injection.

Proof. By assumption, $\ell_1 \neq \ell_2$, and $\ell_1 \parallel \ell$, and $\ell_2 \parallel \ell$. Suppose $\pi_\ell(p) = \pi_\ell(q)$, for points $p, q \in \ell_1$. Let $r = \pi_\ell(p) = \pi_\ell(q)$. We claim: $p = q$.

Figure 8.5: $\ell_1 \parallel \ell_2$



We have $\ell(p, r) \parallel \ell$ and $\ell(q, r) \parallel \ell$. Hence, by transitivity of $\parallel$, $\ell(p, r) \parallel \ell(q, r)$. Hence, either $\ell(p, r) = \ell(q, r)$ or $\ell(p, r) \cap \ell(q, r) = \emptyset$. However, $r \in \ell(p, r)$ and $r \in \ell(q, r)$, and so $\ell(p, r) \cap \ell(q, r) \neq \emptyset$. Therefore, $\ell(p, r) = \ell(q, r)$. Now $\ell(p, r) \not\parallel \ell_1$. Hence, $\ell(p, r) \cap \ell_1 = \{p\}$. Similarly, $\ell(q, r) \cap \ell_1 = \{q\}$. Thus, since $\ell(p, r) = \ell(q, r)$, $\{p\} = \{q\}$, and hence $p = q$. $\square$

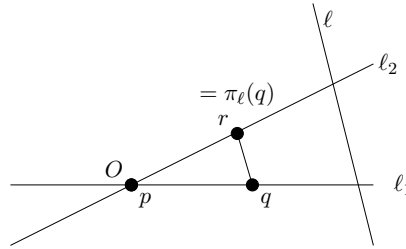
Lemma 8.3.5. Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. Let $\ell_1, \ell_2 \in \mathcal{L}$ be distinct *non-parallel* lines. Let $\ell \in \mathcal{L}$ be a line not parallel to either ℓ_1 or ℓ_2 . Then $\pi_\ell : \ell_1 \rightarrow \ell_2$ is an injection.

Proof. Again, by assumption, $\ell_1 \neq \ell_2$, and $\ell_1 \not\parallel \ell$, and $\ell_2 \not\parallel \ell$. Suppose $\pi_\ell(p) = \pi_\ell(q)$, for points $p, q \in \ell_1$. Let $r = \pi_\ell(p) = \pi_\ell(q)$. We claim: $p = q$. Since ℓ_1 and ℓ_2 are not parallel, let $\ell_1 \cap \ell_2 = \{O\}$.

Case (a) Neither p nor q is equal to O . Then the reasoning is the same as Lemma 8.3.4 above.

Case (b) $p = O$. Thus, $p \in \ell_2$. Then, from Definition 8.3.2, $\pi_\ell(p) = p$. So, $\pi_\ell(p) = p = r = O = \pi_\ell(q)$.

Figure 8.6: $\ell_1 \not\parallel \ell_2$



Towards a contradiction, suppose $q \neq O$. It follows that $\pi_\ell(q) \notin \ell_1$. Hence, $O \notin \ell_1$, a contradiction. Hence $q = O$. Therefore, $p = q$, as claimed.

Case (c) $q = O$. Same as Case (b), *mutatis mutandis*. □

Lemma 8.3.3 tells us that $\pi_\ell : \ell_1 \rightarrow \ell_2$ is surjective. And Lemmas 8.3.4 and 8.3.5, tell us that $\pi_\ell : \ell_1 \rightarrow \ell_2$ is injective, covering both cases of ℓ_1 and ℓ_2 being parallel or not. Therefore:

Theorem 8.3.6 (Parallel projection theorem). Let $(\mathbb{P}, \mathcal{L})$ be an affine plane. Let $\ell_1, \ell_2 \in \mathcal{L}$ be distinct lines. Let $\ell \in \mathcal{L}$ be a line not parallel to either ℓ_1 or ℓ_2 . Then $\pi_\ell : \ell_1 \rightarrow \ell_2$ is an bijection.

Theorem 8.3.7. Let $(\mathbb{P}, \mathcal{L})$ be an affine plane. Let $\ell_1, \ell_2 \in \mathcal{L}$ be lines. Given lines ℓ_1 and ℓ_2 , there exists a bijection

$$\pi : \ell_1 \rightarrow \ell_2 \tag{8.3.3}$$

Proof. If $\ell_2 = \ell_1$, we are done, as the identity map is a bijection. So, suppose $\ell_2 \neq \ell_1$. Pick any line ℓ not parallel to ℓ_1 or ℓ_2 . Then π_ℓ is a bijection. □

The above results yield the theorem:

Theorem 8.3.8. Let $(\mathbb{P}, \mathcal{L})$ be an affine plane. Let $\ell_1, \ell_2 \in \mathcal{L}$ be lines. Then:

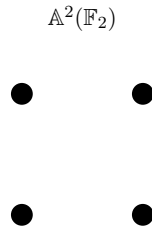
$$|\ell_1| = |\ell_2|. \quad (8.3.4)$$

Definition 8.3.9. Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. The *order* of the affine plane α is the number of any points on each line. (This makes sense because of the previous theorem.) α is said to be *finite*, if its order is finite.

8.4 Affine Planes: Examples

Example 8.4.1. Up to isomorphism, there exists exactly one affine plane of order 2. This is the smallest affine plane, and can be pictured as follows:

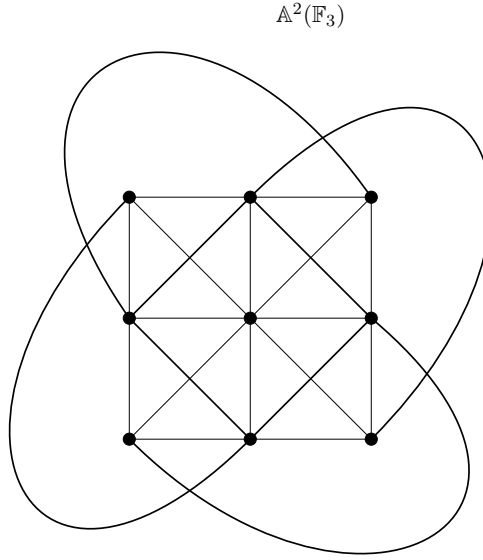
Figure 8.7: Affine Plane of Order 2



This affine plane has four points and six lines. It is usually called $\mathbb{A}^2(\mathbb{F}_2)$ or $\text{AG}(2, 2)$. □

Example 8.4.2. Up to isomorphism there exists exactly one affine plane of order 3:

Figure 8.8: Affine Plane of Order 3



This affine plane has nine points and twelve lines. It is usually called either $\mathbb{A}^2(\mathbb{F}_3)$, or $\text{AG}(2, 3)$. It is also sometimes called *the Hesse configuration*. □

Example 8.4.3. At Definition 4.2.8, we defined the affine plane over $\mathbb{R}$, usually denoted $\mathbb{A}^2(\mathbb{R})$:

$$\mathcal{L}_{\mathbb{R}^2} := \text{the set of lines in } \mathbb{R}^2 \quad (8.4.1)$$

$$\mathbb{A}^2(\mathbb{R}) := (\mathbb{R}^2, \mathcal{L}_{\mathbb{R}^2}) \quad (8.4.2)$$

It is easy to verify that the affine axioms (A1)–(A3) hold in $\mathbb{A}^2(\mathbb{R})$. For given two points $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^2$, there is a unique line in $\mathcal{L}_{\mathbb{R}^2}$ containing both points. And clearly given any line, there are infinitely many points on that line. And furthermore, given a point $\mathbf{x} \in \mathbb{R}^2$, and a line $\ell \in \mathcal{L}_{\mathbb{R}^2}$, there exists a unique line parallel to ℓ and containing $\mathbf{x}$. □

Example 8.4.4. The construction of $\mathbb{A}^2(\mathbb{R})$ can be repeated for the field $\mathbb{Q}$ of rational numbers, *mutatis mutandis*, yielding the affine plane over $\mathbb{Q}$, denoted $\mathbb{A}^2(\mathbb{Q})$. □

8.5 Affine Plane Combinatorics

Definition 8.5.1. Let $(\mathbb{P}, \mathcal{L})$ be an affine plane. Let

$$[\ell] := \{\ell' \in \mathcal{L} \mid \ell' \parallel \ell\} = \text{pencil}_{\parallel}(\ell) \quad (8.5.1)$$

$$\text{pencil}(p) := \{\ell \in \mathcal{L} \mid p \in \ell\} \quad (8.5.2)$$

$$\text{parallel}(\alpha) := \{[\ell]_{\parallel} \mid \ell \in \mathcal{L}\} \quad (8.5.3)$$

Lemma 8.5.2 (Combinatorics of Affine Planes). Let $(\mathbb{P}, \mathcal{L})$ be an affine plane of order n . Then

- (1) $n \geq 2$.
- (2) For each $\ell \in \mathcal{L}$, $|\ell| = n$.
- (3) $|\mathbb{P}| = n^2$.
- (4) $|\mathcal{L}| = n^2 + n$.
- (5) For each $p \in \mathbb{P}$, $|\text{pencil}(p)| = n + 1$.
- (6) For each $\ell \in \mathcal{L}$, $|[\ell]| = n$.
- (7) $|\text{parallel}(\alpha)| = n + 1$.

Proof. For (5), let ℓ be a line not in $\mathcal{L}(p)$. So, $p \notin \ell$. By Lemma ??, there are exactly n points on ℓ , which we call $p_1, \dots, p_n$. Let ℓ' be a line through p . Then either ℓ' intersects ℓ , at some point p_i , or is parallel to ℓ . So, there are n such lines intersecting ℓ ; and there is one line parallel to ℓ . Hence, there are $n + 1$ lines through p . □

So, the number of lines through a point p is $n + 1$. The number of lines in $\text{pencil}_{\parallel}(\ell)$ is n . The number of points in $\mathbb{P}$ is n^2 . And the number of lines in $\mathcal{L}$ is $n^2 + n$. The number of parallel classes in α is $n + 1$.

8.6 Vectors in Affine Planes

Chapter 9

Projective Planes

9.1 Projective planes

Definition 9.1.1 (Projective plane). A *projective plane* is an incidence structure $\pi = (\mathbb{P}, \mathcal{L})$ such that the following conditions hold:¹

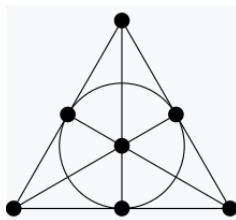
- | | |
|----------|---|
| 110 (P1) | Given any two distinct points, there is exactly one line incident with both of them. |
| (P2) | Given any two distinct lines, there is exactly one point incident with both of them.. |
| (P3) | There are four points such that no line is incident with more than two of them. |

So, (P1) is equivalent to (A1) above for affine planes (the notation $\ell(p, q)$ again denotes the unique line containing the two distinct points $p, q \in \mathbb{P}$).

9.2 Projective Planes: Examples

Example 9.2.1. The smallest projective plane is the so-called Fano Plane:

Figure 9.1: **The Fano Plane**



¹ Bennett (1995): ?? . An equivalent definition appears in Hughes & Piper (1973): ??.

The Fano Plane has seven points and seven lines. It is sometimes denoted $\mathbb{P}^2(\mathbb{F}_2)$ (the projective plane over the field $\mathbb{F}_2$ with two elements) or $\text{PG}(2, 2)$. It has order 2. We'll see in a moment that the Fano Plane is the *projective completion* of the affine plane $\mathbb{A}^2(\mathbb{F}_2)$. □

9.3 Correspondence between affine and projective planes

Definition 9.3.1 (Projective completion). Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. We define two new sets, $\mathbb{P}^+$ and $\mathcal{L}^+$ extending $\mathbb{P}$ and $\mathcal{L}$, respectively.

Let $\ell \in \mathcal{L}$. Let:

$$[\ell] := \{\ell' \in \mathcal{L} \mid \ell' \parallel \ell\} \quad (= \text{the parallel pencil of } \ell) \quad (9.3.1)$$

$$\ell^+ := \ell \cup \{[\ell]\} \quad (9.3.2)$$

We shall treat each $[\ell]$ as a new “point”—called a *point at infinity*. And the set, ℓ_∞ , of these new points is called *the line at infinity*:

$$\ell_\infty := \{[\ell] \mid \ell \in \mathcal{L}\} \quad (9.3.3)$$

$$\mathbb{P}^+ := \mathbb{P} \cup \ell_\infty \quad (9.3.4)$$

$$\mathcal{L}^+ = \{\ell^+ \mid \ell \in \mathcal{L}\} \cup \{\ell_\infty\} \quad (9.3.5)$$

The last condition tells us that $\ell' \in \mathcal{L}^+$ if and only if $\ell' = \ell^+$, for some $\ell \in \mathcal{L}$, or $\ell' = \ell_\infty$.

Lemma 9.3.2. If $\alpha = (\mathbb{P}, \mathcal{L})$ is an affine plane, then $\pi = (\mathbb{P}^+, \mathcal{L}^+)$ is a projective plane. π is called *the projective completion* of α .

Example 9.3.3. Let us consider $\mathbb{A}^2(\mathbb{F}_2)$ and construct its projective completion. Let a specific representation be given by: $\mathbb{P} = \{1, 2, 3, 4\}$ and $\mathcal{L} = \{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}$. There are three parallel classes, given by:

$$\{1, 2\} \parallel \{3, 4\} \quad (9.3.6)$$

$$\{1, 3\} \parallel \{2, 4\} \quad (9.3.7)$$

$$\{1, 4\} \parallel \{2, 3\} \quad (9.3.8)$$

We have three new points (at infinity) be defined by these three parallel classes, and we call them 5, 6, 7:

$$5 := [\{1, 2\}] \quad (9.3.9)$$

$$6 := [\{1, 3\}] \quad (9.3.10)$$

$$7 := [\{1, 4\}] \quad (9.3.11)$$

So, the extended point set is:

$$\mathbb{P}^+ := \{1, 2, 3, 4, 5, 6, 7\} \quad (9.3.12)$$

And we have one new line:

$$\ell_\infty := \{5, 6, 7\} \quad (9.3.13)$$

and so, the extended system of lines is:

$$\mathcal{L}^+ = \{\{1, 2, 5\}, \{1, 3, 6\}, \{1, 4, 7\}, \{2, 3, 7\}, \{2, 4, 6\}, \{3, 4, 5\}, \ell_\infty\} \quad (9.3.14)$$

Then $\pi = (\mathbb{P}^+, \mathcal{L}^+)$ has seven points and seven lines. It is the Fano plane.

Notice that each pair of distinct lines intersects at exactly one point.

□

9.4 The Classification Problem

It is by no means true that, for any $n \in \mathbb{N}$, with $n \geq 2$, there is an affine plane of order n . For example, there are no affine planes of order 6 or 10. See below.

Lemma 9.4.1. An affine plane of order n exists if and only if a projective plane of order n exists (however, the definition of order in these two cases is not the same).

Lemma 9.4.2. There is no projective plane of order 6.

Lemma 9.4.3. There is no projective plane of order 10.

Remark 9.4.4. There is no affine plane of order 6 or order 10 since there are no projective planes of those orders. □

In fact, finite division rings are always fields—this classic result in algebra is called Wedderburn’s Little Theorem. So, for the finite case, we can always focus on finite fields. In fact, it is a basic result in algebra that:

Theorem 9.4.5. Every finite field is $\mathbb{F}_q$, where $q = p^n$, for some prime p , and integer $n \geq 1$.

Proof. Lee, p. ? □

Consequently:

Lemma 9.4.6. For every prime power p^n , there is a corresponding affine plane: i.e., $\mathbb{A}^2(\mathbb{F}_{p^n})$.

The converse claim, i.e., “The order of any affine plane is a prime power”, is an open problem in incidence geometry. Although there exist affine planes of orders 2,3,4,5,7,8,9, and more, there is no affine plane of either order 6 or order 10. This information is recorded in Moorhouse (2017) (p. 11):

Order	2	3	4	5	6	7	8	9	10	...
Number of affine planes	1	1	1	1	0	1	1	7	0	...

Chapter 10

Collinearity and Parallelity Planes

10.1 Collinearity Planes

In this chapter, I follow Szmielew (1983), Chapter 1, fairly closely:

Definition 10.1.1. A *collinearity structure* is a pair $(\mathbb{P}, \mathbf{L})$, with $\mathbf{L} \subseteq \mathbb{P}^3$, such that:¹

$$(\mathbf{Ax}1.1.1) \quad a = b \vee a = c \vee b = c \rightarrow \mathbf{L}(a, b, c). \quad (10.1.1)$$

$$(\mathbf{Ax}1.1.2) \quad a \neq b \wedge \mathbf{L}(a, b, p) \wedge \mathbf{L}(a, b, q) \wedge \mathbf{L}(a, b, r) \rightarrow \mathbf{L}(p, q, r) \quad (10.1.2)$$

Definition 10.1.2 (Line (in a collinearity structure)). Let $(\mathbb{P}, \mathbf{L})$ be a collinearity structure. A subset $\ell \subseteq \mathbb{P}$ is called a *line* if and only if there exists $a, b \in \mathbb{P}$, with $a \neq b$ and, for any $p \in \mathbb{P}$, $p \in \ell$ if and only if $\mathbf{L}(a, b, p)$.

Definition 10.1.3. If the following holds,

$$\dim \quad \exists a \exists b \exists c \neg \mathbf{L}(a, b, c) \quad (10.1.3)$$

then $(\mathbb{P}, \mathbf{L})$ is said to be *non-trivial* or *of dimension at least 2*. If the following holds,

$$\begin{aligned} \text{Dim} \quad \forall a \forall b \forall c \forall d \exists p ((\mathbf{L}(a, b, p) \wedge \mathbf{L}(c, d, p)) \vee (\mathbf{L}(a, c, p) \wedge \mathbf{L}(b, d, p)) \\ \vee (\mathbf{L}(a, d, p) \wedge \mathbf{L}(b, c, p))) \end{aligned} \quad (10.1.4)$$

then $(\mathbb{P}, \mathbf{L})$ is said to be *of dimension 2*.

Definition 10.1.4 (Parallelism relation). Let $(\mathbb{P}, \mathbf{L})$ be a collinearity structure. The *parallelism relation*, $\parallel \subseteq \mathbb{P}^4$, in $(\mathbb{P}, \mathbf{L})$, is defined by (see Ax 2.1.10, page 17):

$$ab \parallel cd \leftrightarrow (a = b \vee \forall p (\mathbf{L}(a, b, p) \rightarrow \mathbf{L}(c, d, p)) \vee \forall p (\mathbf{L}(a, b, p) \rightarrow \neg \mathbf{L}(c, d, p)) \quad (10.1.5)$$

¹ Szmielew (1983): 12.

Lemma 10.1.5. The axioms (Ax 1.1.1), (Ax 1.1.2) imply:

$$\mathbf{L}(a, b, c) \rightarrow \mathbf{L}(b, a, c) \wedge \mathbf{L}(a, c, b). \quad (10.1.6)$$

Lemma 10.1.6. The axioms (Ax 1.1.1), (Ax 1.1.2) are equivalent to:

$$(\mathbf{Ax1.1.1}') \quad \mathbf{L}(a, b, a) \quad (10.1.7)$$

$$(\mathbf{Ax1.1.2}') \quad a \neq b \wedge \mathbf{L}(a, b, c) \wedge \mathbf{L}(a, b, d) \rightarrow \mathbf{L}(a, c, d) \quad (10.1.8)$$

$$(\mathbf{Ax1.1.3}') \quad \mathbf{L}(a, b, c) \rightarrow \mathbf{L}(b, a, c) \quad (10.1.9)$$

Following Szmielew (1983): 13-14), we define the notion of a “*non-trivial collinearity structure*”:

Definition 10.1.7. A *non-trivial collinearity structure* is a pair $(\mathbb{P}, \mathbf{L})$, with $L \subseteq \mathbb{P}^3$, such that

$$(\mathbf{Ax1.1.1}') \quad \mathbf{L}(a, b, a) \quad (10.1.10)$$

$$(\mathbf{Ax1.1.2}') \quad a \neq b \wedge \mathbf{L}(a, b, c) \wedge \mathbf{L}(a, b, d) \rightarrow \mathbf{L}(a, c, d) \quad (10.1.11)$$

$$(\mathbf{Ax1.1.3}') \quad \mathbf{L}(a, b, c) \rightarrow \mathbf{L}(b, a, c) \quad (10.1.12)$$

$$\dim \quad \exists a \exists b \exists c \neg \mathbf{L}(a, b, c) \quad (10.1.13)$$

10.2 Szmielew's Definition of “Affine Plane”

Szmielew (page 16) gives a somewhat different looking definition of “affine plane”.

Definition 10.2.1 (Affine plane: Szmielew (1983)). A non-trivial collinearity structure $(\mathbb{P}, \mathbf{L})$ is said to be an *affine plane* whenever the following two conditions are satisfied:

$$(i) \quad p \notin \ell_1 \rightarrow \exists \ell_2 (p \in \ell_2 \wedge \ell_1 \cap \ell_2 = \emptyset) \quad (10.2.1)$$

$$(ii) \quad p \in \ell_1 \wedge p \in \ell_2 \wedge \ell_3 \cap \ell_1 = \emptyset \wedge \ell_3 \cap \ell_2 = \emptyset \rightarrow \ell_1 = \ell_2 \quad (10.2.2)$$

The resulting definition is, in fact, equivalent to the standard one (e.g., Bennett's) given above.

10.3 Parallelity Planes

Again, we follow Szmielew (1983). Following Szmielew (1983): 19), we next define the notion of a “*parallelity plane*”:

Definition 10.3.1 (Parallelity plane). A *parallelity plane* is a pair $(\mathbb{P}, \parallel)$, with $\parallel \subseteq \mathbb{P}^4$, such that

$$(\mathbf{Ax2.2.0}) \quad ab \parallel ba \quad (10.3.1)$$

$$(\mathbf{Ax2.2.1}) \quad ab \parallel cc \quad (10.3.2)$$

$$(\mathbf{Ax2.2.2}) \quad a \neq b \wedge ab \parallel pq \wedge ab \parallel rs \rightarrow pq \parallel rs \quad (10.3.3)$$

$$(\mathbf{Ax2.2.3}) \quad ab \parallel ac \rightarrow ba \parallel bc \quad (10.3.4)$$

$$(\mathbf{Ax2.2.4}) \quad \exists a \exists b \exists c ab \not\parallel ac. \quad (10.3.5)$$

$$(\mathbf{Ax2.2.5}) \quad \forall a \forall b \forall p \exists q (ab \parallel pq \wedge p \neq q). \quad (10.3.6)$$

$$(\mathbf{Ax2.2.6}) \quad ab \not\parallel cd \rightarrow \exists p (pa \parallel pb \wedge pc \parallel pd). \quad (10.3.7)$$

Definition 10.3.2 (Line relation). Let $(\mathbb{P}, \parallel)$ be a parallelity plane. The *line relation*, $\mathbf{L} \subseteq \mathbb{P}^3$, in $(\mathbb{P}, \parallel)$, is defined by

$$\mathbf{L}(a, b, c) := ab \parallel ac \quad (10.3.8)$$

So, clearly any parallelity plane $(\mathbb{P}, \parallel)$ can be expanded to $(\mathbb{P}, \parallel, \mathbf{L})$.

Szmielew proves two main theorems relating parallelity planes and affine planes:

Theorem 10.3.3. Let $(\mathbb{P}, \parallel)$ be a parallelity plane. Let the relation $\mathbf{L} \subseteq \mathbb{P}^3$ be defined as per Definition 10.3.2. Then $(\mathbb{P}, \mathbf{L})$ is an affine plane in Szmielew's sense. (This is Szmielew (1983), Theorem 2.2.16, p. 22)

Theorem 10.3.4. Let the structure $(\mathbb{P}, \mathbf{L})$ be an affine plane in Szmielew's sense. Let $\parallel \subseteq \mathbb{P}^4$ be defined by (this is Ax 2.1.10, page 17),

$$ab \parallel cd \leftrightarrow (a = b \vee \forall p (\mathbf{L}(a, b, p) \rightarrow \mathbf{L}(c, d, p)) \vee \forall p (\mathbf{L}(a, b, p) \rightarrow \neg \mathbf{L}(c, d, p)) \quad (10.3.9)$$

Then

$$\mathbf{L}(a, b, c) \leftrightarrow ab \parallel ac \quad (10.3.10)$$

holds, and $(\mathbb{P}, \parallel)$ is a parallelity plane.

(This is Szmielew (1983), Theorem 2.2.17, page 22)

Szmielew comments:

I would prefer to stick with the standard definition: i.e., an *affine plane* is an incidence structure $(\mathbb{P}, \mathcal{L})$ satisfying (A1)–(A3), as above. It is not difficult to see that “affine plane in the standard sense” is equivalent to “affine plane in Szmielew's sense”. The notions are inter-convertible by explicit definitions:

Theorem 10.3.5. Let $(\mathbb{P}, \mathcal{L})$ be an affine plane. Let $\mathbf{L}(a, b, c)$ hold if and only if there is a line $\ell \in \mathcal{L}$ such that $a \in \ell$, $b \in \ell$, $c \in \ell$. Then $(\mathbb{P}, \mathbf{L})$ is an affine plane in Szmielew's sense.

Conversely,

Theorem 10.3.6. Let $(\mathbb{P}, \mathbf{L})$ be an affine plane in Szmielew's sense. Let a subset $\ell \subseteq \mathbb{P}$ be called a *line* just if there are at least two distinct $p, q \in \mathbb{P}$ such that, for any $r \in \mathbb{P}$, $\mathbf{L}(p, q, r)$ if and only if $r \in \ell$. Let $\mathcal{L}$ be the set of lines. Then $(\mathbb{P}, \mathcal{L})$ is an affine plane.

Chapter 11

Coordinate Affine and Projective Planes

11.1 Coordinate Planes over Division Rings and Fields

We next show how to build affine planes from *algebraic structures*—viz., *division rings* and *fields*:

Definition 11.1.1 (Coordinate affine plane). Let D be a division ring or a field. We define, for any $a, b, c \in D$:

$$\ell_{a,b,c} := \{(x, y) \in D^2 \mid ax + by + c = 0\} \quad (11.1.1)$$

Then $\ell_{a,b,c}$, so long as at least one of a and b is non-zero, is the line in D^2 with parameters a, b, c . These parameters are sometimes called *homogeneous coordinates* for the line (nb: they’re not unique). We then define:

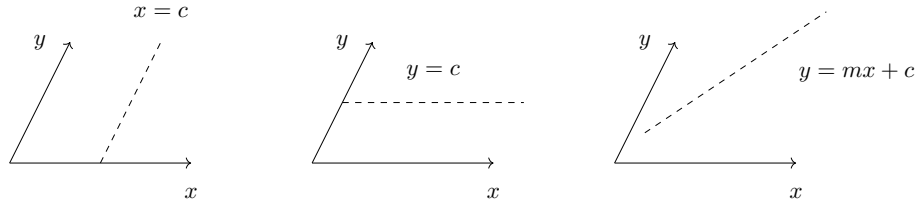
$$\mathcal{L}_{D^2} := \{\ell_{a,b,c} \mid a, b, c \in D \wedge \neg(a = 0 \wedge b = 0)\} \quad (11.1.2)$$

$$\mathbb{A}^2(D) := (D^2, \mathcal{L}_{D^2}). \quad (11.1.3)$$

If D is a division ring, the geometrical structure $\mathbb{A}^2(D)$ is called “*the classical affine plane over the division ring D* ”. If F is a field, the geometrical structure $\mathbb{A}^2(F)$ is called “*the classical affine plane over the field F* ”.

Thinking of D^2 as a kind of Cartesian plane, one may also think of these lines as being either “*vertical*” ones, with equation $x = c$, and “*non-vertical*” ones, with equation $y = mx + c$.

Figure 11.1: **Lines in $\mathbb{A}^2(D)$**



It is possible to see that the line $\{(x, y) \in D^2 \mid x = c\}$ corresponds to $\ell_{1,0,-c}$, and the line $\{(x, y) \in D^2 \mid y = mx + c\}$ corresponds to $\ell_{m,-1,-c}$.

Lemma 11.1.2. Let D be a division ring. Then $\mathbb{A}^2(D)$ is an affine plane.

Lemma 11.1.3. Let F be a field. Then $\mathbb{A}^2(F)$ is an affine plane.

Proof. If F is a field, it is a division ring, and by Lemma 11.1.2, $\mathbb{A}^2(F)$ is an affine plane. $\square$

Chapter 12

The Groups Aut, Dil, Tr and Ht

12.1 The Automorphism Group of an Affine Plane

Definition 12.1.1. Let $(\mathbb{P}_1, \mathcal{L}_1)$ and $(\mathbb{P}_2, \mathcal{L}_2)$ be affine planes. An isomorphism $\phi : (\mathbb{P}_1, \mathcal{L}_1) \rightarrow (\mathbb{P}_2, \mathcal{L}_2)$ is defined to be a bijection $\phi : \mathbb{P}_1 \rightarrow \mathbb{P}_2$ such that, for any $\ell \subseteq \mathbb{P}$, we have:

$$\ell \in \mathcal{L}_1 \leftrightarrow \phi[\ell] \in \mathcal{L}_2 \quad (12.1.1)$$

Recall that we can define the line relation $\mathbf{L}$ given $\alpha = (\mathbb{P}, \mathcal{L})$, and then:

Lemma 12.1.2. Let $\alpha_1 = (\mathbb{P}_1, \mathcal{L}_1)$ and $\alpha_2 = (\mathbb{P}_2, \mathcal{L}_2)$ be affine planes. Let $\mathbf{L}_1$ and $\mathbf{L}_2$ be the corresponding line relations. Let $\phi : \mathbb{P}_1 \rightarrow \mathbb{P}_2$. Then ϕ is an isomorphism if and only if ϕ is a bijection and, for any $p, q, r \in \mathbb{P}$,

$$\mathbf{L}_1(p, q, r) \leftrightarrow \mathbf{L}_2(\phi(p), \phi(q), \phi(r)) \quad (12.1.2)$$

For this reason, isomorphisms between affine planes are usually called *collineations*.

Recall that we can also define the parallelity relation $\parallel$ given $\alpha = (\mathbb{P}, \mathcal{L})$, and then:

Lemma 12.1.3. Let $\alpha_1 = (\mathbb{P}_1, \mathcal{L}_1)$ and $\alpha_2 = (\mathbb{P}_2, \mathcal{L}_2)$ be affine planes. Let $\parallel_1$ and $\parallel_2$ be the corresponding parallelity relations. Let $\phi : \mathbb{P}_1 \rightarrow \mathbb{P}_2$. Then ϕ is an isomorphism if and only if ϕ is a bijection and, for any $p, q, r, s \in \mathbb{P}$,

$$pq \parallel_1 rs \leftrightarrow \phi(p)\phi(q) \parallel_2 \phi(r)\phi(s) \quad (12.1.3)$$

A special kind of collineation is an isomorphism of an affine plane to *itself*, and therefore bijections satisfying the condition (for all points p, q, r, s):

$$pq \parallel rs \leftrightarrow \phi(p)\phi(q) \parallel \phi(r)\phi(s) \quad (12.1.4)$$

These isomorphisms are automorphisms (see Szmielew (1983): 29):

Definition 12.1.4. Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. An *automorphism* of α is an isomorphism from α to α . The set of automorphisms is denoted $\text{Aut}(\alpha)$.

Thus:

$$\phi \in \text{Aut}(\alpha) \leftrightarrow \left[\mathbb{P} \stackrel{\phi}{\cong} \mathbb{P} \wedge (\forall p \in \mathbb{P}) (\forall q \in \mathbb{P}) (\forall r \in \mathbb{P}) (\forall s \in \mathbb{P}) (pq \parallel rs \leftrightarrow \phi(p)\phi(q) \parallel \phi(r)\phi(s)) \right] \quad (12.1.5)$$

Lemma 12.1.5. $\text{Aut}(\alpha)$ is a group (under function composition).

12.2 Dilations, Translations and Homotheties

As Szmielew explains, there are a number of *subgroups* of $\text{Aut}(\alpha)$ which are of interest. The first is the subgroup of dilations:

Definition 12.2.1 (Dilatation). Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. An automorphism $\phi \in \text{Aut}(\alpha)$ is called a *dilation* just if it satisfies the condition, for any $p, q \in \mathbb{P}$:

$$pq \parallel \phi(p)\phi(q) \quad (12.2.1)$$

The set of dilations is denoted $\text{Dil}(\alpha)$. So:

$$\phi \in \text{Dil}(\alpha) \leftrightarrow \left[\mathbb{P} \stackrel{\phi}{\cong} \mathbb{P} \wedge (\forall p \in \mathbb{P}) (\forall q \in \mathbb{P}) (pq \parallel \phi(p)\phi(q)) \right] \quad (12.2.2)$$

As Szmielew notes, the dilation condition (12.2.1) is stronger than the automorphism condition (12.1.4).

Definition 12.2.2 (Fixed point). Let α be an affine plane. Let $\phi \in \text{Aut}(\alpha)$. A point p in α is called a *fixed point* of the automorphism ϕ just if $\phi(p) = p$.

Lemma 12.2.3. Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. Let $p \in \mathbb{P}$. Let $\phi \in \text{Dil}(\alpha)$. Then, if $\phi(p) = p$ then, for any $a \in \mathbb{P}$, $p \in \ell(a, \phi(a))$.

Proof. Let $\phi(p) = p$. Then, for any $a \in \mathbb{P}$, $pa \parallel p\phi(a)$, by (12.2.1). Hence, $\mathbf{L}(p, a, \phi(a))$, by Definition 10.3.2 of $\mathbf{L}$ in an affine plane. Hence $p \in \ell(a, \phi(a))$. $\square$

Theorem 12.2.4. Let α be an affine plane. Let $\phi \in \text{Dil}(\alpha)$, with $\phi \neq I$. Then ϕ has at most one fixed point.

Proof. Let ϕ be a dilation with $\phi \neq I$. Let $\phi(p) = p$ and $\phi(q) = q$. We claim $p = q$. By the above lemma, for any point a , we have $p \in \ell(a, \phi(a))$ and $q \in \ell(a, \phi(a))$. Since $\phi \neq I$, at least one point is moved by ϕ , say c : so, $\phi(c) \neq c$. Let $\ell = \ell(c, \phi(c))$. Thus, $p \in \ell$ and $q \in \ell$. Let d be any point not on ℓ . Let $\ell' = \ell(c, d)$. [INCOMPLETE] $\square$

Definition 12.2.5 (Translation). Let α be an affine plane. Let $\phi \in \text{Aut}(\alpha)$. Then ϕ is called a *translation* of α just if either $\phi = I$, or $\phi \in \text{Dil}(\alpha)$ and ϕ has no fixed points. The set of translations of α is denoted $\text{Tr}(\alpha)$.

Definition 12.2.6 (Homothety). Let α be an affine plane. Let $\phi \in \text{Aut}(\alpha)$. Then ϕ is called a *homothety* of α with centre O just if either $\phi = I$, or $\phi \in \text{Dil}(\alpha)$ and O is a fixed point of ϕ . The set of homotheties of α with centre O is denoted $\text{Ht}_O(\alpha)$.

Chapter 13

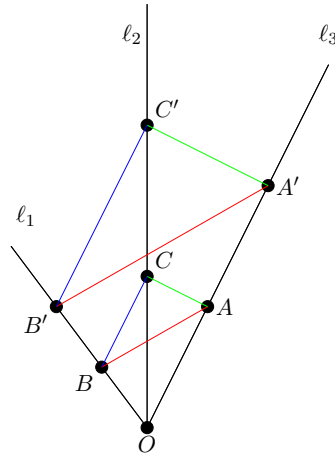
Desarguesian and Pappian Planes

13.1 Desargues's Axiom

We next turn to two central geometrical conditions on planes. These pick out the Desarguesian and Pappian affine planes.

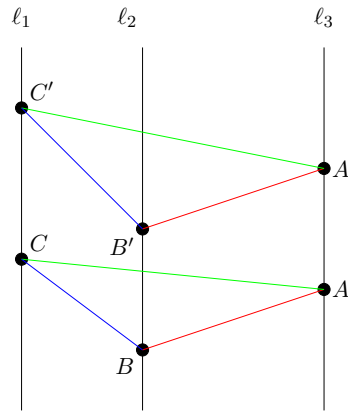
Definition 13.1.1 (Major Affine Desargues's Property). Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. We say that α has *the Major Affine Desargues's Property* just if, for any three copunctual lines ℓ_1, ℓ_2, ℓ_3 , just if the following holds: suppose $A, A' \in \ell_1$, $B, B' \in \ell_2$, and $C, C' \in \ell_3$, as indicated. Let $\triangle ABC$ and $\triangle A'B'C'$ be the corresponding triangles, as indicated. Then, if any two of the coloured edges are parallel, then so is the third. E.g., if $AB \parallel A'B'$ and $AC \parallel A'C'$, then $BC \parallel B'C'$:

Figure 13.1: Major Affine Desargues Property



Definition 13.1.2 (Minor Affine Desargues Property). Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. We say that α has *the Minor Affine Desargues's Property* just if, for any three

parallel lines ℓ_1, ℓ_2, ℓ_3 , just if the following holds: suppose $A, A' \in \ell_1$, $B, B' \in \ell_2$, and $C, C' \in \ell_3$, as indicated. Let $\triangle ABC$ and $\triangle A'B'C'$ be the corresponding triangles, as indicated. Then, if any two of the coloured edges are parallel, then so is the third. E.g., if $AB \parallel A'B'$ and $AC \parallel A'C'$, then $BC \parallel B'C'$:

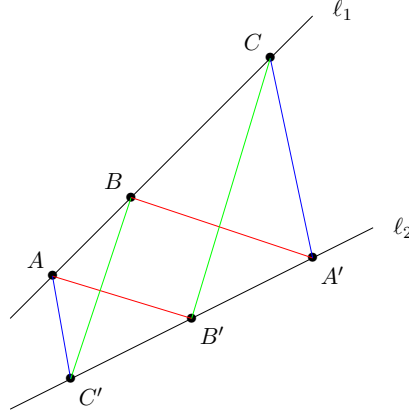
Figure 13.2: **Minor Affine Desargues Property**

Regarding terminology, Blumenthal (1961): 71 refers to the Minor Affine Desargues Property as the First Desargues Property. Goldblatt (1987) refers to it as the Small Affine Desargues Property.

Theorem 13.1.3. The Major Affine Desargues Property implies the Minor one.

13.2 Pappus's Axiom

Figure 13.3: Pappus property



13.3 Pappus Implies Desargues

It can be shown that the Pappus property implies the Desargues property. This is Hessenberg's Theorem (Hessenberg (1905)):

Theorem 13.3.1 (Hessenberg's Theorem). Let α be a Pappian affine plane. Then α is Desarguesian.

Proof. I refer the reader to Blumenthal (1961): 100–103; or Coxeter (1969): 238. Or see Behnke et al. (1984): 68–74 (Hessenberg's Theorem is labelled “Consequence” on page 74).

□

The original proof in Hessenberg (1905) had a small gap. A complete proof appeared in Cronheim (1953). The converse of Hessenberg's Theorem does not hold. Since the issue is the commutativity of the underlying multiplicative group, a counterexample requires a *non-commutative* division ring. Suppose we take $\mathbb{H}$, the division ring of quaternions. Then the corresponding affine plane $\mathbb{A}^2(\mathbb{H})$ is Desarguesian, but not Pappian. (This conclusion relies on the main coordinatization theorem for affine planes, Theorem ?? below.)

Below we prove Hessenberg's Theorem via the Representation theorem for affine geometry.

13.4 Finite Desarguesian Planes are Pappian

The following theorem I shall call Tecklenburg's Theorem.

Theorem 13.4.1 (Tecklenburg (1987)). Any *finite* Desarguesian affine plane is Pappian.

Chapter 14

Coordinatization of Affine Planes

14.1 Geometrical Addition and Multiplication

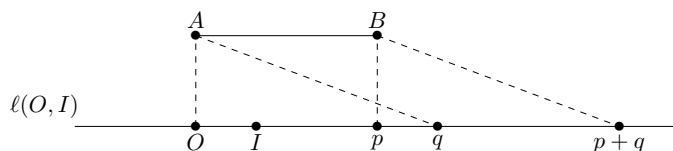
14.1.1 Addition

Above, we explained how we get from an algebraic structure to a geometrical one. In fact, this connection operates in reverse too. The following definitions are explained very clearly in Bennett (1995): for $+$ at p. 48 and for $\times$ at p. 62. Also see Goldblatt (1987): 23–27. We make the following assumptions in the definitions below.

- | | |
|-----------|--|
| 0.510 (i) | Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane. |
| (ii) | Let $O, I \in \mathbb{P}$, with $O \neq I$. |
| (iii) | Let $\ell = \ell(O, I)$. |

Definition 14.1.1. We define a binary operation $+$ on ℓ as follows. Let A be an arbitrarily chosen point not in ℓ . By Playfair, we define the line ℓ' passing through A and parallel to ℓ . By (A1), we construct the line $\ell(O, A)$. By Playfair, we construct the line parallel to $\ell(O, A)$ and passing through p . This line intersects ℓ' at a unique point, B . Now consider the line $\ell(A, q)$. By Playfair, there is a unique line passing through B and parallel to $\ell(A, q)$. This line intersects ℓ as a unique point: we declare this point to be $p + q$:

Figure 14.1: **Addition on a line**

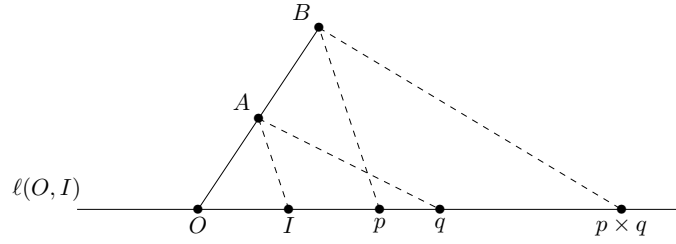


Suppose we have defined $+$ as above, on the line $\ell(O, I)$. Below, we will sketch the proof of the remarkable fact that if α is Desarguesian, then the structure $(\ell, O, +)$ is an *Abelian group*.

14.1.2 Multiplication

Definition 14.1.2. We define an operation $\times$ on ℓ as follows:

Figure 14.2: **Multiplication on a line**



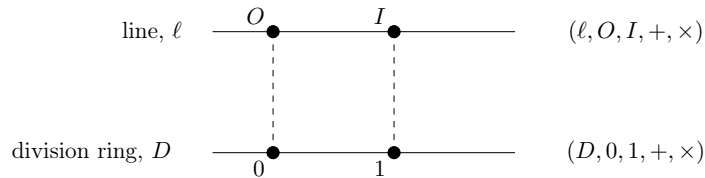
14.2 Line Representation Theorems

For the following results, we make the same assumptions as above, that $\alpha = (\mathbb{P}, \mathcal{L})$ is an affine plane, and $O, I \in \mathbb{P}$, with $O \neq I$, and we let $\ell = \ell(O, I)$. We assume that addition and multiplication are defined (on ℓ) as above. We shall also overload by letting $\ell = (\ell, O, I, +, \times)$.

We now give what I call the two main line representation theorems. The proofs of these are really very complicated and involved, considering many cases, and usually with a fair number of construction points and lines. For details, see Bennett (1995): 48–72, especially Theorem 1, p. 72.

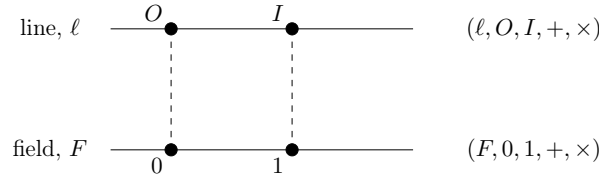
Theorem 14.2.1. If α is Desarguesian, there is a division ring D such that $\ell \cong D$.

Figure 14.3: **Line Representation Theorem (Desarguesian case)**



Theorem 14.2.2. If α is Pappian, there is a field F such that $\ell \cong F$.

Figure 14.4: **Line Representation Theorem (Pappian case)**



14.3 Lines are Isomorphic

14.4 Coordinatization of Affine Planes

The corollary is the following:

Theorem 14.4.1 (Representation Theorem for Synthetic Affine Planes). Let α be an affine plane. Then:

- 0.512 (1) If α is Desarguesian, there exists a division ring D such that $\alpha \cong \mathbb{A}^2(D)$.
- (2) If α is Pappian, there exists a field F such that $\alpha \cong \mathbb{A}^2(F)$.

14.5 The Unity of Algebra and Geometry

An immediate application of Theorem 14.4.1 is an *algebraic* proof of Hessenberg's Theorem:

Theorem 14.5.1 (Hessenberg's Theorem). Any Pappian affine plane is Desarguesian.

Proof. Let $\alpha = (\mathbb{P}, \mathcal{L})$ be a Pappian affine plane. By the Representation theorem 14.4.1, there exists a field F such that $\alpha \cong \mathbb{A}^2(F)$. Now a field F is a division ring. Hence, by Lemma F.3.1, $\mathbb{A}^2(F)$ is Desarguesian, and consequently, so is α . $\square$

Thus, we have proved a geometrical result, via our algebraic representation theorem. We can reason in the opposite direction, so to speak. That is, we can use a geometrical theorem to prove an algebraic theorem. A nice example is that we obtain a *geometrical* proof of Wedderburn's Little Theorem from Tecklenburg's Theorem:

Theorem 14.5.2 (Wedderburn's Little Theorem). Let D be a finite division ring. Then D is a field.

Proof. Let D be a finite division ring. By Lemma F.3.1, $\mathbb{A}^2(D)$ is a finite Desarguesian affine plane, and any line in $\mathbb{A}^2(D)$ is isomorphic to D . By Tecklenburg's Theorem, $\mathbb{A}^2(D)$ is *Pappian*. By the Representation Theorem 14.4.1, there is a field F , such that $\mathbb{A}^2(D) \cong \mathbb{A}^2(F)$. Thus, any line in $\mathbb{A}^2(D)$ is isomorphic to F . Hence, $D \cong F$. And, hence, D is a field. $\square$

This fascinating interplay—“essential unity”, one may call it—between geometrical and algebraic structures is summarized nicely by Leonard Blumenthal, in the Preface to his textbook *A Modern View of Geometry* (Blumenthal (1961)):

Our principal concern is with the postulational geometry of planes, with the greatest emphasis on the coordinatization of affine and projective planes. It is shown in detail how the *algebraic* structure of an abstract “coordinate” set is determined by the *geometric* structure that postulates impose on an abstract “point” set. It is seen that the process can be reversed: a “geometric” entity (*plane*) arises from an initially given “algebraic” entity (*ternary ring*), and the geometric properties of the one are logical consequences of the algebraic properties of the other. The essential unity of algebra and geometry is thus made quite clear. (Blumenthal (1961): vi–vii)

Figure 14.5: **The Unity of Algebra and Geometry**

Algebraic structure	$\Leftrightarrow$	Geometrical structure
Division ring	$\Leftrightarrow$	Desarguesian affine plane.
Field	$\Leftrightarrow$	Pappian affine plane.

Chapter 15

Coordinatization of Projective Planes

Part III

Betweenness Geometry

Chapter 16

Betweenness

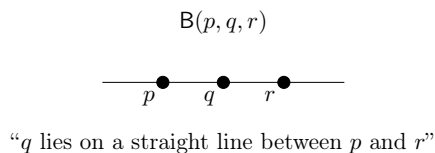
16.1 Pasch and Hilbert

There is a great deal more that can be said about incidence geometry, and finite geometries more generally. But we skip that, and refer the reader to other detailed sources.¹

Likewise, there is a great deal more to be said about affine and projective planes. But we skip that, and refer the reader to other detailed sources.²

We next turn instead to the theory—or, one should say, theories—of *betweenness*:

Figure 16.1: **Betweenness**



Victor Pambuccian writes:

Unlike most concepts of elementary geometry, whose origin is shrouded in the ineluctable fog of all cultural beginnings, the history of the notion of betweenness can be traced back to one person, one book, and one year: Moritz Pasch, *Vorlesungen über neuere Geometrie*, and 1882. (Pambuccian (2011): 24)

The origin of such investigations began with Pasch. These are incorporated as what David Hilbert called “the order axioms” in his celebrated monograph Hilbert (1899).

These pertain to the geometrical relation of “*betweenness*”.

Hilbert notes:

¹ Moorhouse, Moorhouse (2017), Dembowski (1968).

² Bennett (1995), Coxeter (1969), Hughes & Piper (1973), ...

These axioms were first studied in detail by M. Pasch in his *Vorlesungen über neue Geometrie* (Leipzig, 1882; Pasch (1882)). In particular, Axiom II, 4 is essentially due to him. (Hilbert (1899): 4, footnote)

Shortly after Pasch's work, Peano provided an axiomatic formulation in Peano (1889). As Pambuccian records:

Still in the language of points belonging to segments is Peano's (Peano (1889)) axiomatics of ordered geometry, which continues and is influenced by the axiomatic analysis of the notion of order begun by Pasch. ... With Hilbert's Hilbert (1899), the next influential book on the foundations of geometry, after the works of Pasch and Peano, order, in the explicit form of "betweenness", becomes only one aspect of Euclidean geometry, alongside incidence, segment-congruence, and angle-congruence. (Pambuccian (2011): 25)

The betweenness relation is a 3-place relation, expressed by a 3-place predicate:

$$B(p, q, r) \quad \text{(point) } y \text{ lies between (point) } x \text{ and (point) } z$$

To remind the reader, we write out again Hilbert's incidence axioms (for the plane):

Table 16.1: **Hilbert's (plane) incidence axioms**

- I, 1. For every two points A and B there exists a line a that contains each of the points A, B .
- I, 2. For every two points A and B there exists no more than one line that contains each of the points A, B .
- ...
- I, 3. There exist at least two points on a line. There exist at least three points that do not lie on a line. (Hilbert (1899): 3-4)

And then Hilbert's axioms of order are:³

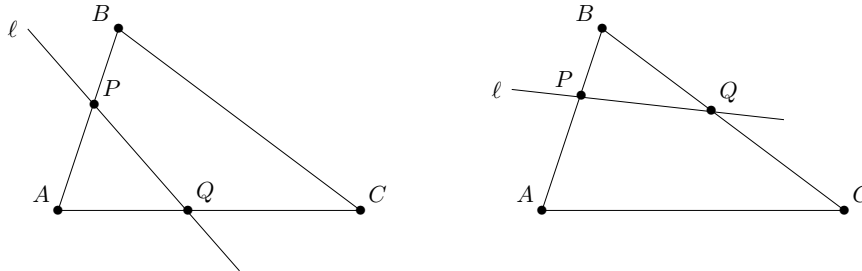
³ See also Goldblatt (1987): 70-71.

Table 16.2: Hilbert's Axioms of Order

- II, 1. If a point B lies between a point A and a point C , then the points A, B, C are three distinct points of a line, and B then also lies between C and A .
- II, 2. For two points A and C , there always exists at least one point B on the line AC such that C lies between A and B .
- II, 3. Of any three points on a line there exists no more than one which lies between the other two.
- II, 4. Let A, B, C be three points that do not lie on a line and let a be a line in the plane ABC which does not meet any of the points A, B, C . If the line a passes through a point of the segment AB , it also passes through a point of the segment AC , or through a point of the segment BC .

As Hilbert notes in the footnote on page 4, the fourth of these axioms is a version of *Pasch's axiom*, or the Classical Pasch axiom, as Szmielew calls it (Szmielew (1983); 151). This was first given in Pasch (1882). It is order axiom II,4 of Hilbert (1899). It says: given three distinct non-collinear points A, B, C (and thus a triangle), if a line ℓ intersects the segment AB , then either ℓ intersects the segment AC or ℓ intersects the segment BC :

Classical Pasch axiom



Remark 16.1.1. Pasch's, Peano's and Hilbert's betweenness relation is *sharp* (or "strict", or "exclusive") betweenness. I write this $B^\#$. Likewise, Veblen's, Huntington's and Coxeter's betweenness in Coxeter (1969) are $B^\#$.

On the other hand, the Tarski-Szmielew-Szczerba-Schwabhauser betweenness relation is blunt (or non-strict, or inclusive). Our betweenness relations will be blunt, like Tarski's. I shall write this B . But these are interdefinable (much as $\leq$ and $<$ are interdefinable). These interdefinitions are:

$$B(p, q, r) := (B^\#(p, q, r) \vee p = q \vee q = r) \quad (16.1.1)$$

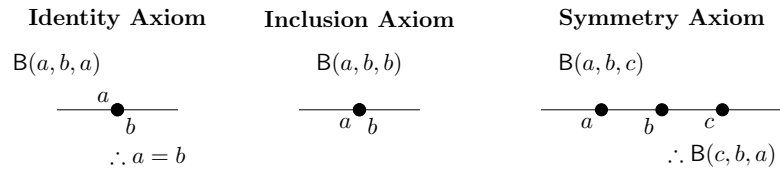
$$B^\#(p, q, r) := (B(p, q, r) \wedge p \neq q \wedge q \neq r) \quad (16.1.2)$$

□

16.2 Betweenness Structures

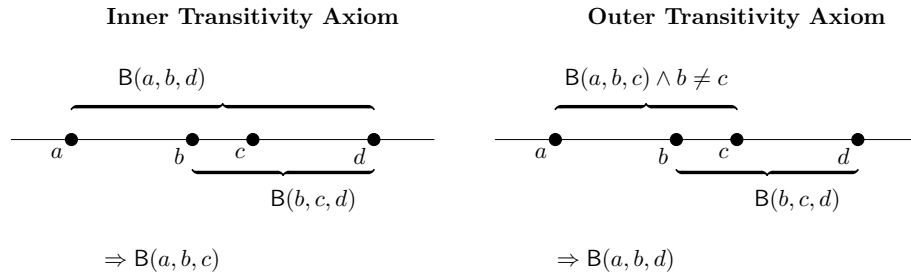
To begin with, I wish to identify six axioms which seem to be “analytic” of the concept of (straight line, inclusive) betweenness. The first three of these are:

Figure 16.2: **Identity, Inclusion, Symmetry**



Next, we have Transitivity and Connectivity, which each have “Outer” and “Inner” forms (see Tarski & Givant (1999)). The Transitivity axioms may be depicted:

Figure 16.3: **Transitivity Axioms**



The Outer and Inner Connectivity axioms may be depicted:

Figure 16.4: **Outer Connectivity Axiom**

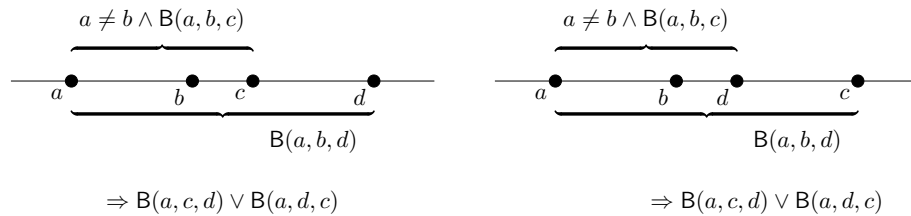
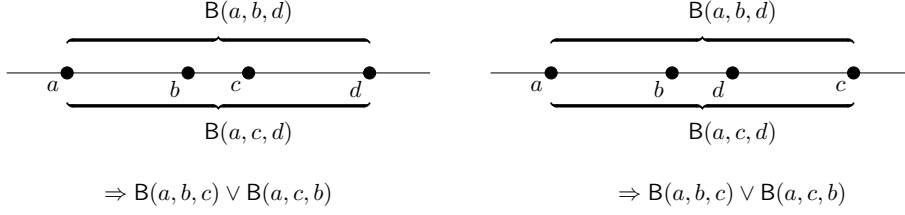


Figure 16.5: Inner Connectivity Axiom



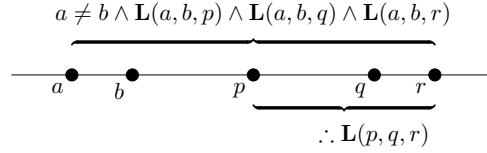
Syntactically, the Inner axioms are a bit simpler than the Outer ones, as the Inner axioms lack the inequality clause. Below, we adopt the two Inner axioms.

Definition 16.2.1 (Collinearity). Let $\mathbb{P}$ be a set and $B \subseteq \mathbb{P}^3$. For $p, q, r \in \mathbb{P}$, we say that p, q, r are *collinear* just if either $B(p, q, r)$ or $B(q, r, p)$ or $B(r, p, q)$. We write this: $\mathbf{L}(p, q, r)$. Thus,

$$\mathbf{L}(p, q, r) := B(p, q, r) \vee B(q, r, p) \vee B(r, p, q) \quad (16.2.1)$$

The sixth axiom is what I call *Szmielew's Collinearity Axiom*:⁴

Figure 16.6: Szmielew's Collinearity Axiom



Definition 16.2.2 (Betweenness structure). Let $\mathbb{P}$ be a set and $B \subseteq \mathbb{P}^3$. The structure $(\mathbb{P}, B)$ is a (blunt) *betweenness structure* just if the following six axioms hold:

Table 16.3: Axioms for a (blunt) betweenness structure

Identity	$B(a, b, a) \rightarrow a = b.$
Inclusion	$B(a, b, b)$
Symmetry	$B(a, b, c) \rightarrow B(c, b, a).$
Inner Transitivity	$B(a, b, d) \wedge B(b, c, d) \rightarrow B(a, b, c).$
Inner Connectivity	$B(a, b, d) \wedge B(a, c, d) \rightarrow (B(a, b, c) \vee B(a, c, b)).$
Szmielew	$a \neq b \wedge \mathbf{L}(a, b, p) \wedge \mathbf{L}(a, b, q) \wedge \mathbf{L}(a, b, r) \rightarrow \mathbf{L}(p, q, r).$

⁴ See Ax 1.1.2 in Szmielew (1983): 12. See also our Definition 10.1.1.

As we noted, the sixth axiom **Szmielew** is what we term Szmielew's *Collinearity Axiom*. Szmielew (1983) (pp 110–111) gives slightly different axioms from our basic ones. We come back to this below in §17.7. Her intention, however, is to cover *one-dimensional betweenness structures*.

Definition 16.2.3 (Line). Let $\beta = (\mathbb{P}, B)$ be a betweenness structure. A *line* in β is a subset $\ell \subseteq \mathbb{P}$ with $|\ell| \geq 2$ and, for any $p, q, r \in \ell$, we have $\mathbf{L}(p, q, r)$.

Definition 16.2.4 ($\ell(p, q)$). Let $\beta = (\mathbb{P}, B)$ be a betweenness structure, and let $p, q \in \mathbb{P}$. We define:

$$\ell(p, q) := \{r \in \mathbb{P} \mid \mathbf{L}(p, q, r)\} \quad (16.2.2)$$

Lemma 16.2.5. Let $\beta = (\mathbb{P}, B)$ be a betweenness structure with $|\mathbb{P}| \geq 2$. Let $p, q \in \mathbb{P}$, with $p \neq q$. Then there exists a line ℓ in β such that $p \in \ell$ and $q \in \ell$.

Proof. By Definition 16.2.1, we have $\mathbf{L}(p, q, p)$ just if either $B(p, q, p)$ or $B(q, p, p)$ or $B(p, p, q)$. By **Inclusion**, we have $B(p, p, q)$. So, $p \in \ell(p, q)$. Similarly, we have $\mathbf{L}(p, q, q)$ just if either $B(p, q, q)$ or $B(q, q, p)$ or $B(q, p, q)$. By **Inclusion**, we have $B(q, q, p)$. So, $q \in \ell(p, q)$. We have $\ell(p, q) \subseteq \mathbb{P}$ and $|\ell(p, q)| \geq 2$. So, $\ell(p, q)$ is a line. $\square$

Thus, any two distinct points in a betweenness structure are joined by at least one line. Thus:

Lemma 16.2.6. If $p \neq q$, then $\ell(p, q)$ is a line.

Lemma 16.2.7. $\ell(p, q) = \ell(q, p)$.

Proof. By Definition 16.2.1, we have: $\mathbf{L}(p, q, r)$ if and only if $\mathbf{L}(q, p, r)$. So, $\ell(p, q) = \{r \in \mathbb{P} \mid \mathbf{L}(p, q, r)\} = \{r \in \mathbb{P} \mid \mathbf{L}(q, p, r)\} = \ell(q, p)$. $\square$

Lemma 16.2.8. $\mathbf{L}(a, b, a)$.

Lemma 16.2.9. $\mathbf{L}(a, b, b)$.

Lemma 16.2.10. $\ell(p, p) = \mathbb{P}$.

Proof. By **Inclusion**, we have, for all $p, r \in \mathbb{P}$, $\mathbf{L}(p, p, r)$. Hence, $\{r \in \mathbb{P} \mid \mathbf{L}(p, p, r)\} = \mathbb{P}$. Hence, $\ell(p, p) = \mathbb{P}$. $\square$

Definition 16.2.11 (Associated incidence structure). Let $\beta = (\mathbb{P}, B)$ be a betweenness structure. Let $\mathcal{L}$ be the set of lines in β . Then $(\mathbb{P}, \mathcal{L})$ is the *associated incidence structure* of β .

Lemma 16.2.12. Let $\beta = (\mathbb{P}, B)$ be a betweenness structure. Let $(\mathbb{P}, \mathcal{L})$ be the associated incidence structure of β . Let $\ell, \ell' \in \mathcal{L}$ be lines such that $\ell \subseteq \ell'$. Then $\ell = \ell'$.

Proof. Let $p, q, a, b \in \mathbb{P}$, with $p \neq q$ and $a \neq b$. Let $\ell = \ell(a, b)$ and let $\ell' = \ell(p, q)$. Suppose $\ell \subseteq \ell'$. Suppose $c \in \ell'$. We claim $c \in \ell$.

Since $c \in \ell'$, we have: $\mathbf{L}(p, q, c)$. And, since $\ell \subseteq \ell'$, for any r ,

$$\mathbf{L}(a, b, r) \rightarrow \mathbf{L}(p, q, r) \quad (16.2.3)$$

Hence, $\mathbf{L}(a, b, a) \rightarrow \mathbf{L}(p, q, a)$, and $\mathbf{L}(a, b, b) \rightarrow \mathbf{L}(p, q, b)$. But $\mathbf{L}(a, ba)$ by Lemma 16.2.8 and $\mathbf{L}(a, b, b)$ by Lemma 16.2.9. So, $\mathbf{L}(p, q, a)$ and $\mathbf{L}(p, q, b)$. Hence,

$$p \neq q \wedge \mathbf{L}(p, q, a) \wedge \mathbf{L}(p, q, b) \wedge \mathbf{L}(p, q, c) \quad (16.2.4)$$

By Szmielew, we conclude $\mathbf{L}(a, b, c)$. Hence, $c \in \ell$. $\square$

Lemma 16.2.13. Let $\beta = (\mathbb{P}, B)$ be a betweenness structure. Let $(\mathbb{P}, \mathcal{L})$ be the associated incidence structure of β . Then, for any two distinct points $p, q \in \mathbb{P}$, there is a *unique line* $\ell \in \mathcal{L}$ such that $p \in \ell$ and $q \in \ell$.

Proof. Let $p, q \in \mathbb{P}$, with $p \neq q$. Let $L = \{r \in \mathbb{P} \mid \mathbf{L}(p, q, r)\}$. So, for any r ,

$$r \in L \leftrightarrow \mathbf{L}(p, q, r). \quad (16.2.5)$$

As above, L is a line. Suppose L' is a line, with $p \in L'$ and $q \in L'$. We claim $L' = L$. By the definition $|L'| \geq 2$ and, for any $a, b, c \in L'$, we have $\mathbf{L}(a, b, c)$. Let $a, b \in L'$ with $a \neq b$. So, for any r ,

$$r \in L' \leftrightarrow \mathbf{L}(a, b, r). \quad (16.2.6)$$

Since $p \in L'$ and $q \in L'$, we have $\mathbf{L}(a, b, p)$ and $\mathbf{L}(a, b, q)$. Now let r be any point in L' . So, we have:

$$\mathbf{L}(a, b, p) \wedge \mathbf{L}(a, b, q) \wedge \mathbf{L}(a, b, r). \quad (16.2.7)$$

By Szmielew, we conclude $\mathbf{L}(p, q, r)$, and therefore, $r \in L$. Thus, $L' \subseteq L$. By Lemma 16.2.12, $L' = L$. $\square$

Thus, any two points in a betweenness structure are joined by exactly one line.

Consulting Definition 7.3.6, we obtain:

Lemma 16.2.14. Let $\beta = (\mathbb{P}, B)$ be a betweenness structure. Let $(\mathbb{P}, \mathcal{L})$ be the associated incidence structure of β . Then $(\mathbb{P}, \mathcal{L})$ is a *linear space*.

We can calculate the properties of small betweenness structures $\beta = (\mathbb{P}, B)$, with cardinality 1, 2 or 3:

Table 16.4: Small betweenness structures

Card	Points	Betweenness relation	Betweenness structures
1	$\mathbb{P} = \{a\}$	$B = \{(a, a, a)\}$	One structure.
2	$\mathbb{P} = \{a, b\}$	$B = \{(a, a, a), (a, a, b), (a, b, b), (b, b, a), (b, a, a), (b, b, b)\}$	One structure.
3	$\mathbb{P} = \{a, b, c\}$	$B = B_0$	Three lines.
3	$\mathbb{P} = \{a, b, c\}$	$B = B_0 \cup \{(a, b, c), (c, b, a)\}$	One line.
3	$\mathbb{P} = \{a, b, c\}$	$B = B_0 \cup \{(b, c, a), (a, c, b)\}$	One line.
3	$\mathbb{P} = \{a, b, c\}$	$B = B_0 \cup \{(c, a, b), (b, a, c)\}$	One line.

where B_0 is the following set of fifteen triples:

$$\begin{aligned}
 &(a, a, a), (a, a, b), (a, a, c), (a, b, b), (a, c, c), (b, a, a) \\
 &\quad (b, b, a), (b, b, b), (b, b, c), (b, c, c), (c, a, a), (c, b, b) \\
 &\quad (c, c, a), (c, c, b), (c, c, c) \quad (16.2.8)
 \end{aligned}$$

Definition 16.2.15. Let $\beta = (\mathbb{P}, B)$ be a betweenness structure, and let $\mathbf{L}$ be defined as above. Then we say that β is *one-dimensional* just if, for any $a, b, c \in \mathbb{P}$:

$$\mathbf{L}(a, b, c) \quad (16.2.9)$$

16.3 Betweenness and Order

It may, at first, seem odd that a betweenness relation is related to an *order*—like a partial order or a linear order. However, one can show that, under certain circumstances, these are interdefinable. The best exposition I know of pertaining to the interdefinability of betweenness and order appears in Wanda Szmielew’s monograph Szmielew (1983), in §???.

Definition 16.3.1. Let $\mathbb{P}$ be a set and let $\leq \subseteq \mathbb{P}^2$. So $L = (\mathbb{P}, \leq)$ is a binary relational structure. (Despite our overloading notation, we make no further assumptions about the relation $\leq$!) Let us define a three-place relation $B_{\leq}$ on $\mathbb{P}$ by:

$$B_{\leq}(p, q, r) := (p \leq q \wedge q \leq r) \vee (r \leq q \wedge q \leq p) \quad (16.3.1)$$

Lemma 16.3.2. Let $A = (\mathbb{P}, \leq)$ be a binary relational structure, and let $B_{\leq}$ be defined as above. Let $\beta = (\mathbb{P}, B_{\leq})$. Then:

$$0.512 (*) \quad A \text{ is a linear order if and only if } \beta \text{ is a one-dimensional betweenness structure.}$$

Lemma 16.3.3. Let $\beta = (\mathbb{P}, B)$ be a betweenness structure. Let $O, I \in \mathbb{P}$, with $O \neq I$. (These are called *parameter points*.) Let $\ell := \ell(O, I) = \{p \in \mathbb{P} \mid \mathbf{L}(O, I, p)\}$. Then: $(\ell, B \upharpoonright_\ell)$ is a one-dimensional betweenness structure.

Definition 16.3.4. Let $\beta = (\mathbb{P}, B)$ be a betweenness structure. Let $O, I \in \mathbb{P}$, with $O \neq I$. Let us define a binary relation $\leq$ on $\mathbb{P}$ by:⁵

$$p \leq q \quad := \quad p = q \vee (B(p, O, I) \wedge \neg B(q, p, I)) \vee (\neg B(p, O, I) \wedge B(O, p, q)) \quad (16.3.2)$$

Lemma 16.3.5. Let $\beta = (\mathbb{P}, B)$ a ternary relational structure. Let O, I be distinct points in $\mathbb{P}$. Let $\leq$ be defined in β as above. Then:

- | | |
|------------|---|
| 0.512 (**) | β is a one-dimensional betweenness structure if and only if $(\ell, \leq)$ is a linear order. |
| 0.510 (2) | If L is an unbounded linear order, then β is an unbounded one-dimensional betweenness structure. |
| (3) | If L is a dense linear order without endpoints (a DLOWE), then β is an unbounded one-dimensional betweenness structure. |
| 0.510 (2) | If $(\mathbb{P}, B)$ is an unbounded one-dimensional betweenness structure, $(\ell, \leq)$ is an unbounded linear order. |
| (3) | If $(\mathbb{P}, B)$ is a dense unbounded one-dimensional betweenness structure, $(\ell, \leq)$ is a DLOWE. |

16.4 Ordered Affine Planes

Definition 16.4.1 (Ordered affine plane). Let $\alpha = (\mathbb{P}, \mathcal{L})$ be an affine plane, as per Definition 8.1.1. Let $B \subseteq \mathbb{P}^3$ be a three-place relation on $\mathbb{P}$. Define $\mathbf{L}$ by: $\mathbf{L}(a, b, c) := B(a, b, c) \vee B(b, c, a) \vee B(c, a, b)$. Then (α, B) is called an *ordered affine plane* just if $(\mathbb{P}, B)$ is a blunt betweenness structure.

Definition 16.4.2 (Ordered affine plane (Szmielew)). Let $\alpha = (\mathbb{P}, \parallel)$ be a parallelity plane, as per Definition 10.3.1. Let the line relation $\mathbf{L} \subseteq \mathbb{P}^3$ be defined by Definition 10.3.2. Let $B \subseteq \mathbb{P}^3$ be a three-place relation on $\mathbb{P}$. Then B is called a *betweenness relation* on the plane α just if:

- | | |
|-----------|--|
| 0.510 (i) | $B \subseteq \mathbf{L}$. |
| (ii) | $B \upharpoonright_\ell$ is a betweenness relation on each line ℓ . |

where, in (ii), “betweenness relation on each line” means that Szmielew’s axioms hold (see Definition 17.7.1).

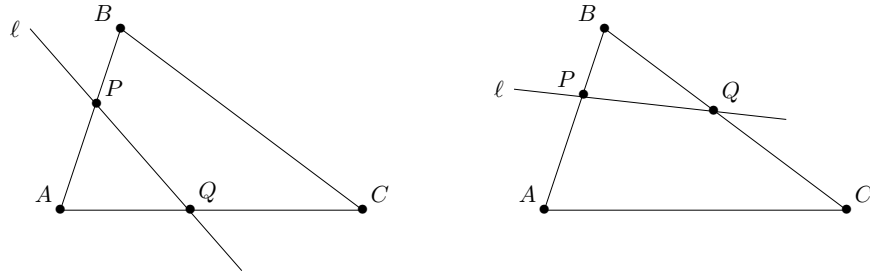
The structure $(\mathbb{P}, \parallel, B)$ is called an *ordered affine plane* in Szmielew’s sense.

⁵ The definition of $\leq$ in terms of betweenness is given in Tarski (1959), in the proof of Theorem 1.

16.5 Adding Geometrical Axioms

As we noted above, the classical Pasch axiom was first given in Pasch (1882), and is order axiom II,4 of Hilbert (1899). It says: given three distinct non-collinear points A, B, C (and thus a triangle), if a line ℓ intersects the segment AB , then either ℓ intersects the segment AC or ℓ intersects the segment BC :

Classical Pasch axiom



Chapter 17

Axiomatic Betweenness Geometry

17.1 The Language of Axiomatic Betweenness Geometry

In this chapter, we study essentially one axiom system, **Bet**, whose models are simply the (blunt) betweenness structures of Definition 16.2.2.

The informal geometric meaning of the expression “ $B(p, q, r)$ ” so far has been “the point q lies on a line inclusively between p and r ”, or “ q lies on the line segment pr ”. Next, we turn to the formal axiomatization of betweenness geometry.

Definition 17.1.1. We define the first-order language $L(\mathbf{B})$ of first-order betweenness geometry as follows:

Table 17.1: Language $L(\mathbf{B})$

Sorts	$\Sigma = \{\text{point}\}$
Variables	$\text{Var}(\text{point}) = \{p, q, r, p_1, p_2, \dots\}$.
Signature	$\sigma = \{\mathbf{B}\}$
$\mathbf{B}$	$\text{point} \Rightarrow \text{point} \Rightarrow \text{point} \Rightarrow \text{bool}$

Thus, $\mathbf{B}$ is a 3-place primitive predicate, and $L(\mathbf{B})$ has atomic formulas of the form of equations, and of the form $\mathbf{B}(p, q, r)$.

These following notions are formally definable using $\mathbf{B}$:¹

Definition 17.1.2. We define, using $\mathbf{B}$,

Sharp betweenness	$\mathbf{B}^\#(p, q, r) :=$	$\mathbf{B}(p, q, r) \wedge p \neq q \wedge q \neq r.$
Collinearity	$\mathbf{L}(p, q, r) :=$	$\mathbf{B}(p, q, r) \vee \mathbf{B}(q, r, p) \vee \mathbf{B}(r, p, q).$

¹ They are similar to those given in Field (1980), Ch. 6.

Coplanarity	$\text{co}_2(p, q, r, s) := \exists u ([\mathbf{L}(u, p, q) \wedge \mathbf{L}(u, r, s)] \vee [\mathbf{L}(u, p, r) \wedge \mathbf{L}(u, q, s)] \vee [\mathbf{L}(u, p, s) \wedge \mathbf{L}(u, q, r)])$
Parallelism	$pq \parallel rs := \text{co}_2(p, q, r, s) \wedge (p = q \vee r = s \vee [p \neq q \wedge r \neq s \wedge (\forall a (\mathbf{L}(p, q, a) \leftrightarrow \mathbf{L}(r, s, a)) \vee \forall a (\mathbf{L}(p, q, a) \rightarrow \neg \mathbf{L}(r, s, a)))]).$

Definition 17.1.3 (Parallelism (betweenness)).

$$pq \parallel rs := \overbrace{\text{co}_2(p, q, r, s)}^{p, q, r, s \text{ are coplanar}} \wedge \forall a \left(\overbrace{(\text{co}_1(p, q, a) \leftrightarrow \text{co}_1(r, s, a))}^{\ell(p, q) = \ell(r, s)} \vee \overbrace{(\text{co}_1(p, q, a) \rightarrow \neg \text{co}_1(r, s, a))}^{\ell(p, q) \cap \ell(r, s) = \emptyset} \right) \quad (17.1.1)$$

17.2 A Simple Conservativeness Lemma

From time to time, theories expressed in $L(\mathbf{B})$ will be extended with some new fixed constants, not present in $L(\mathbf{B})$. In the simplest case, two new constants, O and I , denoting (given a model) two distinct points. In the more general case, $O, I_1, \dots, I_n$. In Burgess & Rosen (1997) call these constants “benchmarks”. In addition, we impose some or other conditions on these constants. In the simplest case, the axiom: $O \neq I$. In the more general case, we impose an axiom corresponding to the claim that $O, I_1, \dots, I_n$ form some special kind of “frame”. Let’s call these “benchmark axioms”. These extensions by “benchmark constants” and “benchmark axioms” are, generally, *conservative extensions*. The basic means of proving this is given by the following lemma:

Lemma 17.2.1. Let σ be a fixed signature and T in $L(\sigma)$ a theory. Let O, I be fresh constants not in σ . So O, I do not appear in any axioms or theorems of T . Expand T with the new axiom $O \neq I$, to

$$T^+ := T + (O \neq I). \quad (17.2.1)$$

Suppose that $T \vdash \exists x \exists y (x \neq y)$. Then T^+ is a conservative extension of T .

Proof. We suppose $\varphi \in L(\sigma)$ and $T + (O \neq I) \vdash \varphi$. We claim: $T \vdash \varphi$. We verify this with a metatheoretic implication stream:²

$$T + (O \neq I) \vdash \varphi \quad (17.2.2)$$

$$\Rightarrow T \vdash (O \neq I) \rightarrow \varphi \quad (\text{DT}) \quad (17.2.3)$$

$$\Rightarrow T \vdash \forall x \forall y ((x \neq y) \rightarrow \varphi) \quad (\text{UGC}) \quad (17.2.4)$$

$$\Rightarrow T \vdash \exists x \exists y (x \neq y) \rightarrow \varphi \quad (\text{“Inverse prenex”}) \quad (17.2.5)$$

$$\Rightarrow T \vdash \varphi \quad (\text{MP}) \quad (\text{since } T \vdash \exists x \exists y (x \neq y)) \quad (17.2.6)$$

□

² (DT) is the Deduction Theorem. (UGC) is Universal Generalization on a Constant. “Inverse prenex” is the opposite operation of prenexing, and involves pushing prenexed initial quantifiers “inside” the subformula. (MP) is Modus Ponens.

17.3 The Axiom System *Bet*

Definition 16.2.2 of a (blunt) betweenness structure can be turned into a *formal system*, denoted *Bet*, as follows:

Definition 17.3.1 (Betweenness axioms). The axioms of *Bet* in $L(\mathbf{B})$ are:³

Table 17.2: Axioms of *Bet*

Inclusion	$\mathbf{B}(a, b, b)$
Identity	$\mathbf{B}(a, b, a) \rightarrow a = b.$
Symmetry	$\mathbf{B}(a, b, c) \rightarrow \mathbf{B}(c, b, a).$
Inner Transitivity	$B(a, b, d) \wedge B(b, c, d) \rightarrow B(a, b, c).$
Inner Connectivity	$B(a, b, d) \wedge B(a, c, d) \rightarrow (B(a, b, c) \vee B(a, c, b)).$
Szmielew	$a \neq b \wedge \mathbf{L}(a, b, p) \wedge \mathbf{L}(a, b, q) \wedge \mathbf{L}(a, b, r) \rightarrow \mathbf{L}(p, q, r).$

Hence:

Lemma 17.3.2. Let $\beta = (\mathbb{P}, B)$. Then β is a blunt betweenness structure if and only if $\beta \models \mathbf{Bet}$.

The following is an assembly of various statements expressible in $L(\mathbf{B})$. First five statements about collinearity:

Table 17.3: Statements about collinearity expressible in $L(\mathbf{B})$

(1) Satz4.11	$\mathbf{L}(a, b, c) \rightarrow (\mathbf{L}(b, c, a) \wedge \mathbf{L}(c, a, b) \wedge \mathbf{L}(c, b, a) \wedge \mathbf{L}(b, a, c) \wedge \mathbf{L}(a, c, b)).$
(2) Satz4.12	$\mathbf{L}(a, a, b).$
(3) Line Uniqueness	$(p \neq q \wedge \mathbf{L}(p, q, r) \wedge \mathbf{L}(p, q, s)) \rightarrow \mathbf{L}(p, r, s).$

Next, a series of statements about betweenness expressible in $L(\mathbf{B})$:

³ As noted, the axiom Szmielew is Ax1.1.2 (Szmielew (1983): 12).

Table 17.4: Statements about betweenness expressible in $L(B)$

(4) Bet to collinear	$B(a, b, c) \rightarrow L(a, b, c).$
(5) Reflexivity	$B(a, a, a).$
(6) Squeeze	$(B(a, b, c) \wedge B(b, a, c)) \rightarrow a = b.$
(7) Restriction	$(B(a, b, d) \wedge B(b, c, d)) \rightarrow B(a, b, c).$
(8) Overlap	$(B(a, b, d) \wedge B(b, c, d)) \rightarrow B(c, b, a).$
(9) Satz3.5z	$(B(a, b, d) \wedge B(b, c, d)) \rightarrow B(a, c, d).$
(10) Satz3.5	$(B(a, b, d) \wedge B(b, c, d)) \rightarrow (B(a, b, c) \wedge B(a, c, d)).$
(11) Satz3.6	$(B(a, b, c) \wedge B(a, c, d)) \rightarrow (B(b, c, d) \wedge B(a, b, d)).$
(12) Satz3.7	$(B(a, b, c) \wedge B(b, c, d) \wedge b \neq c) \rightarrow (B(a, c, d) \wedge B(a, b, d)).$
(13) Satz5.1	$(a \neq b \wedge B(a, b, c) \wedge B(a, b, d)) \rightarrow (B(a, c, d) \vee B(a, d, c)).$
(14) Satz5.2	$(a \neq b \wedge B(a, b, c) \wedge B(a, b, d)) \rightarrow (B(b, c, d) \vee B(b, d, c)).$
(15) Satz5.3	$(B(a, b, d) \wedge B(a, c, d)) \rightarrow (B(a, b, c) \vee B(a, c, b)).$
(16) Inner Transitivity I	$B(a, b, c) \wedge B(a, c, d) \rightarrow B(b, c, d).$
(17) Outer Transitivity	$B(a, b, c) \wedge B(b, c, d) \wedge b \neq c \rightarrow B(a, c, d).$
(18) Hilbert 1	$B^\#(p, q, r) \rightarrow (p \neq q \wedge p \neq r \wedge q \neq r \wedge B^\#(r, q, p)).$
(19) Hilbert 3	$B^\#(p, q, r) \rightarrow (\neg B^\#(q, r, p) \wedge \neg B^\#(q, p, r))$

Next, a series of statements about parallelism expressible in $L(B)$:

 Table 17.5: Statements about parallelism expressible in $L(B)$

(20) Bet to parallel	$B(a, b, c) \rightarrow ab \parallel ac.$
(21) ParallelRef1	$pq \parallel pq$
(22) ParallelRef2	$pq \parallel qp.$
(23) LineDef	$L(a, b, c) \leftrightarrow ab \parallel ac.$
(24) Par	$ab \parallel ba.$
(25) Ax.2.2.1	$ab \parallel cc.$

Finally some statements about coplanarity:

 Table 17.6: Statements about coplanarity expressible in $L(B)$

(26) Copl1	$\text{co}_2(p, q, r, r).$
(27) Copl2	$\text{co}_2(p, q, q, p).$
(28) Copl3	$\text{co}_2(p, q, r, s) \rightarrow \text{co}_2(q, r, s, p) \wedge \text{co}_2(r, s, p, q) \wedge \text{co}_2(s, p, q, r).$
(29) Copl4	$\text{co}_2(p, q, r, s) \rightarrow \text{co}_2(s, r, q, p) \wedge \text{co}_2(r, q, p, s) \wedge \text{co}_2(q, p, s, r).$

In $L(B)$, we can formulate Hilbert's order axioms. The first three are:

Hilbert 1	$B^\#(p, q, r) \rightarrow (p \neq q \wedge p \neq r \wedge q \neq r \wedge B^\#(r, q, p)).$
Hilbert 2	$a \neq c \rightarrow \exists b B^\#(a, c, b).$
Hilbert 3	$B^\#(p, q, r) \rightarrow (\neg B^\#(q, r, p) \wedge \neg B^\#(q, p, r))$

The first and third are (18) and (19) above. The fourth of Hilbert's order axioms is Pasch's Axiom, which we shall return to below.

Theorem 17.3.3. All 29 listed statements in Tables 17.3, 17.4, 17.5, and 17.6, are theorems of Bet.

In particular, Hilbert 1 and Hilbert 3 are theorems of Bet.

17.4 Coplanarity, Cohyperplanarity, ...

We need the definitions of the following geometrical notions:

- 0.512 (1) p, q, r, s are *coplanar*—i.e., they lie in a single plane (I write this as co_2).
- (2) p, q, r, s, w are *cohyperplanar*—i.e., they lie in a single 3-space (I write this as co_3).

These notions are indeed formally definable using $\mathbf{B}$:⁴

Definition 17.4.1. We define co_2 and co_3 as follows:

$$\text{co}_2(p, q, r, s) := \begin{cases} \exists u ([\text{co}_1(u, p, q) \wedge \text{co}_1(u, r, s)] \vee [\text{co}_1(u, p, r) \wedge \text{co}_1(u, q, s)] \\ \vee [\text{co}_1(u, p, s) \wedge \text{co}_1(u, q, r)] \end{cases} \quad (17.4.1)$$

$$\text{co}_3(p, q, r, s, w) := \begin{cases} \exists u ([\text{co}_2(u, p, q, r) \wedge \text{co}_1(u, s, w)] \vee [\text{co}_2(u, p, q, s) \wedge \text{co}_1(u, r, w)] \\ \vee [\text{co}_2(u, p, q, w) \wedge \text{co}_1(u, r, s)] \end{cases} \quad (17.4.2)$$

As we move to higher-dimensions, these definitions become more intricate. But there is recursive pattern to this intricacy, as follows:⁵

Definition 17.4.2. We define the formulas co_n recursively. $\text{co}_1(p, q, r)$ is already defined. Then:

$$\text{co}_n(p_1, \dots, p_{n+2}) := \exists p_0 \bigvee_{\langle p_1, \dots, p_{n+2} \rangle} (\text{co}_{n-1}(p_0, \dots, p_n) \wedge \text{co}_1(p_0, p_{n+1}, p_{n+2})) \quad (17.4.3)$$

where, for a given formula $\Theta(p_1, \dots, p_{n+2})$ with $n + 2$ free variables, the expression

⁴ They are similar to those given in Field (1980), Ch. 6.

⁵ These definitions are due to Kordos (1969). See Szczurba & Tarski (1979): 190. I change notation a little bit from their $\mathbf{L}_n$ to co_n .

$$\bigvee_{\langle p_1, \dots, p_{n+2} \rangle} \Theta(p_1, \dots, p_{n+2}) \quad (17.4.4)$$

is the disjunction of all formulas obtained by substituting permutations of the variables $\{p_1, \dots, p_{n+2}\}$ in the formula $\Theta(p_1, \dots, p_{n+2})$.

As Szczerba & Tarski (1979) note:

120 It is easy to see that co_1 denotes the notion of collinearity, co_2 the notion of coplanarity and, in general, co_n the notion of belonging to the same n -dimensional hyperplane. (Szczerba & Tarski (1979): 190. I change notation from their $\mathbf{I}_n$ to co_n)

Similar definitions,

- | | |
|-----------|---|
| 0.513 (1) | $\mathbf{P}(x_1, x_2, x_3, z)$: equivalent in our notation to $\text{co}_2(x_1, x_2, x_3, z)$. |
| (2) | $\mathbf{TF}(x_1, x_2, x_3, x_4, z)$: equivalent in our notation to $\text{co}_3(x_1, x_2, x_3, x_4, z)$. |
| (3) | $\mathbf{FF}(x_1, x_2, x_3, x_4, x_5, z)$: equivalent in our notation to $\text{co}_4(x_1, x_2, x_3, x_4, x_5, z)$. |

are given in Goldblatt (1987): 164–165.

17.5 *n*-Frames

Definition 17.5.1 (*n*-frame). An *n*-frame is an ordered $n + 1$ -tuple $(O, I_1, \dots, I_n)$ of points such that $\neg \text{co}_{n-1}(O, I_1, \dots, I_n)$.

In particular,

- | | |
|-------------|--|
| 210 2-frame | an ordered triple (O, I_1, I_2) which are not collinear; |
| 3-frame | an ordered quadruple (O, I_1, I_2, I_3) which are not coplanar; |
| 4-frame | an ordered quintuple (O, I_1, I_2, I_3, I_4) which are not co_3 . |

John Burgess refers to such systems of points as “benchmarks”: Burgess & Rosen (1997): 107. For example, in the two-dimensional case, one may imagine marking three non-collinear points O, I_1, I_2 on a bench. This will be a “2-frame”, and will determine a two-dimensional coordinate system, with O at the origin, and $\ell(O, I_1)$ the “ x -axis” and $\ell(O, I_2)$ the “ y -axis”. Goldblatt calls a 2-frame a “3-point”.

17.6 Dimension Axioms in $L(\mathbf{B})$

Equipped with the co_n formulas, we can express *dimension axioms* in $L(\mathbf{B})$:

Definition 17.6.1. In general, upper and lower dimension axioms for n -dimensional betweenness (ordered affine) geometry are expressed as follows:

$$\begin{array}{ll} \text{Lower dimension axiom} & \exists p_1 \dots \exists p_{n+1} \neg \text{co}_{n-1}(p_1, \dots, p_{n+1}) \\ \text{Upper dimension axiom} & \forall p_1 \dots \forall p_{n+2} \text{co}_n(p_1, \dots, p_{n+2}) \end{array}$$

These can be read informally:

$$\begin{array}{ll} \text{Lower dimension axiom} & \text{There are } n+1 \text{ points which are not } \text{co}_{n-1} \\ \text{Upper dimension axiom} & \text{Any } n+2 \text{ points are } \text{co}_n \end{array}$$

17.7 Szmielew's and Huntington's Axioms for One-Dimensional Betweenness

17.7.1 Szmielew

Szmielew (see Szmielew (1983), Chapter 7, §7.1; and §7.2) defines the notion of a *one-dimensional (blunt) betweenness relation* as follows:

Definition 17.7.1 (One-dimensional betweenness relation (Szmielew)). Szmielew's axioms for one-dimensional (blunt) betweenness are:

Table 17.7: Szmielew's axioms for one-dimensional (blunt) betweenness

OneDim	$\forall p \forall q \forall r \mathbf{L}(p, q, r).$
Identity	$B(p, q, p) \rightarrow p = q.$
Symmetry	$B(p, q, r) \rightarrow B(r, q, p).$
Inner Transitivity I	$B(a, b, c) \wedge B(a, c, d) \rightarrow B(b, c, d).$
Outer Transitivity	$B(a, b, c) \wedge B(b, c, d) \wedge b \neq c \rightarrow B(a, c, d).$

(Szmielew's "Outer Transitivity" is almost the same as our Transitivity.)

As Szmielew notes (p. 120), "each of these axioms is independent of the others". In §7.2, Szmielew provides fairly simple, finite, independence models for each. It is trivial to formalize in $L(\mathbf{B})$ Szmielew's axioms for one dimensional betweenness, described above. Call the formalized version of this **SzmBet**. One can show the following two results:⁶

Lemma 17.7.2. $\text{SzmBet} \vdash \text{Bet}.$

⁶ I verified both in the Isabelle ATP.

Lemma 17.7.3. $\text{Bet} + \text{OneDim} \vdash \text{SzmBet}$.

Thus, modulo the one-dimensionality axiom OneDim , the axiom systems Bet and SzmBet are logically equivalent.

17.7.2 Huntington

In (add two other papers) Huntington (1935), Edward Huntington gives the following five postulates for a *one-dimensional sharp betweenness relation*:⁷

Table 17.8: **Huntington's postulates for one dimensional (sharp) betweenness**

Postulate A	$\neq(a, b, c) \rightarrow (\text{B}^\#(a, b, c) \rightarrow \text{B}^\#(c, b, a))$
Postulate B	$\neq(a, b, c) \rightarrow (\text{B}^\#(b, a, c) \vee \text{B}^\#(c, a, b) \vee \text{B}^\#(a, b, c) \vee \text{B}^\#(c, b, a) \vee \text{B}^\#(a, c, b) \vee \text{B}^\#(b, c, a))$
Postulate C	$\neq(a, b, c) \rightarrow (\text{B}^\#(a, b, c) \rightarrow \neg \text{B}^\#(a, c, b))$
Postulate D	$\text{B}^\#(a, b, c) \rightarrow \neq(a, b, c)$
Postulate 9	$\neq(a, b, c, d) \wedge \text{B}^\#(a, b, c) \rightarrow (\text{B}^\#(a, b, d) \vee \text{B}^\#(d, b, c))$

From these postulates, Huntington derives a further 8 postulates (Postulate 1 to Postulate 8), and then provides a list of equivalent (independent) axiomatizations.

It is possible to formalize Huntington's postulate system, for one dimensional betweenness, described above. Call the formalized version of this HuntBet . We extend Bet with the explicit definition of $\text{B}^\#$:

$$\text{B}^\#(p, q, r) \quad := \quad (\text{B}(p, q, r) \wedge p \neq q \wedge q \neq r) \quad (17.7.1)$$

One may show:

Lemma 17.7.4. $\text{Bet} \vdash \text{Postulate A} \wedge \text{Postulate C} \wedge \text{Postulate D}$.

Moreover, if we add the axiom OneDim , we get Postulate B and Postulate 9 too:

Lemma 17.7.5. $\text{Bet} + \text{OneDim} \vdash \text{Postulate B} \wedge \text{Postulate 9}$.

Therefore,

Lemma 17.7.6. $\text{Bet} + \text{OneDim} \vdash \text{HuntBet}$.

Likewise we extend HuntBet with the explicit definition of B :

$$\text{B}(p, q, r) \quad := \quad (\text{B}^\#(p, q, r) \vee p = q \vee q = r) \quad (17.7.2)$$

$$(17.7.3)$$

Then we have:

⁷ Huntington's five postulates are given also in Struve (2024).

Lemma 17.7.7. $\text{HuntBet} \vdash \text{Bet}$.

Thus, modulo the one-dimensionality axiom **OneDim**, the axiom systems **Bet** and **HuntBet** are logically equivalent:

$$\text{SzmBet} \equiv (\text{Bet} + \text{OneDim}) \equiv \text{HuntBet} \quad (17.7.4)$$

In consequence, both **SzmBet** and **HuntBet** can reasonably be called “axioms for a betweenness relation on a line”. Henceforth, we take **Bet** as our basic system of axioms for dimension-free betweenness. Dimensionality axioms can be then be added; and, in the one-dimensional case, we recover Szmielew’s **SzmBet** and **HuntBet**.

17.8 Linear Order From Bet

In §16.3, we showed how a betweenness structure $(\mathbb{P}, B)$ induces a *linear order* on each line. Next, we do this formally, using the betweenness axioms **Bet**.

Definition 17.8.1. We extend the language $L(\mathbf{B})$ to $L(\mathbf{B}, O, I)$, by adding two new constants, O and I , which may be called “origin” and “unit”. We shall use the following abbreviation:

$$p \in \ell(O, I) \quad \text{for} \quad \mathbf{L}(O, I, p) \quad (17.8.1)$$

To simplify notation, we shall write “ ℓ ” instead of “ $\ell(O, I)$ ”. So:

$$(\forall p \in \ell) \varphi(p) \quad \text{is short for} \quad \forall p (\mathbf{L}(O, I, p) \rightarrow \varphi(p)) \quad (17.8.2)$$

Definition 17.8.2 (Order). We define $\leq$ as follows:⁸

$$p \leq q \quad := \quad p = q \vee (\mathbf{B}(p, O, I) \wedge \neg \mathbf{B}(q, p, I)) \vee (\neg \mathbf{B}(p, O, I) \wedge \mathbf{B}(O, p, q)) \quad (17.8.3)$$

And the corresponding sharp order is defined as usual:

$$p < q \quad := \quad p \leq q \wedge p \neq q \quad (17.8.4)$$

Definition 17.8.3. Let Bet^* be the theory $\text{Bet} + (O \neq I)$ in $L(\mathbf{B}, O, I)$.

⁸ As explained above, I have first seen this definition in Tarski (1959), where it appears in the proof of Tarski’s Theorem 1.

Theorem 17.8.4. The following metatheorems hold:

$$\text{Bet}^* \vdash (\forall p \in \ell) (p \leq p). \quad (17.8.5)$$

$$\text{Bet}^* \vdash (\forall p, q, r \in \ell) (p \leq q \leq r \rightarrow p \leq r). \quad (17.8.6)$$

$$\text{Bet}^* \vdash (\forall p, q \in \ell) (p \leq q \leq p \rightarrow p = q). \quad (17.8.7)$$

$$\text{Bet}^* \vdash (\forall p, q \in \ell) (p \leq q \vee q \leq p). \quad (17.8.8)$$

These four formulas clearly express reflexivity, transitivity, antisymmetry and connectedness on the line $\ell(O, I)$.

Definition 17.8.5. Let $\text{LO}^{\ell(a,b)}$ is the $L(\mathbf{B})$ -formula obtained by replacing O and I in the above four formulas by fresh distinct free variables, a, b , and then taking the conjunction of the result:

$$(\forall p \in \ell(a, b)) (p \leq p) \wedge \cdots \wedge (\forall p, q \in \ell(a, b)) (p \leq q \vee q \leq p). \quad (17.8.9)$$

Thus,

Theorem 17.8.6. $\text{Bet} \vdash a \neq b \rightarrow \text{LO}^{\ell(a,b)}$.

Proof. Substitute constant O for a and a constant I for b in $\text{LO}^{\ell(a,b)}$. The result is $\text{LO}^{\ell(O,I)}$. By Theorem 17.8.4, $\text{Bet} \vdash O \neq I \rightarrow \text{LO}^{\ell(O,I)}$. By (UGC), we conclude: $\text{Bet} \vdash a \neq b \rightarrow \text{LO}^{\ell(a,b)}$. $\square$

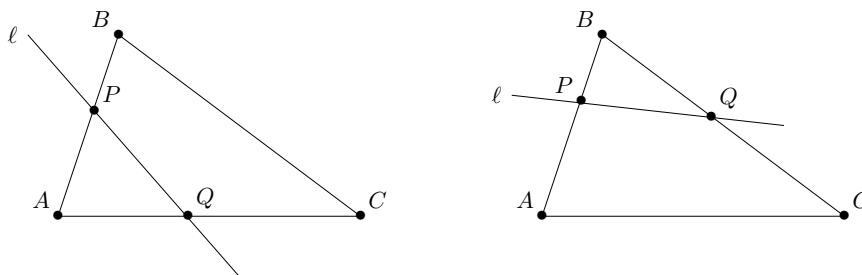
Chapter 18

Pasch's Axiom

18.1 Pasch's Axiom in $L(\mathbf{B})$

We explained the classical Pasch axiom (was first given in Pasch (1882), and is order axiom II,4 of Hilbert (1899)) in §>???

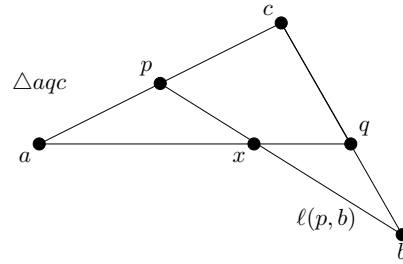
Classical Pasch axiom



Tarski and Givant in Tarski & Givant (1999) formulate a version they call the *Inner Pasch Axiom*:

$$\textbf{Inner Pasch Axiom} \quad \mathbf{B}(a, p, c) \wedge \mathbf{B}(b, q, c) \rightarrow \exists x (\mathbf{B}(p, x, b) \wedge \mathbf{B}(q, x, a)) \quad (18.1.1)$$

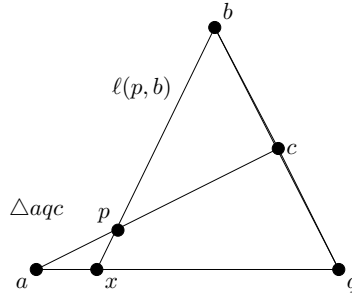
This is listed as Ax.7 on page 179 and depicted:

Figure 18.1: **Inner Pasch Axiom**


Tarski and Givant also formulate a version they call the *Outer Pasch Axiom*:

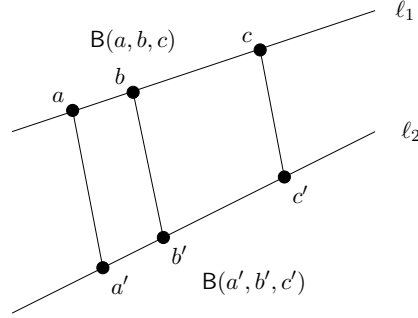
$$\textbf{Outer Pasch Axiom} \quad \mathcal{B}(a, p, c) \wedge \mathcal{B}(q, c, b) \rightarrow \exists x (\mathcal{B}(a, x, b) \wedge \mathcal{B}(b, p, x)) \quad (18.1.2)$$

This is listed as Ax.7₁ on page 180 and depicted:

Figure 18.2: **Outer Pasch Axiom**


Before discussing the machine-code level formalization of Pasch's axiom in $L(\mathcal{B})$, it is useful to draw attention to a certain *invariance* feature of Pasch's axiom. See Szmielew (1983): 147. Let lines ℓ_1 and ℓ_2 be given. Suppose $\mathcal{B}(p, q, r)$ holds on ℓ_1 for three distinct points, and consider projections of p, q, r on the line ℓ_2 . Call these p', q', r' . Then Pasch's axiom implies that $\mathcal{B}(p', q', r')$ holds :

Figure 18.3: Pasch axiom: Invariance Form



Szmielew then formulates what might be called the “*invariance form*” of Pasch’s axiom as follows:

$$\textbf{Pasch} \quad \ell_1 \neq \ell_2 \wedge a, b, c \in \ell_1 \wedge a', b', c' \in \ell_2 \wedge aa' \parallel bb' \parallel cc' \wedge B(a, b, c) \rightarrow B(a', b', c') \quad (18.1.3)$$

As Szmielew notes, “the sentence Pasch states that the relation B is preserved by an arbitrary parallel projection” (p. 148). Szmielew then gives two “particular cases” of Pasch:

$$\textbf{oPasch} \quad \neg L(a, b, b') \wedge L(a, b, c) \wedge L(a', b', c') \wedge bb' \parallel cc' \wedge B(a, b, c) \rightarrow B(a, b', c') \quad (18.1.4)$$

$$\textbf{iPasch} \quad \neg L(a, a', b) \wedge L(a, b, c) \wedge L(a', b, c') \wedge aa' \parallel cc' \wedge B(a, b, c) \rightarrow B(a', b, c') \quad (18.1.5)$$

which Szmielew refers to, respectively, as “Outer invariance” and “Inner invariance”.

Then Szmielew notes, “on the basis of the axioms for $(\mathbb{P}, \parallel, B)$, the laws Pasch, oPasch and iPasch turn out to be equivalent”. Note that this presupposes an underlying *parallelity plane* (i.e., affine plane): these axioms for $(\mathbb{P}, \parallel, B)$ are stated on p. 146:

I could be confused here. Maybe we need Playfair for these equivalences!

Theorem 18.1.1. $\text{Bet} \vdash \textbf{oPasch} \leftrightarrow \textbf{iPasch}$.

Proof. See Theorem 8.1.6 (p. 149) and its proof. $\square$

Theorem 18.1.2. $\text{Bet} \vdash \textbf{Pasch} \leftrightarrow \textbf{oPasch}$.

Proof. See Theorem 8.1.7 (p. 150) and its proof. $\square$

There are a number of syntactically distinct formalizations of Pasch’s axiom in $L(B)$. We shall concentrate on:

Definition 18.1.3. Our official axiom **Pasch** is the following $L(\mathbf{B})$ sentence:

$$111 \text{ Pasch} \quad \mathbf{B}(p, q, r) \wedge \mathbf{B}(s, u, q) \rightarrow \exists x (\mathbf{B}(r, x, s) \wedge \mathbf{B}(p, u, x)).$$

18.1.1 The Axiom System PaschBet

We are now going to extend **Bet** with Pasch's axiom:

Definition 18.1.4 (PaschBet axioms). The axioms of PaschBet in $L(\mathbf{B})$ are:

Table 18.1: Axioms of PaschBet

Identity	$\mathbf{B}(p, q, p) \rightarrow p = q.$
Transitivity	$\mathbf{B}(p, q, r) \wedge \mathbf{B}(q, r, s) \wedge q \neq r \rightarrow \mathbf{B}(p, q, s).$
Connectivity	$\mathbf{B}(p, q, r) \wedge \mathbf{B}(p, q, s) \wedge p \neq q \rightarrow (\mathbf{B}(p, r, s) \vee \mathbf{B}(p, s, r)).$
Symmetry	$\mathbf{B}(p, q, r) \rightarrow \mathbf{B}(r, q, p).$
Extension	$\exists r (r \neq q \wedge \mathbf{B}(p, q, r)).$
Pasch	$\mathbf{B}(p, q, r) \wedge \mathbf{B}(s, u, q) \rightarrow \exists x (\mathbf{B}(r, x, s) \wedge \mathbf{B}(p, u, x)).$
Lower Dim	$\exists p \exists q \exists r \neg \mathbf{L}(p, q, r)$

First of all, we can show that **Bet** is a subsystem of PaschBet:

Theorem 18.1.5. $\text{PaschBet} \vdash \text{Bet}.$

In particular, PaschBet proves Line Uniqueness, as well as Inclusion. So, PaschBet proves each of the fourteen statements in Theorem 17.3.3. In particular, Line Uniqueness expresses the first two of Hilbert's incidence axioms (not the third).

Theorem 18.1.6. PaschBet proves all four of Hilbert's order axioms:

$$\text{PaschBet} \vdash \mathbf{B}^\#(p, q, r) \rightarrow (p \neq q \wedge p \neq r \wedge q \neq r \wedge \mathbf{B}^\#(r, q, p)). \quad (18.1.6)$$

$$\text{PaschBet} \vdash a \neq c \rightarrow \exists b \mathbf{B}^\#(a, c, b). \quad (18.1.7)$$

$$\text{PaschBet} \vdash \mathbf{B}^\#(p, q, r) \rightarrow (\neg \mathbf{B}^\#(q, r, p) \wedge \neg \mathbf{B}^\#(q, p, r)). \quad (18.1.8)$$

$$\text{PaschBet} \vdash \text{Pasch} \quad (18.1.9)$$

Since we have no dimension axioms, we cannot (yet) prove Hilbert's third incidence axiom, which asserts the existence of a triangle.

18.2 Dense Linear Order

Definition 18.2.1. Let PaschBet^* be the theory $\text{PaschBet} + (O \neq I)$ in $L(\mathbf{B}, O, I)$.

Theorem 18.2.2. The following results hold, for PaschBet^* :

$$\text{PaschBet}^* \vdash (\forall p \in \ell) (\exists q \in \ell) (p < q). \quad (18.2.1)$$

$$\text{PaschBet}^* \vdash (\forall p \in \ell) (\exists q \in \ell) (q < p). \quad (18.2.2)$$

These two formulas express unboundedness above & below on the line $\ell(O, I)$. The density axiom is another matter. We now also have the “at least 2-dimensionality” axiom, $\exists p \exists q \exists r \neg \mathbf{L}(p, q, r)$. This is Hilbert’s third incidence axiom.

In the next chapter, we look at Hilbert’s Theorem 3 from Hilbert (1899), which is proved using Pasch’s axiom. In our present context, we can now carry through Hilbert’s Theorem 3 in PaschBet^* :

Theorem 18.2.3. $\text{PaschBet}^{**} \vdash (\forall p, q \in \ell) (p \neq q \rightarrow (\exists r \in \ell) \mathbf{B}^\#(p, r, q))$.

And from this we obtain the density axiom (on the line ℓ):

Theorem 18.2.4. $\text{PaschBet}^* \vdash (\forall p, q \in \ell) (p < q \rightarrow (\exists r \in \ell) (p < r < q))$.

This gives us the density of $\leq$ (on ℓ). These are, of course, the axioms for a *dense linear order without endpoints*—a DLOWE. Let $(\text{DLOWE})^{\ell(O, I)}$ be the conjunction of these seven sentences. So,

Theorem 18.2.5. $\text{PaschBet} \vdash (O \neq I) \rightarrow (\text{DLOWE})^{\ell(O, I)}$.

By UGC, we conclude:

Theorem 18.2.6. $\text{PaschBet} \vdash \forall p \forall q ((p \neq q) \rightarrow (\text{DLOWE})^{\ell(p, q)})$.

Hence, consider a model $\beta = (\mathbb{P}, B) \models \text{PaschBet} + \exists p \exists q \exists r \neg \text{co}_1(p, q, r)$. We know that $(\mathbb{P}, \mathcal{L})$ is a linear space. And we know that if ℓ is a line in β , then ℓ carries a dense linear order without endpoints.

Chapter 19

Desargues's Axiom

19.1 Desargues Axiom in $L(\mathbf{B})$

Definition 19.1.1. We define, using $\mathbf{B}$,

$$\begin{array}{ll} \text{co}_2(p, q, r, s) & \dots \\ \ell(p, q) & \{r \mid \text{co}_1(p, q, r)\}. \\ pq \parallel rs & \text{co}_2(p, q, r, s) \wedge (\ell(p, q) = \ell(r, s) \vee \ell(p, q) \cap \ell(r, s) = \emptyset). \\ \text{pgram}(p, q, s, r) & \neq (p, q, r, s) \wedge \ell(p, q) \parallel \ell(r, s) \wedge \ell(p, r) \parallel \ell(q, s). \end{array}$$

Definition 19.1.2. Desargues's Axiom is the following $L(\mathbf{B})$ sentence:¹

$$\begin{array}{ll} \text{211 Desargues} & [\mathbf{B}(p, a, a') \wedge \mathbf{B}(p, b, b') \wedge \mathbf{B}(p, c, c') \wedge \mathbf{B}(a, b, c'') \wedge \mathbf{B}(a', b', c'') \wedge \\ & \mathbf{B}(b, c, a'') \\ & \wedge \mathbf{B}(b', c', a'') \wedge \mathbf{B}(a, c, b'') \wedge \mathbf{B}(a', c', b'') \wedge \neg \mathbf{B}(p, a, b) \wedge \\ & \neg \mathbf{B}(a, b, p) \\ & \wedge \neg \mathbf{B}(b, p, a) \wedge \neg \mathbf{B}(p, b, c) \wedge \neg \mathbf{B}(b, c, p) \wedge \neg \mathbf{B}(c, p, b) \wedge \neg \mathbf{B}(p, c, a) \\ & \wedge \neg \mathbf{B}(c, a, p) \wedge \neg \mathbf{B}(a, p, c)] \\ & \rightarrow \mathbf{B}(a'', b'', c'') \end{array}$$

¹ This is from Szczerba & Tarski (1979): 159.

Chapter 20

Euclid's Axiom and Playfair's Axiom

20.1 Euclid in $L(\mathbf{B})$

In addition to Pasch's axiom, we can formalize Euclid's axiom:

Definition 20.1.1. Tarski's formulation of Euclid's Axiom is the following $L(\mathbf{B})$ sentence:

$$\begin{array}{ll} 111 \text{ Euclid} & \mathbf{B}(a, d, t) \wedge \mathbf{B}(b, d, c) \wedge a \neq d \rightarrow \exists x \exists y (\mathbf{B}(a, b, x) \wedge \mathbf{B}(a, c, y) \wedge \\ & \mathbf{B}(x, t, y)) \end{array}$$

Chapter 21

Pappus's Axioms

21.1 Pappus's Axiom in $L(\mathbf{B})$

Definition 21.1.1. Pappus's Axiom is the following $L(\mathbf{B})$ sentence:

111 Pappus ...

A clever argument using Continuity establishes that a *continuously ordered* Desarguesian affine plane is Pappian—for the commutativity of multiplication follows from Continuity in this content (add Goldblatt (1987)).

Chapter 22

The Continuity Axiom(s)

22.1 Continuity

Chapter 23

The Axiom System $\text{BG}_0(n)$

23.1 The Axioms of $\text{BG}_0(n)$

We next set out the following one-sorted first-order axiom systems, which I denote $\text{BG}_0(n)$.¹

Definition 23.1.1 (Axiom system $\text{BG}_0(n)$). For $n \geq 2$ a positive integer, the axioms of $\text{BG}_0(n)$ in $L(\text{B})$ are the following nine axioms and one axiom scheme:²

Table 23.1: **Axioms of $\text{BG}_0(n)$**

B-Identity	$\text{B}(p, q, p) \rightarrow p = q.$
B-Transitivity	$\text{B}(p, q, r) \wedge \text{B}(q, r, s) \wedge q \neq r \rightarrow \text{B}(p, q, s).$
B-Connectivity	$\text{B}(p, q, r) \wedge \text{B}(p, q, r') \wedge p \neq q \rightarrow (\text{B}(p, r, r') \vee \text{B}(p, r', r)).$
B-Extension	$\exists p (\text{B}(p, q, r) \wedge p \neq q).$
Pasch	$\text{B}(p, q, r) \wedge \text{B}(s, u, q) \rightarrow \exists x (\text{B}(r, x, s) \wedge \text{B}(p, u, x)).$
Desargues	See Definition 19.1.2.
Euclid	$\text{B}(a, d, t) \wedge \text{B}(b, d, c) \wedge a \neq d \rightarrow \exists x \exists y (\text{B}(a, b, x) \wedge \text{B}(a, c, y) \wedge \text{B}(x, t, y))$
Lower dimension	There exist $n + 1$ points which are not co_{n-1} .
Upper dimension	Any $n + 2$ points are co_n .
Continuity axiom scheme	$\exists r \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow \text{B}(r, p, q)) \rightarrow \exists s \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow \text{B}(p, s, q))$

The first main result we can establish concerns the system **PaschBet** (Definition 18.1.4):

Lemma 23.1.2. Let $n \geq 2$. $\text{BG}_0(n) \vdash \text{PaschBet}$.³

¹ “BG” is short for “betweenness geometry” (some authors call this “ordered affine geometry”). See also Goldblatt (1987): 165 for $\text{BG}_0(4)$, which Goldblatt calls “the first-order theory of ordered affine fourfolds over real-closed fields”. It is the same theory that John Burgess calls **Geom**₄ in Burgess (1984).

² These axioms are given originally in Szczerba & Tarski (1965) and Szczerba & Tarski (1979).

³ The **Symmetry** axiom **PaschBet** can be proved in $\text{BG}_0(4)$. I do not know the exact reference for this.

The system **PaschBet** is therefore a subsystem of each $\text{BG}_0(n)$.

We have seen at Theorem 18.1.6, some of the resulting theorems, including Line Uniqueness and Hilbert’s order axioms. What is more, $\text{BG}_0(4)$ proves the existence of at least three non-collinear points (a triangle):

Lemma 23.1.3. Let $n \geq 2$. $\text{BG}_0(n) \vdash \exists p \exists q \exists r \neg \text{co}_1(p, q, r)$.

See Szczerba & Tarski (1979), pp. 159–160, for the first-order two-dimensional theory GA_2 (for “neutral” or “absolute geometry”), which lacks the Euclid Parallel axiom (which is called (E) in Szczerba & Tarski (1979) and is called (Euclid) above). Their system includes Desargues’s Theorem. But for us, this axiom is no longer required, as it provable from the remaining axioms in dimensions above two (Szczerba & Tarski (1979): 190). The above axiom system is the second-order, four-dimensional theory, and containing (E), i.e., (Euclid).⁴ The relevant representation theorem follows from Theorem 5.12 of Szczerba & Tarski (1979) (see also Example 6.1). The same theorem is stated, somewhat indirectly, on pp. 196–197 of Borsuk & Szmielew (1960). The representation theorem itself goes back to Veblen (1904).

The first-order formulation of these axioms are given originally in Szczerba & Tarski (1965) and Szczerba & Tarski (1979). Their formulation on pp. is for “neutral geometry”, in two dimensions. Their axioms include Desargues’s axiom, but omit the Euclid axiom. We are interested in four dimensions, and in fact Desargues’s axiom can be proved from the others for dimensions above 2. If we change the dimension axioms to those for four dimensions, and add Euclid axiom, we obtain a corresponding first-order theory, which we have called $\text{BG}_0(4)$. See also Goldblatt (1987): 165 for Goldblatt calls this “the first-order theory of ordered affine fourfolds over real-closed fields”.

Informally speaking, the Lower Dimension axiom states “There exist five points which are not in the same three-dimensional space” and the Upper Dimension axiom states, “Any six points are in the same four-dimensional space”.

23.2 The Axioms of $\text{BG}(n)$

$\text{BG}_0(n)$ is a first-order geometry. Correspondingly, there is a *second-order* geometry $\text{BG}(n)$, in the (monadic) second-order language $L_2(\mathbf{B})$. The axioms are obtained by replacing the Continuity axiom scheme by the single Continuity Axiom, and then introducing the Comprehension axiom scheme and the usual Extensionality axiom.

Definition 23.2.1. The (second-order) logical axioms of $\text{BG}(n)$ in $L_2(\mathbf{B})$ are Comprehension and Extensionality. The non-logical axioms of $\text{BG}(n)$ in $L_2(\mathbf{B})$ are the following ten:

I believe the argument uses Continuity.

⁴ In Suppes (2002), Suppes claims to give this axiom system; but, in fact, he gives the *neutral geometry* system, lacking the Euclid axiom.

Table 23.2: Axioms of $\text{BG}(n)$

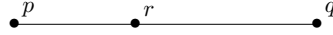
B-Identity	$\text{B}(p, q, p) \rightarrow p = q.$
B-Transitivity	$\text{B}(p, q, r) \wedge \text{B}(q, r, s) \wedge q \neq r \rightarrow \text{B}(p, q, s).$
B-Connectivity	$\text{B}(p, q, r) \wedge \text{B}(p, q, r') \wedge p \neq q \rightarrow (\text{B}(p, r, r') \vee \text{B}(p, r', r)).$
B-Extension	$\exists p (\text{B}(p, q, r) \wedge p \neq q).$
Pasch	$\text{B}(p, q, r) \wedge \text{B}(s, u, q) \rightarrow \exists x (\text{B}(r, x, s) \wedge \text{B}(p, u, x)).$
Desargues	See Definition 19.1.2.
Euclid	$\text{B}(a, d, t) \wedge \text{B}(b, d, c) \wedge a \neq d \rightarrow \exists x \exists y (\text{B}(a, b, x) \wedge \text{B}(a, c, y) \wedge \text{B}(x, t, y))$
Lower Dimension	There exist $n + 1$ points which are not co_{n-1} .
Upper Dimension	Any $n + 2$ points are co_n .
Continuity Axiom	$[\exists r (\forall p \in X) (\forall q \in Y) \text{B}(r, p, q)] \rightarrow \exists s (\forall p \in X) (\forall q \in Y) \text{B}(p, s, q)$

For the purposes of spacetime physics, we are interested in the second-order theory, which we call $\text{BG}(4)$.⁵ In dimensions about 2 (and so for dimension 4), the Desargues's axiom is redundant, and can be proved from the remaining axioms.

23.3 Internal Point Theorem

The development of geometry in the austere systems $\text{BG}(n)$ is quite involved. The Pasch axiom, with several others, allows us to prove that, given a non-trivial segment pq ($p \neq q$), there exists an “internal point” r lying strictly between p and q . This corresponds to THEOREM 3 in Hilbert (1899) (p. 6):

Figure 23.1: Internal point theorem



Lemma 23.3.1. Let $n \geq 2$. Then $\text{BG}_0(n) \vdash \forall p \forall q (p \neq q \rightarrow \exists r \text{B}^\#(p, r, q))$.

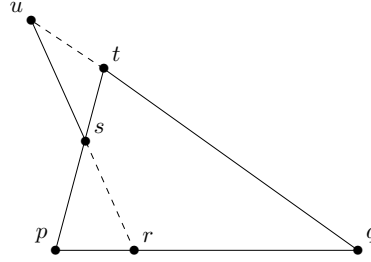
Proof. Assume $p \neq q$. By non-triviality, there exists a point s not on $\ell(p, q)$. By the axiom B-Extension, we extend ps to a point t such that $\text{B}(p, s, t)$ and $s \neq t$. Suppose that $\ell(p, t) = \ell(q, t)$, then p, q both lie on $\ell(p, t)$ and so $\ell(p, q) = \ell(p, t)$. But s lies on $\ell(p, t)$, and hence s lies on $\ell(p, q)$, which is a contradiction. Hence, $\ell(p, t) \neq \ell(q, t)$.

Use B-Extension to extend qt to a point u such that $\text{B}(q, t, u)$ and $u \neq t$. Apply Pasch to $\triangle pqt$ and the line $\ell(s, u)$. By Pasch, there exists some point r on $\ell(s, u)$ such that r is either between q and t , or r is between p and q . But r cannot be on the line $\ell(q, t)$. For, otherwise the line $\ell(s, u)$ would have two points in common with line $\ell(q, t)$, and thus s would be on $\ell(q, t)$, which is impossible.

⁵ One may also find it in Ketland (2023).

Hence, r is on segment pq , with $p \neq q$. It remains to prove that $r \neq p$ and $r \neq q$. We already have $r \neq q$. Note that $\ell(p, t) \neq \ell(q, t)$. Suppose $r = p$. Then, the lines $\ell(p, s)$ and $\ell(r, s)$ coincide. Hence, the lines $\ell(p, t)$ and $\ell(r, s)$ coincide. The line $\ell(r, s)$ is the line $\ell(r, u)$. Hence, the lines $\ell(r, u)$ and $\ell(p, t)$ coincide. Hence, u lies on $\ell(p, t)$. Since $t \neq u$, $\ell(t, u)$ coincides with $\ell(p, t)$. But $\ell(t, u)$ is the line $\ell(q, t)$. Hence, the lines $\ell(p, t)$ and $\ell(q, t)$ coincide, which is a contradiction.

Figure 23.2: **Construction of Internal Point (r , on pq) by Pasch's Axiom**



□

23.4 Core Theorems of $\text{BG}_0(n)$

The core part of the development of geometry in $\text{BG}_0(n)$ requires proof of the axioms we have seen above, starting with the incidence axioms and then moving through to the more complicated axioms (Playfair, Desargues, Pappus). To start with, we have:

Theorem 23.4.1.

$$\text{BG}_0(n) \vdash \forall p \forall q \text{B}(p, q, q). \quad (23.4.1)$$

$$\text{BG}_0(n) \vdash \forall p \forall q \forall r (\text{B}(p, q, r) \rightarrow \text{B}(r, q, p)). \quad (23.4.2)$$

$$\text{BG}_0(n) \vdash \forall p \forall q (p \neq q \rightarrow \exists! \ell (p \in \ell \wedge q \in \ell)). \quad (23.4.3)$$

$$\text{BG}_0(n) \vdash (\exists p, q, r) \neg \text{co}_1(p, q, r). \quad (23.4.4)$$

Theorem 23.4.2. $\text{BG}_0(n) \vdash$ Playfair's axiom.

Theorem 23.4.3. $\text{BG}_0(n) \vdash$ Desargues's axiom.

Theorem 23.4.4. $\text{BG}_0(n) \vdash$ Pappus's axiom.

23.5 The Soundness Theorem for $\text{BG}_0(n)$

Theorem 23.5.1. Let F be a real-closed field. Let $n \geq 2$. Then $\text{BG}_0(n)$ is *sound* in its corresponding coordinate structure:

$$\mathbb{B}^n(F) \models \mathbf{BG}_0(n) \quad (23.5.1)$$

Proof. Step by step, we prove that each axiom of $\mathbf{BG}_0(n)$ holds in the coordinate structure $\mathbb{B}^n(F)$. For example, the first axiom is $\forall p \forall q (\mathbf{B}(p, q, p) \rightarrow p = q)$. This holds in $\mathbb{B}^n(F)$ if and only if, for all $\mathbf{x}, \mathbf{y} \in F^n$, if $B_{F^n}(\mathbf{x}, \mathbf{y}, \mathbf{x})$, then $\mathbf{x} = \mathbf{y}$. □

Definition 23.5.2. Let the relevant real-closed field $F = \mathbb{R}$. Then the *standard* coordinate structure for $\mathbf{BG}(n)$ is $\mathbb{B}^n$.

Theorem 23.5.3. For $n \geq 2$, $\mathbf{BG}(n)$ is sound in its corresponding standard coordinate structure:

$$\mathbb{B}^n \models \mathbf{BG}(n) \quad (23.5.2)$$

Chapter 24

Proportionality

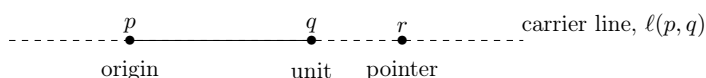
24.1 Theory of Ratio Configurations and Proportion

We shall be extremely interested in the coding of *real numbers* inside $\text{BG}_0(4)$ and $\text{BG}(4)$. The Greeks considered real numbers to be abstracted from geometry, and this view held sway until Descartes’s remarkable inversion, where geometry is encoded inside algebra, via coordinate pairs (and triples, and so on). We wish to understand both of these encodings.

To explain the encoding of reals inside betweenness geometry, we introduce the notion of a *ratio configuration*—or, simply, a *ratio*:¹

Definition 24.1.1 (Ratio Configuration). We say that an ordered triple (p, q, r) of points is a *ratio configuration*—or, simply, a *ratio*—just if $p \neq q$ and p, q, r are collinear:

Figure 24.1: **Ratios**

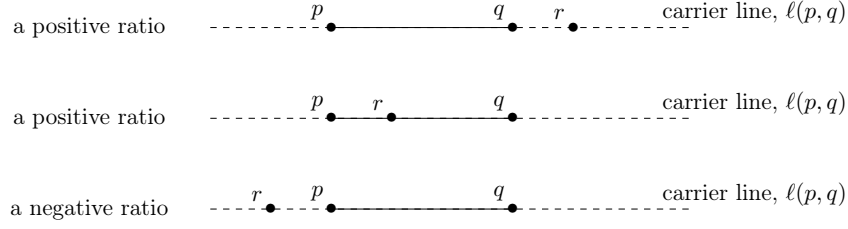


The line $\ell(p, q)$ is called “the carrier line” (of the ratio). The point p is called “the origin”. The point q is called “the unit”. The point r is called “the pointer”.

Definition 24.1.2. We say that a ratio (p, q, r) is a *negative* just if $\text{B}(r, p, q)$. We say that a ratio (p, q, r) is a *positive* just if $\text{B}(p, r, q)$ or $\text{B}(p, q, r)$.

¹ Here I am following the brief introduction in Burgess (1984).

Figure 24.2: Positive and negative ratios



Definition 24.1.3. A ratio (p, q, p) is called *null*, and a ratio (p, q, q) is called a *unit ratio*.

Lemma 24.1.4. If the ratio (p, q, r) is negative, then (q, p, r) is a positive ratio.

Proof. Let (p, q, r) be a negative ratio. Hence, $p \neq q$ and $B(r, p, q)$. Hence, by (Symmetry) $B(q, p, r)$. Hence, either $B(q, r, p)$ or $B(q, p, r)$, and so (q, p, r) is a positive ratio. $\square$

Lemma 24.1.5. If the ratio (p, q, r) is a positive and not- $B(p, r, q)$, then (q, p, r) is a negative ratio.

Proof. Let (p, q, r) be a positive ratio with not- $B(p, r, q)$. So, either $B(p, r, q)$ or $B(p, q, r)$. Hence, by (Symmetry), either $B(p, r, q)$ or $B(r, q, p)$. Hence, $B(r, q, p)$. Therefore, (q, p, r) is a negative ratio. $\square$

Definition 24.1.6. Let (p, q, r) and (a, b, c) be ratios. We say that they are *coplanar ratios* just if their carrier lines are coplanar. I.e., $\text{co}_2(p, q, a, b)$ holds.

24.2 Parallel Copying and Proportionateness

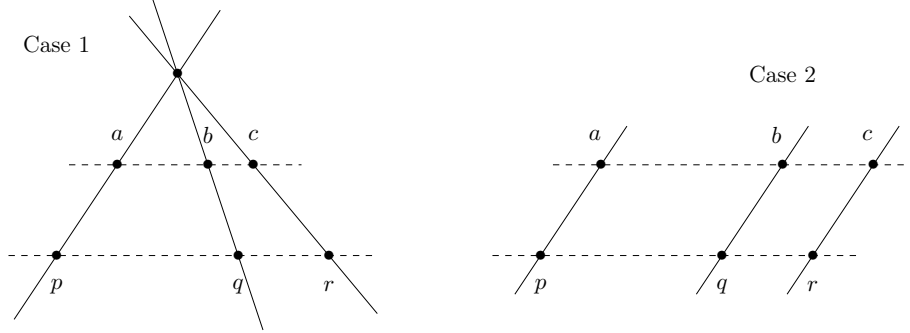
We now wish to define a fundamental geometrical notion (for ratios):

$$(p, q, r) \sim (a, b, c) \quad (24.2.1)$$

Definition 24.2.1. Let (p, q, r) and (a, b, c) be coplanar positive ratios. We say that (p, q, r) is a (*coplanar*) *parallel copy* of (a, b, c) just if $\ell(p, q) \parallel \ell(a, b)$ and $\ell(p, q) \neq \ell(a, b)$; and, either (Case 1 below) $\ell(p, a), \ell(q, b)$ and $\ell(r, c)$ are concurrent, or (Case 2 below) $pa \parallel qb \parallel rc$.

A similar definition is given for coplanar negative ratios.

Figure 24.3: (Coplanar) Parallel Copying

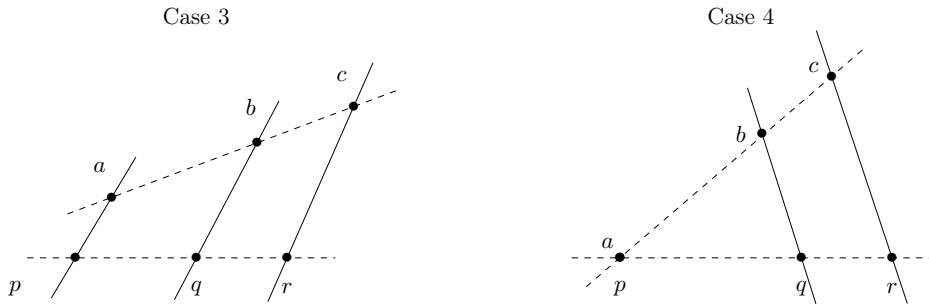


Notice that we rule out the case where $\ell(p, q)$ and $\ell(a, b)$ are the same line.

Definition 24.2.2. Let (p, q, r) and (a, b, c) be coplanar positive ratios. Then we say that (p, q, r) is (coplanar) *positive-proportionate* to (a, b, c) if and only if one of the conditions (1)–(4) holds:

- 0.510 (1) $pq \parallel ab$ and $\ell(p, q) \neq \ell(a, b)$ and (p, q, r) is a parallel copy of (a, b, c) .
- (2) $pq \parallel ab$ and $\ell(p, q) = \ell(a, b)$, and there exists (a', b', c') such that (a', b', c') is a parallel copy of (a, b, c) , and of (p, q, r) .
- (3) $pq \not\parallel ab$ and $p \neq a$ and Case 3 below holds.
- (4) $pq \not\parallel ab$ and $p = a$, and Case 4 below holds.

Figure 24.4: (Coplanar) Positive-Proportionateness



In Burgess & Rosen (1997) (p. 110), Burgess and Rosen list two basic cases, which are our Case 1 and Case 4. In a sense, Case 2 and Case 4 reduce to Case 3 (for Case 2 by taking the configurations parallel; for Case 4, by “sliding” the configuration abc , parallel to the three parallel lines, until a now coincides with p).

A corresponding definition is then given for *negative-proportionateness*.

Definition 24.2.3. Let (p, q, r) and (a, b, c) be coplanar ratios. Then (p, q, r) and (a, b, c) are *coplanar proportionate* if and only if

- 0.512 (1) either both are positive and (p, q, r) and (a, b, c) are coplanar positive-proportionate;
 (2) or both are negative and (p, q, r) and (a, b, c) are coplanar negative-proportionate.

Next, the non-coplanar case. In dimensions three and above, a pair of ratios, (p, q, r) and (a, b, c) may be non-coplanar. However, one can construct a (unique) “copy” (a', b', c') of (a, b, c) such that $a' = p$. In this case, (a', b', c') is then coplanar with (a, b, c) , because ab and $a'b'$ are parallel, and (a', b', c') is coplanar with (p, q, r) (because $a' = p$), and the previous definitions apply.

Definition 24.2.4. Let (p, q, r) and (a, b, c) be ratios. Then (p, q, r) and (a, b, c) are said to be *proportionate* if and only if:

- 0.512 (1) either (p, q, r) and (a, b, c) are coplanar, and Definition 24.2.3 holds;
 (2) or (p, q, r) and (a, b, c) are non-coplanar, and Definition 24.2.3 holds for (p, q, r) and the copy (a', b', c') .

Definition 24.2.5 (Proportionateness). The 6-place formula:

$$(p, q, r) \sim (a, b, c) \quad (24.2.2)$$

is short for “ (p, q, r) and (a, b, c) are ratios, and (p, q, r) is proportionate to (a, b, c) ”. It may also be written:

$$\begin{array}{c} \text{the ratio of } pr \text{ to the unit segment } pq \\ \overbrace{pr : pq} \end{array} = \begin{array}{c} \text{the ratio of } ac \text{ to the unit segment } ab \\ \overbrace{ac : ab} \end{array} \quad (24.2.3)$$

Lemma 24.2.6. The following hold:

$$\text{BG}_0(4) \vdash \text{rat}(p, q, r) \rightarrow (p, q, r) \sim (p, q, r) \quad (24.2.4)$$

$$\text{BG}_0(4) \vdash (\text{rat}(p, q, r) \wedge \text{rat}(a, b, c)) \rightarrow ((p, q, r) \sim (a, b, c) \rightarrow (a, b, c) \sim (p, q, r)) \quad (24.2.5)$$

$$\begin{aligned} \text{BG}_0(4) \vdash & (\text{rat}(p, q, r) \wedge \text{rat}(a, b, c) \wedge \text{rat}(d, e, f)) \\ & \rightarrow (((p, q, r) \sim (a, b, c) \wedge (a, b, c) \sim (d, e, f)) \rightarrow (a, b, c) \sim (d, e, f)) \end{aligned} \quad (24.2.6)$$

That is, proportionateness is an equivalence relation on configurations.

Lemma 24.2.7. Let $O \neq I$. Let (a, b, c) be a configuration. Then there exists a point $r \in \ell(O, I)$ such that (O, I, r) and (a, b, c) are proportionate.

24.3 Encoding Geometry into Algebra

In this section, I show how to formally carry out Descartes’s programme of *algebraicizing geometry*—carrying out geometry “inside algebra”. We shall focus on $\text{BG}_0(2)$, primarily for notational simplicity. The definitions and results can be modified for the case of $\text{BG}_0(n)$, for any $n \geq 2$.

Definition 24.3.1 (The BET formula). The axioms of RCF, in the language $L(0, 1, +, \times, \leq)$ of ordered fields, are given in Appendix ???. We now *define*, in $L(0, 1, +, \times, \leq)$, the following 6-place formula:²

$$\text{BET}(x_1, x_2, y_1, y_2, z_1, z_2) := \exists \alpha (0 \leq \alpha \leq 1 \wedge \bigwedge_{i=1}^2 (y_i - x_i = \alpha(z_i - x_i))) \quad (24.3.1)$$

This may be abbreviated:

$$\text{BET}(\vec{x}, \vec{y}, \vec{z}) := (\exists \alpha \in [0, 1]) (\vec{y} - \vec{x} = \alpha(\vec{z} - \vec{x})) \quad (24.3.2)$$

Definition 24.3.2. We define what I call “*the Cartesian translation*”,

$$(\cdot)^\circ : L(\mathbf{B}) \rightarrow L(0, 1, +, \times, \leq) \quad (24.3.3)$$

as follows:

Table 24.1: **Cartesian translation (dimension = 2)**

$L(\mathbf{B})$ expression	Translation	
Variables	$(p_i)^\circ$	$:= (x_{2i}, x_{2i+1}).$
Atomic (identity)	$(p_i = p_j)^\circ$	$:= (x_{2i} = x_{2j} \wedge x_{2i+1} = x_{2j+1}).$
Atomic (betweenness)	$(\mathbf{B}(p_i, p_j, p_k))^\circ$	$:= \text{BET}((p_i)^\circ, (p_j)^\circ, (p_k)^\circ).$
Boolean	$(\neg \varphi)^\circ$	$:= \neg \varphi^\circ.$
Boolean	$(\varphi \rightarrow \theta)^\circ$	$:= \varphi^\circ \rightarrow \theta^\circ.$
Boolean	$(\varphi \wedge \theta)^\circ$	$:= \varphi^\circ \wedge \theta^\circ.$
Boolean	$(\varphi \vee \theta)^\circ$	$:= \varphi^\circ \vee \theta^\circ.$
Boolean	$(\varphi \leftrightarrow \theta)^\circ$	$:= \varphi^\circ \leftrightarrow \theta^\circ.$
Quantified	$(\forall p_i \varphi(p_i))^\circ$	$:= \forall x_{2i} \forall x_{2i+1} \varphi^\circ.$
Quantified	$(\exists p_i \varphi(p_i))^\circ$	$:= \exists x_{2i} \exists x_{2i+1} \varphi^\circ.$

Next, our main aim is to show that RCF *interprets* first-order betweenness geometry $\text{BG}_0(2)$. That is, under the Cartesian translation $(\cdot)^\circ$, RCF proves the B axioms.

² Subtraction is explicitly defined by: $z = x - y := x = z + y$.

Lemma 24.3.3. For any sentence $\varphi \in L(\mathbf{B})$,

$$\mathbf{BG}_0(2) \vdash \varphi \Rightarrow \mathbf{RCF} \vdash \varphi^\circ \quad (24.3.4)$$

Proof. This requires a proof for each axiom of $\mathbf{BG}_0(2)$, and for each continuity axiom. For example, $\mathbf{RCF}$ proves:

$$\begin{array}{ll} \text{B-Identity} & \mathbf{BET}(\vec{x}_1, \vec{x}_2, \vec{x}_1) \rightarrow \vec{x}_1 = \vec{x}_2. \\ \text{B-Transitivity} & \mathbf{BET}(\vec{x}_1, \vec{x}_2, \vec{x}_3) \wedge \mathbf{BET}(\vec{x}_2, \vec{x}_3, \vec{x}_4) \wedge \vec{x}_2 \neq \vec{x}_3 \rightarrow \mathbf{B}(\vec{x}_1, \vec{x}_2, \vec{x}_4). \end{array}$$

We work in $\mathbf{RCF}$. In the first case, suppose $\mathbf{BET}(\vec{x}_1, \vec{x}_2, \vec{x}_1)$. Hence, for some $\alpha \in [0, 1]$, we have $\vec{x}_2 - \vec{x}_1 = \alpha(\vec{x}_1 - \vec{x}_1)$. Hence, $\vec{x}_2 - \vec{x}_1 = 0$. Hence, $\vec{x}_2 = \vec{x}_1$. □

In the result above, we can replace $\Rightarrow$ by $\Leftrightarrow$, showing that the Cartesian translation $^\circ$ faithfully interprets $\mathbf{BG}_0(2)$ into $\mathbf{RCF}$. That is, for any sentence $\varphi \in L(\mathbf{B})$,

$$\mathbf{BG}_0(2) \vdash \varphi \Leftrightarrow \mathbf{RCF} \vdash (\varphi)^\circ \quad (24.3.5)$$

The proof, however, requires the Representation Theorem (Theorem ??), which we come to in the next Chapter.

24.4 Encoding Algebra into Geometry: The Greek Lemmas

We now “invert” Cartesian programme of algebraicizing geometry, and go back to the Greek programme of geometrizing algebra: we interpret algebra into geometry.

We work inside $\mathbf{BG}_0(n) + (O \neq I)$, with $n \geq 2$. We define the geometric addition, multiplication and order on the line $\ell(O, I)$ as before. So, addition, $p + q$, is defined at Definition 14.1.1; multiplication; $p \times q$, at Definition 14.1.2; and order, $p \leq q$, at Definition 16.3.4 and Definition 17.8.2.

Lemma 24.4.1 (The Greek Lemma 1). $\mathbf{BG}_0(n) + (O \neq I)$ proves the ten “*field axioms*”:

$$\text{BG}_0(n) + (O \neq I) \vdash O \neq I. \quad (24.4.1)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q \in \ell) (p + q = q + p). \quad (24.4.2)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q \in \ell) (p + (q + r) = (p + q) + r). \quad (24.4.3)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q \in \ell) (p + O = p). \quad (24.4.4)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p \in \ell) (\exists q \in \ell) (p + q = O). \quad (24.4.5)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q \in \ell) (p \times q = q \times p). \quad (24.4.6)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q, r \in \ell) (p \times (q \times r) = (p \times q) \times r). \quad (24.4.7)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q \in \ell) (p \times I = p). \quad (24.4.8)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p \in \ell) (p \neq O \rightarrow \exists q (p \times q = I)). \quad (24.4.9)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q, r \in \ell) (p \times (q + r) = (p \times q) + (p \times r)). \quad (24.4.10)$$

Proof. These proofs are formal replicas of the proofs of the field axioms for a *Pappian affine plane*, which we referred to at Theorem ??(2), on the coordinatization of Pappian affine planes. $\square$

To recall, the definition of $\leq$ (Definition 16.3.4 and Definition 17.8.2) is:³

$$p \leq q := p = q \vee (\text{B}(p, O, I) \wedge \neg \text{B}(q, p, I)) \vee (\neg \text{B}(p, O, I) \wedge \text{B}(O, p, q)) \quad (24.4.11)$$

Lemma 24.4.2 (The Greek Lemma 2). The theory $\text{BG}_0(n) + (O \neq I)$ proves the linear order axioms. That is:

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q, r \in \ell) (p \leq q \wedge q \leq r \rightarrow p \leq r). \quad (24.4.12)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q \in \ell) (p \leq q \wedge q \leq p \rightarrow p = q). \quad (24.4.13)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q \in \ell) (p \leq q \vee q \leq p). \quad (24.4.14)$$

Proof. Recall that **PaschBet** is a subsystem of $\text{BG}_0(n)$. These proofs are formal replicas of the proofs of DLOWE axioms in **PaschBet**, which we referred to at Theorem 18.2.6 in §18.2. $\square$

Lemma 24.4.3 (The Greek Lemma 3). The theory $\text{BG}_0(n) + (O \neq I)$ proves the compatibility axioms:

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q, r \in \ell) (p \leq q \rightarrow p + r \leq q + r). \quad (24.4.15)$$

$$\text{BG}_0(n) + (O \neq I) \vdash (\forall p, q, r \in \ell) (O \leq r \rightarrow (p \leq q \rightarrow p \times r \leq q \times r)). \quad (24.4.16)$$

³ The definition of $\leq$ in terms of betweenness is given in Tarski (1959), in the proof of Theorem 1.

Proof. Though it is quite a bit of work. □

Lemma 24.4.4 (The Greek Lemma 4). $\text{BG}_0(n) + (O \neq I)$ proves all instances of the Continuity axiom scheme:

$$\begin{aligned} \text{BG}_0(n) + (O \neq I) \vdash (\exists p \in \ell) \varphi(p) \wedge (\exists p \in \ell) \theta(p) \wedge (\forall p \in \ell) (\forall q \in \ell) (\varphi(p) \wedge \theta(q) \rightarrow p \leq q) \\ \rightarrow (\exists r \in \ell) (\forall p \in \ell) (\forall q \in \ell) (\varphi(p) \wedge \theta(q) \rightarrow (p \leq r \leq q)). \end{aligned} \quad (24.4.17)$$

Given our fixed benchmark constants O, I , let's write $(\text{RCF})^{\ell(O, I)}$ for the formulas referred to above as theorems, in Greek Lemmas 1 to 4. Note that we cannot conjoin $(\text{RCF})^{\ell(O, I)}$ into a single formula, since there are infinitely many continuity axioms. Summarizing the above:

Theorem 24.4.5 (Main Greek Lemma (first-order)). For any positive integer $n \geq 2$,

$$\text{BG}_0(n) + (O \neq I) \vdash (\text{RCF})^{\ell(O, I)} \quad (24.4.18)$$

Suppose $\beta = (\mathbb{P}, B)$ and $\beta \models \text{BG}_0(n)$. Then, on any line in β the axioms of **RCF** hold. In particular, if ℓ is a line in β , then there is a real-closed field F such that $\ell \cong F$.

Lemma 24.4.6 (The Greek Lemma 5). $\text{BG}(n) + (O \neq I)$ proves the (algebraic) Continuity axiom:

$$\text{BG}(n) + (O \neq I) \vdash X, Y \subseteq \ell \wedge X \neq \emptyset \wedge Y \neq \emptyset \wedge X \leq Y \rightarrow (\exists r \in \ell) (X \leq r \leq Y). \quad (24.4.19)$$

Theorem 24.4.7 (Main Greek Lemma (second-order)). For any positive integer $n \geq 2$,

$$\text{BG}(n) + (O \neq I) \vdash (\text{ALG})^{\ell(O, I)} \quad (24.4.20)$$

24.5 Line Isomorphism

We can prove not only that each line in a model β carries some real-closed field. But also that, given a model β , each line in β carries the same real-closed field:

Theorem 24.5.1. Let $\beta = (\mathbb{P}, B)$ and $\beta \models \text{BG}_0(n)$. Let ℓ and ℓ' be lines in β . Then

$$(\ell, B \upharpoonright_\ell) \cong (\ell', B \upharpoonright_{\ell'}). \quad (24.5.1)$$

Proof. This is obtained from Pasch's axiom, as explained in Szmielew (1983), 147–152. □

Chapter 25

Representation Theorems for $\mathbf{BG}_0(n)$ and $\mathbf{BG}(n)$

25.1 Introduction

Theorem 23.5.1 establishes that, if $n \geq 2$ and F is a real-closed field, then:

$$\mathbb{B}^n(F) \models \mathbf{BG}_0(n). \quad (25.1.1)$$

However, we'd like to prove a kind of converse of this. Namely, that if $n \geq 2$ and $(\mathbb{P}, B) \models \mathbf{BG}_0(n)$, then:

$$(\mathbb{P}, B) \cong \mathbb{B}^n(F), \quad (25.1.2)$$

for some real-closed field F .

This is quite difficult. It amounts to the *Representation Theorem* for $\mathbf{BG}_0(n)$.

25.2 Line Representation Theorem for $\mathbf{BG}_0(n)$

Theorem 25.2.1 (Line Representation Theorem for $\mathbf{BG}_0(n)$). Let $\beta = (\mathbb{P}, B)$ be a $L(\mathbf{B})$ -structure. Let $n \geq 2$. Assume that $\beta \models \mathbf{BG}_0(n)$. Suppose $O, I \in \mathbb{P}$, with $O \neq I$. Let $\ell = \ell(O, I)$ be the line determined by O and I . Overload, by letting $\ell = (\ell, O, I, +, \times, \leq)$, where $+$, $\times$ and $\leq$ are geometrized addition, multiplication and order on ℓ . Then there exists a real-closed field F , and an isomorphism:

$$\varphi_{O,I} : \ell \rightarrow F \quad (25.2.1)$$

Proof. We first define the line $\ell = \ell(O, I)$. As we showed above, one can define geometrical operations $+$, $\times$, and a linear order $\leq$, on ℓ (relative to the two fixed parameters O, I). Overload, by letting $\ell = (\ell, O, I, +, \times, \leq)$,

The Main Greek Lemma (Theorem 24.4.5) for first-order betweenness geometry states that, for any positive integer $n \geq 2$,

$$\mathbf{BG}_0(n) + (O \neq I) \vdash (\mathbf{RCF})^{\ell(O,I)} \quad (25.2.2)$$

Inside the model β , this implies that ℓ is an *real-closed ordered field*. Hence, for some real-closed field F , there is an isomorphism $\varphi_{O,I} : \ell \rightarrow F$, as claimed. $\square$

25.3 Line Representation Theorem for $\mathbf{BG}(n)$

The result of basic importance to *physics*, however, is a modification.

Theorem 25.3.1 (Line Representation Theorem for $\mathbf{BG}(n)$). Let $\beta = (\mathbb{P}, B)$ be an $L(\mathbf{B})$ -structure. Assume that β is a *full model* of $\mathbf{BG}(n)$, with $n \geq 2$. Suppose $O, I \in \mathbb{P}$, with $O \neq I$. Let $\ell = \ell(O, I)$ be the line determined by O and I . Then there exists a unique isomorphism:

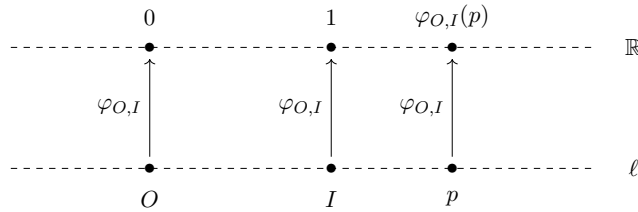
$$\varphi_{O,I} : \ell \rightarrow \mathbb{R} \quad (25.3.1)$$

Proof. The proof is as for $\mathbf{BG}_0(n)$, except that we now have the second-order *Continuity Axiom*. Then the Main Greek Lemma (Theorem 24.4.7) for second-order betweenness geometry states that, for any positive integer $n \geq 2$,

$$\mathbf{BG}(n) + (O \neq I) \vdash (\mathbf{ALG})^{\ell(O,I)} \quad (25.3.2)$$

Define a line $\ell = \ell(O, I)$ as above. By the above, ℓ is a *complete ordered field*. Up to isomorphism, there is exactly one complete ordered field, and this is also rigid. Consequently, there is a (unique) isomorphism $\varphi_{O,I} : \ell \rightarrow \mathbb{R}$, as claimed:

Figure 25.1: Isomorphism from ℓ to $\mathbb{R}$



$\square$

25.4 Representation Theorem for $\mathbf{BG}_0(n)$

The following two theorems are due, primarily, to Veblen (1904);¹ I shall focus on the case of $\mathbf{BG}_0(2)$, for reasons of notation and simplicity, but the same reasoning goes through for any $\mathbf{BG}_0(n)$, with $n \geq 2$.

Definition 25.4.1 (Matching). Suppose that $\beta = (\mathbb{P}, B)$ is an $L(\mathbf{B})$ -structure. Let $\beta \models \mathbf{BG}_0(2)$. Suppose that (O, I_1, I_2) is a 2-frame in β . Let F be a field. Suppose that $\Phi : \mathbb{P} \rightarrow F^4$ is a function. We say that Φ *matches* the 2-frame (O, I_1, I_2) just if:²

$$\Phi(O) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \Phi(I_1) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Phi(I_2) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (25.4.1)$$

Theorem 25.4.2. Let $n \geq 2$. Let $\beta = (\mathbb{P}, B)$. Let $\beta \models \mathbf{BG}_0(2)$. Then

$$\beta \cong \mathbb{B}^2(F), \quad (25.4.2)$$

for some real-closed field F .

Proof. Suppose $\beta = (\mathbb{P}, B)$ and $\beta \models \mathbf{BG}_0(2)$. Thanks to our earlier results, there are four digestible stages in this proof. First recall two prior theorems. Theorem 24.4.5 states:

$$\mathbf{BG}_0(2) \vdash O \neq I \rightarrow (\text{RCF})^{\ell(O, I)}. \quad (25.4.3)$$

Thus, given $\beta \models \mathbf{BG}_0(2)$, on any line ℓ in β the axioms of RCF hold. In particular, if ℓ is a line in β , then there is a real-closed field F such that $\ell \cong F$. And recall Theorem 24.5.1 which says that if $\beta \models \mathbf{BG}_0(2)$, and ℓ and ℓ' are lines in β , then $\ell \cong \ell'$.

Step 1. By the Lower dimension axiom, there is an 2-frame of 3 points $(O, I_1, \dots, I_n)$. Consider the lines $\ell_1 := \ell(O, I_1)$ and $\ell_2 := \ell(O, I_2)$. By the above two theorems, there exists a real-closed field F such that, for each ℓ_i , we have a bijection $\phi_i : \ell_i \rightarrow F$, such that

$$\ell_i \xrightarrow{\phi_i} F. \quad (25.4.4)$$

I.e., each function ϕ_i is an isomorphism from the axis ℓ_i to F .

Step 2. Let a point p be given. Let p_1, p_2 be its ordinates on ℓ_1 and ℓ_2 . We then define:

¹ See also Hilbert (1899), Tarski (1959), Borsuk & Szmielew (1960), Szczerba & Tarski (1965), Szczerba & Tarski (1979)

² A similar definition, *mutatis mutandis*, can be applied to $\mathbf{BG}(n)$ in general, for any $n \geq 2$.

$$x_1 := \phi_1(p_1) \quad (25.4.5)$$

$$x_2 := \phi_2(p_2) \quad (25.4.6)$$

$$\Phi(p) := \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad (25.4.7)$$

This defines a function $\Phi : \mathbb{P} \rightarrow F^2$.

Step 3. We claim that Φ is a bijection.

Step 4. We claim that the coordinate relation $B_{\mathbb{R}^2}$ represents B , with respect to the coordinate system Φ . I.e., for any $p, q, r \in \mathbb{P}$:

$$B(p, q, r) \leftrightarrow B_{F^2}(\Phi(p), \Phi(q), \Phi(r)) \quad (25.4.8)$$

To prove this, let p, q, r be given. Let p_1, p_2 be the ordinates of p on the axes, ℓ_1, ℓ_2 . Similarly, let q_1, q_2 be the ordinates of q on the axes, ℓ_1, ℓ_2 . Similarly, let r_1, r_2 be the ordinates of r on the axes, ℓ_1, ℓ_2 . First, for any p, q, r , we have:

$$B_{F^2}(\Phi(p), \Phi(q), \Phi(r)) \leftrightarrow (B_F(\phi_1(p_1), \phi_1(q_1), \phi_1(r_1)) \wedge B_F(\phi_2(p_2), \phi_2(q_2), \phi_2(r_2))) \quad (25.4.9)$$

Second, for any p, q, r ,

$$B(p, q, r) \leftrightarrow B(p_1, q_1, r_1) \wedge B(p_2, q_2, r_2) \quad (25.4.10)$$

Third, for any a, b, c on axis ℓ_i ,

$$B(a, b, c) \leftrightarrow B_F(\phi_i(a), \phi_i(b), \phi_i(c)) \quad (25.4.11)$$

Hence,

$$B(p, q, r) \leftrightarrow B_F(\phi_1(p_1), \phi_1(q_1), \phi_1(r_1)) \wedge B_F(\phi_2(p_2), \phi_2(q_2), \phi_2(r_2)) \quad (25.4.12)$$

$$\leftrightarrow B_{F^2}(\Phi(p), \Phi(q), \Phi(r)) \quad (25.4.13)$$

as claimed. □

We have the representation theorem for $\mathbf{BG}_0(2)$ as corollary:

Theorem 25.4.3 (Representation Theorem for $\mathbf{BG}_0(2)$). Let $\beta = (\mathbb{P}, B)$ be an $L(\mathbf{B})$ -structure. Then:

$$\beta \models \mathbf{BG}_0(2) \Leftrightarrow \beta \cong \mathbb{B}^2(F), \text{ for some real-closed field } F. \quad (25.4.14)$$

We can now generalize to arbitrary n :

Theorem 25.4.4 (Representation Theorem for $\mathbf{BG}_0(n)$). Let $\beta = (\mathbb{P}, B)$ be an $L(\mathbf{B})$ -structure. Let $n \geq 2$. Then:

$$\beta \models \mathbf{BG}_0(n) \Leftrightarrow \beta \cong \mathbb{B}^n(F), \text{ for some real-closed field } F. \quad (25.4.15)$$

25.5 Representation Theorem for $\mathbf{BG}(n)$

Despite the undoubted mathematical interest of Theorem 25.4.4, it is completely irrelevant to physics, and we shall mostly ignore it. What is of physical importance is the representation theorem for the second-order theories $\mathbf{BG}(n)$:

Theorem 25.5.1 (Representation Theorem for $\mathbf{BG}(n)$). Let $\beta = (\mathbb{P}, B)$ be an $L(\mathbf{B})$ -structure. Let $n \geq 2$. Then:

$$\beta \text{ is a full model of } \mathbf{BG}(n) \Leftrightarrow \beta \cong \mathbb{B}^n. \quad (25.5.1)$$

An immediate corollary is:

Theorem 25.5.2 ($\mathbf{BG}(4)$). For each n , $\mathbf{BG}(n)$ is categorical .

See Veblen (1904).

Chapter 26

Abstraction: Reals

26.1 Extension by Abstraction

In this Chapter, we follow the material set out in Burgess (1984).

Definition 26.1.1 (Equivalence formula). Fix a first-order one sorted signature σ , with variables $p_1, p_2, \dots$, and a theory T in $L(\sigma)$. Now a formula $\Theta(p_1, p_2)$ in $L(\sigma)$ with exactly the displayed variables is called *an equivalence formula* (in T) just if:

$$T \vdash \forall p_1 \Theta(p_1, p_1) \quad (26.1.1)$$

$$T \vdash \forall p_1 \forall p_2 (\Theta(p_1, p_2) \rightarrow \Theta(p_2, p_1)) \quad (26.1.2)$$

$$T \vdash \forall p_1 \forall p_2 \forall p_3 (\Theta(p_1, p_2) \wedge \Theta(p_2, p_3) \rightarrow \Theta(p_1, p_3)) \quad (26.1.3)$$

Definition 26.1.2 (Extension by abstraction). We introduce a new variable sort, with variables, let's say $x_1, x_2, \dots$, say, and a new unary function symbol, ξ , so that we now have terms $\xi(p_i)$.¹ The extended, two-sorted language is then $L(\sigma \cup \{\xi\})$.

T^+ in $L(\sigma \cup \{\xi\})$ is obtained by adding to T the following two abstraction axioms

$$(AB1) \quad \xi(p_1) = \xi(p_2) \leftrightarrow \Theta(p_1, p_2) \quad (26.1.4)$$

$$(AB2) \quad \forall x \exists p (x = \xi(p)) \quad (26.1.5)$$

T^+ is called an *extension of T by abstraction*.

Burgess shows (Theorem 1.3) that:

Theorem 26.1.3 (Burgess (1984), Theorem 1.3). Let

$$\varphi(p_1, \dots, p_n, x_1, \dots, x_k) \quad (26.1.6)$$

¹ Burgess writes as “[x]”: See Burgess (1984): 381.

be a formula of $L(\sigma \cup \{\xi\})$, with the displayed free variables. Then there is a formula

$$\varphi^*(p_1, \dots, p_n, p_{n+1}, \dots, p_{n+k}) \quad (26.1.7)$$

of the sublanguage $L(\sigma)$, with the displayed free variables, such that:

$$T^+ \vdash \bigwedge_{i=1}^k (x_i = \xi(p_{n+i})) \rightarrow [\varphi(p_1, \dots, p_n, x_1, \dots, x_k) \leftrightarrow \varphi^*(p_1, \dots, p_n, p_{n+1}, \dots, p_{n+k})] \quad (26.1.8)$$

Corollary 26.1.4 (Burgess (1984), Theorem 1.3). T^+ in $L(\sigma \cup \{\xi\})$ is a *conservative extension* of T in $L(\sigma)$.

Corollary 26.1.5 (Burgess (1984)). T^+ in $L(\sigma \cup \{\xi\})$ can be *interpreted* into T in $L(\sigma)$.

26.2 Abstraction Axioms for Reals: First-Order Case

Definition 26.2.1 (The language $L(\mathbf{B}, \xi, O, I)$). We extend the language $L(\mathbf{B})$ with a new variable sort, $x_1, x_2, \dots$, and a new 3-place mixed function symbol, ξ (on the original point sort), and two point constants O, I . This yields a number of new atomic formulas:

$$x_i = x_j \quad x_i = \xi(p, q, r) \quad \xi(p, q, r) = x_i \quad \xi(p, q, r) = \xi(a, b, c) \quad (26.2.1)$$

as well as atomic formulas featuring the constants O and I . The new two-sorted language is called $L(\mathbf{B}, \xi, O, I)$.

Definition 26.2.2 (Abstraction axioms). We now define the following (mixed) “abstraction axioms”.²

$$(\text{AR1}) \quad \text{rat}(p, q, r) \wedge \text{rat}(a, b, c) \rightarrow (\xi(p, q, r) = \xi(a, b, c) \leftrightarrow (p, q, r) \sim (a, b, c)) \quad (26.2.2)$$

$$(\text{AR2}) \quad \forall x \exists p \exists q \exists r (\text{rat}(p, q, r) \wedge x = \xi(p, q, r)) \quad (26.2.3)$$

$$(\text{AR3}) \quad \xi(p, p, r) = 0 \quad (26.2.4)$$

Here, $\text{rat}(p, q, r)$ expresses “ (p, q, r) is a ratio” (i.e., $p \neq q \wedge \text{co}_1(p, q, r)$). And the expression “ $(p, q, r) \sim (a, b, c)$ ” expresses the proportionateness relation of ratios, (p, q, r) and (a, b, c) , and is defined above (there are multiple cases) at Definition 24.2.5. In $\mathbf{BG}_0(4)$ this relation can be proved to be an equivalence relation (as stated at Lemma ?).³

² See Burgess (1984), 387, axioms (1) and (2).

³ The axiom (AR3) is for the “don’t care” case.

Definition 26.2.3. The theory $\text{BG}_0^\dagger(4)$ is defined to be the following two-sorted theory:

$$\text{BG}_0^\dagger(4) := \text{BG}_0(4) + \{(O \neq I), (\text{AR1}), (\text{A2}), (\text{A3})\}. \quad (26.2.5)$$

Theorem 26.2.4. $\text{BG}_0^\dagger(4)$ is a conservative extension of $\text{BG}_0(4)$.

Definition 26.2.5. In the language $L(\mathbf{B}, \xi, O, I)$, we introduce the following series of explicit definitions of symbols $0, 1, +, \times, \leq$:⁴

Table 26.1: **Explicit definitions of real number constants and operations**

Definiendum	Definiens
$x = 0$	$:= \exists r (\text{co}_1(O, I, r) \wedge x = \xi(O, I, r) \wedge r = O)$
$x = 1$	$:= \exists r (\text{co}_1(O, I, r) \wedge x = \xi(O, I, r) \wedge r = I)$
$x_3 = x_1 + x_2$	$:= \exists r_1 \exists r_2 \exists r_3 (\text{co}_1(O, I, r_1, r_2, r_3) \wedge x_1 = \xi(O, I, r_1) \wedge x_2 = \xi(O, I, r_2) \wedge x_3 = \xi(O, I, r_3) \wedge r_3 = r_1 + r_2)$
$x_3 = x_1 \times x_2$	$:= \exists r_1 \exists r_2 \exists r_3 (\text{co}_1(O, I, r_1, r_2, r_3) \wedge x_1 = \xi(O, I, r_1) \wedge x_2 = \xi(O, I, r_2) \wedge x_3 = \xi(O, I, r_3) \wedge r_3 = r_1 \times r_2)$
$x_1 \leq x_2$	$:= \exists r_1 \exists r_2 (\text{co}_1(O, I, r_1, r_2) \wedge x_1 = \xi(O, I, r_1) \wedge x_2 = \xi(O, I, r_2) \wedge r_1 \leq r_2)$

These explicitly define two constants $0, 1$, and two function symbols $+$, $\times$, and a predicate $\leq$.

We can now prove that the axioms of the theory RCF in $\text{BG}_0^\dagger(4)$. This depends upon the Greek Lemmas (24.4.1, 24.4.2, 24.4.3 and 24.4.4) above. So, I shall call this the “Greek theorem”:

Theorem 26.2.6 (The Greek Theorem). $\text{BG}_0^\dagger(4) \vdash \text{RCF}$.

Proof. As noted, this depends upon the Greek Lemmas (24.4.1, 24.4.2, 24.4.3 and 24.4.4) above.

We work inside $\text{BG}_0^\dagger(4)$. We define the geometrical addition $+$ (Definition 14.1.1), multiplication $\times$ (Definition 14.1.2) and order $\leq$ (Definition 16.3.4) on the line $\ell(O, I)$ as before. Above, we have the Greek Lemmas (24.4.1, 24.4.2, 24.4.3 and 24.4.4). From these, along with the explicit definitions in Definition 26.2.5, we prove each axiom of RCF. For example, we wish to prove $x + y = y + x$. Let x, y be given. Hence, we have points $r_1, r_2 \in \ell$ such that $x = \xi(O, I, r_1)$ and $y = \xi(O, I, r_2)$. Let $r_3 = r_1 + r_2$ (geometrical addition of points!). Hence, $\xi(O, I, r_3) = x + y$. But note that $r_1 + r_2 = r_2 + r_1$, by Greek Lemma 24.4.1(2). Hence, $\xi(O, I, r_3) = y + x$. Hence, $x + y = y + x$, as claimed. The proofs of the other axioms of RCF are similar, transferring the geometrical result in the Greek Lemmas to the corresponding algebraic one. $\square$

⁴ $\text{co}_1(O, I, r_1, r_2, r_3)$ expresses that O, I, r_1, r_2, r_3 are collinear. I.e., it is defined by: $\text{co}_1(O, I, r_1) \wedge \text{co}_1(O, I, r_2) \wedge \text{co}_1(O, I, r_3)$.

The Greek Theorem 26.2.6 implies tells us that algebra RCF can be *interpreted* into geometry $\text{BG}_0(4)$. We can define this interpretation as follows. It's a bit easier to give a *one-dimensional interpretation* of RCF into the theory $\text{BG}_0(4) + (O \neq I)$ (with two distinct fixed parameter points):

Definition 26.2.7. We define the “Greek translation” $\bullet : L(0, 1, +, \times, \leq) \rightarrow L(\mathbf{B}, O, I)$ as follows:

Table 26.2: **Greek Translation of $L(0, 1, +, \times, \leq)$ into $L(\mathbf{B}, O, I)$**

$(0)^\bullet$	$:= O$
$(1)^\bullet$	$:= I$
$(x_i)^\bullet$	$:= p_i$
$(x_i = x_j)^\bullet$	$:= (\text{co}_1(O, I, p_i) \wedge \text{co}_1(O, I, p_j) \wedge p_i = p_j)$
$(x_3 = x_1 + x_2)^\bullet$	$:= (\text{co}_1(O, I, p_1) \wedge \text{co}_1(O, I, p_2) \wedge \text{co}_1(O, I, p_3) \wedge p_3 = p_1 + p_2)$
$(x_3 = x_1 \times x_2)^\bullet$	$:= (\text{co}_1(O, I, p_1) \wedge \text{co}_1(O, I, p_2) \wedge \text{co}_1(O, I, p_3) \wedge p_3 = p_1 \times p_2)$
$(x_1 \leq x_2)^\bullet$	$:= (\text{co}_1(O, I, p_1) \wedge \text{co}_1(O, I, p_2) \wedge p_1 \leq p_2)$
$(\neg\varphi)^\bullet$	$:= \neg(\varphi)^\bullet$
$(\varphi \rightarrow \theta)^\bullet$	$:= (\varphi)^\bullet \rightarrow (\theta)^\bullet$
$(\forall x_i \varphi)^\bullet$	$:= \forall p_i (\text{co}_1(O, I, p_i) \rightarrow (\varphi)^\bullet).$

The result we obtain (with this one-dimensional translation) is then:

Theorem 26.2.8. If $\text{RCF} \vdash \varphi$, then $\text{BG}_0(4) + (O \neq I) \vdash (\varphi)^\bullet$.

Remark 26.2.9. Below, we consider the surprising completeness theorem for $\text{BG}_0(4)$. This result turns on the equally surprising “RCF Theorem” of Alfred Tarski (Tarski (1948)): that RCF is complete. Below we may first prove a reverse “Cartesian” interpretation, of $\text{BG}_0(4)$ into RCF. Then, the above result, the Greek Theorem 26.2.6, allows us to conclude that the reverse interpretation of $\text{BG}_0(4)$ into RCF is *faithful* (Lemma ??, and Theorem ??). $\square$

26.3 Reals: Second-Order Case

We next repeat all of this for the second-order theory $\text{BG}(4)$ in $L_2(\mathbf{B})$.

Definition 26.3.1. We extend the second-order $\text{BG}(4)$ with a new variable sort, x_i , and a new 3-place mixed function symbol, $\xi(p, q, r)$, and two point constants O, I . We call the extended language $L_2(\mathbf{B}, \xi, O, I)$.⁵ We then add the two “abstraction axioms” A1 and A2 above, and A3, and $O \neq I$. The result is called $\text{BG}^\dagger(4)$.

The second-order theory of real numbers, **ALG**, is defined in Appendix ??, §??. It is also known as the theory of the complete ordered field. We now show how to prove **ALG** in $\text{BG}^\dagger(4)$.

⁵ We do not add additional variables for “sets of ξ -objects”.

Theorem 26.3.2. $\text{BG}^\dagger(4) \vdash \text{ALG}$.

Proof. We explicitly define constants 0, 1, operations $+$, $\times$ and predicate $\leq$ on these new objects as before. The proof of the RCF axioms (ordered field axioms) proceeds just as above. With O, I as fixed parameters, we define a “surrogate membership relation” $\in^*$ as follows:

$$(*) \quad x \in^* X := \exists r (\text{co}_1(O, I, r) \wedge x = \xi(O, I, r) \wedge r \in X) \quad (26.3.1)$$

One then proves that the resulting abstracta satisfy the second-order axioms for a *complete ordered field*. In particular, we can prove the required Comprehension, Extensionality and Continuity Axiom (for reals):

COMPREHENSION	$\exists X \forall x (x \in^* X \leftrightarrow \varphi(x))$
EXTENSIONALITY	$\forall x (x \in^* X \leftrightarrow x \in^* Y) \rightarrow X = Y$
CONTINUITY	$\forall X \forall Y (X \neq \emptyset \wedge Y \neq \emptyset \wedge X \sqsubseteq Y \rightarrow \exists x (X \sqsubseteq x \sqsubseteq Y))$

I will focus on the proof of COMPREHENSION.⁶ Let a formula $\varphi(x)$ be given. Define $\theta(r)$ as follows:

$$(**) \quad \theta(r) := \exists x (\text{co}_1(O, I, r) \wedge x = \xi(O, I, r) \wedge \varphi(x)). \quad (26.3.2)$$

$\text{BG}(4)$ contains a point-set Comprehension axiom scheme. This gives the existence of a set X of points such that:

$$(***) \quad \forall r (r \in X \leftrightarrow \theta(r)) \quad (26.3.3)$$

We claim: $\forall x (x \in^* X \leftrightarrow \varphi(x))$.

Now let x be any real. By (A2), we have some configuration (a, b, c) with $x = \xi(a, b, c)$. Hence, by Lemma 24.2.7, there exists a point $r \in \ell(O, I)$ such that (O, I, r) and (a, b, c) are proportionate. By (A1), $\xi(O, I, r) = \xi(a, b, c)$. So, $x = \xi(O, I, r)$. We claim:

$$x \in^* X \leftrightarrow \varphi(x) \quad (26.3.4)$$

($\Rightarrow$): suppose $x \in^* X$. Then, by the definition $(*)$ above, there exists r such that $x = \xi(O, I, r)$ and $r \in X$. Thus, by $(***)$, $\theta(r)$. Hence, by $(**)$, there exists x' such that $\text{co}_1(O, I, r)$, $x' = \xi(O, I, r)$, and $\varphi(x')$. Thus, $x' = x$, and so $\varphi(x)$.

($\Leftarrow$): suppose $\varphi(x)$. By $(**)$, it follows that $\theta(r)$. By $(***)$, $r \in X$. And, by the definition $(*)$, $x \in^* X$. □

⁶ This corresponds to Burgess (1984), Theorem 2.1, part (c) (page 389). Burgess’s proof is a bit different. He adds five constants, O, I_1, I_2, I_3, I_4 requiring that they form a four-frame. I simply add O, I , with $O \neq I$, and prove the result for the reals on the line $\ell(O, I)$.

Chapter 27

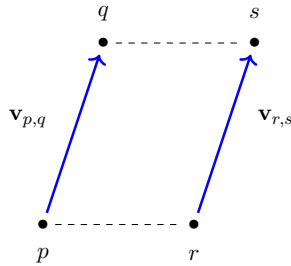
Abstraction: Vectors

27.1 Abstraction Axioms for Vectors

We proceed as above, but by the equivalence relation, on ordered pairs (p, q) , of *equipollence*:

Definition 27.1.1. (p, q) is equipollent to (r, s) just if p, q, s, r is a parallelogram:

Figure 27.1: **Equipollence**



Definition 27.1.2. We extend the second-order $\mathbf{BG}^\dagger(4)$ with a new variable sort, $\mathbf{v}$, and a new 2-place mixed function symbol, $\mathbf{v}_{p,q}$, and (instead of O, I) five new constants O, I_1, I_2, I_3, I_4 . We call the extended language $L_2(\mathbf{B}, \xi, \mathbf{v}, O, I_1, I_2, I_3, I_4)$ (note: we do not add additional variables for “sets of $\mathbf{v}$ s”). We then add the following two “abstraction axioms” A4 and A5:

$$A4 \quad \mathbf{v}_{p,q} = \mathbf{v}_{a,b} \leftrightarrow pq \equiv_{\text{equipollent}} ab \quad (27.1.1)$$

$$A5 \quad \forall \mathbf{v} (\exists p, q) (\mathbf{v} = \mathbf{v}_{p,q}) \quad (27.1.2)$$

and the axiom:

$$\mathbf{4\text{-frame}}(O, I_1, I_2, I_3, I_4) \quad (27.1.3)$$

The result is called $\mathbf{BG}^+(4)$:

$$\mathbf{BG}^4 := \mathbf{BG}(4) + \{\mathbf{4\text{-frame}}(O, I_1, I_2, I_3, I_4), A1, A2, A3, A4, A5\} \quad (27.1.4)$$

Theorem 27.1.3. $\mathbf{BG}^+(4)$ is a conservative extension of $\mathbf{BG}(4)$.

Proof. This is via Burgess's theorem. $\square$

Definition 27.1.4. In the language $L_2(\mathbf{B}, \xi, \mathbf{v}, O, I_1, I_2, I_3, I_4)$, we explicitly define $\mathbf{0}$, $+$, $\cdot$ on these new “ $\mathbf{v}$ -objects” as follows:

$$\mathbf{v} = \mathbf{0} \quad := \quad \exists p (\mathbf{v} = \mathbf{v}_{p,p}) \quad (27.1.5)$$

$$\mathbf{u} = \mathbf{v} + \mathbf{w} \quad := \quad (\exists p, q, r) (\mathbf{v} = \mathbf{v}_{p,q} \wedge \mathbf{w} = \mathbf{v}_{q,r} \wedge \mathbf{u} = \mathbf{v}_{p,r}) \quad (27.1.6)$$

$$\mathbf{w} = x \cdot \mathbf{v} \quad := \quad (\mathbf{v} = \mathbf{0} \wedge \mathbf{w} = \mathbf{0}) \vee (\exists p, q, r) (\mathbf{cnf}g(p, q, r) \wedge \mathbf{v} = \mathbf{v}_{p,q} \wedge \mathbf{w} = \mathbf{v}_{p,r} \wedge x = \xi(p, q, r)) \quad (27.1.7)$$

Lemma 27.1.5. $\mathbf{BG}^+(4)$ proves:

Associativity of vector addition	$\forall \mathbf{v} \forall \mathbf{w} \forall \mathbf{u} (\mathbf{v} + (\mathbf{w} + \mathbf{u}) = (\mathbf{v} + \mathbf{w}) + \mathbf{u})$
Commutativity of vector addition	$\forall \mathbf{v} \forall \mathbf{w} (\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v})$
Identity element	$\forall \mathbf{v} (\mathbf{v} + \mathbf{0} = \mathbf{v})$
Inverse elements	$\forall \mathbf{v} \exists \mathbf{w} (\mathbf{v} + \mathbf{w} = \mathbf{0})$
Compat	$\forall x \forall y \forall \mathbf{v} (x \cdot (y \cdot \mathbf{v}) = (x \times y) \cdot \mathbf{v})$
Identity element	$\forall \mathbf{v} (1 \cdot \mathbf{v} = \mathbf{v})$
Commutativity of vector addition	$\forall \mathbf{v} \forall \mathbf{w} \forall x (x \cdot (\mathbf{v} + \mathbf{w}) = x \cdot \mathbf{v} + x \cdot \mathbf{w})$
Commutativity of vector addition	$\forall \mathbf{v} \forall x \forall y ((x + y) \cdot \mathbf{v} = x \cdot \mathbf{v} + y \cdot \mathbf{v})$

Lemma 27.1.6. $\mathbf{BG}^+(4) \vdash$ “Vector space axioms”.

Definition 27.1.7. We define $\mathbf{e}_1 = \mathbf{v}_{O, I_1}$, and so on.

Then:

Lemma 27.1.8. $\mathbf{BG}^+(4) \vdash \forall \mathbf{v} \exists x_1 \exists x_2 \exists x_3 \exists x_4 (\mathbf{v} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + x_3 \mathbf{e}_3 + x_4 \mathbf{e}_4)$.

Definition 27.1.9. The definition of the action $(p, \mathbf{v}) \mapsto p + \mathbf{v}$ is:

$$q = p + \mathbf{v} \quad := \quad \mathbf{v} = \mathbf{v}_{p,q}. \quad (27.1.8)$$

One may then show:

Theorem 27.1.10. $\mathbf{BG}^+(4) \vdash$ “ $+$ is free and transitive action of $\mathbb{V}$ on $\mathbb{P}$ ”.

The affine space obtained in this way (basically, from the *vector space* $\mathbb{R}^4$, by “forgetting the origin”) is called $\mathbb{A}^4$.

The discussion and constructions above may be summarized in the following three theorems (I follow the usual practice of conflating the name of a structure with the name of its carrier set). According to $\mathbf{BG}^+(4)$, we have:

- | | |
|-----------|--|
| 0.510 (1) | $\mathfrak{R}$ is isomorphic to the complete ordered field $\mathbb{R}$. |
| (2) | $\mathbb{V}$ is isomorphic to the vector space $\mathbb{R}^4$. |
| (3) | $(\mathbb{P}, \mathbb{V}, +)$ is isomorphic to the affine space $\mathbb{A}^4(\mathbb{R})$. |

Chapter 28

The Completeness of Elementary Geometry

28.1 The Cartesian Translation

Definition 28.1.1. We define a (two-dimensional) translation, which we call the “Cartesian translation”:¹

$$^{\circ}: L(\mathbf{B}) \rightarrow L(0, 1, +, \times, \leq). \quad (28.1.1)$$

Table 28.1: **Cartesian translation** $^{\circ}: L(\mathbf{B}) \rightarrow L(0, 1, +, \times, \leq)$

Term/Formula		Translation
p_i	$(p_i)^{\circ}$	x_{2i}, x_{2i+1} .
$p_i = p_j$	$(p_i = p_j)^{\circ}$	$(x_{2i} = x_{2j} \wedge x_{2i+1} = x_{2j+1})$.
$\mathbf{B}(p_i, p_j, p_k)$	$\mathbf{B}(p_i, p_j, p_k)^{\circ}$	$(\exists \alpha \in [0, 1]) (\bigwedge_{n=0}^1 ((x_{2j+n} - x_{2i+n}) = \alpha(x_{2k+n} - x_{2i+n}))$.
$\neg \varphi$	$(\neg \varphi)^{\circ}$	$\neg(\varphi^{\circ})$.
$\varphi \wedge \theta$	$(\varphi \wedge \theta)^{\circ}$	$\varphi^{\circ} \wedge \theta^{\circ}$.
$\forall p_i \varphi$	$(\forall p_i \varphi)^{\circ}$	$\forall x_{2i} \forall x_{2i+1} \varphi^{\circ}$.

Lemma 28.1.2. Let $v : \text{Var}(L(0, 1, +, \times, \leq)) \rightarrow F$ be a valuation of the field variables into F . Then v induces a corresponding valuation $v' : \text{Var}(L(\mathbf{B})) \rightarrow F^2$ of the corresponding geometrical variables into the Cartesian product F^2 .

Proof. Let $v : \text{Var}(L(0, 1, +, \times, \leq)) \rightarrow F$ be a given valuation of field variables. Thus, for each variable x_i in $L(0, 1, +, \times, \leq)$, we have its value $v(x_i) \in F$. Now consider the i th variable p_i of $L(\mathbf{B})$. We define its value as follows:

¹ The generalization to any positive integer dimension (including one) is straightforward.

$$v'(p_i) := (v(x_{2i}), v(x_{2i+1})) \quad (28.1.2)$$

Hence, we have defined a corresponding valuation $v' : \text{Var}(L(\mathbf{B})) \rightarrow F^2$ over the variables of $L(\mathbf{B})$. □

Theorem 28.1.3. Let F be a real-closed field. Let $\varphi(p_1, \dots, p_n)$ be an $L(\mathbf{B})$ -formula. Let $\varphi^\circ(x_1, \dots, x_{2n+1})$ be its algebraic translation, in $L(0, 1, +, \times, \leq)$. Let $v : \text{Var}(L(0, 1, +, \times, \leq)) \rightarrow F$ be a valuation of field variables, and let $v' : \text{Var}(L(\mathbf{B})) \rightarrow F^2$ be the corresponding valuation of geometrical variables as per the above lemma. Then:

$$\mathbb{B}^4(F) \models_{v'} \varphi \Leftrightarrow F \models_v \varphi^\circ \quad (28.1.3)$$

Proof. We prove this by induction on the construction of $L(\mathbf{B})$ -formulas.

Base step: Let φ be the atomic formula $\mathbf{B}(p_1, p_2, p_3)$. Then φ° is the formula

$$(\exists \alpha \in [0, 1]) ((y_1, y_2) - (x_1, x_2) = \alpha((z_1, z_2) - (x_1, x_2))). \quad (28.1.4)$$

Let $v(x_1) = a_1, v(x_2) = a_2$; let $v(y_1) = b_1, v(y_2) = b_2$; let $v(z_1) = c_1, v(z_2) = c_2$, all with $a_1, a_2, b_1, b_2, c_1, c_2 \in F$. So, $v'(p) = \vec{a} = (a_1, a_2)$; let $v'(q) = \vec{b} = (b_1, b_2)$; $v'(r) = \vec{c} = (c_1, c_2)$. Then an equivalence stream gives:

$$\mathbb{B}^4(F) \models_{v'} \varphi \Leftrightarrow \mathbb{B}^4(F) \models_{v'} \mathbf{B}(p, q, r) \quad (28.1.5)$$

$$\Leftrightarrow (\exists \alpha \in [0, 1]) (\vec{b} - \vec{a} = \alpha(\vec{c} - \vec{a})) \quad (28.1.6)$$

$$\Leftrightarrow F \models_v \varphi^\circ \quad (28.1.7)$$

Induction steps. These go through straightforwardly. □

By restricting now to sentences, we have the immediate corollary:

Theorem 28.1.4. Let F be a real-closed field. Let φ be an $L(\mathbf{B})$ -sentence. Let φ° be its algebraic translation, in $L(0, 1, +, \times, \leq)$. Then:

$$\mathbb{B}^4(F) \models \varphi \Leftrightarrow F \models \varphi^\circ \quad (28.1.8)$$

Theorem 28.1.5. Let φ be an $L(\mathbf{B})$ -sentence. Let φ° be its algebraic translation, in $L(0, 1, +, \times, \leq)$. Then:

$$\mathbb{B}^4 \models \varphi \Leftrightarrow \mathbb{R} \models \varphi^\circ \quad (28.1.9)$$

Proof. $\mathbb{R}$ is a real-closed field. Then apply Theorem 28.1.4. $\square$

Theorem 28.1.6. Let F be a real-closed field. Let φ be an $L(\mathbb{B})$ -sentence. Then:

$$\mathbb{B}^4(F) \models \varphi \Leftrightarrow \mathbb{B}^4 \models \varphi \quad (28.1.10)$$

Proof. Let φ be an $L(\mathbb{B})$ -sentence. Let φ° be its algebraic translation, in $L(0, 1, +, \times, \leq)$. Then:

$$\mathbb{B}^4(F) \models \varphi \Leftrightarrow F \models \varphi^\circ \quad (\text{Theorem 28.1.4}) \quad (28.1.11)$$

$$\Leftrightarrow \mathbb{R} \models \varphi^\circ \quad (\text{Theorem C.0.7}) \quad (28.1.12)$$

$$\Leftrightarrow \mathbb{B}^4 \models \varphi \quad (\text{Theorem 28.1.5}) \quad (28.1.13)$$

$\square$

Theorem 28.1.6 can be restated as follows:

Theorem 28.1.7. Let F be a real-closed field. Then, the structures $\mathbb{B}^4(F)$ and $\mathbb{B}^4$ are elementarily equivalent. That is:

$$\mathbb{B}^4(F) \equiv_e \mathbb{B}^4 \quad (28.1.14)$$

From these Representation Theorems, we can show that, for each first-order theory, truth in the standard structure is equivalent to provability in that theory:

Theorem 28.1.8. The following metatheorems hold:

$$\text{BG}_0(4) \vdash \varphi \quad \text{if and only if} \quad \mathbb{B}^4 \models \varphi. \quad (28.1.15)$$

$$\text{EG}_0(3) \vdash \varphi \quad \text{if and only if} \quad \mathbb{E}^3 \models \varphi. \quad (28.1.16)$$

$$\text{Gal}_0(1, 3) \vdash \varphi \quad \text{if and only if} \quad \mathbb{G}^4 \models \varphi. \quad (28.1.17)$$

$$\text{Mink}_0(1, 3) \vdash \varphi \quad \text{if and only if} \quad \mathbb{M}^4 \models \varphi. \quad (28.1.18)$$

Proof. We start with (28.1.15). Suppose φ is an $L(\mathbb{B})$ -sentence such that $\text{BG}_0(4) \vdash \varphi$. Now $\mathbb{B}^4 \models \text{BG}_0(4)$. Thus, $\mathbb{B}^4 \models \varphi$. Conversely, suppose $\mathbb{B}^4 \models \varphi$, but $\text{BG}_0(4) \not\vdash \varphi$. By Completeness Theorem for predicate logic, there exists a model $\beta \models \text{BG}_0(4)$ with $\beta \models \neg\varphi$. By the Representation Theorem 25.4.2, $\beta \cong \mathbb{B}^4(F)$, for some real-closed field F . But, by Theorem 28.1.6, $\mathbb{B}^4(F)$ is elementarily equivalent to $\mathbb{B}^4$. Hence, $\mathbb{B}^4 \models \neg\varphi$, implying $\mathbb{B}^4 \not\models \varphi$, a contradiction.

The proofs for (28.1.16), (28.1.17) and (28.1.18) are analogous. $\square$

The above then yields the *completeness* of each of these synthetic geometrical theories as a corollary.

Theorem 28.1.9. Each of the first-order synthetic geometries $\text{BG}_0(4)$, $\text{EG}_0(3)$, $\text{Gal}_0(1, 3)$ and $\text{Mink}_0(1, 3)$ is complete.

Proof. For the $\text{BG}_0(4)$ case, suppose φ is an $L(\mathbf{B})$ -sentence such that $\text{BG}_0(4) \not\models \varphi$. By the above, $\mathbb{B}^4 \not\models \varphi$. Hence, $\mathbb{B}^4 \models \neg\varphi$. By the above again $\text{BG}_0(4) \vdash \neg\varphi$. Hence, $\text{BG}_0(4)$ is complete. The proofs for the other three cases are analogous. $\square$

A second corollary is *decidability*:

Theorem 28.1.10. Each first-order (“elementary”) synthetic geometry $\text{BG}_0(4)$, $\text{EG}_0(3)$, $\text{Gal}_0(1, 3)$ and $\text{Mink}_0(1, 3)$ is *decidable*.

Proof. The proof is a modification of the proof of Theorem C.0.6 for RCF. Each such theory is computably axiomatized, and also is complete by Theorem 28.1.9. Hence, each such theory is decidable. $\square$

28.2 The Cartesian Interpretation Theorem

Theorem 28.2.1. Let φ be an $L(\mathbf{B})$ -sentence. Then:

$$\text{BG}_0(4) \vdash \varphi \Rightarrow \text{RCF} \vdash \varphi^\circ. \quad (28.2.1)$$

Proof. The translation of each axiom of $\text{BG}_0(4)$ can be proved in RCF. $\square$

Theorem 28.2.2. Let φ be an $L(\mathbf{B})$ -sentence. Then:

$$\text{RCF} \vdash \varphi^\circ \Rightarrow \text{BG}_0(4) \vdash \varphi \quad (28.2.2)$$

Proof. Suppose this fails. Then we have φ , an $L(\mathbf{B})$ -sentence, such that $\text{RCF} \vdash \varphi^\circ$ and $\text{BG}_0(4) \not\models \varphi$. By Completeness, we have a model $\beta \models \text{BG}_0(4)$ such that $\beta \not\models \varphi$. By Representation Theorem 25.4.2, $\beta \cong \mathbb{B}^4(F)$, for some rcf F . Hence, $F \models \text{RCF}$, and $\mathbb{B}^4(F) \not\models \varphi$. Since $\text{RCF} \vdash \varphi^\circ$, it follows that $F \models \varphi^\circ$. However, by Theorem 28.1.4, $F \not\models \varphi^\circ$. Contradiction. $\square$

Combining Theorem 28.2.1 and Theorem 28.2.2, we have the following important theorem:

Theorem 28.2.3 (Cartesian Interpretation Theorem). Let φ be an $L(\mathbf{B})$ -sentence. Then:

$$\text{BG}_0(4) \vdash \varphi \Leftrightarrow \text{RCF} \vdash \varphi^\circ. \quad (28.2.3)$$

Theorem 28.2.4. $\text{BG}_0(4)$ is complete.

Proof. Suppose $\text{BG}_0(4) \not\models \varphi$. We claim $\text{BG}_0(4) \vdash \neg\varphi$. By Theorem 28.2.3, we have $\text{RCF} \not\models \varphi^\circ$. Hence, by the completeness of RCF, $\text{RCF} \vdash \neg\varphi^\circ$. Hence, by the definition of the translation for negated sentences, $\text{RCF} \vdash (\neg\varphi)^\circ$. Then by Theorem 28.2.3 again, we have $\text{BG}_0(4) \vdash \neg\varphi$, as claimed. $\square$

Theorem 28.2.5. Let φ be an $L(\mathbf{B})$ -sentence. Then:

$$\text{BG}_0(4) \vdash \varphi \Leftrightarrow \mathbb{R} \models \varphi^\circ. \quad (28.2.4)$$

Proof. By the Cartesian Interpretation theorem (Theorem 28.2.3), and Tarski's RCF theorem (Theorem C.0.4): $\square$

A corollary of the completeness of $\text{BG}_0(4)$ is *decidability*:

Theorem 28.2.6. $\text{BG}_0(4)$ is *decidable*.

Proof. The proof is a modification of the proof of Theorem C.0.6 for RCF. $\text{BG}_0(4)$ is computably axiomatized, and also is complete by Theorem 28.1.9. Hence, $\text{BG}_0(4)$ is decidable. $\square$

Chapter 29

The Incompleteness of Second-Order Geometry

29.1 PA_2 is Interpretable in $\text{BG}(4)$

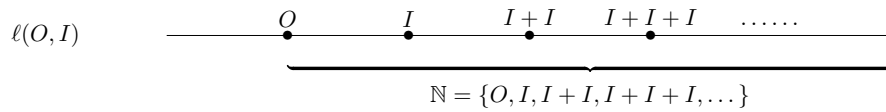
The second-order theory of natural numbers, PA_2 , is defined in Appendix ??, §??. It is also known as second-order arithmetic, or second-order Peano arithmetic. We now show how to interpret PA_2 into $\text{BG}(4)$. The proof is extremely similar to the interpretation of PA_2 into ALG given in Appendix ??, §??.

Theorem 29.1.1. $\text{BG}(4) + (O \neq I) \vdash \text{PA}_2^g(O, I)$.

Proof. We work in $\text{BG}(4) + (O \neq I)$. Let $\ell = \ell(O, I)$ the line containing both O and I . We define the addition $p \oplus_{(O, I)} q$, multiplication $p \otimes_{(O, I)} q$, and order $p \leq_{(O, I)} q$ as before. We shall again the drop the subscripts.

Considering the line ℓ , we may define the discrete sequence $O, I, I + I, I + I + I, \dots$ of points in ℓ . This provides a kind of “geometrical model” of the natural numbers:

Figure 29.1: Geometrical model of the natural numbers



We next give the following four explicit definitions:

$$\begin{aligned}
 S^g(p) &:= p + I \\
 \text{Ind}^g(X) &:= O \in X \wedge (\forall p \in \ell)(p \in X \rightarrow S^g(p) \in X). \\
 \text{nat}^g(p) &:= \forall X (\text{Ind}^g(X) \rightarrow p \in X) \\
 \mathbb{N}^g &:= \{p \in \ell \mid \text{nat}^g(p)\}
 \end{aligned}$$

Since, as we showed before, the complete ordered field axiom system ALG is already contained in $\text{BG}(4)$, we can then prove in $\text{BG}(4) + (O \neq I)$, first, Peano’s axioms:

- (1) $O \in \mathbb{N}^g$
- (2) $(\forall p \in \mathbb{N}^g) (S^g(p) \in \mathbb{N})$
- (3) $(\forall p \in \mathbb{N}^g) (S^g(p) \neq O)$
- (4) $(\forall p, q \in \mathbb{N}^g) (S^g(p) = S^g(q) \rightarrow p = q)$
- (5) $(\forall Y \subseteq \mathbb{N}^g) (\text{Ind}^g(Y) \rightarrow Y = X)$

And also, the recursion clauses:

- (6) $(\forall p \in \mathbb{N}^g) (p + O = p)$
- (7) $(\forall p, q \in \mathbb{N}^g) (p + S^g(q) = S^g(p + q))$
- (8) $(\forall p, q \in \mathbb{N}^g) (p \times O = O)$
- (9) $(\forall p, q \in \mathbb{N}^g) (p \times S^g(q) = (p \times q) + p)$

These are translations of the axioms of PA_2 . Let $\text{PA}_2^g(O, I)$ be the conjunction of the above, (1)–(9). Hence,

$$\text{BG}(4) + (O \neq I) \vdash \text{PA}_2^g(O, I) \quad (29.1.1)$$

□

Remark 29.1.2. Note that PA *cannot* be interpreted into elementary betweenness geometry $\text{BG}_0(4)$. □

29.2 The Incompleteness of BG(4)

The material in this section is closely related to a major objection to Field’s programme, as set out in Field (1980). This objection was raised in Shapiro (1983) and Burgess (1984). Theorem 29.1.1 states that $\text{BG}(4)$ interprets PA_2 . It follows that $\text{BG}(4)$ interprets PA . A corollary of Theorem 29.1.1 is that the second-order theory $\text{BG}(4)$ is *incomplete*. To see why, suppose that T is a computably axiomatized theory in some language L . We know that the syntax of T can be formalized, or arithmetized, in PA . And note also that $\text{BG}(4)$, in $L_2(\mathbb{B})$, is itself computably axiomatized. So, we can arithmetize the syntax of $\text{BG}(4)$ in PA . We suppose the defining arithmetic formulas of $L(0, \mathbb{S}, +, \times)$ are:

Table 29.1: **Arithmetized Syntax**

Formula	Informal meaning
$\text{Form}_{\text{BG}(4)}(n)$	n codes a formula of $L_2(\mathbb{B})$.
$\text{Ax}_{\text{BG}(4)}(n)$	n codes an axiom of $\text{BG}(4)$.
$\text{Proof}_{\text{BG}(4)}(n, k)$	n codes a derivation in $\text{BG}(4)$ of the formula coded by k .
$\text{Prov}_{\text{BG}(4)}(n)$	n codes a formula provable in $\text{BG}(4)$.

But we also know that $\text{BG}(4)$ interprets PA . This implies that we can “*geometrize*” the *syntax* of $\text{BG}(4)$ inside $\text{BG}(4)$. We add two constants O, I to the language of $\text{BG}(4)$. Informally, this fixes two distinct parameter points O, I , which determine a line $\ell(O, I)$. Then define:

Table 29.2: Geometrized Syntax

Formula	Informal meaning
$\text{Form}_{\text{BG}(4)}^g(p, O, I)$	$p \in \ell(O, I)$ codes n , where $\text{Form}_{\text{BG}(4)}(n)$.
$\text{Ax}_{\text{BG}(4)}^g(p, O, I)$	$p \in \ell(O, I)$ codes a number n , where $\text{Ax}_{\text{BG}(4)}(n)$.
$\text{Proof}_{\text{BG}(4)}^g(p, q, O, I)$	$p, q \in \ell(O, I)$, and p codes n , and q codes k , where $\text{Proof}_{\text{BG}(4)}(n, k)$.
$\text{Prov}_{\text{BG}(4)}^g(p, O, I)$	$p \in \ell(O, 1)$, and p codes a number n , such that $\text{Prov}_{\text{BG}(4)}(n)$.

Then we apply Gödel's First Incompleteness Theorem like this:

Theorem 29.2.1. There is a $L_2(\mathbf{B}, O, I)$ -sentence G such that neither G nor $\neg G$ is a theorem of $\text{BG}(4) + (O \neq I)$.

Proof. We let G be a suitable Gödel-Rosser sentence. Then, assuming that $\text{BG}(4) + (O \neq I)$ is consistent, $\text{BG}(4) + (O \neq I)$ does not prove either G or $\neg G$. $\square$

Definition 29.2.2. We define a formula $\text{Con}_{\text{BG}(4)+(O \neq I)}^g(O, I)$, saying that there is no encod derivation of some contradiction in $\text{BG}(4) + (O \neq I)$ on the line $\ell(O, I)$. Then let:

$$\text{Con}^g(\text{BG}(4) + (O \neq I)) := \exists O, I \text{Con}_{\text{BG}(4)+(O \neq I)}^g(O, I). \quad (29.2.1)$$

The sentence $\text{Con}^g(\text{BG}(4) + (O \neq I))$ says that $\text{BG}(4) + (O \neq I)$ is consistent.

Theorem 29.2.3. $\text{BG}(4) + (O \neq I) \not\models \text{Con}^g(\text{BG}(4) + (O \neq I))$.

Proof. This is an application of Gödel's Second Incompleteness Theorem. $\square$

The two parameters points O, I are themselves a calculational convenience ($\text{BG}(4) + (O \neq I)$ conservatively extends $\text{BG}(4)$), and the above can be finessed to:

Theorem 29.2.4. $\text{BG}(4) \not\models \text{Con}^g(\text{BG}(4))$.

Theorem 29.2.5. PA_2 is interpretable in each (second-order) theory $\text{BG}(4), \text{EG}(3), \text{Gal}(4)$ and $\text{Mink}(4)$.

This implies:

Theorem 29.2.6. Let T be any of the second-order synthetic geometries $\text{BG}(4), \text{EG}(3), \text{Gal}(4)$ and $\text{Mink}(4)$. Then there is an $L_2(\mathbf{B})$ -sentence G such that:

$$T \not\models G \quad (29.2.2)$$

$$T \not\models \neg G \quad (29.2.3)$$

Proof. I consider the BG(4) case. The interpretability of PA_2 in BG(4) allows one to follow Gödel's reasoning and formalize the syntactical proof theory of BG(4) in BG(4) itself. This permits the formulation of an $L_2(\mathbf{B})$ -sentence G , expressing its own unprovability in BG(4). Following Gödel's reasoning for the First Incompleteness Theorem, we have:

$$\text{BG}(4) \not\vdash G \quad (29.2.4)$$

$$\text{BG}(4) \not\vdash \neg G \quad (29.2.5)$$

The other cases are analogous. □

Thus, each second-order synthetic geometry BG(4), EG(3), Gal(4) and Mink(4) is incomplete.

Theorem 29.2.7. Let T be any of the second-order synthetic geometries BG(4), EG(3), Gal(4) and Mink(4). Then:

$$T \not\vdash \text{Con}(T) \quad (29.2.6)$$

Proof. I consider the BG(4) case. The interpretability of PA_2 in BG(4) allows one to follow Gödel's reasoning and formalize the syntactical proof theory of BG(4) in BG(4) itself. This permits the formulation of an $L_2(\mathbf{B})$ -sentence $\text{Con}(\text{BG}(4))$, expressing that BG(4) is consistent. Following Gödel's reasoning for the Second Incompleteness Theorem, we have:

$$\text{BG}(4) \not\vdash \text{Con}(\text{BG}(4)) \quad (29.2.7)$$

The other cases are analogous. □

Please note that the Gödel sentence G and the consistency sentence $\text{Con}(\text{BG}(4))$ are second-order sentences of $L_2(\mathbf{B})$, not first-order ones.

Part IV

Euclidean Geometry

Chapter 30

Congruence and Orthogonality: Euclidean Geometry

30.1 Introduction

From Chapter 16 to Chapter 27, we have focused all of our attention on the synthetic geometries, $\text{BG}_0(n)$ and $\text{BG}(n)$: the synthetic betweenness geometries, in various dimensions. I have a background reason for this focus: physics. The reason is that $\text{BG}(4)$ is the crucial geometrical subsystem of Galilean spacetime geometry and Minkowski geometry.

The above methods and results, specific to $\text{BG}_0(n)$ and $\text{BG}(n)$, can be applied equally, with suitable modifications, to the more expressive geometrical theories. In particular, the synthetic Euclidean geometry, of which betweenness geometry is a subsystem.

30.2 Euclid Proposition I

A while back, we gave Euclid's original five postulates:

Euclid's Postulates

Let the following be postulated:

1. To draw a straight line from any point to any point.
2. To produce (extend) a finite straight line continuously in a straight line.
3. To describe a circle with any centre and distance (radius).
4. That all right angles are equal to one another.
5. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which the angles are less than two right angles.

As we have seen, the formalized system $\mathbf{BG}_0(n)$ of betweenness geometry indeed implements Euclid's First, Second and Fifth Postulates (essentially, Playfair's Axiom). However, that leaves the Third and Fourth Postulates. The Third Postulate introduces the notion of a circle, with a given centre and radius. The Fourth Postulate introduces the notion of a right angle. These notions rest on *congruence* and *orthogonality*.

Let's consider Euclid's Proposition I

$$S1 := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\} \quad (30.2.1)$$

$$S2 := \{(x, y) \in \mathbb{R}^2 \mid (x - 1)^2 + y^2 = 1\} \quad (30.2.2)$$

Find $(x, y) \in S1 \cap S2$. Thus,

$$x^2 + y^2 = x^2 - 2x + 1 + y^2 \quad (30.2.3)$$

$$\Rightarrow 0 = -2x + 1 \quad (30.2.4)$$

$$\Rightarrow x = \frac{1}{2} \quad (30.2.5)$$

$$\Rightarrow \frac{1}{4} + y^2 = 1 \quad (30.2.6)$$

$$\Rightarrow y = \pm \frac{\sqrt{3}}{2} \quad (30.2.7)$$

30.3 Euclidean Geometry

30.4 Tarski-Style Axiomatization

First, I give a little history of Tarski-style axiomatization of geometry. We know, from Tarski & Givant, that Tarski's first axiomatic system was developed in the academic year 1926–1927:¹

¹ Tarski & Givant note that Tarski (1967) is reproduced on pp. 289–346 of Tarski [46], volume 4.

120 The axiom set for Euclidean geometry adopted in Schwabhäuser et al. (1983) originates with Tarski. In its original form it was constructed in 1926–27 and presented in his course given that year at the University of Warsaw (see Tarski (1967), footnote 34). It appeared in the paper Tarski (1967), submitted for publication in 1940, but published only in 1967 in a restricted number of copies. (Tarski & Givant (1999): 188)

Tarski's 1927/1948 axiomatization:

Table 30.1: **Axioms of $EG_0^t(2)$**

B-Inclusion	$B(p, q, p) \rightarrow p = q.$
B-Identity	$B(p, q, p) \rightarrow p = q.$
B-Symmetry	$B(p, q, p) \rightarrow B(r, q, p).$
B-Transitivity	$B(p, q, r) \wedge B(q, r, s) \wedge q \neq r \rightarrow B(p, q, s).$
B-Connectivity	$B(p, q, r) \wedge B(p, q, r') \wedge p \neq q \rightarrow (B(p, r, r') \vee B(p, r', r)).$
B-Extension	$\exists p (B(p, q, r) \wedge p \neq q).$
Pasch	$B(p, q, r) \wedge B(s, u, q) \rightarrow \exists x (B(r, x, s) \wedge B(p, u, x)).$
Desargues	See Definition 19.1.2.
Euclid	$B(a, d, t) \wedge B(b, d, c) \wedge a \neq d \rightarrow \exists x \exists y (B(a, b, x) \wedge B(a, c, y) \wedge B(x, t, y))$
Lower dimension	There exist $n + 1$ points which are not co_{n-1} .
Upper dimension	Any $n + 2$ points are co_n .
Continuity axiom scheme	$\exists r \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow B(r, p, q)) \rightarrow \exists s \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow B(p, s, q))$

The original source of this axiomatization is Tarski (1948), Tarski (1951) and Tarski (1959). And later, in Tarski & Givant (1999). See Tarski (1959), pp. 19–20, for a formulation of the first-order two-dimensional theory, with twelve axioms and one axiom scheme (for continuity); and Tarski & Givant (1999) for a simplification down to ten axioms and one axiom scheme (for continuity). The above axiom system is the second-order, four-dimensional theory (i.e., the single Continuity Axiom is the second-order one).

Using the congruence primitive $\equiv$, several new axioms are introduced, one of whose main purposes is to establish that $\equiv$ is an equivalence relation. Four of these new

congruence axioms are:

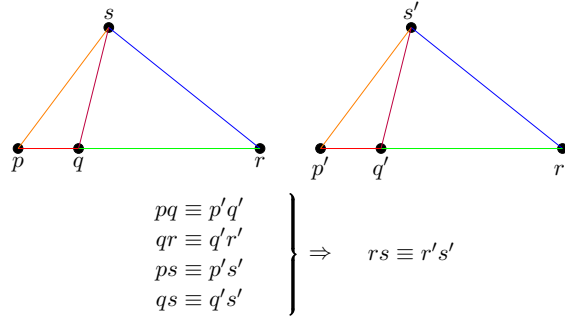
$\equiv$ -Identity	$pq \equiv rr \rightarrow p = q.$
$\equiv$ -Transitivity	$pq \equiv rs \wedge pq \equiv tu \rightarrow rs \equiv tu.$
$\equiv$ -Reflexivity	$pq \equiv qp.$
$\equiv$ -Extension	$\exists r (B(p, q, r) \wedge qr \equiv su).$

There is a fifth new axiom for congruence:

$$\text{5-Segment} \quad (p \neq q \wedge B(p, q, r) \wedge B(p', q', r') \wedge pq \equiv p'q' \wedge qr \equiv q'r' \wedge ps \equiv p's' \wedge qs \equiv q's') \rightarrow rs \equiv r's'.$$

To understand the intuitive meaning of this formal axiom consider the following diagram:

Figure 30.1: Tarski's Five Segment axiom



There are two triangles, Δprs and $\Delta p'r's'$, and each has a distinct point (q or q') lying on the bottom segment pr or $p'r'$. Each triangle then features five segments: pq , qr , ps , qs and rs in the first triangle Δprs ; and five corresponding segments $p'q'$, $q'r'$, $p's'$, $q's'$ and $r's'$ in the second triangle $\Delta p'r's'$. The corresponding segments are coloured the same. Tarski's axiom states that if four of the corresponding pairs are congruent, then the fifth pair are congruent.

This is equivalent to the famous SAS axiom of school geometry.

Definition 30.4.1. The first-order (usually called “elementary”) theory $EG_0(2)$ in $L(B, \equiv)$ is given by the following ten axioms, and one axiom scheme:

B-Identity	$B(p, q, p) \rightarrow p = q.$
$\equiv$ -Identity	$pq \equiv rr \rightarrow p = q.$
$\equiv$ -Transitivity	$pq \equiv rs \wedge pq \equiv tu \rightarrow rs \equiv tu.$
$\equiv$ -Reflexivity	$pq \equiv qp.$
$\equiv$ -Extension	$\exists r (B(p, q, r) \wedge qr \equiv su).$
Pasch	$B(p, q, r) \wedge B(s, u, r) \rightarrow \exists x (B(q, x, s) \wedge B(u, x, p)).$
Euclid	$B(a, d, t) \wedge B(b, d, c) \wedge a \neq d \rightarrow \exists x \exists y (B(a, b, x) \wedge B(a, c, y) \wedge B(x, t, y))$
5-Segment	$(p \neq q \wedge B(p, q, r) \wedge B(p', q', r') \wedge pq \equiv p'q' \wedge qr \equiv q'r' \wedge ps \equiv p's' \wedge qs \equiv q's' \rightarrow rs \equiv r's').$
Lower Dimension	There exist three points which are not collinear.
Upper Dimension	Any four points are coplanar.
Continuity Scheme	$\exists r \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow B(r, p, q)) \rightarrow \exists s \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow B(p, s, q))$

Lemma 30.4.2. PaschBet is a subsystem of $EG_0(2)$.

$$EG_0(2) \vdash \text{PaschBet.} \quad (30.4.1)$$

I.e.

$$EG_0(2) \vdash B(p, q, q). \quad (30.4.2)$$

$$EG_0(2) \vdash B(p, q, r) \rightarrow B(r, q, p). \quad (30.4.3)$$

$$EG_0(2) \vdash B(p, q, r) \wedge B(q, r, s) \wedge q \neq r \rightarrow B(p, q, s). \quad (30.4.4)$$

$$EG_0(2) \vdash B(p, q, r) \wedge B(p, q, r') \wedge p \neq q \rightarrow (B(p, r, r') \vee B(p, r', r)). \quad (30.4.5)$$

$$EG_0(2) \vdash \exists p (B(p, q, r) \wedge p \neq q). \quad (30.4.6)$$

Corollary 30.4.3. $BG_0(2)$ is a subsystem of $EG_0(2)$.

Lemma 30.4.4. $EG_0(3)$ proves the following six sentences:

$$EG_0(2) \vdash pq \equiv pq. \quad (30.4.7)$$

$$EG_0(2) \vdash pq \equiv rs \rightarrow rs \equiv pq. \quad (30.4.8)$$

$$EG_0(2) \vdash pq \equiv rs \wedge rs \equiv tu \rightarrow pq \equiv tu. \quad (30.4.9)$$

$$EG_0(2) \vdash pq \equiv rs \rightarrow pq \equiv sr. \quad (30.4.10)$$

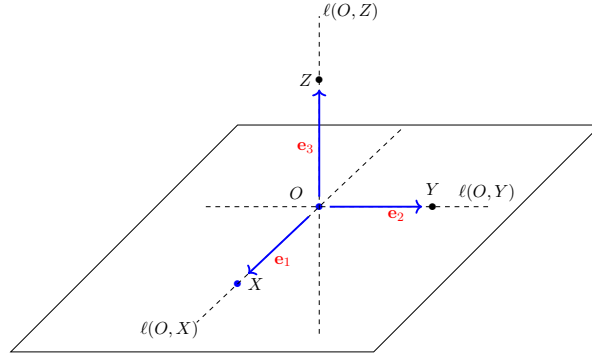
$$EG_0(2) \vdash pq \equiv rs \rightarrow qp \equiv rs. \quad (30.4.11)$$

$$EG_0(2) \vdash pq \equiv rs \rightarrow qp \equiv sr \quad (30.4.12)$$

Definition 30.4.5 (Perpendicularity). In Euclidean geometry, perpendicularity $OX \perp OY$, for three distinct points O, X, Y , is defined as follows: $OX \perp OY$ holds iff $XY \equiv (-X)Y$, where $(-X)$ is the unique point p on $\ell(O, X)$ such that $p \neq X$, and $Op \equiv OX$.

Definition 30.4.6 (Euclidean 3-frame). For Euclidean geometry, a Euclidean 3-frame is an ordered quadruple O, X, Y, Z of points which are not co_2 (i.e., not coplanar), and such that the segments OX, OY, OZ are mutually perpendicular and of equal length.

Figure 30.2: Euclidean 3-Frame



An extremely detailed development of $\text{EG}_0(3)$ is given in Schwabhäuser et al. (1983).

30.5 The Second-Order Theory

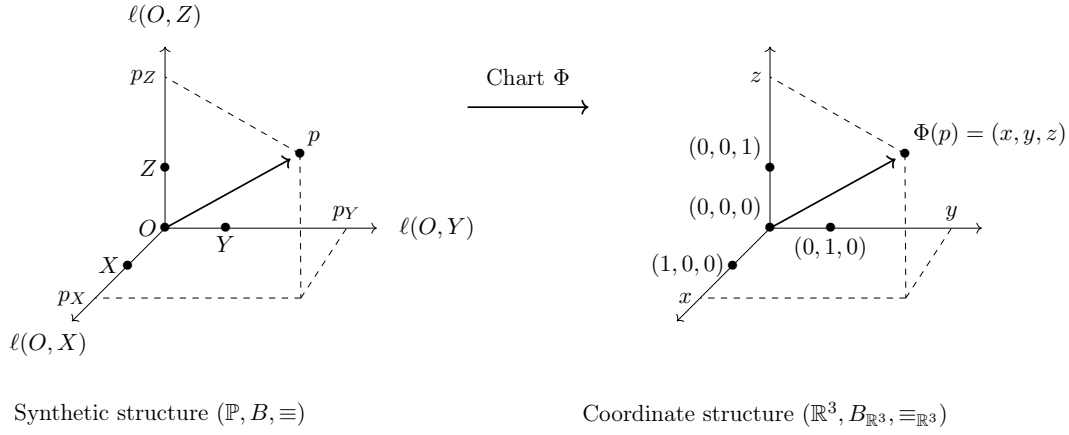
Definition 30.5.1. The non-logical axioms of $\text{EG}(3)$ in $L_2(\mathcal{B}, \equiv)$ are the following eleven:

B-Identity	$\mathcal{B}(p, q, p) \rightarrow p = q.$
$\equiv$ -Identity	$pq \equiv rr \rightarrow p = q.$
$\equiv$ -Transitivity	$pq \equiv rs \wedge pq \equiv tu \rightarrow rs \equiv tu.$
$\equiv$ -Reflexivity	$pq \equiv qp.$
$\equiv$ -Extension	$\exists r (\mathcal{B}(p, q, r) \wedge qr \equiv su).$
Pasch	$\mathcal{B}(p, q, r) \wedge \mathcal{B}(s, u, r) \rightarrow \exists x (\mathcal{B}(q, x, s) \wedge \mathcal{B}(u, x, p)).$
Euclid	$\mathcal{B}(a, d, t) \wedge \mathcal{B}(b, d, c) \wedge a \neq d \rightarrow \exists x \exists y (\mathcal{B}(a, b, x) \wedge \mathcal{B}(a, c, y) \wedge \mathcal{B}(x, t, y))$
5-Segment	$(p \neq q \wedge \mathcal{B}(p, q, r) \wedge \mathcal{B}(p', q', r') \wedge pq \equiv p'q' \wedge qr \equiv q'r' \wedge ps \equiv p's' \wedge qs \equiv q's') \rightarrow rs \equiv r's'.$
Lower Dimension	There exist four points which are not co_2 .
Upper Dimension	Any five points are co_3 .
Continuity Axiom	$[\exists r (\forall p \in X_1) (\forall q \in X_2) \mathcal{B}(r, p, q)] \rightarrow \exists s (\forall p \in X_1) (\forall q \in X_2) \mathcal{B}(p, s, q)$

30.6 Coordinatization

The coordinatization theorem, or representation theorem, for synthetic Euclidean geometry (i.e., the theories $\text{EG}_0(n)$ and $\text{EG}(n)$) is a modification of the corresponding theorem for betweenness geometry (i.e., the theories $\text{BG}_0(n)$ and $\text{BG}(n)$).

Coordinate model of Euclidean geometry (dim = 3)



Fixing on dimension 3, the major modification is the proof of the required isomorphism condition for congruence, $\equiv$:

$$pq \equiv rs \leftrightarrow \Phi(p)\Phi(q) \equiv_{\mathbb{R}^3} \Phi(r)\Phi(s) \quad (30.6.1)$$

where $\equiv_{\mathbb{R}^3}$ is defined as above:

$$\mathbf{x}\mathbf{y} \equiv_{\mathbb{R}^3} \mathbf{z}\mathbf{w} := \sqrt{\sum_i (\mathbf{x}^i - \mathbf{y}^i)^2} = \sqrt{\sum_i (\mathbf{z}^i - \mathbf{w}^i)^2} \quad (30.6.2)$$

First, we given the representation theorem for the elementary theory $\mathbf{EG}_0(2)$. This is indeed Tarski's Theorem 1, in Tarski (1959).

Theorem 30.6.1. Let $M = (\mathbb{P}, B, \equiv)$ be a $L(B, \equiv)$ -structure. Assume that $M \models \mathbf{EG}_0(2)$. Suppose O, I_1, I_2 is a Euclidean 3-frame in M . Then there exists a real-closed field F , and a bijection $\Phi : \mathbb{P} \rightarrow F^3$ such that:

- (a) Φ matches O, I_1, I_2 .
- (b) For all $p, q, r \in \mathbb{P}$: $(p, q, r) \in B \leftrightarrow B_{F^2}(\Phi(p), \Phi(q), \Phi(r))$.
- (c) For all $p, q, r, s \in \mathbb{P}$: $pq \equiv rs \leftrightarrow \Phi(p)\Phi(q) \equiv_{F^2} \Phi(r)\Phi(s)$.

Proof. Theorem 1, Tarski (1959). □

We are interested in the second-order theory $\mathbf{EG}(2)$, and, more generally, the second-order theories $\mathbf{EG}(n)$ (with $n \geq 2$). For the $n = 3$ case, the representation theorem is:

Theorem 30.6.2. Let $M = (\mathbb{P}, B, \equiv)$ be a $L(\mathcal{B}, \equiv)$ -structure. Assume that M is a *full* model of $\mathbf{EG}(3)$ (we sometimes write: $M \models_2 \mathbf{EG}(3)$). Suppose O, I_1, I_2, I_3 is a Euclidean 3-frame in M . Then there exists a bijection $\Phi : \mathbb{P} \rightarrow \mathbb{R}^3$ such that:

- (a) Φ matches O, I_1, I_2, I_3 .
- (b) For all $p, q, r \in \mathbb{P}$: $(p, q, r) \in B \leftrightarrow B_{\mathbb{R}^3}(\Phi(p), \Phi(q), \Phi(r))$.
- (c) For all $p, q, r, s \in \mathbb{P}$: $pq \equiv rs \leftrightarrow \Phi(p)\Phi(q) \equiv_{\mathbb{R}^3} \Phi(r)\Phi(s)$.

The difference between this result and Theorem 30.6.1 (i.e., Theorem 1 of Tarski (1959)) are that Tarski considers the two-dimensional first-order theory, $\mathbf{EG}_0(2)$, with the first-order continuity axiom scheme. The Representation Theorem in Tarski (1959) asserts that, given a model $M \models \mathbf{EG}_0(2)$ and a Euclidean frame, there is a real-closed field F such that the conditions (a), (b), (c) hold. When we strengthen the theory to the second-order Continuity axiom, it follows that this field F is in fact $\mathbb{R}$.

Theorem 30.6.3 (Representation Theorem for $\mathbf{EG}(3)$). Let $M = (\mathbb{P}, B, \equiv)$ be an $L(\mathcal{B}, \equiv)$ structure. Let $\mathbf{EG}^3$ be $(\mathbb{R}^3, B_{\mathbb{R}^3}, \equiv_{\mathbb{R}^3})$, the standard coordinate structure. Then:

$$M \text{ is a full model of } \mathbf{EG}(3) \quad \text{if and only if} \quad M \cong \mathbf{EG}^3. \quad (30.6.3)$$

Part V

Spacetime Geometries

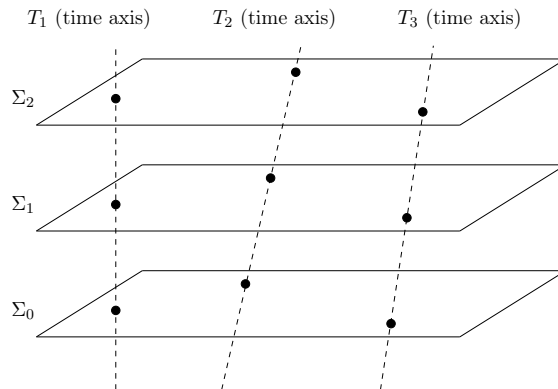
Chapter 31

Galilean Spacetime

31.1 Introduction

In this Chapter, we provide a synthetic (and second-order) axiom system, which I call $\text{Gal}(4)$, which describes Galilean spacetime and does so categorically.¹ Galilean spacetime is a system $\mathbb{P}$ of points on which are defined three physical geometrical primitives, satisfying certain conditions.² Galilean spacetime can be thought of as the background geometry of the system of spacetime events for Newtonian classical mechanics:

Figure 31.1: **Galilean Spacetime**



¹ In Ketland (2023), this theory was called $\text{Gal}(1, 3)$, with the parameters “1” and “3” meaning: “1 time and 3 space dimensions”.

² I think, informally, of a Galilean spacetime *modally*: a *physically possible world* with certain distinguished, or built-in, geometrical (spatio-temporal) relations. Such metaphysical issues, however, don’t matter here, as our whole discussion below is about models of $\text{Gal}(4)$.

31.2 Axiomatization

Definition 31.2.1 ($\text{Gal}(4)$). The signature of the language of $\text{Gal}(4)$ is $\{\mathbf{B}, \sim, \equiv\}$, where $\mathbf{B}$ is a three-place predicate, $\sim$ is a binary predicate, and $\equiv$ is a four-place predicate. The language of $\text{Gal}(4)$ is the (monadic) second-order language $L_2(\mathbf{B}, \sim, \equiv)$. The non-logical axioms of the theory $\text{Gal}(4)$ are:³

Theory	$\text{Gal}(4)$
Gal1	$\mathbf{BG}(4)$ (background four-dimensional geometry)
Gal2	$[\mathbf{EG}(3)]\sim$ (each simultaneity hypersurface is Euclidean)
Gal3	$\sim$ is an equivalence relation.
Gal4	$pq \equiv\sim rs \rightarrow (p \sim q \wedge p \sim r \wedge p \sim s)$.
Gal5	$pq \equiv\sim rs \rightarrow (p + \mathbf{v})(q + \mathbf{v}) \equiv\sim (r + \mathbf{v})(s + \mathbf{v})$.

31.3 Coordinatization

Precisely the same methods apply to give us the coordinatization theorem (or Representation Theorem) for the spacetime geometry $\text{Gal}(4)$.

Theorem 31.3.1 (Representation Theorem for $\text{Gal}(4)$). Let $\gamma = (\mathbb{P}, B, \sim, \equiv)$ be an $L(\mathbf{B}, \sim, \equiv)$ -structure. Let $\mathbb{G}^4 = (\mathbb{R}^3, B_{\mathbb{R}^3}, \sim_{\mathbb{R}^4}, \equiv_{\mathbb{R}^4})$ be the standard coordinate structure (for Galilean spacetime). Then:

$$\mu \text{ is a full model of } \text{Gal}(4) \iff \mu \cong \mathbb{G}^4. \quad (31.3.1)$$

Proof. Ketland (2023), Theorem 55. □

³ The second-order logic axioms are Extensionality ($\forall x (x \in X \leftrightarrow x \in Y) \rightarrow X = Y$) and Comprehension ($\exists X \forall x (x \in X \leftrightarrow \varphi(x))$).

Chapter 32

Aristotelian and Newtonian Spacetimes

32.1 Introduction

In the previous chapter, we formulated categorical synthetic four-dimensional Galilean spacetime geometry $\text{Gal}(4)$ in the (monadic) second-order language $L_2(\mathbf{B}, \sim, \equiv \sim)$.

In this chapter, we extend $\text{Gal}(4)$ in $L_2(\mathbf{B}, \sim, \equiv \sim)$ in two successive ways. First, we add a new unary predicate/property $\mathbf{L}(p)$ expressing “the point p lies on the special geocentric timelike line”, and add the following axiom:

$$(A) \quad (\exists p, q \in \mathbf{L}) [p \not\sim q \wedge \forall r (\mathbf{L}(r) \leftrightarrow \mathbf{L}(p, q, r))] \quad (32.1.1)$$

This will take us to a theory I call $\text{Arist}(4)$ in $L_2(\mathbf{B}, \sim, \equiv \sim, \mathbf{L})$, standing for “Aristotelian spacetime in four dimensions”. Within this theory, we can *define* a binary relation $\text{rest}(p, q)$, which says that points p and q are on the same timelike line, and are “mutually at rest”. This definition is quite simple:

$$\text{rest}(p, q) \quad := \quad p = q \vee \ell(p, q) \parallel \mathbf{L}. \quad (32.1.2)$$

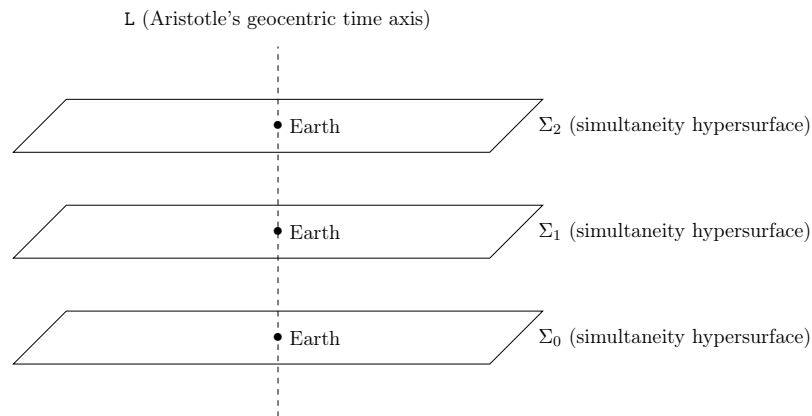
It is useful to consider a useful conservative extension, which we call $\text{Arist}'(4)$ in $L_2(\mathbf{B}, \sim, \equiv \sim, O, I)$ of $\text{Gal}(4)$ obtained by adding two new point constants, O and I , and the single axiom $O \not\sim I$. This theory can easily be shown to be a conservative extension—though not a definitional extension—of both $\text{Gal}(4)$ and $\text{Arist}(4)$.

Second, we can forget the Aristotelian geocentric line predicate $\mathbf{L}$ and introduce a binary primitive $\text{rest}(p, q)$ as primitive, along with a couple of suitable axioms. This will take us a theory I call $\text{Newt}(4)$ in $L_2(\mathbf{B}, \sim, \equiv \sim, \text{Rest})$, standing for “Newtonian spacetime in four dimensions”.

32.2 Informal Pictures of Aristotelian and Newtonian Spacetime

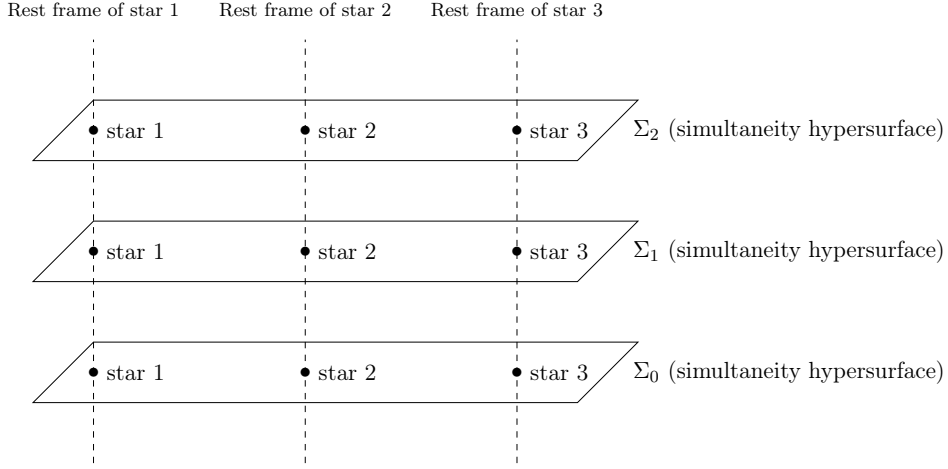
On Aristotle's view, the Earth (at any fixed time) is a kind of *centre* (or is located at that centre) of the universe. As we trace the Earth's world tube through time, we have a special *time axis* (i.e., timelike line), which might be called the *geocentric time axis*. So, Aristotelian spacetime looks like this:

Figure 32.1: Aristotelian Spacetime



On the other hand, Newton abolishes geocentrism, and considers not a single time axis, but instead a *pencil of time axes*. The notion of a pencil of time axes here is four dimensional and would not have been available to Newton! But even so, a pencil is what it is. The world lines of the “fixed stars” can then be assumed to correspond to these timelike lines in this pencil. They are all “at rest”.

Figure 32.2: Newtonian Spacetime



It should be noted, of course, Newton himself certainly did not assume that The Sun is “at rest in Absolute Space”. Indeed, Newton was acutely aware of the fact that, and stated that, the Sun’s being at rest is indistinguishable from any uniform non-accelerated motion of the Sun.

32.3 Standard Coordinate Structures

With the above informal description, we can define standard coordinate relations on $\mathbb{R}^4$ corresponding to both Aristotle’s line L and Newton’s relation **rest**.

Definition 32.3.1. For $\mathbf{x} \in \mathbb{R}^4$, we write:

$$\mathbf{x} = \begin{pmatrix} \vec{x} \\ x_4 \end{pmatrix} \quad (32.3.1)$$

There is a standard coordinate structure for Aristotelian spacetime. Essentially, we just take the basic “ t -axis” in $\mathbb{R}^4$ (i.e., those points (x, y, z, t) with $x = y = z = 0$) as the line to which our new predicate $\mathbb{T}$ is going to refer.

Definition 32.3.2. We let

$$\mathbf{x} \in L_{\mathbb{R}^4} \quad := \quad \vec{x} = 0 \quad (32.3.2)$$

$$(\mathbf{x}, \mathbf{y}) \in \mathbf{rest}_{\mathbb{R}^4} \quad := \quad \vec{x} = \vec{y} \quad (32.3.3)$$

Thus, points $\mathbf{x}$ and $\mathbf{y}$ are “at rest” just if their “spatial” components are identical. Clearly,

$$L_{\mathbb{R}^4} = \{(0, 0, 0, t) \in \mathbb{R}^4 \mid t \in \mathbb{R}\} \quad (32.3.4)$$

Definition 32.3.3. And we define:

$$AS^4 := (\mathbb{R}^4, B_{\mathbb{R}^4}, \sim, \equiv_{\mathbb{R}^4}, L_{\mathbb{R}^4}) \quad (32.3.5)$$

$$NS^4 := (\mathbb{R}^4, B_{\mathbb{R}^4}, \sim, \equiv_{\mathbb{R}^4}, \mathbf{rest}_{\mathbb{R}^4}) \quad (32.3.6)$$

Then AS^4 is called the *standard coordinate structure* for Aristotelian spacetime. NS^4 is called the *standard coordinate structure* for Newtonian spacetime.

It is easy to see that $\mathbf{rest}_{\mathbb{R}^4}$ is definable, given $L_{\mathbb{R}^4}$:

Lemma 32.3.4. $\mathbf{rest}_{\mathbb{R}^4} = \{(\mathbf{x}, \mathbf{y}) \mid \mathbf{x} = \mathbf{y} \vee \ell(\mathbf{x}, \mathbf{y}) \parallel L_{\mathbb{R}^4}\}.$

32.4 Axiomatizing Aristotelian Spacetime

To *axiomatize* Aristotelian spacetime, need only add a single primitive symbol, a unary predicate L , standing for Aristotle's special geocentric time-axis (through the Earth), so that $L(p)$ means “ p lies on the geocentric time-axis”.

And a single axiom for it:

Definition 32.4.1. *Aristotle's Axiom* is the following sentence in $L(B, \sim, \equiv, L)$:

$$(A) \quad (\exists p, q \in L) [p \not\sim q \wedge \forall r (L(r) \leftrightarrow L(p, q, r))] \quad (32.4.1)$$

The *axioms of Aristotelian spacetime geometry* $\mathbf{Arist}^{(1,3)}$ in $L(B, \sim, \equiv, L)$ are simply the result of adding Aristotle's axiom to the Galilean spacetime axiom system $\mathbf{Gal}(4)$, as follows:

$$\mathbf{Arist}(4) := \mathbf{Gal}(4) \wedge (A) \quad (32.4.2)$$

In addition, we add the *explicit definition* of a binary predicate $\mathbf{rest}$:

Definition 32.4.2. The explicit definition of $\mathbf{rest}$ is the following:

$$\mathbf{rest}(p, q) := p = q \vee (p \neq q \wedge \overbrace{\neg \exists r (\mathbf{co}_1(p, q, r) \wedge L(r))}^{\ell(p, q) \parallel L}) \quad (32.4.3)$$

The following three lemmas are fairly easy to prove in :

Lemma 32.4.3. $\mathbf{Arist}(4)$ proves:

- 0.510 (1) L is a time axis.
 (2) rest is an equivalence relation.
 (3) For any points p, q, r, s :

$$p \neq q \wedge r \neq s \wedge \text{rest}(p, q) \rightarrow (\text{rest}(r, s) \leftrightarrow \ell(p, q) \parallel \ell(r, s)). \quad (32.4.4)$$

Next, we prove that $\text{Arist}(4)$ is sound (in the standard structure):

Theorem 32.4.4. $\mathbb{AS}^4 \models \text{Arist}(4)$.

Proof. The reduct $(\mathbb{R}^4, B_{\mathbb{R}^4}, \sim_{\mathbb{R}^4}, \equiv_{\mathbb{R}^4})$ of $\mathbb{AS}^4$, obtained by forgetting Aristotle's line $L_{\mathbb{R}^4}$ (in $\mathbb{R}^4$) is the standard coordinate structure for Galilean spacetime. Hence $M \models \text{Gal}(4)$. We also claim that $\mathbb{AS}^4$ satisfies Aristotle's axiom (A). Aristotle's axiom

$$\exists p \exists q [p \not\sim q \wedge L(p) \wedge L(q) \wedge (\forall r)(L(r) \leftrightarrow r \in \ell(p, q))] \quad (32.4.5)$$

says, relative to $\mathbb{AS}^4$, that there exist points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^4$ such that

$$\mathbf{x} \not\sim_{\mathbb{R}^4} \mathbf{y} \wedge L_{\mathbb{R}^4}(\mathbf{x}) \wedge L_{\mathbb{R}^4}(\mathbf{y}) \quad (32.4.6)$$

and such that, for all $\mathbf{z} \in \mathbb{R}^4$, we have:

$$L_{\mathbb{R}^4}(\mathbf{z}) \leftrightarrow \mathbf{z} \in \ell(\mathbf{x}, \mathbf{y}) \quad (32.4.7)$$

So, we search for two such points in $\mathbb{R}^4$. They are easy to find. Let

$$\mathbf{x} := (0, 0, 0, 0) \quad (32.4.8)$$

$$\mathbf{y} := (0, 0, 0, 1). \quad (32.4.9)$$

Then these points are not simultaneous with respect to $\sim_{\mathbb{R}^4}$. And so $\mathbf{x} \not\sim_{\mathbb{R}^4} \mathbf{y}$ is true. Moreover, both lie in Aristotle's geometric time-axis $L_{\mathbb{R}^4}$. And since they are distinct, they determine a unique line $\ell(\mathbf{x}, \mathbf{y})$ which is precisely the time-axis $L_{\mathbb{R}^4}$. Thus, Aristotle's axiom is true in $\mathbb{AS}^4$. $\square$

32.5 Representation Theorem for Aristotelian Spacetime

Definition 32.5.1. An *Aristotelian 4-frame* (in M) is a quintuple O, I_1, I_2, I_3, I_4 of points (in M) which is a Galilean 4-frame, and also such that $O \in L^M$ and $I_4 \in L^M$ (the points O and I_4 lie in the geocentric line).

Theorem 32.5.2 (Representation Theorem for Aristotelian Spacetime). Let $M = (\mathbb{P}, \mathbf{B}^M, \sim^M, (\equiv\sim)^M, \mathbf{L}^M)$ be a (full) structure for $L_2(\mathbf{B}, \sim, \equiv\sim, \mathbf{L})$.

Let M be a full model of $\mathbf{Arist}(4)$. Then

$$M \cong \mathbb{AS}^4 \quad (32.5.1)$$

Proof. Let $M = (\mathbb{P}, \mathbf{B}^M, \sim^M, (\equiv\sim)^M, \mathbf{L}^M)$. We assume M is a full model of $\mathbf{Arist}(4)$. We claim: $M \cong \mathbb{AS}^4$. Hence, we must construct an isomorphism $\Phi : M \rightarrow \mathbb{AS}^4$.

Let O, I_1, I_2, I_3, I_4 be an Aristotelian 4-frame in M .

Consider the reduct $M_0 := (\mathbb{P}, \mathbf{B}^M, \sim^M, (\equiv\sim)^M)$, obtained by forgetting Aristotle's geocentric line (in M). Since M is a full model of $\mathbf{Arist}(4)$, we infer that M_0 is a full model of $\mathbf{Gal}(4)$, and therefore $M_0 \cong \mathbb{G}^4$. So, M_0 is a model of Galilean spacetime. By our earlier results, the Galilean 4-frame O, I_1, I_2, I_3, I_4 gives us an *inertial frame* on this reduct to Galilean spacetime. I.e., an isomorphism,

$$\Phi : M_0 \rightarrow \mathbb{G}^4 \quad (32.5.2)$$

such that the usual matching conditions holds for the benchmarks (i.e., $\Phi(O) = (0, 0, 0, 0)$, $\Phi(I_2) = (1, 0, 0, 0)$, $\dots$, $\Phi(I_4) = (0, 0, 0, 1)$).

Now since M is a full model of $\mathbf{Arist}(4)$, we have Aristotle's axiom (A) holds in M . So, there are non-simultaneous points $p, q \in \text{dom}(M)$, with $p \in \mathbf{L}^M$, $q \in \mathbf{L}^M$, and $p \not\sim q$, satisfying:

$$\mathbf{L}^M = \ell(p, q) \quad (32.5.3)$$

In general, a Galilean transformation will allow us to find a new inertial frame Φ' , such that, for all points s ,

$$\Phi'(s) = A\Phi(s) + \mathbf{d} \quad (32.5.4)$$

We easily see that, by choosing A and $\mathbf{d}$ correctly, there exists a Galilean transformation to a new inertial frame Φ' such that, for our two given fixed points p, q :

$$\Phi'(p) = (0, 0, 0, 0) \quad \Phi'(q) = (0, 0, 0, 1) \quad (32.5.5)$$

Since p and q have the same spatial coordinates relative to Φ' , it follows that, for any $r \in \mathbb{P}$.

$$r \in \ell(p, q) \iff (\exists \alpha \in \mathbb{R}) (\Phi'(r) = (0, 0, 0, \alpha)) \quad (32.5.6)$$

$$\iff \Phi'(r) \in \mathbf{L}_{\mathbb{R}^4} \quad (32.5.7)$$

(Since the set of points of the form $(0, 0, 0, \alpha)$ is simply $L_{\mathbb{R}^4}$.)
Thus, for any point $r \in \mathbb{P}$, we have:

$$r \in L^M \leftrightarrow \Phi'(r) \in L_{\mathbb{R}^4} \quad (32.5.8)$$

And this establishes that

$$\Phi' : (\mathbb{P}, B^M, \sim^M, (\equiv \sim)^M, L^M) \rightarrow (\mathbb{R}^4, B_{\mathbb{R}^4}, \sim_{\mathbb{R}^4}, \equiv \sim_{\mathbb{R}^4}, L_{\mathbb{R}^4}) \quad (32.5.9)$$

is an isomorphism, as claimed. $\square$

32.6 Newtonian Spacetime and the Notion of “Rest”

The axiomatization of Newtonian spacetime is obtained by an extension of $\text{Gal}(4)$, obtained by adding a primitive binary geometric relation, “**rest**(p, q)”.

Definition 32.6.1. We define the signature of the axiomatic theory of Newtonian spacetime as follows:

$$\sigma_{\text{Newt}} := \{B, \sim, \equiv \sim, \mathbf{rest}\} \quad (32.6.1)$$

Definition 32.6.2. We define the three Newtonian “*rest axioms*” in $L(\sigma_{\text{Newt}})$ as follows:

110 Rest1	rest is an equivalence relation.
Rest2	$\exists p, q (p \not\sim q \wedge \mathbf{rest}(p, q))$.
Rest3	$(p \neq q \wedge r \neq s \wedge \mathbf{rest}(p, q)) \rightarrow (\mathbf{rest}(r, s) \leftrightarrow \ell(p, q) \parallel \ell(r, s))$.

The second axiom states that **rest** is non-trivial.

Definition 32.6.3. The axioms of the theory of Newtonian spacetime $\text{Newt}(4)$ are given as follows:

$$\text{Newt}(4) := \text{Gal}(4) + \{\text{Rest1}, \text{Rest2}, \text{Rest3}\} \quad (32.6.2)$$

Notice that Rest1, Rest2 and Rest3 are first-order axioms. As usual, with all of our axiomatizations, the sole genuine second-order axiom is the Dedekind Continuity Axioms.

32.7 Representation Theorem for Newtonian Spacetime

Theorem 32.7.1 (Representation Theorem for Newtonian Spacetime). Let M be a full structure for $L_2(\sigma_{\text{Newt}})$. Then

$$M \models \text{Newt}(4) \quad \text{iff} \quad (\exists \Phi : M \rightarrow \mathbb{R}^4) (M \cong \overset{\Phi}{\mathbb{NS}}^4). \quad (32.7.1)$$

32.8 Two Theorems Concerning Newtonian Spacetime

Lemma 32.8.1. Extend $\text{Gal}(4)$ with two new constants O, I and the axiom

$$O \not\sim I \quad (32.8.1)$$

Call the result $\text{Gal}^*(4)$. Let ϕ be a formula *not* containing O, I , and such that

$$\text{Gal}^*(4) \vdash \phi \quad (32.8.2)$$

Then

$$\text{Gal}(4) \vdash \phi \quad (32.8.3)$$

Proof. Recall that

$$\text{Gal}(4) \vdash (\exists p, q)(p \not\sim q) \quad (32.8.4)$$

Now let ϕ be a formula not containing O, I , such that

$$\text{Gal}^*(4) \vdash \phi \quad (32.8.5)$$

Reason as follows:

$$\text{Gal}^*(4) \vdash \phi \quad (32.8.6)$$

$$\Rightarrow \text{Gal}(4) + (O \not\sim I) \vdash \phi \quad (32.8.7)$$

$$\Rightarrow \text{Gal}(4) \vdash (O \not\sim I) \rightarrow \phi \quad (32.8.8)$$

$$\Rightarrow \text{Gal}(4) \vdash \forall p, q((p \not\sim q) \rightarrow \phi) \quad (32.8.9)$$

$$\Rightarrow \text{Gal}(4) \vdash (\exists p, q)(p \not\sim q) \rightarrow \phi \quad (32.8.10)$$

$$\Rightarrow \text{Gal}(4) \vdash \phi \quad (32.8.11)$$

The final line is the claim. $\square$

Definition 32.8.2. Define $\text{Gal}^*(4)$ as above. Let

$$\Theta(p, q) := (p = q) \vee \ell(p, q) \parallel \ell(O, I) \quad (32.8.12)$$

Lemma 32.8.3. $\text{Gal}^*(4)$ proves that $\Theta(p, q)$ is an equivalence relation.

Proof. First we claim:

$$\text{Gal}^* \vdash \Theta(p, p) \quad (32.8.13)$$

Obviously, we have $p = p$, which implies $(p = p) \vee (p \not\sim p \wedge \ell(p, p) \parallel \ell(O, I))$ (despite $\ell(p, p)$ being ill-defined!).

Suppose $\Theta(p, q)$ holds. That is,

$$(p = q) \vee \ell(p, q) \parallel \ell(O, I) \quad (32.8.14)$$

We claim $\Theta(p, q)$ holds. That is,

$$(q = p) \vee \ell(q, p) \parallel \ell(O, I) \quad (32.8.15)$$

But this is clearly so, as $\Theta(p, q)$ is symmetric when we flip variables.

Suppose $\Theta(p, q)$ and $\Theta(q, r)$ hold. That is,

$$(p = q) \vee \ell(p, q) \parallel \ell(O, I) \quad (32.8.16)$$

$$(q = r) \vee \ell(q, r) \parallel \ell(O, I) \quad (32.8.17)$$

We claim $\Theta(p, r)$ holds:

$$(p = r) \vee \ell(p, r) \parallel \ell(O, I) \quad (32.8.18)$$

There are four cases to consider.

Case (i). $p = q$ and $q = r$. Then $p = r$, and so $\Theta(p, r)$ holds.

Case (ii). $p = q$ and $q \not\sim r$ and $\ell(q, r) \parallel \ell(O, I)$. Then, by substitution, $\ell(p, r) \parallel \ell(O, I)$. And so, $\Theta(p, r)$ holds.

Case (iii). $p \not\sim q$ and $\ell(p, q) \parallel \ell(O, I)$ and $q = r$. Same as case (ii), with variable relabelling.

Case (iv). $p \not\sim q$ and $\ell(p, q) \parallel \ell(O, I)$ and $q \not\sim r$ and $\ell(q, r) \parallel \ell(O, I)$.

If $p = r$, then $\Theta(p, r)$ holds. So assume $p \neq r$. Since $\ell(p, q) \parallel \ell(O, I)$ and $\ell(q, r) \parallel \ell(O, I)$, we conclude $\ell(p, r) \parallel \ell(O, I)$ by transitivity of $\parallel$. So, $\Theta(p, r)$ holds. $\square$

Lemma 32.8.4. Let $\text{Gal}^*(4)$ and $\Theta(p, q)$ be defined as above. Then:

$$\text{Gal}^* \vdash (\exists p, q)(p \not\sim q \wedge \Theta(p, q)) \quad (32.8.19)$$

Proof. We claim that $\text{Gal}^*(4)$ proves:

$$(\exists p, q)(p \not\sim q \wedge ((p = q) \vee \ell(p, q) \parallel \ell(O, I))) \quad (32.8.20)$$

We already have, since $O \not\sim I$ is an axiom of $\text{Gal}^*(4)$ and $\parallel$ is reflexive:

$$\text{Gal}^*(4) \vdash (O \not\sim I \wedge \ell(O, I) \parallel \ell(O, I)) \quad (32.8.21)$$

Thus,

$$\text{Gal}^*(4) \vdash (\exists p, q)(p \not\sim q \wedge ((p = q) \vee \ell(p, q) \parallel \ell(O, I))) \quad (32.8.22)$$

as claimed. $\square$

Lemma 32.8.5. Let Gal^* and $\Theta(p, q)$ be defined as above. Then:

$$\text{Gal}^*(4) \vdash (p \neq q \wedge r \neq s \wedge \Theta(p, q)) \rightarrow (\Theta(r, s) \leftrightarrow \ell(p, q) \parallel \ell(r, s)) \quad (32.8.23)$$

Proof. We reason inside $\text{Gal}^*(4)$. Let $p \neq q$ and $r \neq s$ and $\Theta(p, q)$. Thus,

$$\ell(p, q) \parallel \ell(O, I) \quad (32.8.24)$$

We claim:

$$(i) \quad \Theta(r, s) \rightarrow \ell(p, q) \parallel \ell(r, s) \quad (32.8.25)$$

$$(ii) \quad \ell(p, q) \parallel \ell(r, s) \rightarrow \Theta(r, s) \quad (32.8.26)$$

So, for (i), suppose $\Theta(r, s)$. Thus,

$$(r = s) \vee \ell(r, s) \parallel \ell(O, I) \quad (32.8.27)$$

Since $r \neq s$, we may infer:

$$\ell(r, s) \parallel \ell(O, I) \quad (32.8.28)$$

Now we have $\ell(p, q) \parallel \ell(O, I)$. And we have $\ell(r, s) \parallel \ell(O, I)$. By transitivity, $\ell(p, q) \parallel \ell(r, s)$, as claimed.

For (ii), we suppose $\ell(p, q) \parallel \ell(r, s)$. We wish to show that $\Theta(r, s)$ holds. That is:

$$(r = s) \vee \ell(r, s) \parallel \ell(O, I) \quad (32.8.29)$$

But $\ell(p, q) \parallel \ell(O, I)$. Hence, $\ell(r, s) \parallel \ell(O, I)$. So, $\Theta(r, s)$ holds. $\square$

Next I will prove two interesting results about the theory $\text{Newt}(4)$ of Newtonian spacetime. The first is a Conservativeness Theorem, saying that the axioms for Newtonian spacetime are conservative over those for Galilean spacetime.

Theorem 32.8.6 (Conservativeness Theorem for Newtonian Spacetime). Let ϕ be a formula not containing the symbol **rest**. Then:

$$\text{Newt}(4) \vdash \phi \Rightarrow \text{Gal}(4) \vdash \phi. \quad (32.8.30)$$

Proof. We shall prove this by first extending $\text{Gal}(4)$ and $\text{Newt}(4)$ with two new constants O, I and the axiom

$$O \not\sim I \quad (32.8.31)$$

to theories $\text{Gal}^*(4)$ and $\text{Newt}^*(4)$.

I can show that there is a translation mapping,

$$\circ: L(\sigma_{\text{Newt}, O, I}) \rightarrow L(\sigma_{\text{Gal}, O, I}) \quad (32.8.32)$$

which will interpret any formula $L(\sigma_{\text{Newt}, O, I})$ containing the predicate **rest** as an $L_2(\sigma_{\text{Gal}, O, I})$ formula (i.e., not containing **rest**); and, under this translation, the *rest* axioms of $\text{Newt}^*(4)$ become *theorems* of $\text{Gal}^*(4)$. Using methods already described, namely

Theorem ??, this translation of $\text{Newt}^*(4)$ axioms into $\text{Gal}^*(4)$ theorems gives us *conservativeness* of $\text{Newt}^*(4)$ over $\text{Gal}^*(4)$ for formulas not containing **rest**. It is then easy to show (Lemma 32.8.1) that $\text{Gal}^*(4)$ is conservative over $\text{Gal}(4)$, which will complete the proof.

So, first, we let the signature $\sigma_{\text{Gal}, O, I}$ be the result of adding the two constant O, I to the Galilean signature σ_{Gal} , and let $\sigma_{\text{Newt}, O, I}$ be the result of adding the two new constants O, I to the Newtonian signature σ_{Newt} . Then we define:

$$\text{Newt}^*(4) := \text{Newt}(4) + (O \not\sim I) \quad (32.8.33)$$

$$\text{Gal}^*(4) := \text{Gal}(4) + (O \not\sim I) \quad (32.8.34)$$

The translation $^\circ$ is defined as follows. It commutes with connectives and quantifiers, and is a homophonic on all primitives, except **rest**, for which the translation (for an atomic formula) is given by:

$$(\text{rest}(p, q))^\circ := (p = q) \vee \ell(p, q) \parallel \ell(O, I) \quad (32.8.35)$$

In other words, its translation is the formula $\Theta(p, q)$ we defined and studied above.

Then we can show that the $^\circ$ -translation of each of the three Newtonian *rest axioms* of $\text{Newt}^*(4)$ is a theorem of $\text{Gal}^*(4)$:

$$\text{Gal}^*(4) \vdash (\text{Rest1})^\circ \quad (32.8.36)$$

$$\text{Gal}^*(4) \vdash (\text{Rest2})^\circ \quad (32.8.37)$$

$$\text{Gal}^*(4) \vdash (\text{Rest3})^\circ \quad (32.8.38)$$

The first of these is Lemma 32.8.3. Then second is Lemma 32.8.4. The third is Lemma 32.8.5.

Using the proof-conversion method used to prove Theorem ??, it follows that, when ϕ is an $L_2(\sigma_{\text{Newt}, O, I})$ -formula *not* containing the predicate **rest**, we have:

$$(A) \quad \text{If } \text{Newt}^*(4) \vdash \phi \text{ then } \text{Gal}^*(4) \vdash \phi \quad (32.8.39)$$

Now suppose that ϕ is an $L_2(\sigma_{\text{Newt}})$ -formula *not* containing the predicate **rest** (i.e., it does not contain O, I also), and such that

$$\text{Newt}(4) \vdash \phi \quad (32.8.40)$$

Then, obviously, we have $\text{Newt}^*(4) \vdash \phi$, since $\text{Newt}^*(4)$ is an *extension* of $\text{NST}(1, 3)$ (by a single axiom). By the result (A) above, we have,

$$\text{Gal}^*(4) \vdash \phi \quad (32.8.41)$$

But $\text{Gal}^*(4)$ is a *conservative extension* of $\text{Gal}(4)$ for formulas not containing the new constants O, I , by Lemma 32.8.1. So, we have,

$$\text{Gal}(4) \vdash \phi \quad (32.8.42)$$

as claimed. □

Second:

Theorem 32.8.7 (Reduction Theorem). Γ is a time axis if and only if there exists a rest axis Γ' , and there exists a velocity $\mathbf{v} \in \mathbb{V}^3$ such that the relative velocity of Γ and Γ' is $\mathbf{v}$.

It is this theorem which shows how to convert Newton's own formulation of the First Law of Motion (aka, The Law of Inertia),

112 (LIa)	A body with no forces acting on it remains at rest, or moves with uniform speed and direction relative to any reference system at rest.
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into the more usual formulation of the Law of Inertia,

112 (LIb)	A body with no forces acting on it moves at constant speed and direction relative to an inertial frame.
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which has nothing to do with *rest* frames.

Chapter 33

Minkowski Spacetime

Chapter 34

Six Main Synthetic Geometries

34.1 Definitions

We have now constructed no less than six main *first-order synthetic geometries*:

Table 34.1: Synthetic geometries (first-order)

Theory	Signature	
$\text{BG}_0(n)$	$\mathbf{B}$	First-order betweenness geometry in $L(\mathbf{B})$ in dimension $n \geq 2$.
$\text{EG}_0(n)$	$\mathbf{B}, \equiv$	First-order Euclidean geometry in $L(\mathbf{B}, \equiv)$ in dimension $n \geq 2$.
$\text{Gal}_0(4)$	$\mathbf{B}, \sim, \equiv \sim$	First-order 4D Galilean spacetime geometry in $L(\mathbf{B}, \sim, \equiv \sim)$.
$\text{Arist}_0(4)$	$\mathbf{B}, \sim, \equiv \sim, \mathbf{L}$	First-order 4D Aristotelian spacetime geometry in $L(\mathbf{B}, \sim, \equiv \sim, \mathbf{L})$.
$\text{Newt}_0(4)$	$\mathbf{B}, \sim, \equiv \sim, \mathbf{rest}$	First-order 4D Newtonian spacetime geometry in $L(\mathbf{B}, \sim, \equiv \sim, \mathbf{rest})$.
$\text{Mink}_0(4)$	$\mathbf{B}, \lambda$	First-order 4D Minkowski spacetime geometry in $L(\mathbf{B}, \lambda)$.

Corresponding to each, we have a *second-order synthetic geometry*:

Table 34.2: Synthetic geometries (second-order)

Theory	Signature	
$\text{BG}(n)$	$\mathbf{B}$	Second-order betweenness geometry in $L_2(\mathbf{B})$ in dimension $n \geq 2$.
$\text{EG}(n)$	$\mathbf{B}, \equiv$	Second-order Euclidean geometry in $L_2(\mathbf{B}, \equiv)$ in dimension $n \geq 2$.
$\text{Gal}(4)$	$\mathbf{B}, \sim, \equiv \sim$	Second-order 4D Galilean spacetime geometry in $L_2(\mathbf{B}, \sim, \equiv \sim)$.
$\text{Arist}(4)$	$\mathbf{B}, \sim, \equiv \sim, \mathbf{L}$	Second-order 4D Aristotelian spacetime geometry in $L(\mathbf{B}, \sim, \equiv \sim, \mathbf{L})$.
$\text{Newt}(4)$	$\mathbf{B}, \sim, \equiv \sim, \mathbf{rest}$	Second-order 4D Newtonian spacetime geometry in $L(\mathbf{B}, \sim, \equiv \sim, \mathbf{rest})$.
$\text{Mink}(4)$	$\mathbf{B}, \lambda$	Second-order 4D Minkowski spacetime geometry in $L_2(\mathbf{B}, \lambda)$.

The axioms for the second-order theories $\text{BG}(4)$, $\text{EG}(3)$ and $\text{Gal}(4)$ may be found in Ketland (2023). Each first-order theory $\text{BG}_0(4)$, $\text{EG}_0(3)$ and $\text{Gal}_0(1, 3)$ is obtained by dropping Comprehension and Extensionality, and replacing the Continuity axiom by the *Continuity axiom scheme*. The axioms for $\text{Mink}_0(1, 3)$ may be found in Goldblatt (1987). The second order theory $\text{Mink}(4)$ is obtained by introducing Comprehension and

Extensionality, and replacing the Continuity axiom scheme by the Continuity axiom. These geometrical axiom systems are listed in Appendix F.

Theorem 34.1.1. Each second-order synthetic theory implies its corresponding underlying first-order theory. For example, $\text{BG}(4) \vdash \text{BG}_0(4)$.

34.2 Soundness

Theorem 34.2.1. Let F be a real-closed field. Then, each first-order theory is *sound* in its corresponding coordinate structure:

$$\mathbb{B}^4(F) \models \text{BG}_0(4) \quad (34.2.1)$$

$$\mathbb{E}^3(F) \models \text{EG}_0(3) \quad (34.2.2)$$

$$\mathbb{G}^4(F) \models \text{Gal}_0(1, 3) \quad (34.2.3)$$

$$\mathbb{M}^4(F) \models \text{Mink}_0(1, 3) \quad (34.2.4)$$

Definition 34.2.2. Let the relevant real-closed field $F = \mathbb{R}$. This defines the *standard* coordinate structure, for each second-order theory:

Table 34.3: Standard structures

Theory	Signature	Standard coordinate structure
$\text{BG}(4)$	$\mathbf{B}$	$\mathbb{B}^4.$
$\text{EG}(3)$	$\mathbf{B}, \equiv$	$\mathbb{E}^3.$
$\text{Gal}(4)$	$\mathbf{B}, \sim, \equiv \sim$	$\mathbb{G}^4.$
$\text{Mink}(4)$	$\mathbf{B}, \lambda$	$\mathbb{M}^4.$

Theorem 34.2.3. Each second-order theory is sound in its corresponding standard coordinate structure:

$$\mathbb{B}^4 \models \text{BG}(4) \quad (34.2.5)$$

$$\mathbb{E}^3 \models \text{EG}(3) \quad (34.2.6)$$

$$\mathbb{G}^4 \models \text{Gal}(4) \quad (34.2.7)$$

$$\mathbb{M}^4 \models \text{Mink}(4) \quad (34.2.8)$$

34.3 Representation Theorems for First-Order Geometries

Next, I give the Representation Theorems for all our first-order synthetic geometries:

Theorem 34.3.1. Let $\beta = (\mathbb{P}, B)$ be an $L(\mathbf{B})$ -structure; let $\epsilon = (\mathbb{P}, B, \equiv)$ be an $L(\mathbf{B}, \equiv)$ -structure; let $\gamma = (\mathbb{P}, B, \sim, \equiv \sim)$ be an $L(\mathbf{B}, \sim, \equiv \sim)$ -structure; and let $\mu = (\mathbb{P}, B, \perp)$ be an $L(\mathbf{B}, \perp)$ -structure. Then:

$$\beta \models \mathbf{BG}_0(4) \Leftrightarrow \beta \cong \mathbb{B}^4(F), \text{ for some real-closed field } F \quad (34.3.1)$$

$$\epsilon \models \mathbf{EG}_0(3) \Leftrightarrow \epsilon \cong \mathbb{E}^3(F), \text{ for some real-closed field } F \quad (34.3.2)$$

$$\gamma \models \mathbf{Gal}_0(1, 3) \Leftrightarrow \gamma \cong \mathbb{G}^4(F), \text{ for some real-closed field } F \quad (34.3.3)$$

$$\mu \models \mathbf{Mink}_0(1, 3) \Leftrightarrow \mu \cong \mathbb{M}^4(F), \text{ for some real-closed field } F \quad (34.3.4)$$

34.4 Representation Theorems for Second-Order Geometries

We next turn to the corresponding *second-order* geometries. Each such synthetic geometry has a “nice” Representation Theorem with respect to *full* models:

Theorem 34.4.1. Let $\beta = (\mathbb{P}, B)$ be an $L(\mathbf{B})$ -structure; let $\epsilon = (\mathbb{P}, B, \equiv)$ be an $L(\mathbf{B}, \equiv)$ -structure; let $\gamma = (\mathbb{P}, B, \sim, \equiv \sim)$ be an $L(\mathbf{B}, \sim, \equiv \sim)$ -structure; and let $\mu = (\mathbb{P}, B, \perp)$ be an $L(\mathbf{B}, \perp)$ -structure.

$$\beta \models_2 \mathbf{BG}(4) \Leftrightarrow \beta \cong \mathbb{B}^4 \quad (34.4.1)$$

$$\epsilon \models_2 \mathbf{EG}(3) \Leftrightarrow \epsilon \cong \mathbb{E}^3 \quad (34.4.2)$$

$$\gamma \models \mathbf{Gal}(4) \Leftrightarrow \gamma \cong \mathbb{G}^4 \quad (34.4.3)$$

$$\mu \models \mathbf{Mink}(4) \Leftrightarrow \mu \cong \mathbb{M}^4 \quad (34.4.4)$$

Proof. The relevant Representation Theorem for $\mathbf{BG}(4)$ follows from Theorem 5.12 of Szczerba & Tarski (1979) (see also Example 6.1). The same theorem is stated, somewhat indirectly, on pp. 196–197 of Borsuk & Sz mielew (1960). The same theorem is stated, also somewhat indirectly, in Goldblatt (1987), p. The Representation Theorem itself goes back to Veblen (1904). $\square$

Chapter 35

Coordinate Systems and Coordination Transformations

Earlier, we sketched the proof of Theorem 5.2.10, characterizing the automorphism (symmetry) groups of the structures defined in Definition 5.1.3 and Definition 5.1.8. Let $n \geq 2$. For the coordinate structures $\mathbb{B}^n$, $\mathbb{E}^n$, $\mathbb{E}_{\text{metric}}^n$, $\mathbb{G}^4$ and $\mathbb{M}^4$ (Definition 5.1.8), the automorphism groups are given by:

$$\text{Aut}(\mathbb{B}^n) \cong \text{Aff}(n) \quad (35.0.1)$$

$$\text{Aut}(\mathbb{E}^n) \cong \text{Sim}(n) \quad (35.0.2)$$

$$\text{Aut}(\mathbb{E}_{\text{metric}}^n) \cong E(n) \quad (35.0.3)$$

$$\text{Aut}(\mathbb{G}^4) \cong \mathcal{G}^e(1, 3) \quad (35.0.4)$$

$$\text{Aut}(\mathbb{M}^4) \cong \mathcal{P}^e(1, 3) \quad (35.0.5)$$

Additionally, we proved Representation Theorems for $\text{BG}(4)$, $\text{EG}(3)$, $\text{Gal}(4)$ and $\text{Mink}(4)$. In summary, writing $\models_2$ to mean “is a *full* model of” (and note: the signature of M changes in each case):

$$\beta \models_2 \text{BG}(4) \Leftrightarrow \beta \cong \mathbb{B}^4 \quad (35.0.6)$$

$$\epsilon \models_2 \text{EG}(3) \Leftrightarrow \epsilon \cong \mathbb{E}^3 \quad (35.0.7)$$

$$\gamma \models_2 \text{Gal}(4) \Leftrightarrow \gamma \cong \mathbb{G}^4 \quad (35.0.8)$$

$$\mu \models_2 \text{Mink}(4) \Leftrightarrow \mu \cong \mathbb{G}^4 \quad (35.0.9)$$

Lemma 35.0.1 (Coordinate Transformations). Given two coordinate systems $\Phi, \Psi : \mathbb{P} \rightarrow \mathbb{R}^4$, on a full model $\beta = (\mathbb{P}, B)$ of $\text{BG}(4)$, they are related as follows: there is a $GL(4)$ matrix A and a translation $\mathbf{d} \in \mathbb{R}^4$ such that, for any point $p \in \mathbb{P}$, we have:

$$\Psi(p) = A\Phi(p) + \mathbf{d} \quad (35.0.10)$$

Proof. This follows from two facts. First, if $\Phi, \Psi : M \rightarrow \mathbb{B}^4$ are isomorphisms, then $\Psi \circ \Phi^{-1} \in \text{Aut}(\mathbb{B}^4)$. Second, we have $\text{Aut}(\mathbb{B}^4) = \text{Aff}(4)$. (This is the result given in Theorem 5.2.10 for the automorphisms of the standard coordinate structure $\mathbb{B}^4$ for $\text{BG}(4)$.) $\square$

A similar result holds for coordinate systems on models of $\text{EG}(3)$, models of $\text{Gal}(4)$ (such a coordinate system is an inertial frame) and models of $\text{Mink}(4)$.

Lemma 35.0.2. Given a 4-frame O, I_1, I_2, I_3, I_4 and two coordinate systems, Φ, Ψ on a model M of $\text{BG}(4)$, both of which match the frame O, I_1, I_2, I_3, I_4 , we have:

$$\Psi = \Phi \tag{35.0.11}$$

Proof. The proof applies the coordinate transformation equation (35.0.10) to the five points, O, I_1, I_2, I_3, I_4 , which gives five specific instances. The first of these implies that $\mathbf{d} = \mathbf{0}$. The remaining four imply that the $GL(4)$ matrix A is the identity matrix. $\square$

Similar reasoning applies in any dimension, and also to the Euclidean case and Galilean cases.

Part VI

Extension With Set Theory

Chapter 36

Extension With Set Theory

36.1 Adding Set Theory to Geometry

Section §3.4, focusing on arithmetic, is a kind of warmup in understanding the issue of how non-conservativeness arises when one “adds set theory”. The original context of Field (1980)—and the important papers Malament (1982), Shapiro (1983) and Burgess (1984) analysing Field’s programme—is *synthetic spacetime geometry*, not arithmetic. The geometrical case is similar, but introduces some differences and is somewhat more complicated.

So far we treated the synthetic geometrical theories as autonomous theories. But now we wish to examine what happens when we *extend* them with the axioms of **AM**. To do this adequately, we extend the geometrical signatures with a unary (urelement) predicate **point**:

Table 36.1: Geometrical Sort Declarations

Primitive	Sort declaration
Point	$\text{point} :: \text{atom} \Rightarrow \text{bool}.$
Betweenness	$B :: \text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{bool}.$
Congruence	$\equiv :: \text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{bool}.$
Simultaneity	$\sim :: \text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{bool}.$
Simcongruence	$\equiv \sim :: \text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{bool}.$
Minkowski light-signal	$\lambda : \text{atom} \Rightarrow \text{atom} \Rightarrow \text{bool}.$
Minkowski orthogonality	$\perp : \text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{bool}.$

And, correspondingly, the following sort axioms:

Table 36.2: Geometrical Sort Axioms

Primitive	Sort axiom
Point	$\forall x (\text{point}(x) \rightarrow \text{atom}(x)).$
Betweenness	$B(x, y, z) \rightarrow (\text{atom}(x) \wedge \text{atom}(y) \wedge \text{atom}(z)).$
Congruence	$xy \equiv zw \rightarrow (\text{atom}(x) \wedge \text{atom}(y) \wedge \text{atom}(z) \wedge \text{atom}(w)).$
Simultaneity	$x \sim y \rightarrow (\text{atom}(x) \wedge \text{atom}(y)).$
Simcongruence	$xy \equiv^{\sim} zw \rightarrow (\text{atom}(x) \wedge \text{atom}(y) \wedge \text{atom}(z) \wedge \text{atom}(w)).$
Minkowski orthogonality	$xy \perp zw \rightarrow (\text{atom}(x) \wedge \text{atom}(y) \wedge \text{atom}(z) \wedge \text{atom}(w)).$
Minkowski light-signal	$x\lambda y \rightarrow (\text{atom}(x) \wedge \text{atom}(y))$

Furthermore, it is convenient to insist that the various geometrical primitive apply only to points. I call these additional axioms “typing axioms”:

Table 36.3: Geometrical Typing Axioms

Primitive	Typing axiom
Betweenness	$B(x, y, z) \rightarrow (\text{point}(x) \wedge \text{point}(y) \wedge \text{point}(z)).$
Congruence	$xy \equiv zw \rightarrow (\text{point}(x) \wedge \text{point}(y) \wedge \text{point}(z) \wedge \text{point}(w)).$
Simultaneity	$x \sim y \rightarrow (\text{point}(x) \wedge \text{point}(y)).$
Simcongruence	$xy \equiv^{\sim} zw \rightarrow (\text{point}(x) \wedge \text{point}(y) \wedge \text{point}(z) \wedge \text{point}(w)).$
Minkowski orthogonality	$xy \perp zw \rightarrow (\text{point}(x) \wedge \text{point}(y) \wedge \text{point}(z) \wedge \text{point}(w)).$
Minkowski light-signal	$x\lambda y \rightarrow (\text{point}(x) \wedge \text{point}(y))$

Since we have made this modification, we now have slightly modified theories, which we denote as follows:

Table 36.4: Synthetic geometries

Theory	Signature	
$\text{BG}_0^*(4)$	point, B	First-order 4D betweenness geometry.
$\text{EG}_0^*(3)$	$\text{point}, B, \equiv$	First-order 3D Euclidean geometry.
$\text{Gal}_0^*(1, 3)$	$\text{point}, B, \sim, \equiv^{\sim}$	First-order (1+3)D Galilean spacetime geometry.
$\text{Mink}_0^*(1, 3)$	$\text{point}, B, \perp$	First-order (1+3)D Minkowski spacetime geometry.
$\text{BG}^*(4)$	point, B	Second-order 4D betweenness geometry.
$\text{EG}^*(3)$	$\text{point}, B, \equiv$	Second-order 3D Euclidean geometry.
$\text{Gal}^*(1, 3)$	$\text{point}, B, \sim, \equiv^{\sim}$	Second-order (1+3)D Galilean spacetime geometry.
$\text{Mink}^*(1, 3)$	point, B, λ	Second-order (1+3)D Minkowski spacetime geometry.

This means that when we consider, say, $\text{AM} + \text{BG}^*(4)$, we have a sort axiom for **point**:

$$\forall x (\text{point}(x) \rightarrow \text{atom}(x)) \quad (36.1.1)$$

Definition 36.1.1. Now, working with AM , and the relevant signature of primitives, we may define set constants:

Table 36.5: Set constants

Set constant			
Point set	$\mathbb{P}$	$:=$	$\{x \mid \text{point}(x)\}.$
Betweenness relation	B	$:=$	$\{(x, y, z) \in \mathbb{P}^3 \mid \mathbf{B}(x, y, z)\}.$
Congruence relation	$\equiv$	$:=$	$\{(x, y, z, w) \in \mathbb{P}^4 \mid xy \equiv zw\}.$
Simultaneity relation	$\sim$	$:=$	$\{(x, y) \in \mathbb{P}^2 \mid x \sim y\}.$
The simcongruence relation	$\equiv\sim$	$:=$	$\{(x, y, z, w) \in \mathbb{P}^4 \mid xy \equiv\sim zw\}.$
The Minkowski light-signal relation	λ	$:=$	$\{(x, y) \in \mathbb{P}^4 \mid x\lambda y\}$

From now on, it simplifies notation if we conflate $\mathbf{BG}^*(4)$ with $\mathbf{BG}(4)$ and so on. Context should always disambiguate whether we mean $\mathbf{BG}^*(4)$ or we mean $\mathbf{BG}(4)$, and so on.

Lemma 36.1.2. $\mathbf{AM} + \mathbf{BG}(4) \vdash \mathbb{P} \subseteq \mathbf{Atom}.$

Proof. From the above sort axiom for **point**, (36.1.1). □

Lemma 36.1.3. $\mathbf{AM} + \mathbf{BG}(4) \vdash \mathbf{set}(\mathbb{P}).$

Proof. Working in $\mathbf{AM} + \mathbf{BG}(4)$, we have the axiom $\mathbf{Atom} \in U$, which is equivalent to $\mathbf{set}(\mathbf{Atom})$. By the above, $\mathbb{P} \subseteq \mathbf{Atom}$. By Separation, we conclude $\mathbf{set}(\mathbb{P})$. □

Lemma 36.1.4. $\mathbf{AM} + \mathbf{BG}(4) \vdash \mathbf{set}(B).$

Proof. Working in $\mathbf{AM} + \mathbf{BG}(4)$, and via the definition above, we have $B \subseteq \mathbb{P}^3$. By the previous lemma, $\mathbf{set}(\mathbb{P})$. Hence, $\mathbf{set}(\mathbb{P}^3)$. By Separation, we conclude $\mathbf{set}(B)$. □

Chapter 37

Internalization

37.1 Internalization I

In 1930, a young research student Kurt Gödel announced his First Incompleteness Theorem (Theorem ?? above) to a meeting of mathematicians in Königsberg. It seems no one in the audience of mathematicians understood his result, except John von Neumann. A few weeks later, Gödel discovered The Second Incompleteness Theorem. The idea is that the first theorem can be proved *inside* PA itself. I.e., Theorem ?? can be internalized inside PA. Inside PRA in fact:

$$\text{PRA} \vdash \text{Con}(T) \rightarrow G \quad (37.1.1)$$

And since G is not a theorem of T , this then gives us:

$$T \not\vdash \text{Con}(T) \quad (37.1.2)$$

which is the Second Incompleteness Theorem.

Above, we have given a model-theoretic representation theorem (Theorem 25.5.1) for $\text{BG}(4)$.

Theorem 37.1.1 (Representation Theorem for $\text{BG}(4)$). Let $M = (\mathbb{P}, B)$ be an $L(\mathbb{B})$ -structure. Then:

$$M \text{ is a full model of } \text{BG}(4) \text{ if and only if } M \cong (\mathbb{R}^4, B_{\mathbb{R}^4}). \quad (37.1.3)$$

But what *mathematical theory* did we prove this Theorem 25.5.1 in? To be sure, all model theorists will agree, “Oh, it can be proved in ZFC. But please ask a proof theorist”; and proof-theorist experts will locate more accurately some relevant much weaker subsystem of ZFC, and very probably a subsystem around the Gödel Hierarchy level of PA_2 or PA_3 (which is the same level as ALG). One thing is clear: Theorem

25.5.1 can certainly be proved in our (very strong, all-purpose) base theory $\mathbf{AM}$. The implication of this result will be:

$$\mathbf{AM} \vdash \overbrace{\mathbf{BG}(4)}^{\text{synthetic}} \leftrightarrow \overbrace{((\mathbb{P}, B) \cong (\mathbb{R}^4, B_{\mathbb{R}^4}))}^{\text{representation-theoretic}} \quad (37.1.4)$$

My aim in this section is to obtain this. First, we need to examine how we “combine” the four-dimensional synthetic geometry $\mathbf{BG}(4)$ with the base theory $\mathbf{AM}$.

37.1.1 Internalized Line Representation Theorem

Definition 37.1.2. We begin with an application signature $\sigma = \{\text{point}, \mathbf{B}\}$ with sort declarations:

Signature	$\sigma = \{\text{point}, \mathbf{B}\}$
$\mathbf{B}$	$\text{atom} \Rightarrow \text{atom} \Rightarrow \text{atom} \Rightarrow \text{bool}$
point	$\text{atom} \Rightarrow \text{bool}$

Here, **point** is a unary predicate on atoms, and $\mathbf{B}$ is a 3-place predicate on atoms. We introduce definitions of the point set $\mathbb{P}$ and the relation B :

$$\mathbb{P} := \{a \mid \text{point}(a)\} \quad (37.1.5)$$

$$B := \{(a, b, c) \in \mathbb{P}^3 \mid \mathbf{B}(a, b, c)\}. \quad (37.1.6)$$

Note that within the theory $\mathbf{AM} + \mathbf{BG}(4)$, the second-order Comprehension axiom scheme of $\mathbf{BG}(4)$ and Extensionality axiom of $\mathbf{BG}(4)$ get absorbed by the ambient set theory $\mathbf{AM}$. In this context, $\mathbf{BG}(4)$ now becomes just the nine axioms given earlier, along with what may be called a typing axiom:

$$\forall x, y, z (\mathbf{B}(x, y, z) \rightarrow (\text{point}(x) \wedge \text{point}(y) \wedge \text{point}(z))) \quad (37.1.7)$$

To begin with, we have:

Lemma 37.1.3. $\mathbf{AM} + \mathbf{BG}(4) \vdash (\mathbb{P} \subseteq \text{Atom}) \wedge (B \subseteq \mathbb{P}^3)$.

The first main result is that we can internalize the Line Representation Theorem:

Theorem 37.1.4 (Internal Line Representation Theorem for $\mathbf{BG}(4)$). To simplify notation, let ℓ be $(\ell(O, I), O, I, +, \times, \leq)$, and let $\mathbb{R}$ be $(\mathbb{R}, 0, 1, +, \times, \leq)$. Then:

$$\mathbf{AM} + \mathbf{BG}(4) + (O \neq I) \vdash \exists! \psi (\ell \stackrel{\psi}{\cong} \mathbb{R}) \quad (37.1.8)$$

Proof. Working inside $\text{AM} + \text{BG}(4) + (O \neq I)$, we give the standard definitions of geometrical $+$, $\times$ and $\leq$, on the fixed line, $\ell(O, I)$, and prove that the resulting system $(\ell(O, I), O, I, +, \times, \leq)$ satisfies the **ALG** axioms. Then, the Isomorphism Theorem for complete ordered fields establishes an isomorphism to $\mathbb{R}$. $\square$

Definition 37.1.5. We then define (for fixed distinct $O, I \in \mathbb{P}$)

$$\psi_{O,I} : \ell \rightarrow \mathbb{R} \quad (37.1.9)$$

to be the function asserted to exist above.

The immediate corollary is:

Corollary 37.1.6. $\text{AM} + \text{BG}(4) + (O \neq I) \vdash \ell \stackrel{\psi_{O,I}}{\cong} \mathbb{R}$.

37.1.2 Configurations and Real Abstraction

Definition 37.1.7. We now define:

$$\begin{aligned} \mathfrak{R} &:= \{\xi(p, q, r) \mid \text{cnfg}(p, q, r)\} \\ x = 0 &:= \exists p \exists q (p \neq q \wedge x = \xi(p, q, p)) \\ x = 1 &:= \exists p \exists q (p \neq q \wedge x = \xi(p, q, q)) \\ x_3 = x_1 + x_2 &:= \exists p \exists q \exists r_1 \exists r_2 \exists r_3 (p \neq q \wedge \text{co}_1(p, q, r_1, r_2, r_3) \wedge x_1 = \xi(p, q, r_1) \wedge x_2 = \xi(p, q, r_2) \wedge x_3 = \xi(p, q, r_3) \wedge r_3 = r_1 \oplus r_2) \\ x_3 = x_1 \times x_2 &:= \exists p \exists q \exists r_1 \exists r_2 \exists r_3 (p \neq q \wedge \text{co}_1(p, q, r_1, r_2, r_3) \wedge x_1 = \xi(p, q, r_1) \wedge x_2 = \xi(p, q, r_2) \wedge x_3 = \xi(p, q, r_3) \wedge r_3 = r_1 \otimes r_2) \\ x_1 \leq x_2 &:= \exists p \exists q \exists r_1 \exists r_2 (p \neq q \wedge \text{co}_1(p, q, r_1, r_2, r_3) \wedge x_1 = \xi(p, q, r_1) \wedge x_2 = \xi(p, q, r_2) \wedge r_2 \leq r_1) \end{aligned}$$

We let $\mathfrak{R}$ abbreviate the structure $(\mathfrak{R}, 0, 1, +, \times, \leq)$.

Theorem 37.1.8. $\text{AM} + \text{BG}(4) \vdash \text{completeorderedfield}(\mathfrak{R})$.

Theorem 37.1.9. $\text{AM} + \text{BG}(4) \vdash (\exists! \phi : \mathfrak{R} \rightarrow \mathbb{R})(\mathfrak{R} \stackrel{\phi}{\cong} \mathbb{R})$.

Definition 37.1.10. $\xi_{O,I}(r) := \xi(O, I, r)$.

Theorem 37.1.11. $\text{AM} + \text{BG}^\dagger(4) \vdash (\exists! \phi : \mathfrak{R} \rightarrow \mathbb{R})(\psi_{O,I} = \phi \circ \xi_{O,I})$

We can then simply identify $\phi \circ \xi_{O,I}$ with $\xi_{O,I}$. To be precise, we introduce “ $\xi'_{O,I}$ ” by explicit definition:

Definition 37.1.12. $\xi'_{O,I}(r) := (\phi \circ \xi_{O,I})(r)$.

It makes notation simpler to conflate ξ and ξ' . The above Theorem 37.1.11 then becomes:

Theorem 37.1.13. $\text{AM} + \text{BG}^\dagger(4) \vdash (\forall r \in \ell(O, I)) (\xi(O, I, r) = \psi_{O,I}(r))$.

Remark 37.1.14. The equation:

$$\xi(O, I, r) = \psi_{O, I}(r) \quad (37.1.10)$$

is tremendously useful in some proofs. $\square$

37.1.3 Internalized Representation Theorem

The above “external” representation theorems are proved (semi-formally) in the meta-theory, and stated model-theoretically (about models and satisfaction). However, because the model-theoretic concept “structure M satisfies theory T ” can be explicitly defined within our base theory **AM**, these proofs can be formalized and carried out inside **AM** itself.

Theorem 37.1.15 (Internal Representation Theorem for **BG(4)**).

$$\text{AM} \vdash \overbrace{\text{BG}(4)}^{\text{synthetic}} \leftrightarrow \overbrace{((\mathbb{P}, B) \cong (\mathbb{R}^4, B_{\mathbb{R}^4}))}^{\text{representation-theoretic}} \quad (37.1.11)$$

Proof. The external Representation Theorem is Theorem 25.5.1. This states, in the model-theoretic metatheory, the following. Suppose M is a $L(\mathbf{B})$ structure. Then M is a *full model* of **BG(4)** if and only if $M \cong (\mathbb{R}^4, B_{\mathbb{R}^4})$. This metatheorem is a theorem of **AM**. Stated as a theorem of **AM**, this becomes:

$$M \in \mathbf{Struct}(L(\mathbf{B})) \rightarrow (\text{sat}_2(M, \ulcorner \text{BG}(4) \urcorner) \leftrightarrow (M \cong (\mathbb{R}^4, B_{\mathbb{R}^4}))) \quad (37.1.12)$$

Where $\mathbf{Struct}(L(\mathbf{B}))$ defines the (class of) $L(\mathbf{B})$ structures, and $\text{sat}_2(X, Y)$ states that X is a *full model* of (the formula encoded by) Y , and $\ulcorner \text{BG}(4) \urcorner$ is the code of the conjunction of the nine axioms of **BG(4)**. Hence, by instantiation:

$$(\mathbb{P}, B) \in \mathbf{Struct}(L(\mathbf{B})) \rightarrow (\text{sat}_2((\mathbb{P}, B), \ulcorner \text{BG}(4) \urcorner) \leftrightarrow ((\mathbb{P}, B) \cong (\mathbb{R}^4, B_{\mathbb{R}^4}))) \quad (37.1.13)$$

Moreover, **AM** proves two required results (And also, by the “disquotational” property of sat_2 , we have:)

$$\text{AM} \vdash (\mathbb{P}, B) \in \mathbf{Struct}(L(\mathbf{B})) \quad (37.1.14)$$

$$\text{AM} \vdash \text{BG}(4) \leftrightarrow \text{sat}_2((\mathbb{P}, B), \ulcorner \text{BG}(4) \urcorner) \quad (37.1.15)$$

Hence,

$$\text{AM} \vdash \text{BG}(4) \leftrightarrow ((\mathbb{P}, B) \cong (\mathbb{R}^4, B_{\mathbb{R}^4})) \quad (37.1.16)$$

$\square$

Thus, AM proves that the synthetic theory $\mathbf{BG}(4)$ is equivalent to the claim that $(\mathbb{P}, B)$ is isomorphic to the corresponding Cartesian space $\mathbb{B}\mathbb{G}^4$. This result corresponds, more or less, to the following comment by John Burgess:

100 Within ZFU one can define the pure sets as those none of whose elements, elements of elements, and so on, is an urelement or non-set. . . . Within $G0 + ZFU$ it can be proved that the system of point surrogates (and sets thereof, and sets of sets thereof) is isomorphic to the system of points (and sets thereof, and sets of sets thereof). (Burgess (1988): 461. Burgess (1988), p. 461)

Finally, we appeal to Hilbert's Beermug and *skeletonize* Theorem 37.1.15. First recall Definition above, which gave the skeleton of $\mathbf{BG}(4)$. We can easily define the general case:¹

Definition 37.1.16. For $n \geq 3$, the geometrical skeleton $\mathbf{bet}.\mathbf{geom}_n(X, R)$ is the conjunction of the following ten formulas:

The skeleton $\mathbf{bet}.\mathbf{geom}_n(X, R)$

$$\begin{array}{l}
 R \subseteq X^3. \\
 (\forall x, y \in X) ((x, y, x) \in R \rightarrow x = y). \\
 (\forall x, y, z, w \in X) ((x, y, z) \in R \wedge (y, z, w) \in R \wedge y \neq z \rightarrow (x, y, w) \in R). \\
 (\forall x, y, z, w \in X) ((x, y, z) \in R \wedge (x, y, w) \in R \wedge p \neq q \rightarrow ((x, z, w) \in R \vee (x, w, z) \in R)). \\
 (\forall y, z \in X) (\exists x \in X) ((x, y, z) \in R \wedge x \neq y). \\
 \hline
 (\forall x, y, z, s, u \in X) ((x, y, z) \in R \wedge (s, u, y) \in R \rightarrow (\exists w \in X) ((z, w, s) \in R \wedge (x, u, w) \in R)). \\
 (\forall a, d, t, b, c, d \in X) ((a, d, t) \in R \wedge (b, d, c) \in R \wedge a \neq d \rightarrow (\exists u, v \in X) ((a, b, u) \in R \wedge (a, c, v) \in R \wedge (u, t, v) \in R)) \\
 \hline
 (\exists x_1, \dots, x_{n+1} \in X) \neg \mathbf{co}_{n-1}(X, R, x_1, \dots, x_{n+1}). \\
 (\forall x_1, \dots, x_{n+2} \in X) \mathbf{co}_n(X, R, x_1, \dots, x_{n+2}). \\
 \hline
 (\forall Z_1, Z_2 \subseteq X) (\exists z (\forall x \in Z_1) (\forall y \in Z_2) ((z, x, y) \in R) \rightarrow \exists z (\forall x \in Z_1) (\forall y \in Z_2) ((x, z, y) \in R))
 \end{array}$$

Theorem 37.1.17 (Betweenness Skeletonization). Let $n \geq 3$. Let $\mathbf{bet}.\mathbf{geom}_n(X, Y)$ be the above skeleton. Then:

¹ For $n \geq 3$. For $n = 2$ we must include a suitably skeletonized form of Desargues's axiom. In addition, the Lower and Upper Dimension axioms must be formalized using $\mathbf{co}_n$ formulas making the point set variable X , and the betweenness variable R , explicit.

$$\text{AM} \vdash \forall X \forall Y [\text{bet. geom}_n(X, Y) \leftrightarrow (X, Y) \cong (\mathbb{R}^n, B_{\mathbb{R}^n})] \quad (37.1.17)$$

37.2 Internalization II

In Section §37.1, we showed how to “internalize” the meta-theoretical Representation Theorem for axiomatic betweenness geometry $\text{BG}(4)$ *inside* AM itself. Consequently, instead of a result that looks like a meta-theoretical claim of interest only to a logician, we obtain a result in *mathematical physics*.

In this Section, we show how precisely the same method works for Euclidean geometry and for Galilean spacetime geometry.

Definition 37.2.1. We define the three geometrical application signatures:

Table 37.1: Geometrical Application Signatures

Geometry	Geometrical application signatures
Betweenness geometry	$\sigma_{\text{BG}} := \{\text{point}, \text{B}\}.$
Euclidean geometry	$\sigma_{\text{EG}} := \{\text{point}, \text{B}, \equiv\}.$
Galilean spacetime geometry	$\sigma_{\text{Gal}} := \{\text{point}, \text{B}, \sim, \equiv\}$

In AM , we define relations:

$$\mathbb{P} := \{p \mid \text{point}(p)\} \quad (37.2.1)$$

$$B := \{(p, q, r) \in \mathbb{P}^3 \mid \text{B}(p, q, r)\} \quad (37.2.2)$$

$$\equiv := \{(p, q, r, s) \in \mathbb{P}^4 \mid pq \equiv rs\}. \quad (37.2.3)$$

$$\sim := \{(p, q) \in \mathbb{P}^2 \mid p \sim q\} \quad (37.2.4)$$

$$\equiv \sim := \{(p, q, r, s) \in \mathbb{P}^4 \mid pq \equiv \sim rs\}. \quad (37.2.5)$$

Theorem 37.2.2 (Internal Representation Theorem for $\text{EG}(3)$).

$$\text{AM} \vdash \overbrace{\text{EG}(3)}^{\text{synthetic axiom(s)}} \leftrightarrow \overbrace{(\mathbb{P}, B, \equiv) \cong (\mathbb{R}^3, B_{\mathbb{R}^3}, \equiv_{\mathbb{R}^3})}^{\text{representation-theoretic axiom}} \quad (37.2.6)$$

Theorem 37.2.3 (Internal Representation Theorem for $\text{Gal}(4)$).

$$\text{AM} \vdash \overbrace{\text{Gal}(4)}^{\text{synthetic axiom(s)}} \leftrightarrow \overbrace{(\mathbb{P}, B, \sim, \equiv \sim) \cong (\mathbb{R}^4, B_{\mathbb{R}^4}, \sim_{\mathbb{R}^4}, \equiv \sim_{\mathbb{R}^4})}^{\text{representation-theoretic axiom}} \quad (37.2.7)$$

Theorem 37.2.4. Internalized geometrical representation theorems:

$$\text{AM} + \text{BG}_0(4) \vdash \exists F (\text{RCF}(F) \wedge [(\mathbb{P}, B) \cong (F^4, B_{F^4})]) \quad (37.2.8)$$

$$\text{AM} + \text{EG}_0(3) \vdash \exists F (\text{RCF}(F) \wedge [(\mathbb{P}, B, \equiv) \cong (F^3, B_{F^3}, \equiv_{F^3})]) \quad (37.2.9)$$

$$\text{AM} + \text{Gal}_0(1, 3) \vdash \exists F (\text{RCF}(F) \wedge [(\mathbb{P}, B, \sim, \equiv) \cong (F^4, B_{F^4}, \sim_{F^4}, \equiv_{F^4})]) \quad (37.2.10)$$

$$\text{AM} + \text{Mink}_0(1, 3) \vdash \exists F (\text{RCF}(F) \wedge [(\mathbb{P}, B, \lambda) \cong (F^4, B_{F^4}, \lambda_{F^4})]) \quad (37.2.11)$$

Theorem 37.2.5. Internalized geometrical representation theorems:

$$\text{AM} + \exists F (\text{RCF}(F) \wedge [(\mathbb{P}, B) \cong (F^4, B_{F^4})]) \vdash \text{BG}_0(4) \quad (37.2.12)$$

$$\text{AM} + \exists F (\text{RCF}(F) \wedge [(\mathbb{P}, B, \equiv) \cong (F^3, B_{F^3}, \equiv_{F^3})]) \vdash \text{EG}_0(3) \quad (37.2.13)$$

$$\text{AM} + \exists F (\text{RCF}(F) \wedge [(\mathbb{P}, B, \sim, \equiv) \cong (F^4, B_{F^4}, \sim_{F^4}, \equiv_{F^4})]) \vdash \text{Gal}_0(1, 3) \quad (37.2.14)$$

$$\text{AM} + \exists F (\text{RCF}(F) \wedge [(\mathbb{P}, B, \perp) \cong (F^4, B_{F^4}, \perp_{F^4})]) \vdash \text{Mink}_0(1, 3) \quad (37.2.15)$$

Theorem 37.2.6. Internalized geometrical representation theorems:

$$\text{AM} \vdash \text{BG}^*(4) \leftrightarrow [(\mathbb{P}, B) \cong (\mathbb{R}^4, B_{\mathbb{R}^4})] \quad (37.2.16)$$

$$\text{AM} \vdash \text{EG}^*(3) \leftrightarrow [(\mathbb{P}, B, \equiv) \cong (\mathbb{R}^3, B_{\mathbb{R}^3}, \equiv_{\mathbb{R}^3})] \quad (37.2.17)$$

$$\text{AM} \vdash \text{Gal}^*(1, 3) \leftrightarrow [(\mathbb{P}, B, \sim, \equiv) \cong (\mathbb{R}^4, B_{\mathbb{R}^4}, \sim_{\mathbb{R}^4}, \equiv_{\mathbb{R}^4})] \quad (37.2.18)$$

$$\text{AM} \vdash \text{Mink}^*(1, 3) \leftrightarrow [(\mathbb{P}, B, \perp) \cong (\mathbb{R}^4, B_{\mathbb{R}^4}, \perp_{\mathbb{R}^4})] \quad (37.2.19)$$

Chapter 38

Conservativeness and Representation

38.1 Conservativeness

We have now reached a point where two of the main results in Shapiro (1983) and Burgess (1984) can be stated easily:

Theorem 38.1.1. Let T be any of the *first-order* synthetic geometries $\text{BG}_0(4)$, $\text{EG}_0(3)$, $\text{Gal}_0(1, 3)$ and $\text{Mink}_0(1, 3)$. Then

$$\text{AM} + T \text{ is a conservative extension.} \quad (38.1.1)$$

Proof. In each case, the first-order synthetic geometry T is an *urelement theory*. By Theorem 3.2.4, $\text{AM} + T$ conservatively extends T . There is also a slightly different proof. By Theorem 28.1.9, each T is *complete*. Hence, $\text{AM} + T$ is either inconsistent or conservative. The consistency of $\text{AM} + T$ follows from the consistency of T and Theorem 3.2.4. $\square$

Theorem 38.1.2. Let T be any of the *second-order* synthetic geometries $\text{BG}(4)$, $\text{EG}(3)$, $\text{Gal}(4)$ and $\text{Mink}(4)$. Then

$$\text{AM} + T \text{ is a non-conservative extension.} \quad (38.1.2)$$

(I.e., of its geometric base, wrt $L_2(\sigma)$ -sentences for the relevant signature σ .)

Proof. I consider the $\text{BG}(4)$ case. We have:

$$\text{BG}(4) \not\vdash \text{Con}(\text{BG}(4)) \quad (38.1.3)$$

$$\text{AM} + \text{BG}(4) \vdash \text{Con}(\text{BG}(4)) \quad (38.1.4)$$

The other cases are analogous. $\square$

38.2 Representation

In this section, I focus entirely on $\text{Gal}(4)$. Although in fact analogous points arise for the other geometries, including Minkowski. Let us define the sentence REP in $L_{\in}(\text{point}, B, \sim, \equiv \sim)$:

$$\text{REP} : \left[\overbrace{(\mathbb{P}, B, \sim, \equiv \sim)}^{\text{synthetic structure}} \cong \overbrace{(\mathbb{R}^4, B_{\mathbb{R}^4}, \sim_{\mathbb{R}^4}, \equiv \sim_{\mathbb{R}^4})}^{\text{standard coordinate structure}} \right] \quad (38.2.1)$$

Thus, by the Internalized Representation Theorems (Theorem 37.2.6):

Theorem 38.2.1. $\text{AM} \vdash \text{Gal}(4) \leftrightarrow \text{REP}$.

This tell us that the mathematicized *physical* theory of Galilean spacetime (i.e., including the applied mathematics base) has two *entirely equivalent axiomatizations*:

$$T_1 := \text{AM} + \text{Gal}(4) \quad (38.2.2)$$

$$T_2 := \text{AM} + \text{REP} \quad (38.2.3)$$

Ignoring Comprehension and Extensionality (which come for free in AM), there are 22 axioms of $\text{Gal}(4)$. But REP is a *single axiom*! Let's introduce (within AM) a definition of a "Galilean spacetime":

Definition 38.2.2 (Galilean spacetime). (X, Y, Z, W) is a Galilean spacetime model if and only if $(X, Y, Z, W) \cong (\mathbb{R}^4, B_{\mathbb{R}^4}, \sim_{\mathbb{R}^4}, \equiv \sim_{\mathbb{R}^4})$.

The single physical axiom REP is equivalent to:

$$\text{REP}' : (\mathbb{P}, B, \sim, \equiv \sim) \text{ is a Galilean spacetime model.} \quad (38.2.4)$$

Theorem 38.2.3. $\text{AM} + \text{Gal}(4) \vdash |\text{Atom}| > \aleph_0$.

Proof. We have: $\text{AM} + \text{Gal}(4) \vdash \text{REP}$. So, $\text{AM} + \text{Gal}(4) \vdash \mathbb{P} \sim \mathbb{R}^4$. And thus, $\text{AM} + \text{Gal}(4) \vdash |\mathbb{P}| = |\mathbb{R}|$. So, $\text{AM} + \text{Gal}(4) \vdash |\mathbb{P}| > \aleph_0$. But (since it is an axiom that each point is an atom), we have $\text{AM} + \text{Gal}(4) \vdash |\mathbb{P}| \leq |\text{Atom}|$. And therefore, $\text{AM} + \text{Gal}(4) \vdash |\text{Atom}| > \aleph_0$. $\square$

Theorem 38.2.4. The following metatheorems hold:

$$\text{AM} + \text{Gal}_0(1, 3) \not\vdash \text{REP}. \quad (38.2.5)$$

$$\text{AM} + \text{Gal}_0(1, 3) \not\vdash \text{Gal}(4) \quad (38.2.6)$$

Proof. Item (38.2.6) follows from (38.2.5), since $\text{AM} \vdash \text{Gal}(4) \leftrightarrow \text{REP}$. For (38.2.5) itself, recall that $\text{Gal}_0(1, 3)$ is an urelement theory. So, by Theorem 3.3.3,

$$\text{AM} + \text{Gal}_0(1, 3) \not\vdash |\text{Atom}| > \aleph_0. \quad (38.2.7)$$

We have $\text{AM} \vdash \mathbb{P} \subseteq \text{Atom}$. Hence, $\text{AM} \vdash |\mathbb{P}| \leq |\text{Atom}|$. Now suppose, for a contradiction, $\text{AM} + \text{Gal}_0(1, 3) \vdash \text{REP}$. Hence, $\text{AM} + \text{Gal}_0(1, 3) \vdash |\mathbb{P}| = |\mathbb{R}|$. Thus, $\text{AM} + \text{Gal}_0(1, 3) \vdash |\text{Atom}| > \aleph_0$. Contradiction. $\square$

Theorem 38.2.5. The following metatheorems hold:

$$\text{AM} + \text{Gal}_0(1, 3) + (\text{Extended continuity scheme}) \vdash \text{Gal}(4). \quad (38.2.8)$$

$$\text{AM} + \text{Gal}_0(1, 3) + (\text{Extended continuity scheme}) \vdash \text{REP}. \quad (38.2.9)$$

Proof. Item (38.2.9) follows from (38.2.8), since $\text{AM} \vdash \text{Gal}(4) \leftrightarrow \text{REP}$. For (38.2.8) itself, we wish to prove each axiom of $\text{Gal}(4)$. The essential idea we use is entirely analogous to the proof of Theorem 3.4.2 above. AM supplies us with Comprehension (for point sets) and Extensionality (for point sets). That leaves the Continuity Axiom. The Continuity *scheme* of $\text{Gal}_0(1, 3)$ supplies an axiom of the form:

$$\exists a \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow \text{B}(a, p, q)) \rightarrow \exists b \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow \text{B}(p, b, q)) \quad (38.2.10)$$

Taking $\varphi(p)$ as $p \in X$, and $\theta(q)$ as $q \in Y$, we obtain the instance:

$$\exists a \forall p \forall q (p \in X \wedge q \in Y \rightarrow \text{B}(a, p, q)) \rightarrow \exists b \forall p \forall q (p \in X \wedge q \in Y \rightarrow \text{B}(p, b, q)) \quad (38.2.11)$$

That is obviously equivalent to the desired Continuity Axiom of $\text{Gal}(4)$. $\square$

38.3 Mereological Synthetic Spacetime Theory

In Field (1980), Field works for the most part with a synthetic spacetime theory equivalent to $\text{Gal}(4)$ —the *second-order* theory of Galilean spacetime. But then it is clear that “adding mathematics” to $\text{Gal}(4)$ is indeed *non-conservative*, because $\text{Gal}(4)$ does not prove $\text{Con}(\text{Gal}(4))$, while the extension $\text{AM} + \text{Gal}(4)$ proves $\text{Con}(\text{Gal}(4))$. Instead, Field proposes a more “nominalistic” theory of Galilean spacetime, in which *pointset* quantification is replaced by *region* quantification, and the system of regions is treated *mereologically*, and points are now understood as mereological atoms. I shall call this theory $\text{Gal}^{\text{mer}}(1, 3)$.

I strongly suspect the following conjectures hold:

120 **Conjecture A.**

$\text{Gal}^{\text{mer}}(1, 3)$ and $\text{Gal}(4)$
are *mutually inter-*
pretable.

Conjecture B.

$\text{Gal}^{\text{mer}}(1, 3)$ and $\text{Gal}(4)$
are *bi-interpretable*.

Conjecture C.

$\text{Gal}^{\text{mer}}(1, 3)$ and $\text{Gal}(4)$
are *definitionally equiv-*
alent

However, I disagree with the claim that $\text{Gal}^{\text{mer}}(1, 3)$ and $\text{Gal}(4)$ are “formulations of the same theory”. The reason I give is that these theories behave differently when extended with AM (see Theorem 38.5.1).

Theorem 38.3.1. $\text{AM} + \text{Gal}^{\text{mer}}(1, 3)$ *conservatively extends* $\text{Gal}^{\text{mer}}(1, 3)$.

Proof. This follows from Theorem 3.2.4 (Field’s conservativeness theorem), since $\text{Gal}^{\text{mer}}(1, 3)$ is an urelement theory. $\square$

Theorem 38.3.2. The following metatheorems hold:

$$\text{AM} + \text{Gal}^{\text{mer}}(1, 3) \not\vdash \text{REP}. \quad (38.3.1)$$

$$\text{AM} + \text{Gal}^{\text{mer}}(1, 3) \not\vdash \text{Gal}(4). \quad (38.3.2)$$

Proof. Item (38.3.2) follows from (38.3.1), since $\text{AM} \vdash \text{Gal}(4) \leftrightarrow \text{REP}$. The proof of (38.3.1) is the same as that for Theorem 38.2.4(1) (using Theorem 3.3.3), since $\text{Gal}^{\text{mer}}(1, 3)$ is an urelement theory. $\square$

Theorem 38.3.3. $\text{AM} + \text{Gal}^{\text{mer}}(1, 3) + (\text{Extended continuity scheme}) \vdash \text{Gal}(4)$.

Proof. This follows from Theorem 38.2.5(1), since $\text{Gal}_0(1, 3)$ is a subtheory of $\text{Gal}^{\text{mer}}(1, 3)$. $\square$

Let us define the sentence **Bridge** in $L_{\in}(\text{point}, \text{region}, \text{B}, \sim, \equiv, \preceq)$:

$$\text{Bridge} : (\forall X \subseteq \mathbb{P}) (X \neq \emptyset \rightarrow \exists! r (\text{region}(r) \wedge \forall p (\text{point}(p) \rightarrow (p \preceq r \leftrightarrow p \in X)))) \quad (38.3.3)$$

Theorem 38.3.4. The following metatheorems hold:

$$\text{AM} + \text{Gal}^{\text{mer}}(1, 3) + \text{Bridge} \vdash \text{Gal}(4). \quad (38.3.4)$$

$$\text{AM} + \text{Gal}^{\text{mer}}(1, 3) + \text{Bridge} \vdash \text{REP}. \quad (38.3.5)$$

Proof. Item (38.3.5) follows from (38.3.4). So, we verify (38.3.4) as follows. First, $\text{Gal}^{\text{mer}}(1, 3)$ contains the $\text{Gal}_0(1, 3)$ axioms. Second, **AM** yields Comprehension (for point sets) and Extensionality (for point sets). That leaves Continuity axiom of $\text{Gal}(4)$, namely,

$$\forall X \forall Y ([\exists a (\forall p \in X) (\forall q \in Y) \mathbf{B}(a, p, q)] \rightarrow [\exists b (\forall p \in X) (\forall q \in Y) \mathbf{B}(p, b, q)]) \quad (38.3.6)$$

The mereological Continuity axiom of $\text{Gal}^{\text{mer}}(1, 3)$ is:

$$\forall r_1 \forall r_2 ([\exists a (\forall p \preceq r_1) (\forall q \preceq r_2) \mathbf{B}(a, p, q)] \rightarrow [\exists b (\forall p \preceq r_1) (\forall r_2 \preceq Y) \mathbf{B}(p, b, q)]) \quad (38.3.7)$$

We now work inside $\text{AM} + \text{Gal}^{\text{mer}}(1, 3) + \text{Bridge}$. Suppose $X, Y \subseteq \mathbb{P}$. If either $X = \emptyset$ or $Y = \emptyset$, then $\forall b (\exists p \in X) (\exists q \in Y) \mathbf{B}(p, b, q)$ is false. Hence, the clause $\exists b (\forall p \in X) (\forall q \in Y) \mathbf{B}(p, b, q)$ is vacuously true. Hence the Continuity axiom holds trivially for those cases. Instead, let $X \neq \emptyset$ and $Y \neq \emptyset$, and $\exists a (\forall p \in X) (\forall q \in Y) \mathbf{B}(a, p, q)$. By **Bridge**, we infer that there exists unique r_1, r_2 such that $\exists a (\forall p \preceq r_1) (\forall q \preceq r_2) \mathbf{B}(a, p, q)$. Then by mereological continuity (38.3.7), we infer $\exists b (\forall p \preceq r_1) (\forall r_2 \preceq Y) \mathbf{B}(p, b, q)$. And therefore, $\exists b (\forall p \in X) (\forall r_2 \in Y) \mathbf{B}(p, b, q)$. That verifies (38.3.6). $\square$

38.4 The Conservativeness/Representation Problem

Return now finally to the two main proposals in Field (1980), which we first sketched way back in §1.3 of Chapter 1:¹

¹ We can now definitely identify **App.Math** with our canonical system **AM** of ambient applied mathematics.

120 Conservativeness

Thesis: Given the ambient mathematical base theory AM for applied mathematics, and a “mathematics-free” theory N restricted to urelements, the mathematicized extension $AM + N$ is *conservative* with respect to mathematics-free sentences. For any such “mathematics-free” sentence φ ,

60 If $AM + N \vdash \varphi$, then

$N \vdash \varphi$.
Nominalization Thesis: Mathematically-formulated scientific theories can be “nominalized”. Given a mathematically-formulated theory $AM + P$, there exists a “mathematics-free” theory N such that:

60 $AM + P \equiv_{df} AM + N$.

In relation to the issue of conservativeness and the provability of (REP), our main results above can now be summarized in the following table, specialized to the case of Galilean spacetime geometry). Let “Cons?” stand for the question “Is its extension with AM conservative?” and let “REP?” stand for the question “Does its extension with AM prove the representation claim (REP)?”:

Table 38.1: The Conservativeness/Representation Problem

Geometrical theory		Cons?	REP?
First-order spacetime geometry	$\text{Gal}_0(1, 3)$	✓	×
First-order mereological spacetime geometry	$\text{Gal}^{\text{mer}}(1, 3)$	✓	×
Second-order spacetime geometry	$\text{Gal}(4)$	×	✓

Consequently, one cannot maintain both demands. If one insists on conservativeness as a constraint on a nominalistic theory, one must then drop the requirement of proving

the Representation Theorem: a nominalist reconstruction *cannot* prove (REP).

38.5 A No-Go Theorem for Nominalization

The considerations above indicate a more general problem for the nominalization thesis. Let P be second-order Galilean spacetime geometry, $\text{Gal}(4)$. In fact, $\text{Gal}(4)$ is not an urelement theory, because it quantifies over point *sets*. We have seen that first-order synthetic geometry $\text{Gal}_0(1, 3)$ and first-order synthetic “mereological” geometry $\text{Gal}^{\text{mer}}(1, 3)$ are urelement theories, but they are not adequate nominalizations: this is because neither $\text{AM} + \text{Gal}_0(1, 3)$ nor $\text{Gal}^{\text{mer}}(1, 3)$ proves $\text{Gal}(4)$, or (REP).

We arrive in the following situation. If Conjecture C above is correct, then $\text{Gal}(4)$ and $\text{Gal}^{\text{mer}}(1, 3)$ are definitionally equivalent. However, when we extend each with the ambient mathematical base AM , that equivalence is broken. That is:

$$\text{Gal}(4) \equiv_{df} \text{Gal}^{\text{mer}}(1, 3) \quad (38.5.1)$$

$$\text{AM} + \text{Gal}(4) \not\equiv_{df} \text{AM} + \text{Gal}^{\text{mer}}(1, 3). \quad (38.5.2)$$

Does there even exist any nominalization at all?

Theorem 38.5.1 (No-go theorem for nominalization). There does not exist an urelement theory N in $L(\sigma)$, with σ an urelement signature, such that:

$$\text{AM} + \text{Gal}(4) \equiv_{df} \text{AM} + N. \quad (38.5.3)$$

Proof. The reason is:

$$\text{AM} + \text{Gal}(4) \vdash |\text{Atom}| > \aleph_0 \quad (\text{Theorem 38.2.3}) \quad (38.5.4)$$

$$\text{AM} + N \not\vdash |\text{Atom}| > \aleph_0 \quad (\text{Theorem 3.3.3}) \quad (38.5.5)$$

Hence, $\text{AM} + \text{Gal}(4)$ and $\text{AM} + N$ are not equivalent. □

Appendix A

Basic Implementation

D1.	$x \notin y$	$:= \neg(x \in y)$
D2.	$\text{atom}(x)$	$:= \exists a (a = x)$
D3.	$\text{class}(x)$	$:= \exists X (X = x)$
D4.	$\text{empty}(x)$	$:= \forall y (y \notin x)$
D5.	$\text{El}(x)$	$:= \exists X (x \in X)$
D6.	$\text{set}(x)$	$:= \text{class}(x) \wedge \text{El}(x)$
D7.	$\text{prop.class}(x)$	$:= \text{class}(x) \wedge \neg \text{El}(x)$
D8.	$(\forall x \in y) \varphi(x)$	$:= \forall x (x \in y \rightarrow \varphi(x))$
D9.	$(\exists x \in y) \varphi(x)$	$:= \exists x (x \in y \wedge \varphi(x))$
D10.	$z = \{x \mid \varphi(x)\}$	$:= \text{class}(z) \wedge \forall x (x \in z \leftrightarrow (\text{El}(x) \wedge \varphi(x)))$
D11.	$\{x \in y \mid \varphi(x)\}$	$:= \{x \mid x \in y \wedge \varphi(x)\}$
D12.	$\{t(\vec{x}) \mid \varphi(\vec{x})\}$	$:= \{z \mid \exists \vec{x} (z = t(\vec{x}) \wedge \varphi(\vec{x}))\}$
D13.	$\emptyset$	$:= \{x \mid x \neq x\}.$
D14.	U	$:= \{x \mid x = x\}.$
D15.	Atom	$:= \{x \mid \text{atom}(x)\}.$
D16.	El	$:= \{x \mid \text{El}(x)\}.$
D17.	Set	$:= \{x \mid \text{set}(x)\}.$
D18.	$X \subseteq Y$	$:= \forall z (z \in X \rightarrow z \in Y).$
D19.	$(\forall Y \subseteq X) \varphi(Y)$	$:= \forall Y (Y \subseteq X \rightarrow \varphi(Y))$
D20.	$(\exists Y \subseteq X) \varphi(Y)$	$:= \exists Y (Y \subseteq X \wedge \varphi(Y))$
D21.	$\{x_1, \dots, x_n\}$	$:= \{z \mid z = x_1 \vee \dots \vee z = x_n\}.$
D22.	$X \cup Y$	$:= \{z \mid z \in X \vee z \in Y\}.$
D23.	$X \cap Y$	$:= \{z \mid z \in X \wedge z \in Y\}.$
D24.	$\overline{X}$	$:= \{z \mid z \notin X\}.$
D25.	$X - Y$	$:= X \cap \overline{Y}.$
D26.	$\bigcup X$	$:= \{z \mid \exists y (z \in y \wedge y \in X)\}.$
D27.	$\bigcap X$	$:= \{z \mid \forall y (y \in X \rightarrow z \in y)\}.$
D28.	X^+	$:= X \cup \{X\}.$
D29.	$\text{Inductive}(X)$	$:= \emptyset \in X \wedge (\forall Y \in X) (Y^+ \in X).$
D30.	(x)	$:= x.$
D31.	(x, y)	$:= \{\{x\}, \{x, y\}\}.$
D32.	$(x_1, \dots, x_n, x_{n+1})$	$:= ((x_1, \dots, x_n), x_{n+1}).$
D33.	$X \times Y$	$:= \{(z, u) \mid z \in X \wedge u \in Y\}.$
D34.	$\mathcal{P}(X)$	$:= \{z \mid \text{set}(z) \wedge z \subseteq X\}.$
D35.	$\text{Rel}(X)$	$:= (\forall w \in X) \exists x \exists y (w = (x, y))$
D36.	$\text{dom}(X)$	$:= \{y \mid \exists x ((y, x) \in X)\}$
D37.	$\text{ran}(X)$	$:= \{y \mid \exists x ((x, y) \in X)\}$
D38.	$\text{fld}(X)$	$:= \text{dom}(X) \cup \text{ran}(X).$
D39.	$\text{functional}(X)$	$:= \forall w \forall y \forall z ((w, y) \in X \wedge (w, z) \in X \rightarrow y = z)$
D40.	$\text{Fun}(X)$	$:= \text{Rel}(X) \wedge \text{functional}(X)$
D41.	$f : X \rightarrow Y$	$:= \text{Fun}(f) \wedge \text{dom}(f) = X \wedge \text{ran}(f) \subseteq Y$
D42.	$f : X \rightharpoonup Y$	$:= \text{Fun}(f) \wedge \text{dom}(f) \subseteq X \wedge \text{ran}(f) \subseteq Y$
D43.	$\text{Def}(f(x))$	$:= \text{Fun}(f) \wedge x \in \text{dom}(f)$
D44.	$y = f(x)$	$:= (\text{Def}(f(x)) \wedge (x, y) \in f) \vee (\neg \text{Def}(f(x)) \wedge y = \emptyset)$
D45.	$\text{Choice.fun}(f, X)$	$:= \text{Fun}(f) \wedge \text{dom}(f) = X \wedge (\forall Y \in X)(Y \neq \emptyset \rightarrow f(Y) \in Y).$

Appendix B

Algebra

B.1 Groups, Rings and Fields

What is modern algebra? It is the theory of groups, rings, fields and vector spaces.¹

Definition B.1.1. Let σ_F be the one-sorted signature $\{0, 1, +, \times\}$, with $0, 1$ constants and $+$ and $\times$ binary function symbols. (We are overloading symbols). Let $L(\sigma_F)$ be the usual first-order language over σ_F . Let F (i.e., “field”) be the axiom system in $L(\sigma_F)$ consisting of the following ten axioms for a field:

Table B.1: Axioms of F

$0 \neq 1.$	
$\forall x \forall y \forall z (x + (y + z) = (x + y) + z)$	$\forall x \forall y \forall z (x \times (y \times z) = (x \times y) \times z).$
$\forall x (x + 0 = x)$	$\forall x (x \times 1 = x).$
$\forall x \exists y (x + y = 0)$	$\forall x (x \neq 0 \rightarrow \exists y (x \times y = 1)).$
$\forall x \forall y (x + y = y + x)$	$\forall x \forall y (x \times y = y \times x)$
$\forall x \forall y \forall z (x \times (y + z) = (x \times y) + (x \times z)).$	

Definition B.1.2. Let $(F, 0, 1, +, \times)$ be an $L(\sigma_F)$ -structure. We’ll call it F . Then F is a field if and only if $F \models F$.

Definition B.1.3 (Ordered field). Let σ_{OF} be the signature $\{0, 1, +, \times, \leq\}$ and let $L(\sigma_{OF})$ be the usual first-order language over σ_{OF} . Let OF (i.e., “ordered field”) be the axiom system in $L(\sigma_{OF})$ consisting of the fifteen axioms for an ordered field:²

¹ A nice textbook on this is Lee (2018). An advanced textbook is Dummit & Foote (2004).

² A slightly different axiom system is given in the standard analysis textbook Apostol (1974), except that Apostol takes $\text{Pos}(x)$ as primitive instead of $x \leq y$. But these are interdefinable: first, via $\text{Pos}(x) \leftrightarrow (x \neq 0 \wedge 0 \leq x)$; second (we define $x - y$ to be $x + (-y)$), via $x \leq y \leftrightarrow \text{Pos}(x - y) \vee x = y$.

Table B.2: Axioms of OF

Field axioms, F	
Antisymmetry	$\forall x \forall y (x \leq y \wedge y \leq x \rightarrow x = y)$
Transitivity	$\forall x \forall y \forall z (x \leq y \leq z \rightarrow x \leq z).$
Connected	$\forall x \forall y (x \leq y \vee y \leq x)$
Translation	$\forall x \forall y \forall z (x \leq y \rightarrow (x + z) \leq (y + z))$
Scaling	$\forall x \forall y \forall z (0 \leq z \wedge x \leq y \rightarrow (x \times z) \leq (y \times z))$

Definition B.1.4. Let $(F, 0, 1, +, \times, \leq)$ be an $L(\sigma_{\text{OF}})$ -structure. We'll call it F . Then F is an ordered field if and only if $F \models \text{OF}$.

Definition B.1.5 (Theory of the complete ordered field, **ALG**). Let **ALG** consist of the ordered field axioms **OF**, along with (monadic) second-order Comprehension, Extensionality and the Dedekind order-completeness axiom:³

Order completeness axiom $(X \neq \emptyset \wedge X \text{ is bounded above}) \rightarrow \exists x (x = \sup X)$ (B.1.1)

The second-order theory **ALG** is the theory of the complete ordered field.

³ So, **ALG** then is (equivalent to) the axiom system in Apostol (1974).

Appendix C

The Completeness of Elementary Algebra

The results in the previous section are, more or less, all due to Kurt Gödel (Gödel (1931)), and explained in standard intermediate logic textbooks such as Enderton (2001) or Boolos et al. (2007). The new results in this section are, more or less, all due to Alfred Tarski (Tarski (1948)), and are usually explained in model theory textbooks, like Marker (2002).

Definition C.0.1. The axioms of

Table C.1: Axioms of RCF

Ordered field axioms (OF)	$0 \neq 1 \dots \dots$
Euclidean field axiom	$\forall x (0 \leq x \rightarrow \exists y (x = y^2))$
For n odd, p a polynomial in x :	$\forall z_0 \dots \forall z_n (z_n \neq 0 \rightarrow \exists x (p(x, z_0, \dots, z_n) = 0))$

Definition C.0.2. Let $(F, 0, 1, +, \times, \leq)$ be a structure for the signature $\{0, 1, +, \times, \leq\}$. We shall call it F , for short. F is called a *real-closed field* if and only if $F \models \text{RCF}$.

Theorem C.0.3. $(\mathbb{R}, 0, 1, +, \times, \leq)$ is a real-closed field.

What I shall call “Tarski’s RCF theorem” is the following remarkable result:

Theorem C.0.4 (Tarski’s RCF theorem). Let $\varphi \in L(0, 1, +, \times, \leq)$. Then:

$$\text{RCF} \vdash \varphi \iff \mathbb{R} \models \varphi. \quad (\text{C.0.1})$$

Proof. This is due to Tarski (1948). □

From Theorem C.0.4, several corollaries follow:

Theorem C.0.5. RCF is complete.

Proof. Let $\text{RCF} \not\models \varphi$. So, by Theorem C.0.4, $(\mathbb{R}, 0, 1, +, \times, \leq) \not\models \varphi$. So, $(\mathbb{R}, 0, 1, +, \times, \leq) \models \neg\varphi$. Therefore, by Theorem C.0.4, $\text{RCF} \vdash \neg\varphi$. $\square$

Theorem C.0.6. RCF is decidable.

Proof. RCF is axiomatically specified by sixteen axioms, and a single axiom scheme, as per Table C.1. Hence, it is *computably axiomatized*—i.e., a computer program exists which checks an input expression of $L(0, 1, +, \times, \leq)$ and decides correctly if it is an axiom of RCF or not. And, by Theorem C.0.5, RCF is complete. Hence, a computer program exists which outputs “Yes” if $\text{RCF} \vdash \varphi$ and outputs “No” if $\text{RCF} \not\models \varphi$. Therefore, RCF is decidable. $\square$

Theorem C.0.7. Let F be a real-closed field. Let φ be a sentence in $L(0, 1, +, \times, \leq)$. Then:

$$F \models \varphi \Leftrightarrow \mathbb{R} \models \varphi. \quad (\text{C.0.2})$$

Proof. Let φ be a sentence in $L(0, 1, +, \times, \leq)$. Then:

$$F \models \varphi \Leftrightarrow F \not\models \neg\varphi \quad (\text{truth definition}) \quad (\text{C.0.3})$$

$$\Leftrightarrow \text{RCF} \not\models \neg\varphi \quad (\text{Definition C.0.2}) \quad (\text{C.0.4})$$

$$\Leftrightarrow \text{RCF} \vdash \varphi \quad (\text{Theorem C.0.5}) \quad (\text{C.0.5})$$

$$\Leftrightarrow \mathbb{R} \models \varphi \quad (\text{Theorem C.0.4}) \quad (\text{C.0.6})$$

$\square$

Appendix D

Affine Spaces

Definition D.0.1 (The affine space $\mathbb{A}^n(F)$). Let $F = (F, 0, 1, +, \times)$ be a field. Let $V = F^n$ (i.e., the vector space whose vectors are elements of F^n and whose scalars are elements of F). Let

$$\mathbb{P} := F^n (\text{the carrier set of } n\text{-tuples}) \quad (\text{D.0.1})$$

Let $\mathbf{v} \in V$. Let $b \in \mathbb{P}$. Let:

$$\ell(b, v) := \{a \in \mathbb{P} \mid (\exists \lambda \in F) (a = b + \lambda \mathbf{v})\} \quad (\text{D.0.2})$$

$$= \{b + \lambda v \mid \lambda \in F\} \quad (\text{D.0.3})$$

Let

$$\mathcal{L} := \{\ell(b, v) \mid b \in \mathbb{P}, v \in V\} \quad (\text{D.0.4})$$

Let

$$\text{AG}(2, F) := (\mathbb{P}, \mathcal{L}) \quad (\text{D.0.5})$$

Appendix E

Coordinate Structures on Real-Closed Fields

Definition E.0.1. Geometrical coordinate relations (and functions). Let F be a real-closed field. Let F^n be the set of n -tuples with coordinates from F .

Geometrical relation (or function)	Definiens
$\mathbf{x} \bullet_{F^n} \mathbf{y}$	$:= \sum_{i=1}^n x_i y_i.$
$\ \mathbf{x}\ _{F^n}$	$:= \sqrt{\mathbf{x} \bullet_{F^n} \mathbf{x}}.$
$\Delta_{F^n}(\mathbf{x}, \mathbf{y})$	$:= \ \mathbf{x} - \mathbf{y}\ _{F^n}.$
$(\mathbf{x}, \mathbf{y}, \mathbf{z}) \in B_{F^n}$	$:= (\exists \alpha \in [0, 1]) ((\mathbf{y} - \mathbf{x}) = \alpha(\mathbf{z} - \mathbf{x})).$
$(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{w}) \in \equiv_{F^n}$	$:= \Delta_{F^n}(\mathbf{x}, \mathbf{y}) = \Delta_{F^n}(\mathbf{z}, \mathbf{w})$
$(\mathbf{x}, \mathbf{y}) \in \sim_{F^n}$	$:= x_n = y_n$
$(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{w}) \in \equiv_{F^n}^{\sim}$	$:= \Delta_{F^n}(\mathbf{x}, \mathbf{y}) = \Delta_{F^n}(\mathbf{z}, \mathbf{w}) \wedge x_n = y_n = z_n = w_n$
$(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{w}) \in \perp_{F^n}^m$	$:= (x_n - y_n)(z_n - w_n) = \sum_{i=1}^{n-1} (x_i - y_i)(z_i - w_i).$

Definition E.0.2. Let F be a real-closed field. For each geometrical signature, we define a class of corresponding coordinate structures:

Table E.1: Coordinate structures

Signature	Coordinate structure
$\mathbf{B}$	$\mathbb{B}^4(F) := (F^4, B_{F^4}).$
$\mathbf{B}, \equiv$	$\mathbb{E}^3(F) := (F^3, B_{F^3}, \equiv_{F^3}).$
$\mathbf{B}, \sim, \equiv^{\sim}$	$\mathbb{G}^4(F) := (F^4, B_{F^4}, \sim_{F^4}, \equiv_{F^4}^{\sim})$
$\mathbf{B}, \perp$	$\mathbb{M}^4(F) := (F^4, B_{F^4}, \perp_{F^4}).$

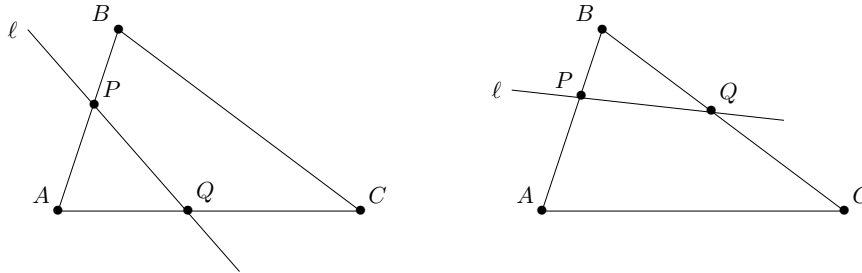
Appendix F

Geometrical Axioms

F.1 Pasch's Axiom

We next turn to the Classical Pasch axiom, as Szmielew calls it. This was first given in Pasch (1882), and is order axiom of Hilbert (1899). It says: given three distinct non-collinear points A, B, C (and thus a triangle), if a line ℓ intersects the segment AB , then either ℓ intersects the segment AC or ℓ intersects the segment BC :

Classical Pasch axiom



There are a number of syntactically distinct formalizations of Pasch's axiom in $L(\mathbf{B})$. We shall concentrate on:

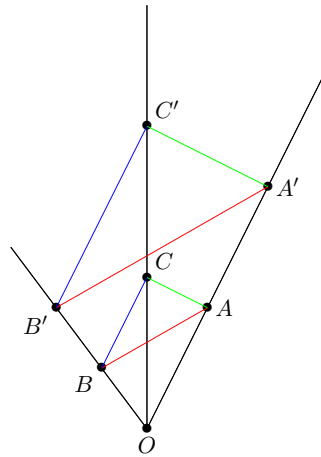
Definition F.1.1. Pasch's Axiom is the following $L(\mathbf{B})$ sentence:

$$Pasch \quad \mathbf{B}(p, q, r) \wedge \mathbf{B}(s, u, q) \rightarrow \exists x (\mathbf{B}(r, x, s) \wedge \mathbf{B}(p, u, x)). \quad (\text{F.1.1})$$

F.2 Playfair's Axiom (a version of Euclid)

F.3 Desargues's Axiom

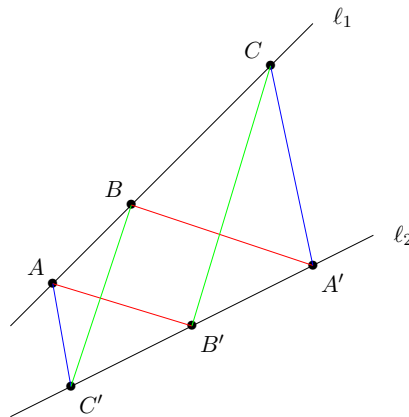
Desargues's property



Lemma F.3.1. Let D be a division ring. Then $\mathbb{A}^2(D)$ is Desarguesian.

F.4 Pappus's Axiom

Pappus property



Appendix G

Geometrical Axiom Systems

G.1 Betweenness (Ordered Affine) Geometry

Definition G.1.1. The axioms of the first-order theory $\text{BG}_0(4)$ of four-dimensional betweenness geometry in $L(\mathbf{B})$ are the following eight and one axiom scheme:¹

B1.	B-Identity	$\mathbf{B}(p, q, p) \rightarrow p = q.$
B2.	B-Transitivity	$\mathbf{B}(p, q, r) \wedge \mathbf{B}(q, r, s) \wedge q \neq r \rightarrow \mathbf{B}(p, q, s).$
B3.	B-Connectivity	$\mathbf{B}(p, q, r) \wedge \mathbf{B}(p, q, r') \wedge p \neq q \rightarrow (\mathbf{B}(p, r, r') \vee \mathbf{B}(p, r', r)).$
B4.	B-Extension	$\exists p (\mathbf{B}(p, q, r) \wedge p \neq q).$
B5.	Pasch	$\mathbf{B}(p, q, r) \wedge \mathbf{B}(s, u, q) \rightarrow \exists x (\mathbf{B}(r, x, s) \wedge \mathbf{B}(p, u, x)).$
B6.	Euclid	$\mathbf{B}(a, d, t) \wedge \mathbf{B}(b, d, c) \wedge a \neq d \rightarrow \exists x \exists y (\mathbf{B}(a, b, x) \wedge \mathbf{B}(a, c, y) \wedge \mathbf{B}(x, t, y))$
B7.	Lower dimension	There exist five points which are not co_3 .
B8.	Upper dimension	Any six points are co_4 .
Continuity axiom scheme		$[\exists a \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow \mathbf{B}(a, p, q))] \rightarrow [\exists b \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow \mathbf{B}(p, b, q))]$

Definition G.1.2. The axioms of the second-order theory $\text{BG}(4)$ in $L_2(\mathbf{B})$ are as follows:

B1-8.	Same as for $\text{BG}_0(4)$	
Cont	Continuity axiom	$[\exists a (\forall p \in X) (\forall q \in Y) \mathbf{B}(a, p, q)] \rightarrow [\exists b (\forall p \in X) (\forall q \in Y) \mathbf{B}(p, b, q)]$
Comp	Comprehension scheme	$\exists X \forall p (p \in X \leftrightarrow \varphi(p))$
Ext	Extensionality	$\forall p (p \in X \leftrightarrow p \in Y) \rightarrow X = Y.$

¹ The $\text{BG}_0(4)$ axioms are given originally in Szczerba & Tarski (1965) and Szczerba & Tarski (1979). See Szczerba & Tarski (1979), pp. 159–160, for the first-order two-dimensional theory GA_2 (for “neutral” or “absolute geometry”), which lacks the Euclid Parallel axiom (which is called (E) in Szczerba & Tarski (1979) and is called (Euclid) above). Their system includes Desargues’s Theorem. But for us, this axiom is no longer required, as it provable from the remaining axioms in dimensions above two (Szczerba & Tarski (1979): 190). For $\text{BG}_0(4)$, see also Goldblatt (1987): 165. Goldblatt calls $\text{BG}_0(4)$ “the first-order theory of ordered affine fourfolds over real-closed fields”.

G.2 Euclidean Geometry

Definition G.2.1. The axioms of the first-order theory $\text{EG}_0(3)$ of three-dimensional Euclidean geometry in $L(\mathbf{B}, \equiv)$ are the following:²

E1.	B-Identity	$\mathbf{B}(p, q, p) \rightarrow p = q.$
E2.	$\equiv$ -Identity	$pq \equiv rr \rightarrow p = q.$
E3.	$\equiv$ -Transitivity	$pq \equiv rs \wedge pq \equiv tu \rightarrow rs \equiv tu.$
E4.	$\equiv$ -Reflexivity	$pq \equiv qp.$
E5.	$\equiv$ -Extension	$\exists r (\mathbf{B}(p, q, r) \wedge qr \equiv su).$
E6.	Pasch	$\mathbf{B}(p, q, r) \wedge \mathbf{B}(s, u, r) \rightarrow \exists x (\mathbf{B}(q, x, s) \wedge \mathbf{B}(u, x, p)).$
E7.	Euclid	$\mathbf{B}(a, d, t) \wedge \mathbf{B}(b, d, c) \wedge a \neq d \rightarrow \exists x \exists y (\mathbf{B}(a, b, x) \wedge \mathbf{B}(a, c, y) \wedge \mathbf{B}(x, t, y))$
E8.	5-Segment	$(p \neq q \wedge \mathbf{B}(p, q, r) \wedge \mathbf{B}(p', q', r') \wedge pq \equiv p'q' \wedge qr \equiv q'r' \wedge ps \equiv p's' \wedge qs \equiv q's') \rightarrow rs$
E9.	Lower dimension	There exist four points which are not co_2 .
E10.	Upper dimension	Any five points are co_3 .
Continuity axiom scheme		$[\exists a \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow \mathbf{B}(a, p, q))] \rightarrow [\exists b \forall p \forall q (\varphi(p) \wedge \theta(q) \rightarrow \mathbf{B}(p, b, q))]$

Definition G.2.2. The axioms of the second-order theory $\text{EG}(3)$ of three-dimensional Euclidean geometry in $L_2(\mathbf{B}, \equiv)$ are as follows:

E1-10.	Same as for $\text{EG}_0(3)$	
Cont	Continuity axiom	$[\exists a (\forall p \in X) (\forall q \in Y) \mathbf{B}(a, p, q)] \rightarrow [\exists b (\forall p \in X) (\forall q \in Y) \mathbf{B}(p, b, q)]$
Comp	Comprehension scheme	$\exists X \forall p (p \in X \leftrightarrow \varphi(p))$
Ext	Extensionality	$\forall p (p \in X \leftrightarrow p \in Y) \rightarrow X = Y.$

G.3 Synthetic Galilean Spacetime Geometry

Definition G.3.1. The axioms of the first-order theory $\text{Gal}_0(1, 3)$ of Galilean spacetime in $L(\mathbf{B}, \sim, \equiv)$ are the following:

$\text{BG}_0(4)$	$\text{BG}_0(4)$ axioms	
$[E1 - E10] \sim$	Relativized Euclidean axiom	
G1	$\sim$	$\sim$ is an equivalence relation
G2	Restriction	$\equiv \sim$ is restricted to simultaneous points
G3	Translation invariance	$pq \equiv \sim rs \rightarrow (p + \mathbf{v})(q + \mathbf{v}) \equiv \sim (r + \mathbf{v})(s + \mathbf{v})$

Definition G.3.2. The axioms of the second-order theory $\text{Gal}(4)$ of Galilean spacetime in $L_2(\mathbf{B}, \sim, \equiv)$ are the following:

² The original source of this axiomatization is Tarski (1959) and Tarski & Givant (1999). See Tarski (1959), pp. 19–20, for a formulation of the first-order two-dimensional theory, with twelve axioms and one axiom scheme (for continuity); and Tarski & Givant (1999) for a simplification down to ten axioms and one axiom scheme (for continuity). A more complicated version (for two dimensions) appears in Tarski (1948).

BG(4)	As for BG(4)	
[E1–E10]~	Relativized Euclidean axiom	
Comp	Comprehension scheme	$\exists X \forall p (p \in X \leftrightarrow \varphi(p))$
Ext	Extensionality	$\forall p (p \in X \leftrightarrow p \in Y) \rightarrow X = Y.$

G.4 Synthetic Minkowski Spacetime Geometry

Definition G.4.1. The axioms of the first-order theory $\text{Mink}_0(1, 3)$ of Minkowski spacetime in $L(\mathbf{B}, \perp)$ are the following:

BG ₀ (4)	BG ₀ (4) axioms	
OS1.		
OS2.		
OS3.		
OS4.		
OS5.		
M1		
M2		
M3		

Definition G.4.2. The axioms of the second-order theory $\text{Mink}(4)$ of Minkowski spacetime in $L_2(\mathbf{B}, \lambda)$ are the following:

BG(4)	BG(4) axioms	
OS1–OS5.	As above	
M1–M3	As above	

General Index

Euclid's Axiom (Tarskian), 117

Fano Plane, 64

Field's Programme, 4

Hesse Configuration, 60

Inner Pasch Axiom, 109

Outer Pasch Axiom, 110

Index of Names

- Aristotle, 180
- Bennett, Mary, 50, 54, 63, 81, 82, 89
Blumenthal, Leonard, 52, 78, 79, 84
Borsuk, Karol, 10, 124, 193
Burgess, John, 100, 104, 123, 129, 131,
143, 147, 150, 199, 207, 211
- Church, Alonzo, 4
Cieśliński, Cezary, 18
Classical Pasch Axiom, 109
Coxeter, Harold Scott MacDonald, 79,
89, 91
- Dedekind, Richard, 14
Dembowski, Peter, 89
Desargues, Gerard, 77, 115
Descartes, René, 25, 27
- Enderton, Herbert, 7
Euclid, 117, 165
- Fano, Gino, 64
Feferman, Solomon, 4
Feynman, Richard, 7
Field, Hartry, 3, 4, 18, 99, 199
Frege, Gottlob, 4
- Gödel, Kurt, 3, 4, 203
Givant, Steven, 109
Goldblatt, Robert, 30, 50, 78, 81, 123,
189, 193
- Halbach, Volker, 18
Heath, Sir Thomas, 42
- Hesse, Ludwig, 60
Hessenberg, Karl, 79
Hilbert, David, 4, 44, 89, 91, 102, 109,
113, 125
Huntington, Edward V., 91, 106
- Kordos, M., 103
- Longair, Malcolm, 39
- Malament, David, 199
Minkowski, Hermann, 189
Moorhouse, Eric, 66, 89
Mundy, Brent, 189
- Newton, Sir Isaac, 178
- Pambuccian, Victor, 89
Pappus of Alexandria, 77, 119
Pasch, Moritz, 89, 91, 98, 109
Peano, Guiseppe, 90, 91
Poincaré, Henri, 4
Putnam, Hilary, 4
- Quine, W.V., 4
- Rindler, Wolfgang, 39
Robb, Alfred, 189
Rosen, Gideon, 131
- Schwabhauser, Wolfram, 91
Shapiro, Stewart, 159, 199, 211
Suppes, Patrick, 124
Szcerba, L., 123, 193
Szcerba, L.W., 103

- | | |
|--|------------------------------|
| Szmielew, Wanda, 10, 56, 67–69, 73, 74,
91, 93, 94, 96, 97, 105, 110,
111, 115, 124, 193 | Tong, David, 26 |
| Tao, Terrence, 5 | Veblen, Oswald, 91, 124, 193 |
| Tarski, Alfred, 91, 103, 107, 109, 117,
123, 167, 171, 193 | von Neumann, John, 203 |
| Tecklenberg, Helga, 79 | Witten, Edward, 5 |
| | Zermelo, Ernst, 17 |

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